

# iDRAC10 Version 1.20.xx User's Guide

1.20.xx Series

## Notes, cautions, and warnings

 **NOTE:** A NOTE indicates important information that helps you make better use of your product.

 **CAUTION:** A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

 **WARNING:** A WARNING indicates a potential for property damage, personal injury, or death.

# Revision history

Table 1. Revision history

| Date        | Document revision | Description of changes     |
|-------------|-------------------|----------------------------|
| June, 2025  | A01               | iDRAC10 version 1.20.50.50 |
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# Overview of iDRAC

The Integrated Dell Remote Access Controller (iDRAC) is designed to make you more productive as a system administrator and improve the overall availability of Dell servers. iDRAC alerts you to system issues, helps you to perform remote management, and reduces the need for physical access to the system.

iDRAC technology is part of a larger data center solution that increases the availability of business-critical applications and workloads. The technology allows you to deploy, monitor, manage, configure, update, and troubleshoot Dell systems from any location without using any agents or an operating system.

 **NOTE:** iDRAC behavior may not be consistent when used with Non-Dell hardware.

Several products work with the iDRAC to simplify and streamline IT operations. The following are some of the tools:

- OpenManage Enterprise
- OpenManage Power Center Plugin
- OpenManage Integration for VMware vCenter
- Dell Repository Manager

iDRAC is available in the following variants:

- iDRAC Core—Available by default on all servers
- iDRAC Enterprise—Supported on all server models
- iDRAC Datacenter—Supported on all server models

## Topics:

- [Benefits of using iDRAC](#)
- [Key features](#)
- [New features added](#)
- [Deprecated features](#)
- [Features supported or not supported in iDRAC10 releases](#)
- [How to use this guide](#)
- [Supported web browsers](#)
- [iDRAC licenses](#)
- [Licensed features in iDRAC10](#)
- [Interfaces and protocols to access iDRAC](#)
- [iDRAC port information](#)
- [Other documents you may need](#)
- [Contacting Dell](#)
- [Accessing documents from the Dell Support site](#)
- [Accessing Redfish API](#)

## Benefits of using iDRAC

The benefits include:

- **Increased Availability**—Early notification of potential or actual failures that help prevent a server failure or reduce recovery time after failure.
- **Improved Productivity and Lower Total Cost of Ownership (TCO)**—Extending the reach of administrators to larger numbers of distant servers can make IT staff more productive while driving down operational costs such as travel.
- **Secure Environment**—By providing secure access to remote servers, administrators can perform critical management functions while maintaining server and network security.

# Key features

The key features of iDRAC include:

**i NOTE:** Some features are available only with iDRAC Enterprise or Datacenter license. For information about the features available for each license, see [Licensed features in iDRAC10](#). For the list of features that are not supported in the iDRAC10 version 1.20.50.50, see the [Features supported or not supported in iDRAC10 releases](#) section.

## Inventory and Monitoring

- Telemetry data streaming.
- View managed server health.
- Inventory and monitor network interface controllers and storage subsystem (PERC and direct attached storage) without any operating system agents.
- View and export system inventory.
- View sensor information such as temperature, voltage, and intrusion.
- Monitor CPU state, processor automatic throttling, and predictive failures.
- View memory information.
- Monitor and control power usage.
- Support for SNMPv3 gets and alerts.
- View network interfaces available on host operating systems.
- iDRAC10 provides improved monitoring and management functionality with Quick Sync 2. You need the OpenManage Mobile app that is configured in your Android or iOS mobile device.

## Deployment

- Manage iDRAC network settings.
- Configure and use virtual console and virtual media.
- Deploy operating systems using remote file share, and virtual media.
- Enable autodiscovery.
- Perform server configuration changes using Server Configuration Profile (SCP) feature. For more information, see the [SCP Reference Guide](#).
- Configure a persistence policy for virtual addresses, initiator, and storage targets.
- Remotely configure storage devices that are attached to the system at run-time.
- Perform the following operations for storage devices:
  - Physical disks: Assign or unassign a physical disk as a global hot spare.
  - Virtual disks:
    - Create virtual disks.
    - Edit virtual disks cache policies.
    - Check virtual disk consistency.
    - Initialize virtual disks.
    - Encrypt virtual disks.
    - Assign or unassign dedicated hot spare.
    - Delete virtual disks.
  - Controllers:
    - Configure controller properties.
    - Import or autoimport foreign configuration.
    - Clear foreign configuration.
    - Reset the controller configuration.
    - Create or change security keys.
  - PCIe SSD devices:
    - Inventory and remotely monitor the health of PCIe SSD devices in the server.
    - Prepare the PCIe SSD to be removed.
    - Securely erase the data.
  - Set the backplane mode (unified or split mode).
  - Blink or unblink component LEDs.
  - Apply the device settings immediately, at the next system reboot, at a scheduled time, or as a pending operation to be applied as a batch as part of the single job.

## Update



- Manage iDRAC licenses.
- Update or rollback iDRAC firmware and Lifecycle Controller firmware using a single firmware image.
- Manage staged updates.
- Access iDRAC interface over direct USB connection.
- Configure iDRAC using Server Configuration Profiles on a USB device.

### **Maintenance and Troubleshooting**

- Perform power-related operations and monitor power consumption.
- Optimize system performance and power consumption by modifying the thermal settings.
- Log event data: Lifecycle and RAC logs.
- Set email alerts, IPMI alerts, remote system logs, WS Eventing logs, Redfish event, and SNMP traps (v1, v2c, and v3) for events and improved email alert notification.
- Capture last system crash image.
- View boot and crash capture videos.
- Out-of-band monitor and alert the performance index of CPU, memory, and I/O modules.
- Configure a warning threshold for inlet temperature and power consumption.
- Use iDRAC Service Module to:
  - View operating system information.
  - Replicate Lifecycle Controller logs to operating system logs.
  - Automate system recovery options.
  - Enable or disable the status of Full Power Cycle for all System components except the PSU.
  - Remotely hard-reset iDRAC
  - Enable in-band iDRAC SNMP alerts.
  - Access iDRAC using host operating system (experimental feature)
  - Populate Windows Management Instrumentation (WMI) information.
  - Integrate with the SupportAssist collection. This is applicable only if iDRAC Service Module Version 2.0 or later is installed.

### **Dell Best Practices regarding iDRAC**

- Dell iDRAC's are intended to be on a separate management network; they are not designed nor intended to be placed on or connected directly to the Internet. Doing so could expose the connected system to security and other risks for which Dell is not responsible.
- Dell Technologies recommends using the Dedicated Gigabit Ethernet port available on rack and tower servers. This interface is not shared with the host operating system and routes the management traffic to a separate physical network, enabling it to be separated from the application traffic. This option implies that iDRAC's dedicated network port routes its traffic separately from the server's LOM or NIC ports. The Dedicated option allows iDRAC to be assigned an IP address from the same subnet or different subnet in comparison to the IP addresses assigned to the Host LOM or NICs.
- Along with locating iDRACs on a separate management subnet, users should isolate the management subnet/vLAN with technologies such as firewalls, and limit access to the subnet/vLAN to authorized server administrators.

### **Secure Connectivity**

Securing access to critical network resources is a priority. iDRAC implements a range of security features that includes:

- Custom signing certificate for Secure Socket Layer (SSL) certificate.
- Signed firmware updates.
- User authentication through Microsoft Active Directory, generic Lightweight Directory Access Protocol (LDAP) Directory Service, or locally administered user IDs and passwords.
- Two-factor authentication using the Smart-Card login feature. The two-factor authentication is based on the physical smart card and the smart card PIN.
- Public Key Authentication.
- Role-based authorization, to configure specific privileges for each user.
- SNMPv3 authentication for user accounts stored locally in the iDRAC. It is recommended to use this, but it is disabled by default.
- User ID and password configuration.
- Default login password modification.
- Set user passwords and BIOS passwords using one-way hash format for improved security.
- FIPS 140-2 Level 1 capability.
- Session time-out configuration (in seconds).
- Configurable IP ports (for HTTP, HTTPS, SSH, Virtual Console, and Virtual Media).
- Secure Shell (SSH) that uses an encrypted transport layer for higher security.

- Login failure limits per IP address, with login blocking from that IP address when the limit is exceeded.
- Limited IP address range for clients connecting to iDRAC.
- Dedicated Gigabit Ethernet adapter available on rack and tower servers (additional hardware may be required).

## New features added

This section provides the list of new features added for each release of iDRAC10.

### Firmware version 1.20.50.50

The new features that are supported in iDRAC10 1.20.50.50 are:

- Installation of operating system using Server Configuration Profile (SCP) and Lifecycle Controller UI.
- Firmware updates for network cards using the Redfish Multipart method, supporting Mellanox ConnectX-6 and ConnectX-7 cards.
- Display of temperature sensor details for all Network Interface Card (NIC), InfiniBand (IB), Fibre Channel (FC), and Data Processing Unit (DPU) cards.
- Support for zero touch configuration.
- Support for TPM device ID in SCV hardware certificate.
- Energy pack updates for controllers of both PERC12 and PERC13.
- Support for PERC13 (H975i Front only).
- Support for sliced virtual disks on both PERC12 and PERC13.
- Energy pack updates for controllers of both PERC12 and PERC13.
- Support for PCIe switch.
- Support for 802.1x security.
- Support Reboot-less updates for PERC 12 and PERC13 controllers.
- Support for Redfish DIMM inventory.
- Support for CXL IPMI sensor for BIOS link and protocol error reporting.
- Improved CMOS battery failure handling and messaging.
- Support for Dell Connectivity Client (DCC) plugin.

#### NOTE:

- For more details on the licenses that support the features, see the [Licensed features in iDRAC10](#) section.
- For more information about the features that are supported in this release, see [Features supported or not supported in iDRAC10 releases](#).

### Firmware version 1.20.25.00

In the iDRAC10 1.20.25.00 release, several features that were not supported in the previous release are now supported.

For more details, see [Features supported or not supported in iDRAC10 releases](#).

The new features that are supported are:

- Added basic support for Inventory, monitoring, and closed loop for Compute Express Link (CXL) devices.
- Support for Intel 6th generation Xeon scalable server processors.
- Introduced **Cold Redundancy** and **Cold Redundancy PSU Rotation** policies for efficient power management.
- Added the alert group **Liquid Cooling System Default Action UnGraceful Power Off** for diligent leak detection and management.

 **NOTE:** For more details on the licenses that support the features, see the [Licensed features in iDRAC10](#) section.

### Firmware version 1.10.17.00

In iDRAC10 1.10.17.00 release, the following features are added in iDRAC:

- Enhanced Security with Dedicated Security Processor:
  - Improved security performance

- Integrated Root-of-Trust (RoT), device-level attestation, and encryption
  - Stronger encryption algorithms
- Refreshed User Interface:
  - Consistent user experience across all Dell Technologies consoles
  - Simplified interface
  - Easier navigation
- Simplified license structure in PowerEdge 17th Generation
- Create custom iDRAC user roles
- AC Power Recovery controlled by iDRAC (previous generations BIOS managed this setting)

## Deprecated features

The following features are deprecated in iDRAC10:

- Group Manager
- WS-Man
- TLS 1.1
- vFlash
- Extended schema
- Software RAID
- Memory NVDIMM
- Local User- Reserved account
- Two-factor authentication (2FA) - Simple (email)

## Features supported or not supported in iDRAC10 releases

The following table lists the features that are supported or not supported for the iDRAC10 releases:

**Table 2. Features supported or not supported**

| Function                    | Feature                                            | Version 1.10.17.00 | Version 1.20.25.00 | Version 1.20.50.50                                                                                                                                                                                                                          |
|-----------------------------|----------------------------------------------------|--------------------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| iDRAC UI                    | iDRAC UI support for other international languages | No                 | Yes                | Yes                                                                                                                                                                                                                                         |
|                             | Local RACADM (operating system in-band)            | No                 | Yes                | Yes                                                                                                                                                                                                                                         |
|                             | Remote RACADM                                      | No                 | Yes                | Yes                                                                                                                                                                                                                                         |
|                             | iDRAC Tools                                        | No                 | Yes                | Yes                                                                                                                                                                                                                                         |
|                             | Serial- RAC Serial                                 | No                 | Yes                | Yes                                                                                                                                                                                                                                         |
|                             | Serial- IPMI Serial                                | No                 | Yes                | Yes                                                                                                                                                                                                                                         |
| Lifecycle Controller UI     | Operating System installation                      | No                 | No                 | Yes<br> <b>NOTE:</b> Use Lifecycle Controller UI (LC UI) only for Operating System installation. Other features of LC UI are not supported in iDRAC10. |
| Networking and Connectivity | IP Blocking                                        | No                 | Yes                | Yes                                                                                                                                                                                                                                         |
|                             | IP Range                                           | No                 | Yes                | Yes                                                                                                                                                                                                                                         |

**Table 2. Features supported or not supported (continued)**

| Function                   | Feature                                                                                                                                                                                                                                                                                | Version 1.10.17.00 | Version 1.20.25.00 | Version 1.20.50.50 |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|--------------------|--------------------|
|                            | iDRAC Direct (USB) SCP configuration                                                                                                                                                                                                                                                   | No                 | Yes                | Yes                |
|                            | Quick Sync                                                                                                                                                                                                                                                                             | No                 | Yes                | Yes                |
|                            | Network connection view                                                                                                                                                                                                                                                                | No                 | Yes                | Yes                |
|                            |  <b>NOTE:</b> PLDM sensors, such as Power and Temperature, are not supported for PLDM-enabled NIC devices in the current release of iDRAC for the Dell PowerEdge 17 <sup>th</sup> generation servers. |                    |                    |                    |
| Remote Management          | Serial-over-LAN with SSH                                                                                                                                                                                                                                                               | Yes                | Yes                | Yes                |
|                            | Virtual folders                                                                                                                                                                                                                                                                        | No                 | Yes                | Yes                |
|                            | Remote File Share                                                                                                                                                                                                                                                                      | Yes                | Yes                | Yes                |
|                            | Quality and Bandwidth control                                                                                                                                                                                                                                                          | No                 | Yes                | Yes                |
|                            | Virtual Console collaboration (up to six simultaneous users)                                                                                                                                                                                                                           | No                 | Yes                | Yes                |
|                            | Virtual Console chat                                                                                                                                                                                                                                                                   | No                 | Yes                | Yes                |
|                            | Virtual clipboard                                                                                                                                                                                                                                                                      | No                 | Yes                | Yes                |
|                            | Asset Tracking                                                                                                                                                                                                                                                                         | No                 | Yes                | Yes                |
|                            | Market Usage Tracking (MUT)                                                                                                                                                                                                                                                            | No                 | Yes                | Yes                |
|                            | Software Development Kit (SDK)                                                                                                                                                                                                                                                         | No                 | Yes                | Yes                |
|                            | Virtual Media – Single Session                                                                                                                                                                                                                                                         | Yes                | Yes                | Yes                |
|                            |                                                                                                                                                                                                                                                                                        |                    |                    |                    |
| iDRAC Service Module (iSM) | iDRAC Hard Reset and other features                                                                                                                                                                                                                                                    | No                 | Yes                | Yes                |
|                            | Network Information                                                                                                                                                                                                                                                                    | No                 | Yes                | Yes                |
|                            | IBIA                                                                                                                                                                                                                                                                                   | No                 | Yes                | Yes                |
|                            | Virtual Power cycle                                                                                                                                                                                                                                                                    | No                 | Yes                | Yes                |
|                            | Prepare to remove                                                                                                                                                                                                                                                                      | No                 | Yes                | Yes                |
|                            | iDRAC SSO                                                                                                                                                                                                                                                                              | No                 | No                 | No                 |
|                            | Operating System details                                                                                                                                                                                                                                                               | No                 | Yes                | Yes                |
|                            | Chipset SATA Monitoring                                                                                                                                                                                                                                                                | No                 | Yes                | Yes                |
|                            | Lifecycle Controller Logs Replication                                                                                                                                                                                                                                                  | No                 | Yes                | Yes                |
|                            | In Band SNMP                                                                                                                                                                                                                                                                           | No                 | Yes                | Yes                |
|                            | Dell SupportAssist Integration                                                                                                                                                                                                                                                         | No                 | Yes                | Yes                |
|                            | SDS Event Corelation                                                                                                                                                                                                                                                                   | No                 | Yes                | Yes                |
|                            | iSM Installation                                                                                                                                                                                                                                                                       | No                 | Yes                | Yes                |
|                            |                                                                                                                                                                                                                                                                                        |                    |                    |                    |
| Alerts                     | Redfish LifeCycle Events (RLCE)                                                                                                                                                                                                                                                        | No                 | Yes                | Yes                |
|                            | Remote Syslog-Non-Secure                                                                                                                                                                                                                                                               | No                 | Yes                | Yes                |

**Table 2. Features supported or not supported (continued)**

| Function            | Feature                                                                                                                                                                                              | Version 1.10.17.00 | Version 1.20.25.00 | Version 1.20.50.50 |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|--------------------|--------------------|
|                     | Remote Syslog-Secure                                                                                                                                                                                 | No                 | Yes                | Yes                |
|                     | Telemetry- Streaming                                                                                                                                                                                 | No                 | Yes                | Yes                |
|                     | Telemetry- Metric Reports                                                                                                                                                                            | No                 | Yes                | Yes                |
|                     |  <b>NOTE:</b> Telemetry reports for CPUMemMetrics and SystemUsage are not supported in the current iDRAC10 release. |                    |                    |                    |
|                     | Telemetry Triggers                                                                                                                                                                                   | No                 | Yes                | Yes                |
|                     | iSM operating system<br>Metric Injection features for<br>Telemetry                                                                                                                                   | No                 | Yes                | Yes                |
| Platform Management | Memory SDPM                                                                                                                                                                                          | No                 | No                 | No                 |
|                     | Out-of-band performance monitoring                                                                                                                                                                   | No                 | No                 | No                 |
|                     | Idle server detection                                                                                                                                                                                | No                 | No                 | No                 |
|                     | Liquid Cooling monitoring                                                                                                                                                                            | No                 | Yes                | Yes                |
| Power and Thermal   | Power capping                                                                                                                                                                                        | No                 | Yes                | Yes                |
|                     | Power monitoring - Real-time power meter                                                                                                                                                             | No                 | No                 | No                 |
|                     | Real-time power graphing                                                                                                                                                                             | No                 | No                 | No                 |
|                     | Historical power counters                                                                                                                                                                            | No                 | No                 | No                 |
|                     | OpenManage Enterprise Power manager plug-in integration                                                                                                                                              | No                 | Yes                | Yes                |
|                     | Input Current Limit                                                                                                                                                                                  | No                 | No                 | No                 |
|                     | ASHERE                                                                                                                                                                                               | No                 | Yes                | Yes                |
|                     | Customized exhaust temperature                                                                                                                                                                       | No                 | Yes                | Yes                |
|                     | Temperature graphing                                                                                                                                                                                 | No                 | Yes                | Yes                |
|                     | Soundcap 1.0                                                                                                                                                                                         | No                 | Yes                | Yes                |
|                     | System Airflow Consumption, Custom PCIe inlet temperature                                                                                                                                            | No                 | Yes                | Yes                |
|                     | PCIe airflow customization                                                                                                                                                                           | No                 | Yes                | Yes                |
|                     |                                                                                                                                                                                                      |                    |                    |                    |
| Storage Management  | External Enclosure Management                                                                                                                                                                        | No                 | No                 | No                 |
|                     | Secure Enterprise Key Manager (SEKM) and iDRAC Local Key Management (iLKM)                                                                                                                           | No                 | Yes                | Yes                |
|                     | HBA - Realtime Inventory and Monitoring                                                                                                                                                              | No                 | No                 | No                 |
|                     | HBA - Staged Inventory                                                                                                                                                                               | No                 | No                 | No                 |
|                     | PERC Rebootless updates                                                                                                                                                                              | No                 | No                 | Yes                |

**Table 2. Features supported or not supported (continued)**

| Function                                   | Feature                                                    | Version 1.10.17.00 | Version 1.20.25.00 | Version 1.20.50.50 |
|--------------------------------------------|------------------------------------------------------------|--------------------|--------------------|--------------------|
| Communication Management                   | DPU- Realtime Inventory, Monitoring, and Configuration     | No                 | Yes                | Yes                |
|                                            | DPU - Staged Inventory and Configuration                   | No                 | Yes                | Yes                |
|                                            | Virtual Address Management                                 | No                 | Yes                | Yes                |
|                                            | DPU/ SmartNIC features                                     | No                 | Yes                | Yes                |
|                                            | FC - Realtime Inventory and Monitoring                     | No                 | Yes                | Yes                |
|                                            | FC - Staged Inventory and Configuration                    | No                 | Yes                | Yes                |
| Accelerator Management                     | GPU - Staged Inventory                                     | No                 | Yes                | Yes                |
|                                            | GPU - Realtime Inventory, Power, and Temperature           | No                 | Yes                | Yes                |
| Server Configuration and Deployment        | Driver pack                                                | No                 | Yes                | Yes                |
|                                            | Server Configuration Profile - Repository update           | No                 | Yes                | Yes                |
|                                            | Server Configuration Profile - operating system deployment | No                 | No                 | No                 |
|                                            | Local configuration                                        | No                 | Yes                | Yes                |
|                                            | Zero touch configuration                                   | No                 | Yes                | Yes                |
| System Update                              | Catalog update                                             | No                 | Yes                | Yes                |
|                                            | Embedded update tools                                      | No                 | No                 | No                 |
|                                            | Auto Update                                                | No                 | Yes                | Yes                |
| Factory, Diagnostics, Service, and Logging | iDRAC reset to default                                     | Yes                | Yes                | Yes                |
|                                            | Secure Erase                                               | Yes                | Yes                | Yes                |
|                                            | Crash Screen video - with agent                            | No                 | No                 | No                 |
|                                            | Crash Screen Capture                                       | No                 | Yes                | Yes                |
|                                            | Serial Data Log                                            | No                 | Yes                | Yes                |
|                                            | Crash video capture - agent free                           | No                 | Yes                | Yes                |
|                                            | Work Notes                                                 | No                 | Yes                | Yes                |
|                                            | Remote syslog for alerts                                   | No                 | Yes                | Yes                |
| Authentication and Authorization           | Directory services                                         | No                 | Yes                | Yes                |
|                                            | Active Directory integration                               | No                 | Yes                | Yes                |
|                                            | Single Sign-On (SSO)                                       | No                 | No                 | No                 |
|                                            | Two-factor authentication (2FA)- Smart Card                | No                 | No                 | No                 |
|                                            | Secure Shell Public Key Authentication (SSH PKA)           | No                 | Yes                | Yes                |
|                                            | RSA MultiFactor Authentication (RSA MFA)                   | No                 | Yes                | Yes                |

**Table 2. Features supported or not supported (continued)**

| Function           | Feature                                         | Version 1.10.17.00 | Version 1.20.25.00 | Version 1.20.50.50 |
|--------------------|-------------------------------------------------|--------------------|--------------------|--------------------|
| Security           | Oauth                                           | No                 | Yes                | Yes                |
|                    | OpenID Connect                                  | No                 | No                 | No                 |
|                    | SSL certificate-Custom Signing Certificate      | No                 | Yes                | Yes                |
|                    | SSL certificate- Auto Certificate Enrollment    | No                 | Yes                | Yes                |
|                    | FIPS 140-2/140-3                                | No                 | Yes                | Yes                |
|                    | Custom Security Policy Banner                   | No                 | Yes                | Yes                |
|                    | IP blocking/ Denial-of-Service (DOS) protection | No                 | Yes                | Yes                |
|                    | QuickSync 2.0 additional security               | No                 | Yes                | Yes                |
|                    | System Lockdown mode                            | No                 | Yes                | Yes                |
|                    | IP range filtering                              | No                 | Yes                | Yes                |
|                    | Custom cipher string                            | No                 | Yes                | Yes                |
|                    | Local configuration                             | No                 | Yes                | Yes                |
|                    | iDRAC network- 802.1x security                  | No                 | No                 | No                 |
|                    | BIOS Livescan                                   | No                 | Yes                | Yes                |
| Thermal Management | PCIe airflow customization (LFM)                | No                 | Yes                | Yes                |
|                    | Custom exhaust control                          | No                 | Yes                | Yes                |
|                    | Custom delta-T control                          | No                 | Yes                | Yes                |
|                    | Custom PCIe inlet temperature                   | No                 | Yes                | Yes                |
|                    | System airflow consumption                      | No                 | Yes                | Yes                |
| Other              | Adaptive ICA                                    | No                 | Yes                | Yes                |
|                    | Basic support for CXL devices                   | No                 | Yes                | Yes                |

## How to use this guide

### About this task

The contents of this users guide enable you to perform various tasks using:

### Steps

1. iDRAC web interface—Only the task-related information is provided here. For information about the fields and options, see the **iDRAC Online Help** that you can access from the web interface.
2. RACADM—The RACADM command or the object that you must use is provided here. For more information, see the **iDRAC10 RACADM CLI Guide** available on the Dell Support site.
3. iDRAC Settings Utility—Only the task-related information is provided here. For information about the fields and options, see the **iDRAC Settings Utility Online Help** that you can access when you click **Help** in the iDRAC Settings UI (press <F2> during boot, and then click **iDRAC Settings** on the **System Setup Main Menu** page).

4. Redfish—Only the task-related information is provided here. For information about the fields and options, see the **iDRAC10 Redfish API guide** on the [developer portal](#).

## Supported web browsers

For the list of supported versions, see the **iDRAC10 Release Notes** on the Dell Support site.

## Supported operating systems and hypervisors

For the list of supported versions of operating systems and hypervisors, see the **iDRAC10 Release Notes** on the Dell Support site.

## iDRAC licenses

iDRAC features are available based on the type of the license. iDRAC Core license is installed by default. iDRAC Enterprise license is available as an upgrade and can be purchased anytime. Only licensed features are available in the interfaces that enable you to configure or use iDRAC. For more information, see [Licensed features in iDRAC10](#).

### Types of licenses

The standard license iDRAC is available by default on your system. iDRAC Enterprise and Datacenter licenses include all the licensed features and can be purchased at any time. The types of upsell that is offered are:

- 30-day evaluation—Evaluation licenses are duration-based and the timer runs when power is applied to the system. This license cannot be extended.
- Perpetual—The license is bound to the Service Tag and is permanent.

The following table lists the default licenses available on the systems:

**Table 3. Default Licenses**

| iDRAC Core License                                                                   | iDRAC Enterprise License                                                                                                                                                  | iDRAC Datacenter License                                                                                                                                                                                                                   |
|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Basic instrumentation with iDRAC interfaces and serviceability                       | iDRAC10 Core with the following features: <ul style="list-style-type: none"><li>• Enterprise-class management capabilities</li><li>• Virtual media capabilities</li></ul> | iDRAC10 Enterprise with the following features: <ul style="list-style-type: none"><li>• Dell advanced iDRAC Telemetry</li><li>• High-end server accelerator GPU and DPU options</li><li>• Granular power and thermal management.</li></ul> |
| Sold separately: Secure Enterprise Key Management with Secure Component Verification | Included in the bundled licenses: Secure Enterprise Key Management with Secure Component Verification                                                                     | Included in the bundled licenses: Secure Enterprise Key Management with Secure Component Verification                                                                                                                                      |
| Recommended for entry-level and small-medium business                                | Recommended for Enterprise users who require full remote capabilities.                                                                                                    | Recommended for Datacenter users with hardware accelerated performance analytics, and granular power and thermal management.                                                                                                               |

 **NOTE:** For information about the license that supports the iDRAC10 features, see [Licensed features in iDRAC10](#).

## Methods for acquiring licenses

Use any of the following methods to acquire the licenses:

- Dell Digital Locker—Dell Digital Locker allows you to view and manage your products, software, and licensing information in one location. A link to the Dell Digital Locker is available in iDRAC web interface, go to **Configuration > Licenses**.



 **NOTE:** To know more about Dell Digital Locker, refer to [FAQ](#) on the website.

- Email—License is attached to an email that is sent after requesting it from the technical support center.
- Point-of-sale—License is acquired while placing the order for a system.

 **NOTE:** To manage licenses or purchase new licenses, go to the [Dell Digital Locker](#).

## Acquiring license key from Dell Digital Locker

### Prerequisites

To obtain the license key from your account, you must first register your product using the registration code that is sent in the order confirmation email. This code must be entered in the **Product Registration** tab after logging into Dell Digital Locker.

From the left pane, click the **Products** or **Order History** tab to view the list of your products. Subscription-based products are listed under **Billing accounts** tab.

### About this task

To download the license key from your Dell Digital Locker account:

### Steps

1. Sign in to your Dell Digital Locker account.
2. From the left pane, click **Products**.
3. Click the product that you want to view.
4. Click the product name.
5. On the **Product management** page, click **Get Key**.
6. Follow the instructions on the screen to obtain the license key.

### Next steps

 **NOTE:** If you do not have a Dell Digital Locker account, create an account using the email address provided during your purchase.

 **NOTE:** To generate multiple license keys for new purchases, follow the instructions under **Tools > License Activation > Unactivated licenses**.

## License operations

Before you perform the license management tasks, ensure that you acquire the licenses. For more information, see the [Methods for acquiring licenses](#).

 **NOTE:** If you have purchased a system with all the licenses preinstalled, then license management is not required.

 **NOTE:** To ensure that iDRAC UI reflects licenses installed using RACADM, Redfish, SCV factory, or other interfaces, log in to iDRAC UI again or refresh the browser.

You can perform the following licensing operations using iDRAC, RACADM, and Redfish for one-to-one license management, and Dell License Manager for one-to-many license management:

- View—View the current license information.
- Import—After acquiring the license, store the license in a local storage and import it into iDRAC using one of the supported interfaces. The license is imported if it passes the validation checks.

 **NOTE:** Although you can export the factory-installed license, you cannot import it. To import the license, download the equivalent license from the Digital Locker or retrieve it from the email you received when you purchased the license.

- Export—Exports the installed license. For more information, see the **iDRAC Online Help**.
- Delete—Deletes the license. For more information, see the **iDRAC Online Help**.
- Learn More—Learn more about an installed license, or the licenses available for a component installed in the server.



**NOTE:** For the Learn More option to display the correct page, ensure that \*.**dell.com** is added to the list of Trusted Sites in the Security Settings. For more information, see the Internet Explorer help documentation.

Following are the user privilege requirements for different license operations:

- License View and Export: Login privilege.
- License Import and Delete: Login + Configure iDRAC + Server Control privilege.

## Managing licenses using RACADM

License management

### About this task

#### Steps

1. To manage licenses using RACADM, use the **license** sub-command.
2. For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

## Managing licenses using iDRAC web interface

### About this task

To manage the licenses using the iDRAC web interface, go to **Configuration > Licenses**.

#### Steps

The **Licensing** page displays the licenses that are associated to devices, or the licenses that are installed but the device is not present in the system. For more information on importing, exporting, or deleting a license, see the **iDRAC Online Help**.

## Licensed features in iDRAC10

The following table lists iDRAC10 features that are enabled on iDRAC10 based on the license purchased:

**Table 4. Licensed features in on iDRAC10**

| Feature                            | iDRAC 10 Core | iDRAC 10 Enterprise | iDRAC 10 Datacenter |
|------------------------------------|---------------|---------------------|---------------------|
| <b>Interfaces and Standards</b>    |               |                     |                     |
| iDRAC RESTful API and Redfish      | Yes           | Yes                 | Yes                 |
| IPMI 2.0                           | Yes           | Yes                 | Yes                 |
| DCMI 1.5                           | Yes           | Yes                 | Yes                 |
| Web-based GUI                      | Yes           | Yes                 | Yes                 |
| RACADM command line (local/remote) | Yes           | Yes                 | Yes                 |
| SSH                                | Yes           | Yes                 | Yes                 |
| Serial Redirection                 | Yes           | Yes                 | Yes                 |
| Network Time Protocol (NTP)        | Yes           | Yes                 | Yes                 |
| <b>Connectivity</b>                |               |                     |                     |
| Shared NIC (LOM)                   | Yes           | Yes                 | Yes                 |
| Dedicated NIC                      | Yes           | Yes                 | Yes                 |

**Table 4. Licensed features in on iDRAC10 (continued)**

| Feature                                                      | iDRAC 10 Core | iDRAC 10 Enterprise     | iDRAC 10 Datacenter     |
|--------------------------------------------------------------|---------------|-------------------------|-------------------------|
| VLAN tagging                                                 | Yes           | Yes                     | Yes                     |
| IPv4                                                         | Yes           | Yes                     | Yes                     |
| IPv6                                                         | Yes           | Yes                     | Yes                     |
| DHCP                                                         | Yes           | Yes                     | Yes                     |
| DHCP with zero touch                                         | No            | Yes                     | Yes                     |
| Dynamic DNS                                                  | Yes           | Yes                     | Yes                     |
| OS pass-through                                              | Yes           | Yes                     | Yes                     |
| iDRAC Direct-Front panel USB                                 | Yes           | Yes                     | Yes                     |
| Connection View                                              | Yes           | Yes                     | Yes                     |
| Data Processing Unit (DPU)                                   | No            | Yes                     | Yes                     |
| <b>Security</b>                                              |               |                         |                         |
| Role-based authority                                         | Yes           | Yes                     | Yes                     |
| Local users                                                  | Yes           | Yes                     | Yes                     |
| SSL encryption                                               | Yes           | Yes                     | Yes                     |
| Secure Enterprise Key Management and iDRAC Local Key Manager | No            | Yes (with SEKM license) | Yes (with SEKM license) |
| IP address blocking                                          | No            | Yes                     | Yes                     |
| Directory services (AD, LDAP)                                | No            | Yes                     | Yes                     |
| Two-factor authentication (smart card)                       | No            | Yes                     | Yes                     |
| PK authentication (for SSH)                                  | No            | Yes                     | Yes                     |
| OAuth integration with Web-based Authentication services     | No            | Yes                     | Yes                     |
| Port-based Network Access Control (IEEE 802.1x)              | No            | Yes                     | Yes                     |
| FIPS 140-2                                                   | Yes           | Yes                     | Yes                     |
| Secure UEFI boot - certificate management                    | Yes           | Yes                     | Yes                     |
| System lock down                                             | No            | Yes                     | Yes                     |
| System lock down - GPU                                       | No            | No                      | Yes                     |
| Unique iDRAC default password                                | Yes           | Yes                     | Yes                     |
| Customizable Security Policy Banner - login page             | Yes           | Yes                     | Yes                     |
| Easy Multi Factor Authentication                             | No            | Yes                     | Yes                     |
| Auto Certificate Enrollment (SSL Certs)                      | No            | Yes                     | Yes                     |
| iDRAC Quick Sync 2 - optional auth for read operations       | Yes           | Yes                     | Yes                     |

**Table 4. Licensed features in on iDRAC10 (continued)**

| <b>Feature</b>                                               | <b>iDRAC 10 Core</b> | <b>iDRAC 10 Enterprise</b> | <b>iDRAC 10 Datacenter</b> |
|--------------------------------------------------------------|----------------------|----------------------------|----------------------------|
| iDRAC Quick Sync 2 - add mobile device number to LCL         | Yes                  | Yes                        | Yes                        |
| Secure Erase                                                 | Yes                  | Yes                        | Yes                        |
| BIOS Live Scanning                                           | No                   | No                         | Yes                        |
| RSA SecureID Two Factored Authentication (2FA)               | No                   | Yes                        | Yes                        |
| Device RoT                                                   | Yes                  | Yes                        | Yes                        |
| Intrusion detection                                          | Yes                  | Yes                        | Yes                        |
| USB access control                                           | Yes                  | Yes                        | Yes                        |
| Secure Component Verification                                | No                   | Yes                        | Yes                        |
| <b>Remote Presence</b>                                       |                      |                            |                            |
| Power control                                                | Yes                  | Yes                        | Yes                        |
| Boot control                                                 | Yes                  | Yes                        | Yes                        |
| Serial-over-LAN                                              | Yes                  | Yes                        | Yes                        |
| Virtual Media                                                | No                   | Yes                        | Yes                        |
| Virtual Folders                                              | No                   | Yes                        | Yes                        |
| Remote File Share                                            | No                   | Yes                        | Yes                        |
| HTML5 access to Virtual Console                              | No                   | Yes                        | Yes                        |
| Virtual Console                                              | No                   | Yes                        | Yes                        |
| Virtual Clipboard                                            | No                   | Yes                        | Yes                        |
| VNC connection to OS                                         | No                   | Yes                        | Yes                        |
| Quality/bandwidth control                                    | No                   | Yes                        | Yes                        |
| Virtual Console collaboration (up to six simultaneous users) | No                   | Yes                        | Yes                        |
| Virtual Console chat                                         | No                   | Yes                        | Yes                        |
| HTTP / HTTPS support along with NFS/CIFS                     | Yes                  | Yes                        | Yes                        |
| <b>Power and Thermal</b>                                     |                      |                            |                            |
| Advanced thermal management                                  | No                   | Yes                        | Yes                        |
| Real-time power meter                                        | No                   | No                         | No                         |
| Power thresholds and alerts                                  | Yes                  | Yes                        | Yes                        |
| Real-time power graphing                                     | No                   | Yes                        | Yes                        |
| Historical power counters                                    | No                   | No                         | No                         |
| Power capping                                                | No                   | Yes                        | Yes                        |
| Power Center integration                                     | No                   | Yes                        | Yes                        |
| Temperature graphing                                         | No                   | Yes                        | Yes                        |

**Table 4. Licensed features in on iDRAC10 (continued)**

| Feature                                       | iDRAC 10 Core | iDRAC 10 Enterprise | iDRAC 10 Datacenter |
|-----------------------------------------------|---------------|---------------------|---------------------|
| PCIe airflow customization (LFM)              | No            | No                  | Yes                 |
| Custom Exhaust Control                        | No            | No                  | Yes                 |
| Custom Delta-T control                        | No            | No                  | Yes                 |
| System Airflow Consumption                    | No            | No                  | Yes                 |
| Custom PCIe inlet temperature                 | No            | No                  | Yes                 |
| <b>Health</b>                                 |               |                     |                     |
| Full agent-free monitoring                    | Yes           | Yes                 | Yes                 |
| Temperature monitoring                        | Yes           | Yes                 | Yes                 |
| Predictive failure monitoring                 | Yes           | Yes                 | Yes                 |
| SNMPv1, v2, and v3 (traps and gets)           | Yes           | Yes                 | Yes                 |
| Email Alerting                                | No            | Yes                 | Yes                 |
| Configurable thresholds                       | Yes           | Yes                 | Yes                 |
| Fan monitoring                                | Yes           | Yes                 | Yes                 |
| Power Supply monitoring                       | Yes           | Yes                 | Yes                 |
| Memory monitoring                             | Yes           | Yes                 | Yes                 |
| GPU - Staged Inventory                        | Yes           | No                  | No                  |
| GPU - Realtime Inventory, Power, Temperature  | No            | No                  | Yes                 |
| GPU - Thermal configuration                   | No            | No                  | Yes                 |
| CPU monitoring                                | Yes           | Yes                 | Yes                 |
| CPU and GPU leak detection                    | Yes           | Yes                 | Yes                 |
| Storage Controllers                           | Yes           | Yes                 | Yes                 |
| Network Interface Card (NIC) monitoring       | Yes           | Yes                 | Yes                 |
| Optic Inventory                               | Yes           | Yes                 | Yes                 |
| Optic Statistics                              | No            | No                  | Yes                 |
| Hard Drive monitoring                         | Yes           | Yes                 | Yes                 |
| Out of Band Performance Monitoring            | No            | Yes                 | Yes                 |
| Alerts for excessive SSD wear                 | Yes           | Yes                 | Yes                 |
| Customizable settings for Exhaust Temperature | Yes           | Yes                 | Yes                 |
| Serial data logs                              | No            | Yes                 | Yes                 |
| SMART logs for storage drives                 | No            | Yes                 | Yes                 |
| Idle Server detection                         | No            | Yes                 | Yes                 |
| Telemetry                                     | No            | No                  | Yes                 |

**Table 4. Licensed features in on iDRAC10 (continued)**

| <b>Feature</b>                                                                                                                                                                            | <b>iDRAC 10 Core</b> | <b>iDRAC 10 Enterprise</b> | <b>iDRAC 10 Datacenter</b> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|----------------------------|----------------------------|
| Health Rollup                                                                                                                                                                             | Yes                  | Yes                        | Yes                        |
| <b>Update</b>                                                                                                                                                                             |                      |                            |                            |
| Remote agent-free update                                                                                                                                                                  | Yes                  | Yes                        | Yes                        |
| Embedded update tools                                                                                                                                                                     | Yes                  | Yes                        | Yes                        |
| Update from repository                                                                                                                                                                    | Yes                  | Yes                        | Yes                        |
| Schedule update from repository (Auto-Update)                                                                                                                                             | Yes                  | Yes                        | Yes                        |
| Firmware updates                                                                                                                                                                          | Yes                  | Yes                        | Yes                        |
| <b>Deployment and Configuration</b>                                                                                                                                                       |                      |                            |                            |
| Local configuration using F2                                                                                                                                                              | Yes                  | No                         | No                         |
| Embedded OS deployment tools                                                                                                                                                              | Yes                  | Yes                        | Yes                        |
| Embedded configuration tools                                                                                                                                                              | Yes                  | Yes                        | Yes                        |
| Autodiscovery                                                                                                                                                                             | Yes                  | No                         | No                         |
| Remote OS deployment                                                                                                                                                                      | No                   | Yes                        | Yes                        |
| Embedded driver pack                                                                                                                                                                      | Yes                  | Yes                        | Yes                        |
| Full configuration inventory                                                                                                                                                              | Yes                  | Yes                        | Yes                        |
| Inventory export                                                                                                                                                                          | Yes                  | Yes                        | Yes                        |
| Remote configuration                                                                                                                                                                      | Yes                  | Yes                        | Yes                        |
| Zero-touch configuration                                                                                                                                                                  | No                   | Yes                        | Yes                        |
| System Retire or Repurpose                                                                                                                                                                | Yes                  | Yes                        | Yes                        |
| Export Server Configuration Profile                                                                                                                                                       | Yes                  | Yes                        | Yes                        |
| Import Server Configuration Profile                                                                                                                                                       | No                   | Yes                        | Yes                        |
| Preview Server Configuration Profile                                                                                                                                                      | No                   | Yes                        | Yes                        |
| iDRAC amd BIOS configuration using configuration job                                                                                                                                      | Yes                  | Yes                        | Yes                        |
| BIOS configuration using Server Configuration Profile (SCP)                                                                                                                               | No                   | Yes                        | Yes                        |
| iDRAC configuration using SCP                                                                                                                                                             | No                   | Yes                        | Yes                        |
| Configuration of the following using configuration job: <ul style="list-style-type: none"> <li>• Storage</li> <li>• Network Interface Card (NIC)</li> <li>• Fibre Channel (FC)</li> </ul> | No                   | Yes                        | Yes                        |
| Configuration of the following using SCP:                                                                                                                                                 | No                   | Yes                        | Yes                        |

**Table 4. Licensed features in on iDRAC10 (continued)**

| Feature                                                                                                                                                                                                             | iDRAC 10 Core | iDRAC 10 Enterprise | iDRAC 10 Datacenter |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|---------------------|---------------------|
| <ul style="list-style-type: none"> <li>Storage</li> <li>NIC</li> <li>FC</li> </ul>                                                                                                                                  |               |                     |                     |
|  <b>NOTE:</b> To configure devices that are currently not supported by iDRAC, use F2 System Setup or device tools such as perccli. |               |                     |                     |
| <b>Diagnostics, Service, and Logging</b>                                                                                                                                                                            |               |                     |                     |
| Asset tracking                                                                                                                                                                                                      | Yes           | Yes                 | Yes                 |
| Embedded diagnostic tools                                                                                                                                                                                           | Yes           | Yes                 | Yes                 |
| Part Replacement                                                                                                                                                                                                    | Yes           | Yes                 | Yes                 |
| Easy Restore (system configuration)                                                                                                                                                                                 | Yes           | Yes                 | Yes                 |
| LED Health status indicators                                                                                                                                                                                        | Yes           | Yes                 | Yes                 |
| iDRAC Quick Sync 2 (BLE/Wi-Fi hardware)                                                                                                                                                                             | Yes           | Yes                 | Yes                 |
| iDRAC Direct (front USB management port)                                                                                                                                                                            | Yes           | Yes                 | Yes                 |
| iDRAC Service Module (iSM) embedded                                                                                                                                                                                 | Yes           | Yes                 | Yes                 |
| iSM to in-band alert forwarding to consoles                                                                                                                                                                         | Yes           | Yes                 | Yes                 |
| SupportAssist Collection (embedded)                                                                                                                                                                                 | Yes           | Yes                 | Yes                 |
| Crash screen capture                                                                                                                                                                                                | Yes           | Yes                 | Yes                 |
| Crash video capture                                                                                                                                                                                                 | No            | Yes                 | Yes                 |
| Agent Free Crash Video Capture (Windows only)                                                                                                                                                                       | No            | No                  | Yes                 |
| Boot capture                                                                                                                                                                                                        | Yes           | Yes                 | Yes                 |
| Remote reset for iDRAC (requires iSM)                                                                                                                                                                               | Yes           | Yes                 | Yes                 |
| Virtual NMI                                                                                                                                                                                                         | Yes           | Yes                 | Yes                 |
| Two-factor authentication (2FA) - Smart card                                                                                                                                                                        | No            | No                  | Yes                 |
| Device inventory and monitoring                                                                                                                                                                                     | Yes           | Yes                 | Yes                 |
| Work Notes                                                                                                                                                                                                          | Yes           | Yes                 | Yes                 |
| OS watchdog                                                                                                                                                                                                         | Yes           | Yes                 | Yes                 |
| System Event Log                                                                                                                                                                                                    | Yes           | Yes                 | Yes                 |
| Lifecycle Log                                                                                                                                                                                                       | Yes           | Yes                 | Yes                 |
| Enhanced Logging in Lifecycle Controller Log                                                                                                                                                                        | Yes           | Yes                 | Yes                 |
| Remote syslog                                                                                                                                                                                                       | No            | Yes                 | Yes                 |

 **NOTE:** For the list of features that are not supported in this release, see [Features not supported in this release](#)

# Interfaces and protocols to access iDRAC

The following table lists the interfaces to access iDRAC.

 **NOTE:** Using more than one interface simultaneously may generate unexpected results.

**Table 5. Interfaces and protocols to access iDRAC**

| Interface or Protocol         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| iDRAC Settings Utility (F2)   | Use the iDRAC Settings utility to perform pre-OS operations. It has a subset of the features that are available in the iDRAC web interface along with other features. To access the iDRAC Settings utility, press <F2> during boot, and then click <b>iDRAC Settings</b> on the <b>System Setup Main Menu</b> page.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| iDRAC Web Interface           | Use the iDRAC web interface to manage iDRAC and monitor the managed system. The browser connects to the web server through the HTTPS port. Data streams are encrypted using 128-bit SSL to provide privacy and integrity. Any connection to the HTTP port is redirected to HTTPS. Administrators can upload their own SSL certificate through an SSL CSR generation process to secure the web server. The default HTTP and HTTPS ports can be changed. The user access is based on user privileges.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| RACADM                        | Use this command-line utility to perform iDRAC and server management. You can use RACADM locally and remotely. <ul style="list-style-type: none"><li>• The local RACADM command-line interface runs on the managed systems that have Server Administrator installed. Local RACADM communicates with iDRAC through its in-band IPMI host interface. Since it is installed on the local managed system, users are required to log in to the operating system to run this utility. A user must have a full administrator privilege or be a root user to use this utility.</li><li>• Remote RACADM is a client utility that runs on a management station. It uses the out-of-band network interface to run RACADM commands on the managed system and uses the HTTPs channel. The <b>-r</b> option runs the RACADM command over a network.</li><li>• Firmware RACADM is accessible by logging in to iDRAC using SSH. You can run the firmware RACADM commands without specifying the iDRAC IP, user name, or password.</li><li>• You do not have to specify the iDRAC IP, user name, or password to run the firmware RACADM commands. After you enter the RACADM prompt, you can directly run the commands without the racadm prefix.</li></ul> |
| iDRAC RESTful API and Redfish | The Redfish Scalable Platforms Management API is a standard defined by the Distributed Management Task Force (DMTF). Redfish is a next-generation systems management interface standard, which enables scalable, secure, and open server management. It is a new interface that uses RESTful interface semantics to access data that is defined in model format to perform out-of-band systems management. It is suitable for a wide range of servers ranging from stand-alone servers to rackmount environments and for large-scale cloud environments. Redfish provides the following benefits over existing server management methods: <ul style="list-style-type: none"><li>• Increased simplicity and usability</li><li>• High data security</li><li>• Programmable interface that can be easily scripted.</li><li>• Follows widely used standards.</li></ul> See the <a href="#">iDRAC Redfish API</a> guide.                                                                                                                                                                                                                                                                                                                        |
| Type-C USB port               | Access iDRAC using the Type-C USB port that is labeled.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| SSH                           | Use SSH to either run RACADM commands or start a console redirection session. The SSH service is enabled by default on iDRAC. The SSH service can be disabled in iDRAC. iDRAC only supports SSH version 2 with the RSA host key algorithm. A unique 1024-bit RSA host key is generated when your power-up iDRAC for the first time.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| IPMITool                      | Use the IPMITool to access the remote system's basic management features through iDRAC. The interface includes local IPMI, IPMI over LAN, IPMI over Serial, and Serial over LAN. For more information about IPMITool, see the <b>Dell OpenManage Baseboard Management Controller Utilities User's Guide</b> .<br> <b>NOTE:</b> IPMI version 1.5 is not supported.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| NTLM                          | iDRAC allows NTLM to provide authentication, integrity, and confidentiality to the users. NT LAN Manager (NTLM)) is a suite of Microsoft security protocols and it works in a Windows network.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |



**Table 5. Interfaces and protocols to access iDRAC (continued)**

| Interface or Protocol | Description                                                                                                                                                                                                  |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SMB                   | iDRAC supports the Server Message Block (SMB) protocol. This is a network file-sharing protocol. The SMB versions that are supported are 2.0 - 3.11. SMBv1 is no longer supported.                           |
| NFS                   | iDRAC supports Network File System (NFS). This is a distributed file system protocol that enables users to mount remote directories on the servers.                                                          |
| TFTP                  | iDRAC uses Trivial File Transfer Protocol (TFTP) for firmware updates and certificate installations.                                                                                                         |
| CIFS                  | iDRAC uses the Common Internet File System (CIFS) protocol to share files remotely. CIFS mounts ISO or IMG image files from a network share to the Operating System of the managed server as virtual drives. |
| HTTP and HTTPS        | iDRAC supports both Hyper-Text Transfer Protocol (HTTP) and Hyper-Text Transfer Protocol Secure (HTTPS) for remote management of Dell servers.                                                               |

## iDRAC port information

The following table lists the ports that are required to remotely access iDRAC through firewall. These are the default ports iDRAC listens to for connections. Optionally, you can modify most of the ports. To modify ports, see [Configuring Services](#).

**Table 6. Ports iDRAC listens for connections**

| Port number | Type | Function     | Configurable port | Maximum Encryption Level |
|-------------|------|--------------|-------------------|--------------------------|
| 22          | TCP  | SSH          | Yes               | 256-bit SSL              |
| 80          | TCP  | HTTP         | Yes               | None                     |
| 161         | UDP  | SNMP Agent   | Yes               | None                     |
| 443         | TCP  | HTTPS        | Yes               | 256-bit SSL              |
| 623         | UDP  | RMCP/RMCP+   | No                | 128-bit SSL              |
| 5000        | TCP  | iDRAC to iSM | No                | 256-bit SSL              |
| 5901        | TCP  | VNC          | Yes               | 128-bit SSL              |

 **NOTE:** Port 5901 opens when the VNC feature is enabled.

The following table lists the ports that iDRAC uses as a client:

**Table 7. Ports iDRAC uses as client**

| Port number | Type | Function                           | Configurable port | Maximum Encryption Level |
|-------------|------|------------------------------------|-------------------|--------------------------|
| 25          | TCP  | SMTP                               | Yes               | None                     |
| 53          | UDP  | DNS                                | No                | None                     |
| 68          | UDP  | DHCP-assigned IP address           | No                | None                     |
| 69          | TFTP | TFTP                               | No                | None                     |
| 123         | UDP  | Network Time Protocol (NTP)        | No                | None                     |
| 162         | UDP  | SNMP trap                          | Yes               | None                     |
| 445         | TCP  | Common Internet File System (CIFS) | No                | None                     |
| 636         | TCP  | LDAP Over SSL (LDAPS)              | No                | 256-bit SSL              |
| 2049        | TCP  | Network File System (NFS)          | No                | None                     |
| 3269        | TCP  | LDAPS for global catalog (GC)      | No                | 256-bit SSL              |

**Table 7. Ports iDRAC uses as client (continued)**

| Port number | Type | Function      | Configurable port | Maximum Encryption Level |
|-------------|------|---------------|-------------------|--------------------------|
| 5353        | UDP  | mDNS          | No                | None                     |
| 514         | UDP  | Remote syslog | Yes               | None                     |

## Other documents you may need

The iDRAC UI interface supports integrated **Online Help** that can be accessed by clicking the **Help & Feedback** tab. The **Online Help** provides detailed information about the fields available on the web interface and the descriptions for the same. In addition, the following documents are available on the Dell Support website at [dell.com/support](https://dell.com/support) that provide additional information about the setup and operation of iDRAC in your system.

- The **iDRAC Redfish API Guide** provides information about Redfish API.
- The **Integrated Dell Remote Access Controller RACADM CLI Guide** provides information about the RACADM subcommands, supported interfaces, and iDRAC property database groups and object definitions.
- The *Systems Management Overview Guide* provides brief information about the various software available to perform systems management tasks.
- The **Dell Systems Software Support Matrix** provides information about the various Dell systems, the operating systems that are supported by these systems, and the Dell OpenManage components that can be installed on these systems.
- The **iDRAC Service Module User's Guide** provides information to install the iDRAC Service Module.
- The **Dell OpenManage Server Administrator Installation Guide** contains instructions to help you install Dell OpenManage Server Administrator.
- The **Dell OpenManage Management Station Software Installation Guide** contains instructions to help you install Dell OpenManage management station software that includes Baseboard Management Utility, DRAC Tools, and Active Directory Snap-In.
- The **Dell OpenManage Baseboard Management Controller Management Utilities User's Guide** has information about the IPMI interface.
- The **Release Notes** provides last-minute updates to the system or documentation or advanced technical reference material that is intended for experienced users or technicians.
- The **Integrated Dell Remote Access Controller 10 Attribute Registry** provides the details about the groups and objects in the iDRAC property database.

The following system documents are available to provide more information:

- The safety instructions that came with your system provide important safety and regulatory information. For additional regulatory information, see the Regulatory Compliance home page at [dell.com/regulatory\\_compliance](https://dell.com/regulatory_compliance). Warranty information may be included within this document or as a separate document.
- The **Rack Installation Instructions** included with your rack solution describe how to install your system into a rack.
- The *Getting Started Guide* provides an overview of system features, setting up your system, and technical specifications.
- The *Installation and Service Manual* provides information about system features and describes how to troubleshoot the system and install or replace system components.

## Contacting Dell

Dell provides several online and telephone-based support and service options. Availability varies by country or region and product, and some services may not be available in your area. To contact Dell for sales, technical support, or customer service issues, go to [Contact Dell](#).

 **NOTE:** If you do not have an active Internet connection, you can find contact information about your purchase invoice, packing slip, bill, or Dell product catalog.

## Accessing documents from the Dell Support site

Click the following links to access the documents from the Dell Support site:

- [Enterprise Systems Management and OpenManage Connections documents](#)
- [OpenManage documents](#)

- [iDRAC and Lifecycle Controller documents](#)
- [Serviceability Tools documents](#)
- [Client Command Suite Systems Management documents](#)

## Accessing documents using the product search

1. Go to the [Dell support](#) site.
2. In the **Enter a Service Tag, Serial Number...** search box, type the product name. For example, **PowerEdge** or **iDRAC**. A list of matching products is displayed.
3. Select your product and click the search icon or press enter.
4. Click **DOCUMENTATION**.
5. Click **MANUALS AND DOCUMENTS**.

## Accessing documents using product selector

You can also access documents by selecting your product.

1. Go to [Dell support](#).
2. Click **Browse all products**.
3. Click the required product category, such as Servers, Software, Storage, and so on.
4. Click the required product and then click the version if applicable.

 **NOTE:** For some products, you may have to navigate through the subcategories.

5. Click **DOCUMENTATION**.
6. Click **MANUALS AND DOCUMENTS**.

## Accessing Redfish API

The Redfish API guide is now available at the Dell API Marketplace

### Steps

1. Go to [Developer portal](#).
2. Click **Explore APIs**, and then click **APIs**.
3. Under iDRAC10 Redfish API, click **View More**.

# Setting up iDRAC IP address

To enable communication to and from iDRAC, configure the initial network settings on the iDRAC Settings utility based on your network infrastructure.

## Topics:

- [Setting up iDRAC IP using iDRAC settings utility](#)
- [Modifying local administrator account settings](#)

## Setting up iDRAC IP using iDRAC settings utility

### About this task

To set up the iDRAC IP address:

### Steps

1. Turn on the managed system.
2. Press <F2> during Power-on Self-test (POST).
3. In the **System Setup Main Menu** page, click **iDRAC Settings**.  
The **iDRAC Settings** page is displayed.
4. Click **Network**.  
The **Network** page is displayed.
5. Specify the following settings:
  - Network Settings
  - Common Settings
  - IPv4 Settings
  - IPv6 Settings
  - IPMI Settings
  - VLAN Settings
6. Click **Back**, click **Finish**, and then click **Yes**.  
The network information is saved and the system reboots.

## Configuring the network settings

### About this task

To configure the network settings:

 **NOTE:** For information about the options, see the **iDRAC Settings Utility Online Help**.

### Steps

1. Under **Enable NIC**, select **Enabled**.
2. From the **NIC Selection** list, select one of the following ports based on the network requirement:
  - **Dedicated** — Enables the remote access device to use the dedicated network interface available on the Remote Access Controller (RAC). This interface is not shared with the host operating system and routes the management traffic to a separate physical network, enabling it to be separated from the application traffic.

 **NOTE:** This option implies that iDRAC's dedicated network port routes its traffic separately from the server's LOM or NIC ports. The Dedicated option allows iDRAC to be assigned an IP address from the same subnet or different subnet in comparison to the IP addresses assigned to the Host LOM or NICs to manage the network traffic.

- LOM1
- LOM2
- LOM3
- LOM4

**NOTE:** In the case of rack and tower servers, two LOM options (LOM1 and LOM2) or all four LOM options are available depending on the server model.

- From the **NIC Selection** drop-down menu, select the port from which you want to access the system remotely, following are the options:

**NOTE:** You can select either the dedicated network interface card or from a list of LOMs available in the Quad port or Dual port mezzanine cards.

- **For Quad port cards—LOM1-LOM16**
- **For Dual port cards—LOM1, LOM2, LOM5, LOM6, LOM9, LOM10, LOM13, LOM14.**

- From the **Failover Network** drop-down menu, select one of the remaining LOMs. If a network fails, the traffic is routed through the failover network.

For example, to route the iDRAC network traffic through LOM2 when LOM1 is down, select **LOM1** for **NIC Selection** and **LOM2** for **Failover Network**.

**NOTE:** This option is disabled if **NIC Selection** is set to **Dedicated**.

**NOTE:** When using the **Failover network** settings, it is recommended that all the LOM ports to be connected to the same network.

For more details, refer to the section [Modifying network settings using web interface](#).

- Under **Auto Negotiation**, select **On** if iDRAC must automatically set the duplex mode and network speed. This option is available only for dedicated mode. If enabled, iDRAC sets the network speed to 10, 100, or 1000 Mbps based on the network speed.
- Under **Network Speed**, select either 10 Mbps or 100 Mbps.
 

**NOTE:** You cannot manually set the Network Speed to 1000 Mbps. This option is available only if **Auto Negotiation** option is enabled.
- Under **Duplex Mode**, select **Half Duplex** or **Full Duplex** option.
 

**NOTE:** This option is disabled if **Auto Negotiation** is set to **Enabled**.

**NOTE:** If network teaming is configured for the host OS using the same network adapter as NIC Selection, then the Failover Network should also be configured. NIC Selection and Failover Network should use the ports that are configured as a part of the network team. If more than two ports are used as part of the network team, then the Failover Network selection should be "All".
- Under **NIC MTU**, enter the Maximum Transmission Unit size on the NIC.
 

**NOTE:** The default and maximum limit for MTU on NIC is 1500, and the minimum value is 576. An MTU value of 1280 or greater is required if IPv6 is enabled.

## Common settings

### About this task

If network infrastructure has DNS server, register iDRAC on the DNS. These are the initial settings requirements for advanced features such as Directory services—Active Directory or LDAP.

To register iDRAC:

### Steps

- Enable **Register DRAC on DNS**.

2. Enter the **DNS DRAC Name**.
3. Select **Auto Config Domain Name** to automatically acquire domain name from DHCP. Else, provide the **DNS Domain Name**.  
For **DNS iDRAC Name** field, the default name format is **idrac-Service\_Tag**, where Service\_Tag is the service tag of the server. The maximum length is 63 characters and the following characters are supported:
  - A-Z
  - a-z
  - 0-9
  - Hyphen (-)

## Configuring the IPv4 settings

### About this task

To configure the IPv4 settings:

#### Steps

1. Select **Enabled** option under **Enable IPv4**.
2. Select **Enabled** option under **Enable DHCP**, so that DHCP can automatically assign the IP address, gateway, and subnet mask to iDRAC. Else, select **Disabled** and enter the values for:
  - Static IP Address
  - Static Gateway
  - Static Subnet Mask

The options **Use DHCP to Obtain DNS Server Addresses**, **Use DHCPv6 to Obtain DNS Server Addresses**, and **Auto Config Domain Name** are enabled by default.

## Configuring the IPv6 settings

### About this task

Based on the infrastructure setup, you can use IPv6 address protocol.

To configure the IPv6 settings:

 **NOTE:** If IPv6 is set to static, ensure that you configure the IPv6 gateway manually, which is not needed in dynamic IPV6. Failing to configure manually in static IPv6 results in loss of communication.

#### Steps

1. Select **Enabled** option under **Enable IPv6**.
2. For the DHCPv6 server to automatically assign the IP address and prefix length to iDRAC, select the **Enabled** option under **Enable Auto-configuration**.

 **NOTE:** You can configure both static IP and DHCP IP simultaneously.

3. In the **Static IP Address 1** box, enter the static IPv6 address.
4. In the **Static Prefix Length** box, enter a value between 1 and 128.
5. In the **Static Gateway** box, enter the gateway address.

 **NOTE:** If you configure static IP, the current IP address 1 displays static IP and the IP address 2 displays dynamic IP. If you clear the static IP settings, the current IP address 1 displays dynamic IP.

The options **Use DHCP to Obtain DNS Server Addresses**, **Use DHCPv6 to Obtain DNS Server Addresses**, and **Auto Config Domain Name** options are enabled by default.

6. Configure the following if required:
  - In the **Static Preferred DNS Server** box, enter the static DNS server IPv6 address.
  - In the **Static Alternate DNS Server** box, enter the static alternate DNS server.

7. When DNS information is not obtainable by either DHCPv6 or static configuration, you can use RFC 8106 "IPv6 Router Advertisement Options for DNS Configuration. It is identified by IPv6 Router. Using RA DNS configuration does not impact existing DNS configurations (either DHCPv6 or static).
  - The iDRAC can obtain DNS name server and DNS search domain information from IPv6 Router Advertisement messages. Please see RFC 8106 and your IPv6 router's user guide for details on how to configure the router to advertise this information.
  - If DNS information is available from both the DHCPv6 server and the IPv6 Router Advertisement, the iDRAC uses both. In conflict, the DHCPv6 server's DNS information takes precedence in the iDRAC's /etc/resolv.conf settings.

 **NOTE:** For iDRAC to use RA DNS information, IPv6. Enable and IPv6.Autoconfig must be enabled. If Auto-configuration is disabled, the iDRAC does not process IPv6 RA messages, and uses only static DNS settings as configured.

## Configuring the IPMI settings

### About this task

To enable the IPMI Settings:

#### Steps

1. Under **Enable IPMI Over LAN**, select **Enabled**.
2. Under **Channel Privilege Limit**, select **Administrator**, **Operator**, or **User**.
3. In the **Encryption Key** box, enter the encryption key in the format 0 to 40 hexadecimal characters (without any blanks characters.) The default value is all zeros.

## VLAN settings

### About this task

You can configure iDRAC into the VLAN infrastructure. To configure VLAN settings, perform the following steps:

#### Steps

1. Under **Enable VLAN ID**, select **Enabled**.
2. In the **VLAN ID** box, enter a valid number from 1 to 4094.
3. In the **Priority** box, enter a number from 0 to 7 to set the priority of the VLAN ID.

 **NOTE:** After enabling VLAN, the iDRAC IP is not accessible for some time.

## Port-based Network Access Control (IEEE 802.11)

iDRAC provides Port-based network access control (IEEE802.1x). It provides a secured authentication mechanism to devices to attach to a LAN.

This feature requires an iDRAC Datacenter license.

You can access this feature using iDRAC UI by navigating to- **iDRAC Settings > Connectivity > Network > Advanced Network Settings > 802.1x Security**. You can enable or disable the option using the drop-down list. The feature is enabled by default.

 **NOTE:** The effect of 802.1x does not work in Shared LOM mode with VLAN enabled.

Port-based Network Access Control has three ways of configuring the authentication certificates:

- **Default LDevID**—This is the default iDRAC certificate that is installed in the factory.
- **Custom Signing LDevID**—Using this option to define a Certificate Signing Request (CSR), which is signed by the uploaded LDevID Signing Certificate.
- **Custom LDevID**—Using this option, you can upload a custom certificate of choice.

There is option to enable or disable **Authentication Server Certificate**, for providing the necessary information to validate the certificate. This option is disabled by default. The 802.1X Authentication Server Certificate validity message is displayed only during the iDRAC boot up.

 **NOTE:**

- This feature is disabled by default in modular servers.
- Any change in 802.1x configuration, including certificate uploads and enabling/disabling settings, takes effect on the next iDRAC boot.
- Switching iDRAC network between 802.1x enabled switch and Non 802.1x switch needs iDRAC reboot.
- If the Ethernet Switch ports connected to the server's LOM ports are enabled for 802.1X security, all downstream devices on those ports must also be enabled for 802.1X security. This means that the host is impacted if it has not been enabled for 802.1X security.

## Modifying local administrator account settings

### About this task

After setting the iDRAC IP address, you can modify the local administrator account settings (that is, user 2) using the iDRAC Settings utility.

### Steps

1. In the iDRAC Settings utility, go to **User Configuration**.  
The **iDRAC Settings User Configuration** page is displayed.
2. Specify the details for **User Name**, **LAN User Privilege**, **Serial Port User Privilege**, and **Change Password**.  
For information about the options, see the **iDRAC Settings Utility Online Help**.
3. Click **Back**, click **Finish**, and then click **Yes**.  
The local administrator account settings are configured.



# Logging in to iDRAC

You can log in to iDRAC as an iDRAC user, a Microsoft Active Directory user, or a Lightweight Directory Access Protocol (LDAP) user.

To improve security, each system is shipped with a unique password for iDRAC, which is available on the system information tag. This unique password improves the security of iDRAC and your server. The default user name is **root**.

While ordering the system, you can choose to retain the legacy password—calvin—as the default password. If you choose to retain the legacy password, the password is not available on the system information tag.

In this version, DHCP is enabled by default and the iDRAC IP address is assigned dynamically.

## NOTE:

- You must have login to iDRAC privilege to log in to iDRAC.
- iDRAC UI does not support browser buttons such as **Back**, **Forward**, or **Refresh**.
- After the iDRAC IP address is detected, it may take a maximum of five minutes to log in to iDRAC. If you are unable to log in, contact the Technical Support team.

## Customizable security banner

You can customize the security notice that is displayed on the login page. You can use SSH, RACADM, or Redfish to customize the notice. Depending on the language you use, the notice can be either 1024 or 512 UTF-8 characters long.

### Topics:

- [Logging in to iDRAC as local user, Active Directory user, or LDAP user](#)
- [Force Change of Password \(FCP\)](#)
- [Accessing iDRAC using remote RACADM](#)
- [Accessing iDRAC using Local RACADM](#)
- [Accessing iDRAC using firmware RACADM](#)
- [RSA SecurID 2FA](#)
- [Viewing system health](#)
- [Logging in to iDRAC using public key authentication](#)
- [Multiple iDRAC sessions](#)
- [Secure default password](#)
- [Changing the default login password](#)
- [Enabling or disabling default password warning message](#)
- [IP Blocking](#)
- [Enabling or disabling an operating system to iDRAC Pass-through using web interface](#)
- [Enabling or disabling alerts using RACADM](#)

## Logging in to iDRAC as local user, Active Directory user, or LDAP user

### About this task

Before you log in to iDRAC using the web interface, ensure that you have configured a supported web browser and the user account is created with the required privileges.

To configure the user, use a unique user name. It should contain 16 characters including whitespace. The following characters are supported:

### **NOTE:**

- The user name is not case-sensitive for an Active Directory user. The password is case-sensitive for all users.
- In addition to Active Directory, openLDAP, openDS, Novell eDir, and Fedora-based directory services are supported.
- LDAP authentication with OpenDS is supported. The DH key must be larger than 768 bits.
- The RSA feature can be configured and enabled for LDAP users, but RSA does not support if LDAP is configured on Microsoft active directory. Hence, LDAP user login fails. RSA is supported only for OpenLDAP.

To log in to iDRAC as local user, Active Directory user, or LDAP user:

#### Steps

1. Open a supported web browser.
2. In the **Address** field, type `https://[iDRAC-IP-address]` and press Enter.

 **NOTE:** If the default HTTPS port number (port 443) changes, enter: `https://[iDRAC-IP-address]:[port-number]` where `[iDRAC-IP-address]` is the iDRAC IPv4 or IPv6 address and `[port-number]` is the HTTPS port number.

The **Login** page is displayed.

3. For a local user:
  - In the **Username** and **Password** fields, enter your iDRAC user name and password.
  - From the **Domain** drop-down menu, select **This iDRAC**.
4. For an Active Directory user, in the **User name** and **Password** fields, enter the Active Directory user name and password. If you have specified the domain name as a part of the username, select **This iDRAC** from the drop-down menu. The format of the user name can be: `<domain>\<username>`, `<domain>/<username>`, or `<user>@<domain>`.

For example, `dell.com\john_doe`, or `JOHN_DOE@DELL.COM`.

Active Directory domain from the **Domain** drop-down menu displays the last used domain.

5. For an LDAP user, in the **Username** and **Password** fields, enter your LDAP user name and password. Domain name is not required for LDAP login. By default, **This iDRAC** is selected in the drop-down menu.
6. Click **Submit**. You are logged in to iDRAC with the required user privileges.

If you log in with Configure Users privileges and the default account credentials, and if the default password warning feature is enabled, the **Default Password Warning** page is displayed allowing you to easily change the password.

## Force Change of Password (FCP)

The 'Force Change of Password' feature prompts you to change the factory default password of the device. The feature can be enabled as part of factory configuration.

The FCP screen appears after successful user authentication and cannot be skipped. Only after the user enters a password, normal access and operation will be allowed. The state of this attribute will not be affected by a 'Reset Configuration to Defaults' operation.

 **NOTE:** To set or reset the FCP attribute, you must have Login privilege and User configuration privilege.

 **NOTE:** When FCP is enabled, 'Default Password Warning' setting is disabled after changing the default user password.

 **NOTE:** When root user logs in via Public Key Authentication (PKA), FCP is bypassed.

When FCP is enabled, the following actions are not allowed:

- Login to iDRAC through any UI except IPMIpower-LAN interface which uses CLI with default user credentials.
- Login to iDRAC through OMM app using Quick Sync-2

## Accessing iDRAC using remote RACADM

You can use remote RACADM to access iDRAC using RACADM utility.

For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

If the management station has not stored the iDRAC SSL certificate in its default certificate storage, a warning message is displayed when you run the RACADM command. However, the command is performed successfully.

**NOTE:** The iDRAC certificate is the certificate iDRAC sends to the RACADM client to establish the secure session. This certificate is either issued by a CA or self-signed. In either case, if the management station does not recognize the CA or signing authority, a warning is displayed.

## Validating CA certificate to use remote RACADM on Linux

### About this task

Before running remote RACADM commands, validate the CA certificate that is used for secure communications.

To validate the certificate for using remote RACADM:

### Steps

1. Convert the certificate in DER format to PEM format (using openssl command-line tool):

```
openssl x509 -inform pem -in [yourdownloadedderformatcert.crt] -outform pem -out [outcertfileinpemformat.pem] -text
```

2. Find the location of the default CA certificate bundle on the management station. For example, the location for RHEL 64 bit is **/etc/pki/tls/cert.pem**.
3. Append the PEM formatted CA certificate to the management station CA certificate.  
For example, use the `cat` command: `cat testcacert.pem >> cert.pem`
4. Generate and upload the server certificate to iDRAC.

## Accessing iDRAC using Local RACADM

Remote or Local RACADM, which are part of RACADM CLI, are unavailable in iDRAC10 1.10.17.00 version. However, you can use RACADM CLI through the iDRAC SSH interface. For information to access iDRAC using local RACADM, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

## Accessing iDRAC using firmware RACADM

You can use SSH interface to access iDRAC and run firmware RACADM commands. For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

## RSA SecurID 2FA

iDRAC can be configured to authenticate with a single RSA AM server at a time. The global settings on RSA AM server apply to all iDRAC local users, AD, and LDAP users.

**NOTE:** RSA SecurID 2FA feature is available only on Datacenter license.

Following are the pre-requisites before you configure iDRAC to enable RSA SecurID:

- Configure Microsoft Active Directory server.
- If you try to enable RSA SecurID on all AD users, add the AD server to the RSA AM server as an Identity Source.
- Ensure you have a generic LDAP server.
- For all LDAP users, the Identity Source to the LDAP server must be added in RSA AM server.

To enable RSA SecurID on iDRAC, the following attributes from the RSA AM server are required:

1. **RSA Authentication API URL** — The URL syntax is: `https://<rsa-am-server-hostname>:<port>/mfa/v1_1`, and by default the port is 5555.

2. **RSA Client-ID** — By default, the RSA client ID is the same as the RSA AM server hostname. Find the RSA client ID at RSA AM server's authentication agent configuration page.
3. **RSA Access Key** — The Access Key can be retrieved on RSA AM by navigating to **Setup > System Settings > RSA SecurID > Authentication API** section, which is usually displayed as 198cv5x195fdi86u43jw0q069byt0x37umlfwxc2gnp4s0xk11ve2lffum4s8302. To configure the settings through iDRAC GUI:

- Go to **iDRAC Settings > Users**.
- From **Local Users** section, select an existing local user and click **Edit**.
- Scroll down to the bottom of the Configuration page.
- In **RSA SecurID** section, Click the link **RSA SecurID Configuration** to view or edit these settings.

You can also configure the settings as follows:

- Go to **iDRAC Settings > Users**.
- From **Directory Services** section, select **Microsoft Active Service** or **Generic LDAP Directory Service**, and click **Edit**.
- In **RSA SecurID** section, Click the link **RSA SecurID Configuration** to view or edit these settings.

#### 4. RSA AM server certificate (chain)

You can login to iDRAC using RSA SecurID token via iDRAC GUI and SSH.

## RSA SecurID Token App

You need to install RSA SecurID Token app on you system or on smart phone. When you try to log in to iDRAC, you are asked to input the passcode shown in the app.

If a wrong passcode is entered, the RSA AM server challenges the user to provide the "Next Token." This may happen even though the user may have entered the correct passcode. This entry proves that the user owns the right Token that generates the right passcode.

You get the **Next Token** from RSA SecurID Token app by clicking on **Options**. Check **Next Token**, and the next passcode is available. Time is critical in this step. Otherwise, iDRAC may fail the verification of the next token. If the iDRAC user login session times out, it requires another attempt to log in

If a wrong passcode is entered, the RSA AM server will challenge the user to provide the "Next Token." This challenge happens even though the user may have later entered the correct passcode. This entry proves that the user owns the right Token that generates the right passcodes.

To get the next token from RSA SecurID Token app, click on **Options** and check **Next Token**. A new token is generated. Time is critical in this step. Otherwise, iDRAC may fail the verification of the next token. If the iDRAC user login session times out, it requires another attempt to log in.

## Viewing system health

Before you perform a task or trigger an event, you can use RACADM to check if the system is in a suitable state. To view the remote service status from RACADM, use the `getremoteservicesstatus` command.

 **NOTE:** The Real Time Status is displayed as **Not Applicable** if there are no real time capable controllers present on the system.

**Table 8. Possible values for system status**

| Host System                                                                                                                                                                                                                                                                                     | Lifecycle Controller (LC)                                                                                                                                                   | Real Time Status                                                                                         | Overall Status                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Powered Off</li> <li>• In POST</li> <li>• Out of POST</li> <li>• Collecting System Inventory</li> <li>• Automated Task Execution</li> <li>• Lifecycle Controller</li> <li>• Server has halted at F1/F2 error prompt because of a POST error</li> </ul> | <ul style="list-style-type: none"> <li>• Ready</li> <li>• Not Initialized</li> <li>• Reloading data</li> <li>• Disabled</li> <li>• In Recovery</li> <li>• In Use</li> </ul> | <ul style="list-style-type: none"> <li>• Ready</li> <li>• Not Ready</li> <li>• Not Applicable</li> </ul> | <ul style="list-style-type: none"> <li>• Ready</li> <li>• Not Ready</li> </ul> |

**Table 8. Possible values for system status (continued)**

| Host System                                                                                                                                                                                                                         | Lifecycle Controller (LC) | Real Time Status | Overall Status |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|------------------|----------------|
| <ul style="list-style-type: none"> <li>Server has halted at F1/F2/F11 prompt because there are no bootable devices available</li> <li>Server has entered F2 setup menu</li> <li>Server has entered F11 Boot Manager menu</li> </ul> |                           |                  |                |
| <ol style="list-style-type: none"> <li>1. Read/Write: Read Only</li> <li>2. User Privilege: Login User</li> <li>3. License Required: iDRAC Core or iDRAC Enterprise</li> <li>4. Dependency: None</li> </ol>                         |                           |                  |                |

## Logging in to iDRAC using public key authentication

You can log in to the iDRAC over SSH without entering a password. You can also send a single RACADM command as a command line argument to the SSH application. The command line options behave like remote RACADM since the session ends after the command is completed.

For example:

### Logging in:

```
ssh username@<domain>
```

or

```
ssh username@<IP_address>
```

where IP\_address is the IP address of the iDRAC.

### Sending RACADM commands:

```
ssh username@<domain> racadm getversion
```

```
ssh username@<domain> racadm getsel
```

## Multiple iDRAC sessions

The following table provides the number of iDRAC sessions that are possible using the various interfaces.

**Table 9. Multiple iDRAC sessions**

| Interface           | Number of Sessions  |
|---------------------|---------------------|
| iDRAC Web Interface | 8                   |
| Remote RACADM       | 4                   |
| Firmware RACADM     | SSH - 4, Serial - 1 |

iDRAC allows multiple sessions for the same user. After a user has created the maximum number of allowed sessions, other users cannot log in to the iDRAC. This can cause a **Denial of Service** for a legitimate administrator user.

In case of session exhaustion, perform the following remedies:

- If webserver-based sessions are exhausted, you can still login via SSH or local RACADM.
- An administrator can then terminate existing sessions using racadm commands (`racadm getssninfo`; `racadm closessn -i <index>`).

# Secure default password

All supported systems are shipped with a unique default password for iDRAC, unless you choose to set **calvin** as the password while ordering the system. The unique password helps improve the security of iDRAC and your server. To further enhance security, it is recommended that you change the default password.

The unique password for your system is available on the system information tag. To locate the tag, see the documentation for your server at [Dell Support](#) page.

 **NOTE:** Resetting iDRAC to the factory default settings reverts the default password to the one that the server was shipped with.

If you have forgotten the password and do not have access to the system information tag, there are a few methods to reset the password locally or remotely.

## Resetting default iDRAC password locally

If you have physical access to the system, you can reset the password using the following:

- iDRAC Setting utility (System Setup)
- Local RACADM
- OpenManage Mobile
- Server management USB port
- USB-NIC

## Resetting default iDRAC password remotely

If you do not have physical access to the system, you can reset the default password remotely.

# Changing the default login password

The warning message that allows you to change the default password is displayed if:

- You log in to iDRAC with Configure User privilege.
- The default password warning feature is enabled.
- The default iDRAC user name and password are provided on the system information tag.

A warning message is also displayed when you log in to iDRAC using SSH, remote RACADM, or the Web interface. For Web interface and SSH, a single warning message is displayed for each session. For remote RACADM, the warning message is displayed for each command.

## Changing the default login password using web interface

### About this task

When you log in to the iDRAC web interface, if the **Default Password Warning** page is displayed, you can change the password. To do this:

### Steps

1. Select the **Change Default Password** option.
2. In the **New Password** field, enter the new password.

 **NOTE:** For information on recommended characters for user names and passwords, see [Recommended characters in user names and passwords](#).

3. In the **Confirm Password** field, enter the password again.
4. Click **Continue**.

The new password is configured and you are logged in to iDRAC.

 **NOTE:** **Continue** is enabled only if the passwords entered in the **New Password** and **Confirm Password** fields match.  
For information about the other fields, see the **iDRAC Online Help**.

## Changing the default login password using RACADM

To change the password, run the following RACADM command:

```
racadm set iDRAC.Users.<index>.Password <Password>
```

where, <index> is a value from 1 to 16 (indicates the user account) and <password> is the new user defined password.

 **NOTE:** The index for the default account is 2.

For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

 **NOTE:** For information on recommended characters for user names and passwords, see [Recommended characters in user names and passwords](#).

## Changing the default login password using iDRAC settings utility

### About this task

To change the default login password using iDRAC settings utility:

### Steps

1. In the iDRAC Settings utility, go to **User Configuration**.  
The **iDRAC Settings User Configuration** page is displayed.
2. In the **Change Password** field, enter the new password.

 **NOTE:** For information on recommended characters for user names and passwords, see [Recommended characters in user names and passwords](#).

3. Click **Back**, click **Finish**, and then click **Yes**.  
The details are saved.

## Enabling or disabling default password warning message

You can enable or disable the display of the default password warning message. To do this, you must have Configure Users privilege.

## IP Blocking

You can use IP blocking to dynamically determine when excessive login failures occur from an IP address and block or prevent the IP address from logging into the iDRAC10 for a preselected time span. IP blocking includes:

- The number of allowable login failures.
- The timeframe in seconds when these failures must occur.
- The amount of time, in seconds, when the IP address is prevented from establishing a session after the total allowable number of failures is exceeded.

As consecutive login failures accumulate from a specific IP address, they are tracked by an internal counter. When the user logs in successfully, the failure history is cleared and the internal counter is reset.

**NOTE:** When consecutive login attempts are refused from the client IP address, some SSH clients may display the following message:

```
ssh_exchange_identification: Connection closed by remote host
```

**NOTE:** IP blocking feature supports upto 5 IP ranges. You can see / set these only via RACADM.

**Table 10. Login Retry Restriction Properties**

| Property                     | Definition                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| iDRAC.IPBlocking.BlockEnable | Enables the IP blocking feature. When consecutive failures<br><br>iDRAC.IPBlocking.FailCount<br><br>from a single IP address are encountered within a specific amount of time<br><br>iDRAC.IPBlocking.FailWindow<br><br>all further attempts to establish a session from that address are rejected for a certain timespan<br><br>iDRAC.IPBlocking.PenaltyTime |
| iDRAC.IPBlocking.FailCount   | Sets the number of login failures from an IP address before the login attempts are rejected.                                                                                                                                                                                                                                                                  |
| iDRAC.IPBlocking.FailWindow  | The time, in seconds during which the failed attempts are counted. When the failures occur beyond this time period, the counter gets reset.                                                                                                                                                                                                                   |
| iDRAC.IPBlocking.PenaltyTime | Defines the timespan in seconds when all login attempts from an IP address with excessive failures are rejected.                                                                                                                                                                                                                                              |

## Enabling or disabling an operating system to iDRAC Pass-through using web interface

### About this task

To enable operating system to iDRAC Pass-through using Web interface:

### Steps

- Go to **iDRAC Settings > Connectivity > Network > OS to iDRAC Pass-through**. The **OS to iDRAC Pass-through** page is displayed.
- Change the State to **Enabled**.
- Select any of the following options for Pass-through Mode:
  - LOM**—The operating system to iDRAC pass-through link between the iDRAC and the host operating system is established through the LOM or OCP.
  - USB NIC**—The operating system to iDRAC pass-through link between the iDRAC and the host operating system is established through the internal USB bus.

**NOTE:** If you set the pass-through mode to LOM, ensure that:

  - The operating system and iDRAC are on the same subnet.
  - NIC selection in Network Settings is set to LOM.
- If the server is connected in shared LOM mode, then the **OS IP Address** field is disabled.



**NOTE:** If the VLAN is enabled on the iDRAC, the LOM-Passthrough functions only in shared LOM mode with VLAN taggings that are configured on the host.

**NOTE:**

- When Pass-through mode is set to LOM, it is not possible to launch iDRAC from the host operating system after cold boot.
- The LOM Pass-through is removed using the Dedicated mode feature.

5. If you select **USB NIC** as the pass-through configuration, enter the IP address of the USB NIC.

The default value is 169.254.1.1. It is recommended to use the default IP address. However, if this IP address conflicts with an IP address of other interfaces of the host system or the local network, you must change it.

Do not enter 169.254.0.3 and 169.254.0.4 IPs. These IPs are reserved for the USB NIC port on the front panel when Type-C USB cable is used.

**NOTE:** If IPv6 is preferred, the default address is fde1:53ba:e9a0:de11::1. If needed, this address can be modified in the idrac. OS-BMC.UsbNicULA setting. If IPv6 is not wanted on the USB-NIC, it can be disabled by changing the address to ":::"

**NOTE:** When you modify the static IP address of the USB NIC, it automatically adjusts the DHCP address range to align with the new static IP. For instance, if you set the static IP to 169.254.1.1, the DHCP address updates to 169.254.1.2. This change is compatible with the network manager, Wicked, which accepts the new DHCP address.

6. Click **Apply**.

7. Click **Test Network Configuration** to check if the IP is accessible and the link is established between the iDRAC and the host operating system.

## Enabling or disabling alerts using RACADM

Use the following command:

```
racadm set iDRAC.IPMILan.AlertEnable <n>
```

n=0 — Disabled

n=1 — Enabled

# Open Server Manager 3.0.x

You can update to Open Server Manager (OSM) 3.0.x from iDRAC10.

## Topics:

- [Preparing the system for an OSM update](#)
- [Updating OSM on the system](#)

## Preparing the system for an OSM update

Ensure that the prerequisites are met and the system is prepared for an OSM update.

### Prerequisites

- Check if OSM is supported on the target platform.
- OSM license is available.
- OSM DUP is available.

### Steps

1. Ensure that the latest iDRAC, FPGA, and BIOS firmware are running on your system.
2. Upload the OSM license procured from Dell.
3. Ensure that OSM is available in the firmware Inventory of Redfish URI (**redfish/v1/UpdateService/FirmwareInventory**) and iDRAC UI (**System > Inventory > Firmware Inventory**).

## Updating OSM on the system

### Steps

1. Upload the OSM DUP and initiate an update.  
The host automatically turns off before the update begins. If the host does not shut down, use the available interface (UI, Redfish, IPMI) to shut down the host manually so that the update can proceed.
2. After the update is completed, iDRAC reboots and boots with OSM.
3. Go through the latest [Dell Open Server Manager built on OpenBMC™ User's Guide](#) for steps to access the OSM UI and ssh.

# Setting up managed system

If you need to run local RACADM or enable Last Crash Screen capture, install the following from the **Dell Systems Management Tools and Documentation** DVD:

- Local RACADM
- Server Administrator

 **NOTE:** For any update requiring iDRAC reset/ reboot or in case iDRAC is rebooted, it is recommended to check if iDRAC is fully ready by waiting for few seconds of interval with maximum timeout of 5 minutes before using any other command.

## Topics:

- [Setting up managed system location](#)
- [Optimizing system performance and power consumption](#)
- [Setting up management station](#)
- [Configuring supported web browsers](#)
- [Viewing localized versions of web interface](#)
- [Firmware updates](#)
- [Viewing and managing staged updates](#)
- [Rolling back device firmware](#)
- [Easy Restore](#)
- [Monitoring iDRAC using other Systems Management tools](#)
- [Support Server Configuration Profile—Import and Export](#)
- [Secure Boot Configuration from BIOS Settings or F2](#)
- [BIOS recovery](#)
- [Recovering iDRAC](#)

## Setting up managed system location

You can specify the location details of the managed system in the data center using the iDRAC Web interface or iDRAC Settings utility.

### Setting up managed system location using web interface

#### About this task

To specify the system location details:

#### Steps

1. In the iDRAC web interface, go to **System > Details > System Details**.  
The **System Details** page is displayed.
2. Under **System Location**, enter the location details of the managed system in the data center.  
For information about the options, see the **iDRAC Online Help**.
3. Click **Apply**. The system location details are saved in iDRAC.

### Setting up managed system location using RACADM

To specify the system location details, use the `System.Location` group objects.

For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

# Setting up managed system location using iDRAC settings utility

## About this task

To specify the system location details:

## Steps

1. In the iDRAC Settings utility, go to **System Location**.  
The **iDRAC Settings System Location** page is displayed.
2. Enter the location details of the managed system in the data center. For information about the options, see the **iDRAC Settings Utility Online Help**.
3. Click **Back**, click **Finish**, and then click **Yes**.  
The details are saved.

# Optimizing system performance and power consumption

The power required to cool a server can contribute a significant amount to the overall system power. Thermal control is the active management of system cooling through fan speed and system power management to make sure that the system is reliable while minimizing system power consumption, airflow, and system acoustic output. You can adjust the thermal control settings and optimize against the system performance and performance-per-Watt requirements.

Using the iDRAC Web interface, RACADM, or the iDRAC Settings Utility, you can change the following thermal settings:

- Optimize for performance
- Optimize for minimum power
- Set the maximum air exhaust temperature
- Increase airflow through a fan offset, if required
- Increase airflow through increasing minimum fan speed

Following are the list of features in thermal management:

- **System Airflow Consumption:** Displays the real-time system airflow consumption (in CFM), allowing airflow balancing at rack and datacenter level.
- **Custom Delta-T:** Limit air temperature rise from inlet air to exhaust to right-size your infrastructure level cooling.
- **Exhaust Temperature Control:** Specify the temperature limit of the air exiting the server to match your datacenter needs.
- **Custom PCIe inlet temperature:** Choose the right input inlet temperature to match 3rd party device requirements.
- **PCIe Airflow settings:** Provides a comprehensive PCIe device cooling view of the server and allows cooling customization of 3rd party cards.

# Modifying thermal settings using iDRAC web interface

## About this task

To modify the thermal settings:

## Steps

1. In the iDRAC Web interface, go to **Configuration > System Settings > Hardware Settings > Cooling Configuration**.
2. Specify the following:
  - **Thermal Profile Optimization** — Select the thermal profile:
    - **Default Thermal Profile Settings (Minimum Power)** — Implies that the thermal algorithm uses the same system profile settings that are defined under **System BIOS > System BIOS Settings > System Profile Settings** page. By default, this option is set to **Default Thermal Profile Settings**. You can also select a custom algorithm, which is independent of the BIOS profile. The options available are:
      - **Maximum Performance (Performance Optimized)** :
        - Reduced probability of memory or CPU throttling.
        - Increased probability of turbo mode activation.

- Generally, higher fan speeds at idle and stress loads.

**NOTE:** The fan speed does not change even when Maximum Performance is selected for a few specific configurations.

- **Minimum Power (Performance per Watt Optimized):**
  - Optimized for the lowest system power consumption based on optimum fan power state.
  - Generally, lower fan speeds at idle and stress loads.
- **Sound Cap** — Sound Cap provides reduced acoustical output from a server at the expense of some performance. Enabling Sound Cap may include temporary deployment or evaluation of a server in an occupied space, but it should not be used during benchmarking or performance sensitive applications.

**NOTE:** Selecting **Maximum Performance** or **Minimum Power**, overrides thermal settings that are associated to the System Profile setting under **System BIOS > System BIOS Settings.System Profile Settings** page.

- **Maximum Exhaust Temperature Limit** — From the drop-down menu, select the maximum exhaust air temperature. The values are displayed based on the system.

**NOTE:** The default value is **Default, 70°C (158 °F)**.

This option allows the system fans speeds to change such that the exhaust temperature does not exceed the selected exhaust temperature limit. This cannot always be guaranteed under all system operating conditions due to dependency on system load and system cooling capability.

- **Fan Speed Offset** — Selecting this option allows additional cooling to the server. In case hardware is added (example, new PCIe cards), it may require additional cooling. A fan speed offset causes fan speeds to increase (by the offset % value) over baseline fan speeds calculated by the Thermal Control algorithm. Possible values are:
  - **Low Fan Speed** — Drives fan speeds to a moderate fan speed.
  - **Medium Fan Speed** — Drives fan speeds close to medium.
  - **High Fan Speed** — Drives fan speeds close to full speed.
  - **Max Fan Speed** — Drives fan speeds to full speed.
  - **Off** — Fan speed offset is set to off. This is the default value. When set to off, the percentage does not display. The default fan speed is applied with no offset. Conversely, the maximum setting results in all fans running at maximum speed.

**NOTE:**

- The fan speed offset is dynamic and based on the system. The fan speed increase for each offset is displayed next to each option.
- The fan speed offset increases all fan speeds by the same percentage. Fan speeds may increase beyond the offset speeds based on individual component cooling needs. The overall system power consumption is expected to increase.
- Fan speed offset allows you to increase the system fan speed with four incremental steps. These steps are equally divided between the typical baseline speed and the maximum speed of the server system fans. Some hardware configurations results in higher baseline fan speeds, which results in offsets other than the maximum offset to achieve maximum speed.
- The most common usage scenario is nonstandard PCIe adapter cooling. However, the feature can be used to increase system cooling for other purposes.

- **Thresholds**

- **Maximum PCIe Inlet Temperature Limit** — Default value is 55°C. Select the lower temperature of 45°C for third party PCIe cards which require a lower inlet temperature.
- **Exhaust Temperature Limits** — By modifying the values for the following you can set the exhaust temperature limits:
  - **Set Maximum Exhaust Temperature Limit**
  - **Set Air Temperature Rise Limit**
- **Minimum Fan Speed in PWM (% of Max)** — Select this option to fine-tune the fan speed. Using this option, you can set a higher baseline system fan speed or increase the system fan speed if other custom fan speed options are not resulting in the required higher fan speeds.
  - **Default** — Sets minimum fan speed to a default value as determined by the system cooling algorithm.
  - **Custom** — Enter the percentage by which you want to change the fan speed. The range is between 9-100.

**NOTE:**

- The allowable range for minimum fan speed PWM is dynamic based on the system configuration. The first value is the idle speed and the second value is the configuration max (Depending on the system configuration, the maximum speed may be up to 100%).
- For all SAS/SATA storage configurations, fan speed is capped down to 95%.
- System fans can run higher than this speed as per thermal requirements of the system but not lower than the defined minimum speed. For example, setting Minimum Fan Speed at 35% limits the fan speed to never go lower than 35% PWM.
- 0% PWM does not indicate that the fan is off. It is the lowest fan speed that the fan can achieve.
- On servers with multiple zones, a fan failure in any of the zones such as Host Processor Module (HPM) and PCIe Baseboard (PCB) results in all fans running at a maximum speed.

The settings are persistent, which means that once they are set and applied, they do not automatically change to the default setting during system reboot, power cycling, iDRAC, or BIOS updates. The custom cooling options may not be supported on all servers. If the options are not supported, they are not displayed or you cannot provide a custom value.

3. Click **Apply** to apply the settings.

The following message is displayed:

It is recommended to reboot the system when a thermal profile change has been made. This is to ensure all power and thermal settings are activated.

4. Click **Reboot Later** or **Reboot Now**.

**Next steps**

**NOTE:** The fan enablement depends on the relevant thermal configuration (open loop) getting enabled, which again depends on the respective hardware configurations present on the setup. Example: The required rear HDD drives.

**NOTE:** You must reboot the system for the settings to take effect.

## Modifying thermal settings using RACADM

To modify the thermal settings, use the objects in the **system.thermalsettings** group with the **set** sub command as provided in the following table.

**Table 11. Thermal Settings**

| Object         | Description                                                  | Usage                                                                                                                                                                                                                                                                                       | Example                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AirExhaustTemp | Allows you to set the maximum air exhaust temperature limit. | Set to any of the following values (based on the system): <ul style="list-style-type: none"><li>• 0 — Indicates 40°C</li><li>• 1 — Indicates 45°C</li><li>• 2 — Indicates 50°C</li><li>• 3 — Indicates 55°C</li><li>• 4 — Indicates 60°C</li><li>• 255 — Indicates 70°C (default)</li></ul> | <ul style="list-style-type: none"><li>• To check the existing setting on the system:<pre>racadm get system.thermalsettings.AirExhaustTemp</pre></li><li>• The output is:<pre>AirExhaustTemp=70</pre></li><li>• This output indicates that the system is set to limit the air exhaust temperature to 70°C. To set the exhaust temperature limit to 60°C:<pre>racadm set system.thermalsettings.AirExhaustTemp 4</pre></li></ul> |

**Table 11. Thermal Settings (continued)**

| Object                     | Description                                                                                                                                                                                                                                                              | Usage             | Example                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            |                                                                                                                                                                                                                                                                          |                   | <ul style="list-style-type: none"> <li>The output is:<br/> <pre>Object value modified successfully.</pre> </li> <li>If a system does not support a particular air exhaust temperature limit, then when you run the following command:<br/> <pre>racadm set system.thermalsettings.AirExhaustTemp 0</pre> </li> <li>The following error message is displayed:<br/> <pre>ERROR: RAC947: Invalid object value specified.</pre> </li> <li>Make sure to specify the value depending on the type of object. For more information, see RACADM help. To set the limit to the default value:<br/> <pre>racadm set system.thermalsettings.AirExhaustTemp 255</pre> </li> </ul> |
| FanSpeedHighOffsetVal<br>1 | <ul style="list-style-type: none"> <li>Getting this variable reads the fan speed offset value in %PWM for High Fan Speed Offset setting.</li> <li>This value depends on the system.</li> <li>Use FanSpeedOffset object to set this value using index value 1.</li> </ul> | Values from 0-100 | <pre>racadm get system.thermalsettings FanSpeedHighOffsetVal</pre> <p>A numerical value, for example 66, is returned. This value indicates that when you use the following command, it applies a fan speed offset of High (66% PWM) over the baseline fan speed</p> <pre>racadm set system.thermalsettings FanSpeedOffset 1</pre>                                                                                                                                                                                                                                                                                                                                    |
| FanSpeedLowOffsetVal       | <ul style="list-style-type: none"> <li>Getting this variable reads the fan speed offset value in %PWM for Low Fan Speed Offset setting.</li> <li>This value depends on the system.</li> <li>Use FanSpeedOffset object to set this value using index value 0.</li> </ul>  | Values from 0-100 | <pre>racadm get system.thermalsettings FanSpeedLowOffsetVal</pre> <p>This returns a value such as "23". This means that when you use the following command, it applies a fan speed offset of Low (23% PWM) over baseline fan speed.</p> <pre>racadm set system.thermalsettings FanSpeedOffset 0</pre>                                                                                                                                                                                                                                                                                                                                                                |

**Table 11. Thermal Settings (continued)**

| Object                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Usage                                                                                                                                                                                    | Example                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FanSpeedMaxOffsetVal    | <ul style="list-style-type: none"> <li>Getting this variable reads the fan speed offset value in %PWM for Max Fan Speed Offset setting.</li> <li>This value depends on the system.</li> <li>Use FanSpeedOffset to set this value using index value 3</li> </ul>                                                                                                                                                                                                                  | Values from 0-100                                                                                                                                                                        | <pre>racadm get system.thermalsettings FanSpeedMaxOffsetVal</pre> <p>This returns a value such as “100”. This means that when you use the following command, it applies a fan speed offset of Max (meaning full speed, 100% PWM). Usually, this offset results in fan speed increasing to full speed.</p> <pre>racadm set system.thermalsettings FanSpeedOffset 3</pre> |
| FanSpeedMediumOffsetVal | <ul style="list-style-type: none"> <li>Getting this variable reads the fan speed offset value in %PWM for Medium Fan Speed Offset setting.</li> <li>This value depends on the system.</li> <li>Use FanSpeedOffset object to set this value using index value 2</li> </ul>                                                                                                                                                                                                        | Values from 0-100                                                                                                                                                                        | <pre>racadm get system.thermalsettings FanSpeedMediumOffsetVal</pre> <p>This returns a value such as “47”. This means that when you use the following command, it applies a fan speed offset of Medium (47% PWM) over baseline fan speed</p> <pre>racadm set system.thermalsettings FanSpeedOffset 2</pre>                                                              |
| FanSpeedOffset          | <ul style="list-style-type: none"> <li>Using this object with get command displays the existing Fan Speed Offset value.</li> <li>Using this object with set command allows setting the required fan speed offset value.</li> <li>The index value decides the offset that is applied and the FanSpeedLowOffsetVal, FanSpeedMaxOffsetVal, FanSpeedHighOffsetVal, and FanSpeedMediumOffsetVal objects (defined earlier) are the values at which the offsets are applied.</li> </ul> | Values are: <ul style="list-style-type: none"> <li>0 — Low Fan Speed</li> <li>1 — High Fan Speed</li> <li>2 — Medium Fan Speed</li> <li>3 — Max Fan Speed</li> <li>255 — None</li> </ul> | <p>To view the existing setting:</p> <pre>racadm get system.thermalsettings.Fan SpeedOffset</pre> <p><b>NOTE:</b> To set the fan speed offset to High value (as defined in FanSpeedHighOffsetVal)</p> <pre>racadm set system.thermalsettings.Fan SpeedOffset 1</pre>                                                                                                    |
| MFSMaximumLimit         | Read Maximum limit for MFS                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Values from 1 — 100                                                                                                                                                                      | <p>To display the highest value that can be set using MinimumFanSpeed option:</p> <pre>racadm get system.thermalsettings.MFS MaximumLimit</pre>                                                                                                                                                                                                                         |



**Table 11. Thermal Settings (continued)**

| Object                   | Description                                                                                                                                                                                                                                                                                                                                  | Usage                                                                                                                                   | Example                                                                                                                                                                                                                                                  |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MFSMinimumLimit          | Read Minimum limit for MFS                                                                                                                                                                                                                                                                                                                   | Values from 0 to MFSMaximumLimitDefault is 255 (means None)                                                                             | To display the lowest value that can be set using MinimumFanSpeed option.<br><br><pre>racadm get system.thermalsettings.MFSMinimumLimit</pre>                                                                                                            |
| MinimumFanSpeed          | <ul style="list-style-type: none"> <li>Allows configuring the Minimum Fan speed that is required for the system to operate.</li> <li>It defines the baseline (floor) value for fan speed and system allows fans to go lower than this defined fan speed value.</li> <li>This value is %PWM value for fan speed.</li> </ul>                   | Values from MFSMinimumLimit to MFSMaximumLimitWhen get command reports 255, it means user configured offset is not applied.             | To make sure that the system minimum speed does not decrease lower than 45% PWM (45 must be a value between MFSMinimumLimit to MFSMaximumLimit):<br><br><pre>racadm set system.thermalsettings.MinimumFanSpeed 45</pre>                                  |
| ThermalProfile           | <ul style="list-style-type: none"> <li>Allows you to specify the Thermal Base Algorithm.</li> <li>Allows you to set the system profile as required for thermal behavior associated to the profile.</li> </ul>                                                                                                                                | Values: <ul style="list-style-type: none"> <li>0 — Auto</li> <li>1 — Maximum performance</li> <li>2 — Minimum Power</li> </ul>          | To view the existing thermal profile setting:<br><br><pre>racadm get system.thermalsettings.ThermalProfile</pre><br><p><b>NOTE:</b> To set the thermal profile to Maximum Performance:</p> <pre>racadm set system.thermalsettings.ThermalProfile 1</pre> |
| ThirdPartyPCIFanResponse | <ul style="list-style-type: none"> <li>Thermal overrides for third-party PCI cards.</li> <li>Allows you to disable or enable the default system fan response for detected third-party PCI cards.</li> <li>You can confirm the presence of third-party PCI card by viewing the message ID PCI3018 in the Lifecycle Controller log.</li> </ul> | Values: <ul style="list-style-type: none"> <li>1 — Enabled</li> <li>0 — Disabled</li> </ul> <p><b>NOTE:</b> The default value is 1.</p> | To disable any default fan speed response set for a detected third-party PCI card:<br><br><pre>racadm set system.thermalsettings.ThirdPartyPCIFanResponse 0</pre>                                                                                        |

## Modifying thermal settings using iDRAC settings utility

### About this task

To modify the thermal settings:

### Steps

1. In the iDRAC Settings utility, go to **Thermal**.  
The **iDRAC Settings Thermal** page is displayed.

2. Specify the following:

- Thermal Profile
- Maximum Exhaust Temperature Limit
- Fan Speed Offset
- Minimum Fan Speed

The settings are persistent, which means that once they are set and applied, they do not automatically change to the default setting during system reboot, power cycling, iDRAC, or BIOS updates. A few Dell servers may or may not support some or all of these custom user cooling options. If the options are not supported, they are not displayed or you cannot provide a custom value.

3. Click **Back**, click **Finish**, and then click **Yes**.  
The thermal settings are configured.

## Modifying PCIe airflow settings using iDRAC web interface

### About this task

Use the PCIe airflow settings when increased thermal margin is wanted for custom high powered PCIe cards.

 **NOTE:** PCIe airflow settings are not available for M.2 drives that are connected through direct risers or BOSS.

To modify the PCIe airflow settings:

### Steps

1. In the iDRAC Web interface, go to **Configuration > System Settings > Hardware Settings > Cooling Configuration**. The **PCIe Airflow Settings** page is displayed in the fan settings section.
2. Specify the following:
  - **LFM Mode**—Select the **Custom** mode to enable the custom LFM option.
  - **Custom LFM**—Enter the LFM value.

3. Click **Apply** to apply the settings.

The following message is displayed:

It is recommended to reboot the system when a thermal profile change has been made. This is to ensure all power and thermal settings are activated.

Click **Reboot Later** or **Reboot Now**.

 **NOTE:** Reboot the system to apply the settings.

## Setting up management station

### About this task

A management station is a computer used for accessing iDRAC interfaces to remotely monitor and manage the PowerEdge server(s).

To set up the management station:

### Steps

1. Install a supported operating system. For more information, see the release notes.
2. Install and configure a supported Web browser. For more information, see the release notes.
3. From the **Dell Systems Management Tools and Documentation** DVD, install Remote RACADM VMCLI from the SYSMGMT folder. Else, run **Setup** on the DVD to install Remote RACADM by default and other OpenManage software. For more information about RACADM, see [Integrated Dell Remote Access Controller RACADM CLI Guide](#).
4. Install the following based on the requirement:
  - SSH client
  - TFTP
  - Dell OpenManage Essentials

## Accessing iDRAC remotely

To remotely access iDRAC UI from a management station, set the iDRAC NIC to **Dedicated** or **LOM1** to ensure that the management station is in the same network as iDRAC.

To access the managed system's console from a management station, use Virtual Console through iDRAC Web interface.

## Configuring supported web browsers

**NOTE:** For information about the supported browsers and their versions, see the **Release Notes** available at [iDRAC Manuals](#).

Most features of iDRAC web interface can be accessed using these browsers with default settings. For certain feature to work, you must change a few settings. These settings include disabling pop-up blockers, enabling eHTML5 plug-in support and so on.

If you are connecting to iDRAC web interface from a management station that connects to the Internet through a proxy server, configure the web browser to access the Internet through this server.

**NOTE:** If you use Firefox to access the iDRAC web interface, you may need to configure certain settings as described in this section. You can use other supported browsers with their default settings.

**NOTE:** Blank proxy settings are treated equivalent to No proxy.

## Configuring Mozilla Firefox

Configure Firefox to ensure that you can access and use all the features of the iDRAC web interface.

**NOTE:** Mozilla Firefox browser may not have a scroll bar for the iDRAC Online Help page.

## Configuring web browsers to use virtual console

### About this task

**NOTE:** Virtual console uses only eHTML5. Java, and ActiveX are no longer supported.

To use virtual console on your management station:

### Steps

1. Ensure that a supported version of the browser (Microsoft Edge, Mozilla Firefox (Windows or Linux), Google Chrome, Safari) is installed.

**NOTE:** In the RHEL operating system with Mozilla browser, the following are observed during the network drop (removing and re-inserting Network cable):

- Reconnecting message may not be seen in vConsole until the network is up.
- You may see **Login denied** message instead of **Failed to Reconnect** error if the network is down for more than 180s.

**NOTE:** While using the Safari browser, it is recommended to clear the **NSURLSession WebSocket** options if they are selected, and then open the vConsole. For disabling **NSURLSession WebSocket** in Safari by clearing **Safari > Develop > Experimental Features > Experimental Features**.

For more information about the supported browser versions, see the **Release Notes** available at [iDRAC Manuals](#).

**NOTE:** It is recommended to disable the virtual search feature in the Edge browser as a best practice. It is enabled by default. There is a risk of images being searched without your knowledge. Hence, you can disable this behavior by configuring the Edge browser settings.

2. To use Microsoft Edge, set the browser to **Run As Administrator**.

3. Configure the web browser to use eHTML5 plug-in.
4. Import the root certificates on the managed system to avoid the pop-ups that prompt you to verify the certificates.
5. Install the **compat-libstdc++-33-3.2.3-61** related package.  
 **NOTE:** On Windows, the `compat-libstdc++-33-3.2.3-61` related package may be in the .NET framework package or the operating system package.
6. If you are using MAC operating system, select the **Enable access for assistive devices** option in the **Universal Access** window.  
For more information, see the MAC operating system documentation.

## Importing CA certificates to management station

When you launch Virtual Console or Virtual Media, prompts are displayed to verify the certificates. If you have custom Web Server certificates, you can avoid these prompts by importing the CA certificates to the trusted certificate store.

For more information about Automatic Certificate Enrollment (ACE), see [Automatic Certificate Enrollment](#).

## Importing CA certificate to Java trusted certificate store

### About this task

To import the CA certificate to the Java trusted certificate store:

### Steps

1. Launch the **Java Control Panel**.
2. Click **Security** tab and then click **Certificates**.  
The **Certificates** dialog box is displayed.
3. From the Certificate type drop-down menu, select **Trusted Certificates**.
4. Click **Import**, browse, select the CA certificate (in Base64 encoded format), and click **Open**.  
The selected certificate is imported to the Web start trusted certificate store.
5. Click **Close** and then click **OK**. The **Java Control Panel** window closes.

## Viewing localized versions of web interface

iDRAC web interface is supported in the following languages:

- English (en-us)
- French (fr)
- German (de)
- Spanish (es)
- Japanese (ja)
- Simplified Chinese (zh-cn)

The ISO identifiers in parentheses denote the supported language variants. For some supported languages, resizing the browser window to 1024 pixels wide is required to view all features.

iDRAC Web interface is designed to work with localized keyboards for the supported language variants. Some features of iDRAC Web interface, such as Virtual Console, may require additional steps to access certain functions or letters. Other keyboards are not supported and may cause unexpected problems.

 **NOTE:** See the browser documentation on how to configure or setup different languages and view localized versions of iDRAC Web interface.

# Firmware updates

Using iDRAC, you can update the iDRAC itself, the BIOS, and all device firmware components in the system. This enhances performance, security, and compatibility with new software or hardware.

The following firmware and devices are supported for updates:

- Fibre Channel (FC) cards
- Diagnostics
- Operating System Driver Pack
- Network Interface Card (NIC)
- RAID Controller
- Power Supply Unit (PSU)
- Accelerator (GPU)
- NVMe PCIe devices
- SAS/SATA hard drives
- Backplane update for internal and external enclosures
-  **NOTE:** When you update the firmware on the CX7 card, the cards in other slots within the same mux also run in an open loop mode.

 **NOTE:** When upgrading or downgrading firmware for channel devices or hard drives, see the product documentation for brand-specific limitations, including flash limits.

## Firmware image files and supported tools

The following firmware image file extensions are supported for iDRAC updates:

- .exe: Windows-based Dell Update Package (DUP).
- .d10: Contains both iDRAC and Lifecycle Controller firmware.
-  **NOTE:** Only users with Administrator or Operator roles can use these image file types. This is applicable for RACADM update, Redfish Simple update, and iDRAC UI update.

There are multiple tools and interfaces that can be used to update the iDRAC firmware. The following table is applicable only to iDRAC firmware. The table lists the supported interfaces, image-file types, and whether the Lifecycle Controller must be in an enabled state for the firmware to be updated.

**Table 12. Image file types and dependencies**

| .D10 Image                                                                                                                                                                                                                  |           |                     | iDRAC DUP |                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------------|-----------|---------------------|
| Interface                                                                                                                                                                                                                   | Supported | Requires LC enabled | Supported | Requires LC enabled |
| RACADM update (new)                                                                                                                                                                                                         | Yes       | Yes                 | Yes       | Yes                 |
| iDRAC UI                                                                                                                                                                                                                    | Yes       | Yes                 | Yes       | Yes                 |
| In-band operating system DUP                                                                                                                                                                                                | No        | N/A                 | Yes       | No                  |
|  <b>NOTE:</b> After you perform an in-band DUP update, if the update does not get staged to iDRAC, a Multipart: Rollback job is created. |           |                     |           |                     |
| Redfish                                                                                                                                                                                                                     | Yes       | N/A                 | Yes       | N/A                 |
| Diagnostics                                                                                                                                                                                                                 | No        | No                  | No        | No                  |
| Operating system Driver Pack                                                                                                                                                                                                | No        | No                  | No        | No                  |
| iDRAC                                                                                                                                                                                                                       | Yes       | No                  | No*       | Yes                 |

**Table 12. Image file types and dependencies (continued)**

| .D10 Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |           |                     | iDRAC DUP |                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------------|-----------|---------------------|
| Interface                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Supported | Requires LC enabled | Supported | Requires LC enabled |
| BIOS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Yes       | Yes                 | Yes       | Yes                 |
| RAID Controller                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Yes       | Yes                 | Yes       | Yes                 |
| BOSS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Yes       | Yes                 | Yes       | Yes                 |
| NVDIMM                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | No        | Yes                 | Yes       | Yes                 |
| Backplanes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Yes       | Yes                 | Yes       | Yes                 |
|  <b>NOTE:</b> For Expander (Active) backplanes, a system restart is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |           |                     |           |                     |
| Enclosures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Yes       | Yes                 | No        | Yes                 |
| NIC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Yes       | Yes                 | Yes       | Yes                 |
| Power Supply Unit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Yes       | Yes                 | Yes       | Yes                 |
|  <b>NOTE:</b> When a manual reboot is performed or when an update is performed from an operating system, PSU update requires a cold reboot for the update to start.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |           |                     |           | Yes                 |
| FPGA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | No        | Yes                 | Yes       | Yes                 |
|  <b>NOTE:</b> After the FPGA firmware upgrade is complete, iDRAC restarts automatically.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |                     |           |                     |
|  <b>NOTE:</b> Repo update is not supported for FPGA while performing FPGA update alone or FPGA stacked with other updates.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |           |                     |           |                     |
|  <b>NOTE:</b> When an A/C cycle is performed, wait approximately 20 seconds for the power drain to complete to ensure a proper reset of the system. Otherwise, you may encounter the following errors: <ul style="list-style-type: none"> <li>  <b>SWC9016:</b> Unable to authenticate FPGA due to unsuccessful cryptographic authentication or integrity issues.             </li> <li>  <b>SWC9018:</b> Unable to recover the FPGA autorecovery operation due to an internal error.             </li> </ul> |           |                     |           |                     |
| FC Cards                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Yes       | Yes                 | Yes       | Yes                 |
| NVMe PCIe SSD drives                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Yes       | No                  | Yes       | No                  |
| SAS/SATA hard drives                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | No        | Yes                 | Yes       | No                  |
| TPM                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | No        | Yes                 | Yes       | Yes                 |
| Non-SDL Software and Peripherals Application                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | No        | No                  | No        | No                  |

## Considerations for PSU firmware updates

The following are the considerations for firmware updates on a PSU:

-  **CAUTION:** Updating the PSU firmware may take several minutes. Do not interrupt the process or power on the system during the update to avoid damaging the PSU.
- Ensure that all servers in the same chassis are powered off before the update.

## Methods for firmware updates

You can perform firmware updates by using the following methods:

- Upload a supported image type from a local system or a network share.

**NOTE:** When updating device firmware using an image file hosted on a local system or a supported network share, a network speed of at least 5 Mbps is recommended.

- Connect to an FTP, TFTP, HTTP, HTTPS site, or a network repository that contains Windows DUPs and a corresponding catalog file.

**NOTE:** You can create custom repositories by using the Dell Repository Manager. For more information, see the *Dell Repository Manager Data Center User's Guide*. iDRAC can provide a difference report between the BIOS and firmware that is installed on the system and the updates available in the repository. All applicable updates that are contained in the repository are applied to the system. This feature is available with iDRAC Enterprise or Datacenter license.

**NOTE:** Firmware update using FTP fails if the HTTP proxy used is configured without any authentication. Ensure that you change the proxy configuration to allow the CONNECT method to use non-SSL ports. For example, while using a squid proxy, remove the line "http\_access deny CONNECT !SSL\_ports" that restricts using the CONNECT method on non-SSL ports.

**NOTE:** HTTP/HTTPS only supports with either digest authentication or no authentication.

- Schedule recurring automated firmware updates by using the catalog file. Custom repository is also supported.

## LC logs during firmware updates

The following is the behavior of Lifecycle Controller (LC) logs during firmware updates:

- When a firmware update is tried on a hot plugged disk, a duplicate PR7 message is displayed in the LC logs.
- When the job status is in the **Running** and there is no status update from the update modules, the job times out after 6 hours and it is marked as failed.
- If the job status is in the **Running** state, the firmware update job may be marked as failed after an iDRAC reboot.
- After upgrading the iDRAC firmware, you may notice a difference in the timestamp that is displayed in the LC logs. The time that is displayed in the LC log may differ from the NTP/Bios time for a few logs during an iDRAC reset.
- After updating the PERC12 or HBA12 controller firmware using the host operating system, iDRAC may generate the ENC12 and PDR8 messages for drives that are associated with these controllers.
- LC logs are displayed when communication is lost and restored during a GPU firmware update.

## iLKM and SEKM considerations

The following are iLKM and SEKM considerations during firmware updates:

- When iLKM mode is enabled on a controller, iDRAC firmware downgrade or upgrade fails when attempted from an iLKM to a noniLKM iDRAC version. iDRAC firmware upgrade or downgrade passes when performed with the iLKM versions.
- When SEKM mode is enabled on a controller, iDRAC firmware downgrade or upgrade fails when attempted from a SEKM to a non-SEKM iDRAC version. iDRAC firmware upgrade or downgrade passes when performed with the SEKM versions.
- PERC firmware downgrade fails when SEKM or iLKM is enabled.

## Updating firmware using iDRAC web interface

Use the firmware images available on the local system or from a repository on a network share (CIFS, NFS, HTTP, HTTPS, or FTP).

### About this task

Before updating the firmware using single device update method, ensure that you have downloaded the firmware image to a location on the local system. Ensure that the file name for the single component DUP does not have any blank space..

To update single device firmware using iDRAC web interface:

## Steps

1. Go to **Maintenance > System Update**. The **Firmware Update** page is displayed.
2. On the **Update** tab, select **Local** as the **Location Type**.
  - NOTE:** If you select Local, ensure that you download the firmware image to a location on the local system. Select one file to be staged to iDRAC for update. You can select additional files one file at a time, for uploading to iDRAC. The files are uploaded to a temporary space on iDRAC and is limited to approximately 300 MB.
3. Click **Browse**, select the firmware image file for the required component, and then click **Upload**. The required firmware is uploaded to iDRAC.
4. After the upload is complete, the **Update Details** section displays each firmware file that is uploaded to iDRAC and its status.
  - NOTE:** If the firmware image file is valid and was successfully uploaded, the **Contents** column displays a plus icon (  ) icon next to the firmware image file name. Expand the name to view the **Device Name**, **Current**, and **Available firmware version** information.
5. Select the required firmware file and do one of the following:
  - For firmware images that do not require a host system reboot, click **Install** (only available option). For example, the iDRAC firmware file.
  - For firmware images that require a host system reboot, click **Install and Reboot** or **Install Next Reboot**. Updates that require a system reboot are staged and committed to run on the next system reboot. Only one system reboot is required to perform all updates.
  - To cancel the firmware update, click **Cancel**.

When you click **Install**, **Install and Reboot**, or **Install Next Reboot**, the message `Updating Job Queue` is displayed. By default, the auto reboot job is configured for a graceful and forceful shutdown, which may take around 10 minutes to complete.
6. To display the **Job Queue** page, click **Job Queue**. Use this page to view and manage the staged firmware updates or click **OK** to refresh the current page and view the status of the firmware update.
  - NOTE:**
    - If you navigate away from the page without saving the updates, an error message is displayed and all the uploaded content is lost.
    - You cannot proceed further if the session gets expired after uploading the firmware file. This issue can only be resolved by `RACADM reset`.
    - After the firmware update is completed, an error message is displayed - `RAC0508: An unexpected error occurred. Wait for few minutes and retry the operation. If the problem persists, contact service provider..` This is expected. You can wait for sometime and refresh the browser. Then you are redirected to the login page.
7. If the job status is **Completed, Virtual AC Pending**, perform the physical AC power cycle or virtual AC power cycle of the server.
  - NOTE:** Use the Redfish URI (`redfish/v1/Chassis/System.Embedded.1/Actions/Oem/DellOemChassis.ExtendedReset` with `ResetType=PowerCycle` and `FinalState=On/Off` (to either power on the system after the VAC is completed or leave it in the Off state) to perform the virtual power cycle. If any other method is used to perform the AC power cycle, then the real-time job may fail. But the updated version still reflects on the AC power cycle.

## Scheduling automatic firmware updates

You can create a periodic recurring schedule for iDRAC to check for new firmware updates. At the scheduled date and time, iDRAC connects to the specified destination, checks for new updates, and applies or stages all applicable updates. A log file is created on the remote server, which contains information about server access and staged firmware updates.

It is recommended that you create a repository using Dell Repository Manager (DRM) and configure iDRAC to use this repository to check for and perform firmware updates. Using an internal repository enables you to control the firmware and versions available to iDRAC and helps avoid any unintended firmware changes.

- NOTE:** For more information about DRM, see [OpenManage Manuals > Repository Manager](#).



You can schedule automatic firmware updates using the iDRAC web interface or RACADM.

 **NOTE:** IPv6 address is not supported for scheduling automatic firmware updates.

## Updating device firmware using RACADM

To update device firmware using RACADM, use the `update` subcommand. For more information, see the **Integrated Dell Remote Access Controller RACADM CLI** available at [iDRAC Manuals](#).

Examples:

- Upload the update file from a remote HTTP share:

```
racadm update -f <updatefile> -u admin -p mypass -l http://1.2.3.4/share
```

- Upload the update file from a remote HTTPS share:

```
racadm update -f <updatefile> -u admin -p mypass -l https://1.2.3.4/share
```

- To generate a comparison report using an update repository:

```
racadm update -f catalog.xml -l //192.168.1.1 -u test -p passwd --verifycatalog
```

- To perform all applicable updates from an update repository using `myfile.xml` as a catalog file and perform a graceful reboot:

```
racadm update -f "myfile.xml" -b "graceful" -l //192.168.1.1 -u test -p passwd
```

- To perform all applicable updates from an FTP update repository using `Catalog.xml` as a catalog file:

```
racadm update -f "Catalog.xml" -t FTP -e 192.168.1.20/Repository/Catalog
```

## Updating firmware using DUP

### About this task

Before you update firmware using Dell Update Package (DUP), make sure to:

- Install and enable the IPMI and managed system drivers.
- Enable and start the Windows Management Instrumentation (WMI) service if your system is running the Windows operating system,

 **NOTE:** While updating the iDRAC firmware using the DUP utility in Linux, if you see error messages such as `usb 5-2: device descriptor read/64, error -71` displayed on the console, ignore them.

- If the system has an ESX hypervisor installed, then for the DUP file to run, ensure that the "usbarbitrator" service is stopped using the command: `service usbarbitrator stop`

Some versions of DUPs are constructed in ways that conflict with each other. This happens over time as new versions of the software are created. A newer version of software may drop support for legacy devices. Support for new devices may be added. Consider, for example, the two DUPs `Network_Firmware_NDT09_WN64_21.60.5.EXE` and `Network_Firmware_8J1P7_WN64_21.60.27.50.EXE`. The devices supported by these DUPs fit into three groups.

- Group A—legacy devices supported only by NDT09.
- Group B—devices that are supported by both NDT09 and 8J1P7.
- Group C—new devices supported only by 8J1P7.

Consider a server that has one or more devices from each of Groups A, B, and C. If the DUPs are used one at a time, they should be successful. Using NDT09 by itself updates the devices in group A and group B. Using 8J1P7 by itself updates devices in group B and group C. However, if you try to use both DUPs at the same time that may attempt to create two updates for the Group B devices at the same time. That may fail with a valid error: "Job for this device is already present". The update software is unable to resolve the conflict of two valid DUPs attempting two valid updates on the same devices simultaneously. Simultaneously both DUPs are required to support Group A and Group C devices. The conflict extends to performing rollbacks on the devices too. As a best practice, it is recommended to use each DUP individually.

To update iDRAC using DUP:

## Steps

1. Download the DUP based on the installed operating system and run it on the managed system.
2. Run the DUP.  
The firmware is updated. A system restart is not required after firmware update is complete.

# Updating firmware using remote RACADM

## Steps

1. Download the firmware image to the TFTP or FTP server. For example, C:\downloads\firmimg.d10
2. Run the following RACADM command:

TFTP server:

- Using fwupdate command:

```
racadm -r <iDRAC IP address> -u <username> -p <password> fwupdate -g -u -a <path>
```

**path**

the location on the TFTP server where firmimg.d10 is stored.

- Using update command:

```
racadm -r <iDRAC IP address> -u <username> -p <password> update -f <filename>
```

FTP server:

- Using fwupdate command:

```
racadm -r <iDRAC IP address> -u <username> -p <password> fwupdate -f <ftpserver IP>  
<ftpserver username> <ftpserver password> -d <path>
```

**path**

the location on the FTP server where firmimg.d10 is stored.

- Using update command:

```
racadm -r <iDRAC IP address> -u <username> -p <password> update -f <filename>
```

For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

## Rebootless updates

iDRAC supports rebootless updates. This feature enables you to perform firmware update from iDRAC without rebooting the host server to initiate and perform the update in a pre-OPERATING SYSTEM environment. To identify whether the DUP supports sideband update, there are tags to identify if the DUP firmware supports sideband direct update (PLDM, NVMe-MI, so forth) and/or UEFI FMP update method and type of payload present in the DUP.

When iDRAC inventories the components, it decides if the particular component supports direct sideband update or legacy UEFI FMP-based update, and requires host reboot or not.

 **NOTE:** Some devices like Network adapters may need a power cycle for a firmware update.

Two properties specific to PLDM firmware update capabilities are listed in the software inventory -

**PLDMCapabilitiesDuringUpdate** and **PLDMFDPCapabilitiesDuringUpdate**. These parameters are available only for devices that support PLDM firmware update.

 **NOTE:** The PLDM-based update feature is supported only on Platforms with iDRAC 1GB memory.

The iDRAC/ LC update modules handle the reboot or rebootless methods based on the support. Following are the different update methods:

- Sideband Direct update reboot identified at runtime
- Sideband Direct updates no reboot.

- SSM (UEFI FMP based)/ SMA-based update
- Redstone FPGA update from iDRAC
- FPGA update from iDRAC

**NOTE:** In case of repository updates, the application updates that do not require host reboot must be performed immediately.

**NOTE:** For direct updates (realtime FW updates) from iDRAC, there is an LCLOG (SUP200, SUP0518, SUP516) with a device description (friendly FQDD information) instead of the Product description.

**NOTE:** When NVMe drives are behind PERC (in the front) and directly attached in the rear backplane or the opposite way, then rebootless update does not work for the direct attached drives.

## Viewing and managing staged updates

You can view and delete the scheduled jobs including configuration and update jobs. This is a licensed feature. All jobs queued to run during the next reboot can be deleted.

**NOTE:** When any update or other tasks and jobs are in-progress, do not reboot or shutdown or AC power cycle the host or iDRAC by any mode (manual or "Ctrl+Alt+Del" keys or other through iDRAC interfaces). The system (host and iDRAC) should always be rebooted/shutdown gracefully when no tasks or jobs are running in iDRAC or host. Ungraceful shutdown or interrupting an operation can cause unpredictable results such as firmware corruption, generate core files, RSODs, YSODs, error events in LCL, so on.

**NOTE:** For any update requiring iDRAC reset/ reboot or in case iDRAC is rebooted, it is recommended to check if iDRAC is fully ready by waiting for few seconds of interval with maximum timeout of 5 minutes before using any other command.

## Viewing and managing staged updates using RACADM

To view the staged updates using RACADM, use `jobqueue` sub-command. For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

## Viewing and managing staged updates using iDRAC web interface

To view the list of scheduled jobs using iDRAC web interface, go to **Maintenance > Job Queue**. The **Job Queue** page displays the status of jobs in the Lifecycle Controller job queue. For information about the displayed fields, see the **iDRAC Online Help**.

To delete job(s), select the job(s) and click **Delete**. The page is refreshed and the selected job is removed from the Lifecycle Controller job queue. You can delete all the jobs queued to run during the next reboot. You cannot delete active jobs, that is, jobs with the status **Running** or **Downloading**.

You must have Server Control privilege to delete jobs.

## Rolling back device firmware

You can roll back the firmware for iDRAC even if the upgrade was previously performed using another interface. You can perform firmware rollback for multiple devices with one system reboot.

It is recommended to keep the firmware updated to ensure that you have the latest features and security updates. Roll back an update or install an earlier version if you encounter any issues after an update.

You can perform firmware rollback for the following components:

- Integrated Dell Remote Access Controller
- BIOS
- Network Interface Card (NIC)
- Power Supply Unit (PSU)
- Storage Controller
- Backplane

- Software Defined Persistent Memory (SDPM)

 **NOTE:** You cannot perform firmware rollback for Diagnostics, Driver Packs, and FGPA.

Before rolling back the firmware, ensure that:

- You have Configure privilege to roll back iDRAC firmware.
- Change the NIC mode to **Dedicated** if the mode is set as **Shared LOM**.

You can roll back the firmware to the previously installed version using any of the following methods:

- iDRAC web interface
- iDRAC RACADM CLI
- Redfish API

## Rollback firmware using iDRAC web interface

### About this task

To roll back device firmware:

#### Steps

1. In the iDRAC Web interface, go to **Maintenance > System Update > Rollback**.  
The **Rollback** page displays the devices for which you can rollback the firmware. You can view the device name, associated devices, currently installed firmware version, and the available firmware rollback version.
2. Select one or more devices for which you want to rollback the firmware.
3. Based on the selected devices, click **Install and Reboot** or **Install Next Reboot**. If only iDRAC is selected, then click **Install**.  
When you click **Install and Reboot** or **Install Next Reboot**, the message “Updating Job Queue” is displayed.
4. Click **Job Queue**.  
The **Job Queue** page is displayed, where you can view and manage the staged firmware updates.

 **NOTE:**

- While in rollback mode, the rollback process continues in the background even if you navigate away from this page.

An error message appears if:

- You do not have Server Control privilege to rollback any firmware other than the iDRAC or Configure privilege to rollback iDRAC firmware.
- Firmware rollback is already in-progress in another session.
- Updates are staged to run or already in running state.

If Lifecycle Controller is disabled or in recovery state and you try to perform a firmware rollback for any device other than iDRAC, an appropriate warning message is displayed along with steps to enable Lifecycle Controller.

## Rollback firmware using RACADM

### Steps

1. Check the rollback status and the FQDD using the `swinventory` command:

```
racadm swinventory
```

For the device for which you want to rollback the firmware, the `Rollback Version` must be `Available`. Also, note the FQDD.

2. Rollback the device firmware using:

```
racadm rollback <FQDD>
```

For more information, see [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

# Easy Restore

Easy Restore uses the Easy Restore flash memory to back up the data. When you replace the Secure Control Module (SCM) and power on the system, the BIOS queries the iDRAC and prompts you to restore the backed-up data. The first BIOS screen prompts you to restore the Service Tag, licenses, and UEFI diagnostic application. The second BIOS screen prompts you to restore the system configuration settings. If you choose not to restore data on the first BIOS screen and do not set the Service Tag by another method, then the first BIOS screen is displayed again. The second BIOS screen is displayed only once.

## NOTE:

- System configurations settings are backed-up only when Collect System Inventory On Reboot (CSIOR) is enabled. Ensure that Lifecycle Controller and CSIOR are enabled.
- Easy Restore does not back up other data such as firmware images or add-in cards data.
- Replacing the Host Processor Memory Module (HPM) does not initiate the Easy Restore process.

 **NOTE:** While replacing SCM, you have to manually select **Liquid cooled** or **Air cooled**. Incorrect selection of these options leads to thermal issues in the platform and in such cases contact Dell Tech Support for the recovery.

After you replace the system board on your server, Easy Restore allows you to automatically restore the following data:

- System Service Tag
- Asset Tag
- Licenses data
- UEFI Diagnostics application
- System configuration settings—BIOS, iDRAC
- System Event Log (SEL)
- OEM ID Module

The following are the time duration details that are required for some restoring actions:

- Restoring system contents such as Diagnostics, System Event Log (SEL), and OEM ID Module typically takes less than a minute.
- Restoring system configuration data (iDRAC, BIOS) may take several minutes (sometimes around 10 minutes) to complete.

 **NOTE:** During this time, there is no indication or progress bar and the server might be rebooted a couple of times to finish restoring the configuration.

# Monitoring iDRAC using other Systems Management tools

You can discover and monitor iDRAC using OpenManage Enterprise. You can also use Dell Remote Access Configuration Tool (DRACT) to discover iDRACs, update firmware, and set up Active Directory. For more information, see the respective user's guides.

# Support Server Configuration Profile—Import and Export

Server Configuration Profile (SCP) allows you to import and export server configuration files.

 **NOTE:** You need admin privileges to perform Export and Import SCP task.

 **NOTE:** Enterprise or Datacenter license is required to perform SCP import. With Core license you can perform only SCP export.

You can import and export from a local management station, and from a Network Share using CIFS, NFS, HTTP, or HTTPS. Using SCP, you can select and import or export component-level configurations for BIOS, NIC, and RAID. You can import and export SCP to the local management station or to a CIFS, NFS, HTTP, or HTTPS network share. You can either import and export individual profiles of iDRAC, BIOS, NIC, and RAID, or all of them together as a single file.

You can specify preview import or export of the SCP where the job is running and a configuration result is generated but none of the configuration has applied.

A job is created once the import or export is initiated through the UI. The status of the jobs can be viewed on the Job Queue page.

**NOTE:**

- Only Host Name or IP Addresses are accepted for destination address.
- You can browse to a specific location to import the server configuration files. Select the correct server configuration file that you want to import. For example, import.xml.
- Depending on the exported file format (that you selected), the extension is added automatically. For example, export\_system\_config.xml.
- While exporting, the SCP file name may change. For example con.xml to \_con.xml.
- SCP applies the full configuration in a single job with a minimal number of reboots. However, in a few system configurations some attributes change the operation mode of a device or may create subdevices with new attributes. When this occurs, SCP may be unable to apply all settings during a single job. Review the ConfigResult entries for the job to resolve any pending configuration settings.
- To ensure that user roles are imported accurately, set the **PasswordHashSaltset** value, and set the **Set On Import** value to **True** in the SCP export file.

SCP allows you to perform operating system deployment (OSD) using a single xml/json file across multiple systems. You can also perform existing operations such as configurations and repository updates all at once.

SCP also allows to export and import SSH public keys for all iDRAC users. There are four SSH public keys for all users.

The following are the steps for operating system deployment using SCP:

1. Export the SCP file. The SCP file contains all the suppressed attributes that are required to perform OSD.
2. Edit or update the OSD attributes, and then perform the import operation.
3. SCP orchestrator validates these OSD attributes.
4. SCP orchestrator performs the configuration and repository updates that are specified in the SCP file.
5. After the configuration and updates are done, the host operating system shutdowns.

**NOTE:** Only CIFS and NFS share are supported for hosting operating system media.

6. SCP orchestrator initiates the OSD by attaching the drivers for the selected operating system, and then initiates one-time boot to the operating system media present in the CIFS or NFS share.
7. LCL shows the progress of the job.
8. After the BIOS boots to the operating system media, the SCP job shows as Complete.
9. The attached operating system media is automatically detached after 65,535 seconds or after the duration specified in the OSD.1#ExposeDuration attribute.

For detailed information about the overall feature along with deployment workflow, see [Using Server Configuration Profiles to Deploy Operating Systems to Dell PowerEdge Servers](#).

To know more about SCP, see the [Server Configuration Profiles: Reference Guide](#).

**NOTE:** For detailed instructions on how to import iDRAC SCP into OpenManage Enterprise, see the OpenManage Enterprise documentation available on the Dell Support site.

## Importing server configuration profile using iDRAC web interface

### Prerequisites

Before importing the SCP file, it is recommended to perform the Import Preview operation. This operation identifies any potential formatting issues or invalid attribute settings without impacting the state of the server.

### About this task

To import the server configuration profile:

## Steps

1. Go to **Configuration > Server Configuration Profile**.  
The **Server Configuration Profile** page is displayed.
2. Select one of the following to specify the location type:
  - **Local** to import the configuration file saved in a local drive.
  - **Network Share** to import the configuration file from CIFS or NFS share.
  - **HTTP or HTTPS** to import the configuration file from a local file using HTTP or HTTPS file transfer.

 **NOTE:** Depending on the location type, you must enter the Network Settings or HTTP or HTTPS settings. If the proxy is configured for HTTP or HTTPS, proxy settings are also required.
3. Select the components listed in **Import Components** option.
4. Select the **Shutdown** type.
5. Select the **Maximum wait time** to specify the wait time before the system shuts down after the import is complete.
6. Click **Import**.

## Exporting server configuration profile using iDRAC web interface

### About this task

To export the server configuration profile:

## Steps

1. Go to **Configuration > Server Configuration Profile**.  
The **Server Configuration Profile** page is displayed.
2. Click **Export**.
3. Select one of the following to specify the location type:
  - **Local** to save the configuration file on a local drive.
  - **Network Share** to save the configuration file on a CIFS or NFS share.
  - **HTTP or HTTPS** to save the configuration file to a local file using HTTP/HTTPS file transfer.

 **NOTE:** Depending on the location type, you must enter the Network Settings or HTTP/HTTPS settings. If the proxy is configured for HTTP/HTTPS, proxy settings are also required.
4. Select the components that you must back up the configuration for.
5. Select the **Export type**, following are the options:
  - **Basic:** Creates a nondestructive snapshot of the configuration.
  - **Replacement Export :** Replaces the server settings with new settings or restores server settings to a known baseline.
  - **Clone Export:** Clones the settings from one server to another server with identical hardware. All settings except I/O identity are updated. The settings in this export are destructive when uploaded to another system.
6. Select an **Export file format**.
7. Select **Additional export items**.
8. Click **Export**.

## Secure Boot Configuration from BIOS Settings or F2

UEFI Secure Boot is a technology that eliminates a major security void that may occur during a handoff between the UEFI firmware and UEFI operating system (operating system). In UEFI Secure Boot, each component in the chain is validated and authorized against a specific certificate before it is allowed to load or run. Secure Boot removes the threat and provides software identity checking at every step of the boot—Platform firmware, Option Cards, and operating system BootLoader.

The Unified Extensible Firmware Interface (UEFI) Forum—an industry body that develops standards for pre-boot software—defines Secure Boot in the UEFI specification. Computer system vendors, expansion card vendors, and operating system providers collaborate on this specification to promote interoperability. As a portion of the UEFI specification, Secure Boot represents an industry-wide standard for security in the pre-boot environment.

When enabled, UEFI Secure Boot prevents the unsigned UEFI device drivers from being loaded, displays an error message, and does not allow the device to function. You must disable Secure Boot to load the unsigned device drivers.

## Acceptable file formats

The Secure Boot policy contains only one key in PK, but multiple keys may reside in KEK. Ideally, either the platform manufacturer or platform owner maintains the private key corresponding to the public PK. Third parties (such as operating system providers and device providers) maintain the private keys corresponding to the public keys in KEK. In this way, platform owners or third parties may add or remove entries in the db or dbx of a specific system.

The Secure Boot policy uses db and dbx to authorize pre-boot image file execution. To run an image file, associate it with a key or hash value in db, and not associate with a key or hash value in dbx. Any attempts to update the contents of db or dbx must be signed by a private PK or KEK. Any attempts to update the contents of PK or KEK must be signed by a private PK.

**Table 13. Acceptable file formats**

| Policy Component  | Acceptable File Formats                                                                                   | Acceptable File Extensions               | Max records allowed |
|-------------------|-----------------------------------------------------------------------------------------------------------|------------------------------------------|---------------------|
| <b>PK</b>         | X.509 Certificate (binary DER format only)                                                                | 1. .cer<br>2. .der<br>3. .crt            | One                 |
| <b>KEK</b>        | X.509 Certificate (binary DER format only) Public Key Store                                               | 1. .cer<br>2. .der<br>3. .crt<br>4. .pbk | More than one       |
| <b>DB and DBX</b> | X.509 Certificate (binary DER format only) EFI image (system BIOS will calculate and import image digest) | 1. .cer<br>2. .der<br>3. .crt<br>4. .efi | More than one       |

The Secure Boot Settings feature can be accessed by clicking System Security under System BIOS Settings. To go to System BIOS Settings, press F2 when the company logo is displayed during POST.

- By default, Secure Boot is Disabled and the Secure Boot policy is set to Standard. To configure the Secure Boot Policy, you must enable Secure Boot.
- When the Secure Boot mode is set to Standard, it indicates that the system has default certificates and image digests or hash loaded from the factory. This caters to the security of standard firmware, drivers, option-roms, and boot loaders.
- To support a new driver or firmware on a server, the respective certificate must be enrolled into the database of Secure Boot certificate store. Therefore, the Secure Boot Policy must be configured to Custom.

When the Secure Boot Policy is configured as Custom, it inherits the standard certificates and image digests loaded in the system by default, which you can modify. Secure Boot Policy configured as Custom allows you to perform operations such as View, Export, Import, Delete, Delete All, Reset, and Reset All. Using these operations, you can configure the Secure Boot Policies.

Configuring the Secure Boot Policy to Custom enables the options to manage the certificate store by using various actions such as Export, Import, Delete, Delete All, Reset, and Rest All on PK, KEK, database, and DBX. You can select the policy (PK/KEK/DB/DBX) on which you want to make the change and perform appropriate actions by clicking the respective link. Each section has links to perform the Import, Export, Delete, and Reset operations. Links are enabled based on what is applicable, which depends on the configuration at the time. Delete All and Reset All are the operations that have an impact on all the policies. Delete All deletes all the certificates and image digests in the Custom policy, and Rest All restores all the certificates and image digests from the Standard or Default certificate store.



# BIOS recovery

The BIOS recovery feature allows you to manually recover the BIOS from a stored image. The BIOS is checked when the system is powered on and if a corrupt or compromised BIOS is detected, an error message is displayed. You can then initiate the process of BIOS recovery using RACADM and Redfish. To perform a manual BIOS recovery, see the iDRAC RACADM Command Line Interface Reference Guide available at [iDRAC Manuals](#).

## Recovering iDRAC

iDRAC supports two operating system images so that a bootable iDRAC is always available. During an unforeseen catastrophic error when you lose both boot paths, the iDRAC bootloader detects that there is no bootable image. The Bootloader displays the early boot video messages in the connected monitor.

### About this task

Perform the following steps to recover iDRAC:

### Steps

1. Format the USB drive with FAT32 using the Windows operating system, or EXT3 or EXT4 using Linux operating system.
2. Copy the **firmimg.d10** or DUP.exe to the USB drive location **/scm/images/** .
3. Insert the USB drive to the rear-top USB port of the server and perform the AC Power Cycle on the server. The bootloader detects the USB drive, reads the payload, reprograms iDRAC, and then reboots iDRAC.

# Data Processing Unit (DPU)

A Data Processing Unit (DPU) is a system on a chip that consists of ARM cores, a NIC ASIC, and acceleration engines. A DPU is programmable and potentially capable of running an operating system. DPUs combine network connectivity with CPU cores independent from the Hypervisor or operating system, allowing acceleration and offload services. DPUs distinguish themselves from traditional offload engines by their flexibility, programmability, and ability to host a wide variety of services. Mellanox BF3 and Intel IPU cards are supported.

 **NOTE:** DPUs require an iDRAC10 Enterprise or Datacenter license.

Use of DPU offers the following advantages:

- Isolates infrastructure services from the host operating system and applications.
- Enables an environment to deliver new services independent of the host application environment.
- Enables hardware acceleration to perform data-intensive operations at wire-speed.
- Free up server/x86 CPU cores for enabling customer applications single socket and small form factor edge platforms

After the DPU operating system is booted, additional PCIe Functions can be initialized. So, the BIOS PCIe enumeration (and host Operating System/Hypervisor boot process) shall happen only after DPU operating system has booted or ready.

Points to consider about DPU:

- On an unsupported DPU platform, while performing a system erase, the DPU LC logs display the SYS560 message (*some of the DPU devices failed to reset*). On a supported DPU platform, if the DPU is not present and a system erase is performed, the logs display the SYS564 message (*unable to perform system erase of DPU because there is no DPU available in the system*).
- When the NVIDIA BF3 cards are disabled from the BIOS configuration, the **Status** in the **Network Devices** page. (**System > Overview > Network Devices > Summary**) in iDRAC UI is displayed in Green.
- When the NVIDIA BF3 DPU cards PCIe slots are **Boot Driver Disabled** in the BIOS configuration (**System BIOS Settings > Integrated Devices > Slot Disablement**), PR7 and PR8 logs are displayed in the iDRAC LC logs.
- The device hardware inventory is displayed for the NVIDIA BF3 cards when the PCIe slot is **Boot Drive Disabled** from the BIOS configuration.

The following are the features of DPU:

- Other PCIe functions that are exposed by DPU operating system are enumerated by BIOS and reported in the iDRAC Hardware Inventory.

## Inventory and Monitoring of DPUs

iDRAC system inventory provides the make and model of the DPU while monitoring the health of the DPU cores, peripherals, and the installed operating system. GET is used to retrieve inventory information. This action ensures that no unauthorized devices are installed maliciously. Using the GET operation, you can periodically check the DPU health. If the system is healthy, it returns a payload response and status update to provide health updates.

To detect malicious or accidental DPU operating system installations use GET operation. Using the GET operation that you can retrieve operating system name, vendor name, version, and status of the DPU operating system.

You can view the installed DPU from iDRAC UI (**System > Inventory > Firmware**) **Inventory**

## Redfish

Using Redfish, you can set a one-time boot configuration which is used to boot the DPU with the configured value once on reboot. On the next reboot, the DPU boot is based on the Boot Order that is configured. Redfish also allows ARM-UEFI and BMC firmware updates. For more information, see [developer.dell.com](https://developer.dell.com).

## Serial Console

To access serial control using RACADM, log in to iDRAC SSH and run the command `Racadm> console dpu`

# Plugin Management

Plugins are software components that extend the functionality of iDRAC. Plugins are individually packaged in a DUP. Plugins are not deleted on iDRAC reboot, reset, or AC cycles. The iDRAC sanitize operation or LC wipe operation is used to remove the plugins. You can enable or disable the plugins.

To manage plugins from the iDRAC UI, go to **iDRAC Settings > Settings > Plugins**.

 **NOTE:** You must have login, control and configure privileges to install, update, and remove the plugins. You can only view the installed plugins with login privileges.

## Topics:

- [Installing a plugin](#)
- [Uninstalling a plugin](#)
- [Restarting a plugin](#)
- [Enabling or disabling a plugin](#)
- [Viewing the plugin details](#)

## Installing a plugin

Install a plugin if you want to extend the functionality of iDRAC. Some plugins are pre-installed into iDRAC from Dell factory for PowerEdge servers.

### About this task

 **NOTE:** For more information about the Dell Connectivity Client plugin, see the Dell Connectivity Client Overview document at the [Dell Support](#) site.

When a Non-Standard Device List (Non-SDL) card is installed, iDRAC cannot detect an SDK plugin. Manually search and install the SDK plug-in. iDRAC firmware update, downgrade, or rollback does not have an impact on the functionality of plug-ins.

### Steps

1. Download the plugin from Dell.com.
2. Go to **iDRAC Settings > Settings > Plugins**.
3. Click **Add/Update**.
4. Select the **Location Type** and click **Choose File** and select the plugin file.
5. Click **Upload**.  
If a plugin is valid, a success message is displayed after the plugin installed. If the hardware is not present, an LC message is logged indicating that the plugin is not started. If the plugin is invalid, an error message is displayed.

## Uninstalling a plugin

### Steps

1. Go to **iDRAC Settings > Settings > Plugins**.
2. Select the plugin and click **Uninstall**.  
The selected plugin is uninstalled.

## Restarting a plugin

You can restart a plugin that is installed in iDRAC.

### Steps

1. Go to **iDRAC Settings > Settings > Plugins**.
2. Click **Restart**.

## Enabling or disabling a plugin

You can enable or disable a plugin.

### Steps

1. Go to **iDRAC Settings > Settings > Plugins**
2. Select the plugin and click **Enable** or **Disable**.

## Viewing the plugin details

You can view the details of the installed plugins.

### Steps

1. Go to **iDRAC Settings > Settings > Plugins**.
2. Select the plugin and click **Details**.  
The details of the plugin are displayed.

# Configuring iDRAC

## About this task

iDRAC enables you to configure iDRAC properties, set up users, and set up alerts to perform remote management tasks.

Before you configure iDRAC, make sure that the iDRAC network settings and a supported browser is configured, and the required licenses are updated. For more information about the licensable feature in iDRAC, see [iDRAC licenses](#).

You can configure iDRAC using:

- iDRAC Web Interface
- RACADM
- IPMITool (see **Baseboard Management Controller Management Utilities User's Guide**)

**NOTE:** When any task or job is in-progress, do not reboot or shutdown or AC power cycle the host or iDRAC by any mode (manual or "Ctrl+Alt+Del" keys or options provided through iDRAC interfaces). The system (host and iDRAC) should always be rebooted or shutdown gracefully when no tasks or jobs are running in iDRAC or host. Ungraceful shutdown or interrupting an operation can cause unpredictable results such as firmware corruption, generate core files, RSODs, YSODs, error events in LCL, and so on.

To configure iDRAC:

## Steps

1. Log in to iDRAC.
2. Modify the network settings if required.
 

**NOTE:** If you have configured iDRAC network settings, using iDRAC Settings utility during iDRAC IP address setup, then ignore this step.
3. Configure interfaces to access iDRAC.
4. Configure front panel display.
5. Configure System Location if required.
6. Configure time zone and Network Time Protocol (NTP) if required.
7. Establish any of the following alternate communication methods to iDRAC:
  - IPMI or RAC serial
  - IPMI serial over LAN
  - IPMI over LAN
  - SSH
8. Obtain the required certificates.
9. Add and configure iDRAC users with privileges.
10. Configure and enable email alerts, SNMP traps, or IPMI alerts.
11. Set the power cap policy if required.
12. Enable the Last Crash Screen.
13. Configure virtual console and virtual media if required.
14. Set the first boot device if required.
15. Set the OS to iDRAC Pass-through if required.

## Topics:

- [Viewing iDRAC information](#)
- [Modifying network settings](#)
- [Cipher suite selection](#)
- [FIPS mode](#)
- [Configuring services](#)

- Using VNC client to manage remote server
- Configuring time zone and NTP
- Setting first boot device
- Enabling or disabling OS to iDRAC Pass-through
- Obtaining certificates
- Configuring multiple iDRACs using RACADM
- Disabling access to modify iDRAC configuration settings on host system

## Viewing iDRAC information

You can view the basic properties of iDRAC.

### Viewing iDRAC information using web interface

In the iDRAC Web interface, go to **iDRAC Settings > Overview** to view the following information related to iDRAC. For information about the properties, see **iDRAC Online Help**.

#### iDRAC Details

- Device Type
- Hardware Version
- Firmware Version
- Firmware Update
- RAC time
- IPMI version
- Number of Possible Sessions
- Number of Current Sessions
- IPMI Version

#### iDRAC Service Module

- Status

#### Connection View

- State
- Switch Connection ID
- Switch Port Connection ID

#### Current Network Settings

- iDRAC MAC Address
- Active NIC Interface
- DNS Domain Name

#### Current IPv4 Setting

- IPv4 Enabled
- DHCP
- Current IP Address
- Current Subnet Mask
- Current Gateway
- Use DHCP to Obtain DNS Server Address
- Current Preferred DNS Server
- Current Alternate DNS Server

#### Current IPv6 Settings

- IPv6 Enable
- Autoconfiguration
- Current IP Address
- Current IP Gateway
- Link Local Address

- Use DHCPv6 to obtain DNS
- Current Preferred DNS Server
- Current Alternate DNS Server

## Viewing iDRAC information using RACADM

To view iDRAC information using RACADM, see `getsysinfo` or `get` sub-command details provided in the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

## Modifying network settings

After configuring the iDRAC network settings using the iDRAC Settings utility, you can also modify the settings through the iDRAC Web interface, RACADM, and Server Administrator (after booting to the operating system). For more information about the tools and privilege settings, see the respective user's guides.

To modify the network settings using iDRAC Web interface or RACADM, you must have **Configure** privileges.

 **NOTE:** Changing the network settings may terminate the current network connections to iDRAC.

## Modifying network settings using local RACADM

To generate a list of available network properties, use the command

```
racadm get iDRAC.Nic
```

To use DHCP to obtain an IP address, use the following command to write the object `DHCPEnable` and enable this feature.

```
racadm set iDRAC.IPv4.DHCPEnable 1
```

The following example shows how the command may be used to configure the required LAN network properties:

```
racadm set iDRAC.Nic.Enable 1
racadm set iDRAC.IPv4.Address 192.168.0.120
racadm set iDRAC.IPv4.Netmask 255.255.255.0
racadm set iDRAC.IPv4.Gateway 192.168.0.120
racadm set iDRAC.IPv4.DHCPEnable 0
racadm set iDRAC.IPv4.DNSFromDHCP 0
racadm set iDRAC.IPv4.DNS1 192.168.0.5
racadm set iDRAC.IPv4.DNS2 192.168.0.6
racadm set iDRAC.Nic.DNSRegister 1
racadm set iDRAC.Nic.DNSRacName RAC-EK00002
racadm set iDRAC.Nic.DNSDomainFromDHCP 0
racadm set iDRAC.Nic.DNSDomainName MYDOMAIN
```

 **NOTE:** If `iDRAC.Nic.Enable` is set to **0**, the iDRAC LAN is disabled even if DHCP is enabled.

## Modifying network settings using web interface

### About this task

To modify the iDRAC network settings:

### Steps

1. In the iDRAC Web interface, go to **iDRAC Settings > Connectivity > Network > Network Settings**. The **Network** page is displayed.
2. Specify the network settings, common settings, IPv4, IPv6, IPMI, and/or VLAN settings as per your requirement and click **Apply**.

If you select **Auto Dedicated NIC** under **Network Settings**, when the iDRAC has its NIC Selection as shared LOM (1, 2, 3, or 4) and a link is detected on the iDRAC dedicated NIC, the iDRAC changes its NIC selection to use the dedicated NIC. If no link is detected on the dedicated NIC, then the iDRAC uses the shared LOM. The switch from shared to dedicated time-out is five seconds and from dedicated to shared is 30 seconds. You can configure this time-out value using RACADM.

For information about the various fields, see the **iDRAC Online Help**.

**NOTE:** If the iDRAC is using DHCP and has obtained a lease for its IP address, it is released back to the DHCP server's address pool when NIC or Ipv4 or DHCP is disabled.

## Cipher suite selection

Cipher Suite Selection can be used to limit the ciphers in iDRAC or client communications and determine how secure the connection will be. It provides another level of filtering the effective in-use TLS Cipher Suite. These settings can be configured through iDRAC web interface and RACADM command line interfaces.

### Configuring cipher suite selection using RACADM

To configure cipher suite selection using RACADM, use any one of the following commands:

- `racadm set idrac.webServer.customCipherString ALL:!DHE-RSA-AES256-GCM-SHA384:!DHE-RSA-AES256-GCM-SHA384`
- `racadm set idrac.webServer.customCipherString ALL:-DHE-RSA-CAMELLIA256-SHA`
- `racadm set idrac.webServer.customCipherString ALL:!DHE-RSA-AES256-GCM-SHA384:!DHE-RSA-AES256-SHA256:+AES256-GCM-SHA384:-DHE-RSA-CAMELLIA256-SHA`

For more information about these objects, see **iDRAC RACADM Command Line Interface Reference Guide** available on the [iDRAC Manuals](#) page.

### Configuring Cipher Suite selection using iDRAC web interface

#### Prerequisites

**CAUTION:** Using the OpenSSL Cipher command to parse strings with invalid syntax may lead to unexpected errors.

**NOTE:** This is an advanced security option. Before you configure this option, ensure that you have thorough knowledge of the following:

- The OpenSSL Cipher String Syntax and its use.
- Tools and Procedures to validate the resultant Cipher Suite configuration to ensure that the results align with the expectations and requirements.

**NOTE:** Before you configure the Advanced Settings for TLS Cipher Suites, ensure that you are using a supported web browser.

**NOTE:** Irrespective of the TLS version configured in iDRAC, the Firefox browser in RHEL allows you to launch the iDRAC UI.

**NOTE:** To obtain the list of ciphers for a specific port, run the nmap tool.

#### About this task

To add custom cipher strings:

#### Steps

1. In the iDRAC web interface, go to **iDRAC Settings > Services > Web Server**.
2. Click **Set Cipher String** under the **Customer Cipher String** option.  
The **Set Custom Cipher String** page is displayed.



3. In the **Custom Cipher String** field, enter a valid string and click **Set Cipher String**.

 **NOTE:**

- For more information about cipher strings, see the [OpenSSL](#) page.
- TLS 1.3 is not supported.

4. Click **Apply**.

Setting the custom cipher string terminates the current iDRAC session. Wait for a few minutes before you open a new iDRAC session.

## FIPS mode

FIPS is a computer security standard that United States government agencies and contractors must use. iDRAC supports enabling FIPS mode.

iDRAC will be officially certified to support FIPS mode in the future.

## Difference between FIPS-mode supported and FIPS-validated

Software that has been validated by completing the Cryptographic Module Validation Program is referred to as FIPS-validated. Because of the time it takes to complete FIPS-validation, not all versions of iDRAC are validated. For information about the latest status of FIPS-validation for iDRAC, see the Cryptographic Module Validation Program page on the NIST website.

## Disabling FIPS mode

To disable FIPS mode, you must reset iDRAC to the factory-default settings.

## Enabling FIPS Mode

 **CAUTION:** Enabling FIPS mode resets iDRAC to factory-default settings. If you want to restore the settings, back up the server configuration profile (SCP) before you enable FIPS mode, and restore the SCP after iDRAC restarts.

## Configuring services

You can configure and enable the following services on iDRAC:

|                                        |                                                                                                                                                                           |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Local Configuration</b>             | Disable access to iDRAC configuration (from the host system) using Local RACADM and iDRAC Settings utility.                                                               |
| <b>Web Server</b>                      | Enable access to iDRAC web interface. If you disable the web interface, remote RACADM also gets disabled. Use local RACADM to re-enable the web server and remote RACADM. |
| <b>SEKM Configuration</b>              | Enables secure enterprise key management functionality on iDRAC using a client server architecture.                                                                       |
| <b>SSH</b>                             | Access iDRAC through firmware RACADM.                                                                                                                                     |
| <b>Remote RACADM</b>                   | Remotely access iDRAC.                                                                                                                                                    |
| <b>SNMP Agent</b>                      | Enables support for SNMP queries (GET, GETNEXT, and GETBULK operations) in iDRAC.                                                                                         |
| <b>Automated System Recovery Agent</b> | Enable Last System Crash Screen.                                                                                                                                          |
| <b>Redfish</b>                         | Enables support for Redfish RESTful API.                                                                                                                                  |
| <b>VNC Server</b>                      | Enable VNC server with or without SSL encryption.                                                                                                                         |

## Configuring services using RACADM

To enable and configure services using RACADM, use the `set` command with the objects in the following object groups:

- iDRAC.LocalSecurity
- iDRAC.LocalSecurity
- iDRAC.SSH
- iDRAC.Webserver
- iDRAC.Racadm
- iDRAC.SNMP

For more information about these objects, see [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

## Configuring services using web interface

### About this task

To configure the services using iDRAC Web interface:

### Steps

1. In the iDRAC Web interface, go to **iDRAC Settings > Services**.  
The **Services** page is displayed.

2. Specify the required information and click **Apply**.

For information about the various settings, see the **iDRAC Online Help**.

**NOTE:** Do not select the **Prevent this page from creating additional dialogs** check-box. Selecting this option prevents you from configuring services.

You can configure **SEKM** from iDRAC Settings page. Click **iDRAC Settings > Services > SEKM Configuration**.

**NOTE:** For detailed step by step procedure for configuring SEKM, see the **iDRAC Online Help**.

**NOTE:** When **Security (Encryption)** mode is changed from **None** to **SEKM**, Real-Time job is not available. But it will be added to Staged job list. However, Real-Time job is successful when the mode is changed from **SEKM** to **None**.

Verify the following when changing the value of the **Username** Field in Client Certificate section on the KeySecure server (for ex: changing the value from **Common Name (CN)** to **User ID (UID)**)

- a. While using an existing account:
  - Verify in the iDRAC SSL certificate that, instead of the **Common Name** field, the **User name** field now matches the existing username on the KMS. If they don't, then you will have to set the username field and regenerate the SSL certificate again, get it signed on KMS and re-upload to iDRAC.
- b. While using a new user account:
  - Make sure the **User name** string matches the username field in the iDRAC SSL certificate.
  - If they don't match, then you will need to reconfigure the iDRAC KMS attributes Username and Password.
  - After the certificate is verified to contain the username, then the only change that needs to be made is to change the key ownership from the old user to the new user to match the newly created KMS username.

While using Vormetric Data Security Manager as KMS, ensure that the Common Name (CN) field in iDRAC SSL certificate matches with the host name added to Vormetric Data Security Manager. Otherwise, the certificate may not import successfully.

- NOTE:**
- **Rekey** option will be disabled when `racadm sekm getstatus` reports as **Failed**.
  - SEKM only supports **Common name**, **User ID**, or **Organization Unit** for **User Name** field under Client certificate.
  - If you are using a third party CA to sign the iDRAC CSR, ensure that the third party CA supports the value **UID** for **User Name** field in Client certificate. If it is not supported, use **Common Name** as the value for **User Name** field.
  - If you are using Username and Password fields, ensure that KMS server supports those attributes.

**NOTE:** For a KeySecure key management server, while creating an SSL certificate request, you must include at least one of the IP addresses or DNS name of the key management server in the **Subject Alternative Name** field. Ensure that the IP address is in the format IP:xxx.xxx.xxx.xxx.

## iLKM functionalities

### About this task

iDRAC Local Key Management (iLKM) is a security solution much like the Secure Enterprise Key Management (SEKM). This solution is ideal for users who do not plan to use SEKM but would like to secure devices using iDRAC. However, customers can migrate to SEKM at a later point in time. For more information, see the [Enable iLKM on Dell PowerEdge Servers whitepaper](#).

When using iLKM, iDRAC acts as key manager and generates authentication keys that are used to secure storage devices. To use iLKM as key management system, go to **iDRAC Settings > Services > iDRAC Key Management** and select iLKM from the drop-down menu.

**NOTE:** iLKM requires a combination of SEKM license and iDRAC Enterprise, or SEKM license and iDRAC data center license.

Provide a Passphrase and a Key ID to enable iLKM. Both Passphrase and Key ID lengths should be a maximum of 255 characters.

- NOTE:**
- iLKM can be viewed and configured through iDRAC UI, RACADM, and Redfish interfaces.
  - It is possible to enable or disable security on supported NVMe SED when iDRAC is in the iLKM security mode.
  - It is not possible to enable, disable, or rekey iLKM in the System Lockdown mode.
  - iLKM currently supports direct-attached NVMe SED that supports TCG Opal 2.0 protocol and above.
  - iLKM provides a rekey option, where you must provide the passphrase and key ID for authentication.

### Auto Secure Security Capable Drives

- Option to request iDRAC to auto secure non-PERC attached NVMe SED and SAS SED behind a security enabled SAS HBA. Drives are auto secured on a host reboot or on a drive hot plug.
- Option does not auto enable security on controllers such as PERC and SAS HBA.
- Option is enabled by default—you can be disabled by using the RACADM command.
- Disable the option before repurposing a drive by using the cryptographic erase option (or PSID revert option) if the drive is no longer required to be secured by iDRAC.

**NOTE:** NVMe direct-attached drives can be encryption capable and encryption not-capable drives. SEKM 044 logs are generated for encryption not-capable drives because the plug-in status is verified for both the NVMe direct-attached drives during the autosecure operation.

**NOTE:** PSID-based revert operations can be performed only on the locked or foreign drives. PSID-based revert operations cannot be performed on the drives which are connected to PERC controller.

**NOTE:** Do not run power cycle on the host system immediately after enabling the **Auto Secure Security Capable Drives** option. This may interrupt security enablement on the drives and might put the drives in an undefined security state.

### Steps

#### Transition from iLKM to SEKM

You must provide an iLKM passphrase to authenticate the transition along with the SEKM configuration details. If the authentication is successful, SEKM is enabled on iDRAC and the previous iLKM key ID is deleted. To transition from iLKM to SEKM, do the following:

1. Set up the CSA certificate.
2. Configure SEKM settings.
3. Perform the iLKM to SEKM transition.

## SEKM Functionalities

The following are the SEKM functionalities available in iDRAC:

1. **SEKM Key Purge Policy**—iDRAC provides a policy setting that allows you to configure iDRAC to purge old unused keys at the Key Management Server (KMS) when Rekey operation is performed. You can set the iDRAC read-writable attribute `KMSKeyPurgePolicy` to one of the following values:
  - Keep All Keys – This is the default setting and is the existing behavior where iDRAC leaves all the keys on the KMS untouched while performing Rekey operation.
  - Keep N and N-1 keys – iDRAC deletes all keys at the KMS except the current (N) and previous key (N-1) when performing Rekey operation.
2. **KMS Key Purge on SEKM Disable**—As part of the Secure Enterprise Key Manager (SEKM) solution, iDRAC allows you to disable SEKM on the iDRAC. Once SEKM is disabled, the keys that are generated by iDRAC at the KMS are unused and remain at the KMS. This feature is for allowing iDRAC to delete those keys when SEKM is disabled. iDRAC provides a new option “-purgeKMSKeys” to existing legacy command “racadm sekm disable” which will let you purge keys at the KMS when SEKM is disabled on iDRAC.

**NOTE:** If SEKM is already disabled and you want to purge old keys, you must re-enable SEKM, then disable passing in option -purgeKMSKeys.

3. **Key Creation Policy**—As part of this release, iDRAC has been preconfigured with a Key Creation Policy. Attribute `KeyCreationPolicy` is read-only and set to "Key per iDRAC" value.

- iDRAC read-only attribute `iDRAC.SEKM.KeyIdentifierN` reports the Key Identifier that is created by the KMS.

```
racadm get iDRAC.SEKM.KeyIdentifierN
```

- iDRAC read-only attribute `iDRAC.SEKM.KeyIdentifierNMinusOne` reports the previous Key Identifier after performing a Rekey operation.

```
racadm get iDRAC.SEKM.KeyIdentifierNMinusOne
```

4. **SEKM Rekey**—iDRAC provides the following two options from the UI to rekey your SEKM solution, either Rekey iDRAC or PERC. It is recommended to rekey the iDRAC since this rekeys all SEKM Secure capable and enabled devices.
  - **SEKM iDRAC Rekey [ Rekey on iDRAC.Embedded.1 FQDD ]**—When performing `racadm sekm rekey iDRAC.Embedded.1`, all SEKM Secure capable/Enabled devices are Rekeyed with a new key from KMS and this is common key to all SEKM enabled devices. iDRAC Rekey operation can also be performed from iDRAC UI- **iDRAC Settings > Services > SEKM Configuration > Rekey**. After performing this operation, the change in the Key can be validated by reading `KeyIdentifierN` and `KeyIdentifierNMinusOne` attributes.
  - **SEKM PERC Rekey ( Rekey On Controller [ Example RAID.Slot.1-1 ] FQDD )**—When performing `racadm sekm rekey <controller FQDD>`, the corresponding SEKM enabled controller gets rekeyed to the currently active iDRAC common key created from KMS. Storage Controller Rekey operation can also be performed from iDRAC UI- **Storage > Controllers > <controller FQDD> > Actions > Edit > Security > Security(Encryption) > Rekey**.

**NOTE:** When you run Rekey on PERC while the controller and iDRAC keys are in sync, you may get a **config job failure**, or configuration job may succeed but the key is not changed when you run the job. You can use iDRAC Rekey option to fix this issue.

5. **SEKM Rekey only from Redfish:** The following two SEKM Rekey options are supported from Redfish:
  - **SEKM iDRAC Schedule Rekey**—Sends a new key generation request from iDRAC for automatic change of the SEKM keys based on a recurrence interval that is configured by the user.
  - **SEKM iDRAC Periodic Sync with Key Management Server (KMS)**—Enables automatic change of the SEKM keys based on the recurrence interval configured on the KMS Server. iDRAC polls for any new key that is generated by the KMS server.

For detailed information about all supported SEKM features and deployment workflow, see the white paper - [Enable OpenManage Secure Enterprise Key Manager \(SEKM\) on Dell PowerEdge Servers](#)

**NOTE:** When SEKM is enabled on PERC, a CTL136 log is generated. However, in PERC 12 while performing rekey, the CTL136 log is not generated. This is because the controller does not create a key request as keys are provided as part of the rekey command.

## Enabling or disabling HTTPS redirection

If you do not want automatic redirection from HTTP to HTTPS due to certificate warning issue with default iDRAC certificate or as a temporary setting for debugging purpose, you can configure iDRAC such that redirection from http port (default is 80) to https port (default is 443) is disabled. By default, it is enabled. You have to log out and log in to iDRAC for this setting to take effect. When you disable this feature, a warning message is displayed.

You must have Configure iDRAC privilege to enable or disable HTTPS redirection.

An event is recorded in the Lifecycle Controller log file when this feature is enabled or disabled.

To disable the HTTP to HTTPS redirection:

```
racadm set iDRAC.Webserver.HttpsRedirection Disabled
```

To enable HTTP to HTTPS redirection:

```
racadm set iDRAC.Webserver.HttpsRedirection Enabled
```

To view the status of the HTTP to HTTPS redirection:

```
racadm get iDRAC.Webserver.HttpsRedirection
```

## Using VNC client to manage remote server

You can use a standard open VNC client to manage the remote server using both desktop and mobile devices such as Dell Wyse PocketCloud. When servers in data centers stop functioning, the iDRAC or the operating system sends an alert to the console on the management station. The console sends an email or SMS to a mobile device with required information and launches VNC viewer application on the management station. This VNC viewer can connect to OS/Hypervisor on the server and provide access to keyboard, video, and mouse of the host server to perform the necessary remediation. Before launching the VNC client, you must enable the VNC server and configure the VNC server settings in iDRAC such as password, VNC port number, SSL encryption, and the time-out value. You can configure these settings using iDRAC Web interface or RACADM.

 **NOTE:** VNC feature is licensed and is available in the iDRAC Enterprise or Datacenter license.

You can choose from many VNC applications or Desktop clients such as the ones from RealVNC or Dell Wyse PocketCloud.

Two VNC client sessions can be activated simultaneously. Second session is in Read-Only mode.

If a VNC session is active, you can only launch the Virtual Media using Launch Virtual Console and not the Virtual Console Viewer.

If video encryption is disabled, the VNC client starts RFB handshake directly, and an SSL handshake is not required. During VNC client handshake (RFB or SSL), if another VNC session is active or if a Virtual Console session is open, the new VNC client session is rejected. After completion of the initial handshake, VNC server disables Virtual Console and allows only Virtual Media. After termination of the VNC session, VNC server restores the original state of Virtual Console (enabled or disabled).

 **NOTE:**

- While launching a VNC session, if you get an RFB protocol error, change the VNC client settings to High quality and then relaunch the session.
- When iDRAC NIC is in shared mode and the host system is power cycled, the network connection is lost for a few seconds. During this time, if you perform any action in the active VNC client, the VNC session may close. You must wait for timeout (value configured for the VNC Server settings in the **Services** page in iDRAC Web interface) and then re-establish the VNC connection.
- If the VNC client window is minimized for more than 60 seconds, the client window closes. You must open a new VNC session. If you maximize the VNC client window within 60 seconds, you can continue to use it.

## Configuring VNC server using iDRAC web interface

### About this task

To configure the VNC server settings:

### Steps

1. In the iDRAC Web interface, go to **Configuration > Virtual Console**.  
The **Virtual Console** page is displayed.
2. In the **VNC Server** section, enable the VNC server, specify the password, port number, and enable or disable SSL encryption.  
For information about the fields, see the **iDRAC Online Help**.
3. Click **Apply**.  
The VNC server is configured.

## Configuring VNC server using RACADM

To configure the VNC server, use the `set` command with the objects in `VNCserver`.

For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

## Setting up VNC viewer with SSL encryption

### About this task

While configuring the VNC server settings in iDRAC, if the **SSL Encryption** option was enabled, then the SSL tunnel application must be used along with the VNC Viewer to establish the SSL encrypted connection with iDRAC VNC server.

 **NOTE:** Most of the VNC clients do not have built-in SSL encryption support.

To configure the SSL tunnel application:

### Steps

1. Configure SSL tunnel to accept connection on `<localhost>:<localport number>`. For example, `127.0.0.1:5930`.
2. Configure SSL tunnel to connect to `<iDRAC IP address>:<VNC server port Number>`. For example, `192.168.0.120:5901`.
3. Start the tunnel application.  
To establish connection with the iDRAC VNC server over the SSL encrypted channel, connect the VNC viewer to the localhost (link local IP address) and the local port number (`127.0.0.1:<local port number>`).

## Setting up VNC viewer without SSL encryption

In general, all Remote Frame Buffer (RFB) compliant VNC Viewers connect to the VNC server using the iDRAC IP address and port number that is configured for the VNC server. If the SSL encryption option is disabled when configuring the VNC server settings in iDRAC, then to connect to the VNC Viewer do the following:

In the **VNC Viewer** dialog box, enter the iDRAC IP address and the VNC port number in the **VNC Server** field.

The format is `<iDRAC IP address>:VNC port number>`

For example, if the iDRAC IP address is `192.168.0.120` and VNC port number is `5901`, then enter `192.168.0.120:5901`.

## Configuring time zone and NTP

You can configure the time zone in iDRAC and synchronize the iDRAC time using Network Time Protocol (NTP) instead of BIOS or host system times. You must have Configure privilege to configure time zone or NTP settings. After you have updated the NTP settings, log out of all current sessions, and then log in to iDRAC.

When the time zone is changed in iDRAC, the time and time zone offset in the timestamp are automatically adjusted. Select the same time zone between Real-Time Clock (RTC) and iDRAC because time synchronization issues between BIOS and iDRAC can lead to unexpected behavior. Enable the NTP settings in iDRAC to ensure timestamp accuracy.

 **NOTE:** If NTP is enabled in iDRAC, and different time zones are set for Real-Time Clock (RTC) and iDRAC:

- iDRAC synchronizes with the NTP server to obtain Coordinated Universal Time (UTC).

- The timestamp displays <date><local time><iDRAC time zone offset> .Where <local time>= UTC with time zone offset

**NOTE:** If NTP is disabled in iDRAC, and different time zones are set for RTC and iDRAC:

- iDRAC synchronizes with RTC time as is. RTC can be either UTC or local depending on the operating system and its settings.
- The timestamp displays <date><RTC><iDRAC time zone>

## Configuring time zone and NTP using iDRAC web interface

When you enable or disable NTP Server Settings, log out from all the current sessions, and then log in to iDRAC.

### About this task

To configure time zone and NTP using iDRAC web interface:

#### Steps

1. Go to **iDRAC Settings > Settings > Time zone and NTP Settings**.  
The **Time zone and NTP** page is displayed.
2. To configure the time zone, from the **Time Zone** drop-down menu, select the required time zone, and then click **Apply**.
3. To configure NTP, enable NTP, enter the NTP server addresses, and then click **Apply**.  
For information about the fields, see **iDRAC Online Help**.

## Configuring time zone and NTP using RACADM

To configure time zone and NTP, use the `set` command with the objects in the `iDRAC.Time` and `iDRAC.NTPConfigGroup` group.

For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

**NOTE:** iDRAC syncs the time with the host (local time). Hence it is recommended to configure both iDRAC and host with the same time zone so that the time sync is proper. If you want to change a time zone, you need to change it on both host and iDRAC and then the host needs to reboot.

## Setting first boot device

You can set the first boot device for the next boot only or for all subsequent reboots. If you set the device to be used for all subsequent boots, it remains as the first boot device in the BIOS boot order until it is changed again either from the iDRAC web interface or from the BIOS boot sequence.

You can set the first boot device to one of the following:

- Normal Boot
- PXE
- BIOS Setup
- Local Floppy/Primary Removable Media
- Local CD/DVD
- Hard Drive
- Virtual Floppy
- Virtual CD/DVD/ISO
- Lifecycle Controller
- BIOS Boot Manager
- UEFI Device Path
- UEFI HTTP
- Virtual Network File 1
- Virtual Network File 2

### **NOTE:**

- BIOS Setup (F2), Lifecycle Controller (F10), and BIOS Boot Manager (F11) cannot be set as permanent boot device.
- The first boot device setting in iDRAC Web Interface overrides the System BIOS boot settings.

## Setting first boot device using web interface

### About this task

To set the first boot device using iDRAC Web interface:

### Steps

1. Go to **Configuration > System Settings > Hardware Settings > First Boot Device**. The **First Boot Device** page is displayed.
  2. Select the required first boot device from the drop-down list, and click **Apply**. The system boots from the selected device for subsequent reboots.
  3. To boot from the selected device only once on the next boot, select **Boot Once**. Thereafter, the system boots from the first boot device in the BIOS boot order.
- For more information about the options, see the [iDRAC Online Help](#).

## Setting first boot device using RACADM

- To set the first boot device, use the `iDRAC.ServerBoot.FirstBootDevice` object.
- To enable boot once for a device, use the `iDRAC.ServerBoot.BootOnce` object.

For more information about these objects, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

## Setting first boot device using virtual console

### About this task

You can select the device to boot from as the server is being viewed in the Virtual Console viewer before the server runs through its boot-up sequence. Boot-once is supported by all devices listed in [Setting first boot device](#).

To set the first boot device using Virtual Console:

### Steps

1. Launch Virtual Console.
2. In the Virtual Console Viewer, from the **Next Boot** menu, set the required device as the first boot device.

## Enabling or disabling OS to iDRAC Pass-through

In servers that have Open Compute Project (OCP) cards or embedded LAN On Motherboard (LOM) devices, you can enable the OS to iDRAC Pass-through feature. This feature provides a high-speed bi-directional in-band communication between iDRAC and the host operating system through a shared LOM, a dedicated NIC, or through the USB NIC. This feature is available for iDRAC Enterprise or Datacenter license.

 **NOTE:** iDRAC Service Module (iSM) provides more features for managing iDRAC through the operating system. For more information, see the iDRAC Service Module User's Guide available on the [iDRAC Service Module](#) page.

When enabled through dedicated NIC, you can launch the browser in the host operating system and then access the iDRAC Web interface.

Switching between dedicated NIC or shared LOM does not require a reboot or reset of the host operating system or iDRAC.

You can enable this channel using:

- iDRAC web interface
- RACADM (post operating system environment)



- iDRAC Settings utility (pre-operating system environment)

If the network configuration is changed through iDRAC Web interface, you must wait for at least 10 seconds before enabling OS to iDRAC Pass-through.

If you are configuring the server using a Server Configuration Profile through RACADM or Redfish and if the network settings are changed in this file, then you must wait for 15 seconds to either enable OS to iDRAC Pass-through feature or set the OS Host IP address.

Before enabling OS to iDRAC Pass-through, make sure that:

- iDRAC is configured to use dedicated NIC or shared mode (that is, NIC selection is assigned to one of the LOMs).
- Host operating system and iDRAC are in the same subnet and same VLAN.
- Host operating system IP address is configured.
- A card that supports OS to iDRAC Pass-through capability is installed.
- You have the Configure privilege.

When you enable this feature:

- In shared mode, the host operating system's IP address is used.
- In dedicated mode, you must provide a valid IP address of the host operating system. If more than one LOM is active, enter the first LOM's IP address.

If the OS to iDRAC Pass-through feature does not work after it is enabled, ensure that you check the following:

- The iDRAC dedicated NIC cable is connected properly.
- At least one LOM is active.

**NOTE:** Use the default IP address. Ensure that the IP address of the USB NIC interface is not in the same network subnet as the iDRAC or host OS IP addresses. If this IP address conflicts with an IP address of other interfaces of the host system or the local network, you must change it.

**NOTE:** If you launch iDRAC Service Module while USB NIC is in disabled state, the iDRAC Service Module changes the USB NIC IP address to 169.254.0.1.

**NOTE:** Do not use 169.254.0.3 and 169.254.0.4 IP addresses. These IP addresses are reserved for the USB NIC port on the front panel when Type-C USB cable is used.

**NOTE:** iDRAC may not be accessible from the host server using LOM-Passthrough when NIC teaming is enabled. Then, iDRAC can be accessed from the host server OS using the iDRAC USB NIC or through the external network, via the iDRAC dedicated NIC.

## Supported operating systems for USB NIC

For list of operating systems supported for USB NIC, see the respective release note at [iDRAC Release Versions and Release notes](#).

For Linux operating systems, configure the USB NIC as DHCP on the host operating system before enabling USB NIC.

For vSphere, you must install the VIB file before enabling USB NIC.

**NOTE:**

- If you disable USB-NIC on the iDRAC while iSM is running in the OS, iSM service module status changes to "Running(limited functionality)".
- If you install iSM in the OS while USB NIC is disabled in the iDRAC, iSM automatically enables USB NIC in the iDRAC to finish the installation. If needed, disable USB NIC after the installation is completed.

**NOTE:** To configure USB NIC as DHCP in Linux operating system or XenServer, refer to the operating system or hypervisor documentation.

## Installing VIB file

### About this task

For vSphere operating systems, before enabling the USB NIC, you must install the VIB file.

To install the VIB file:

### Steps

1. Using Win-SCP, copy the VIB file to /tmp/ folder of the ESX-i host operating system.
2. Go to the ESXi prompt and run the following command:

```
esxcli software vib install -v /tmp/ iDRAC_USB_NIC-1.0.0-799733X03.vib --no-sig-check
```

The output is:

```
Message: The update completed successfully, but the system needs to be rebooted for
the changes to be effective.
Reboot Required: true
VIBs Installed: Dell_bootbank_iDRAC_USB_NIC_1.0.0-799733X03
VIBs Removed:
VIBs Skipped:
```

3. Reboot the server.
4. At the ESXi prompt, run the command: `esxcfg-vmknics -l`.  
The output displays the usb0 entry.

## Enabling or disabling OS to iDRAC Pass-through using RACADM

To enable or disable OS to iDRAC Pass-through using RACADM, use the objects in the `iDRAC.OS-BMC` group.

For more information, see the Integrated Dell Remote Access Controller Attribute Registry available on the [iDRAC Manuals](#) page.

## Enabling or disabling OS to iDRAC Pass-through using iDRAC settings utility

### About this task

To enable or disable OS to iDRAC Pass-through using iDRAC Settings Utility:

### Steps

1. In the iDRAC Settings utility, go to **Communications Permissions**.  
The **iDRAC Settings.Communications Permissions** page is displayed.
2. Select any of the following options to enable OS to iDRAC pass-through:
  - **LOM** — The OS to iDRAC pass-through link between the iDRAC and the host operating system is established through the LOM or NDC.
  - **USB NIC** — The OS to iDRAC pass-through link between the iDRAC and the host operating system is established through the internal USB bus.

**NOTE:** If you set the pass-through mode to LOM, ensure that:

- OS and iDRAC are on the same subnet
- NIC selection in Network Settings is set to a LOM

To disable this feature, select **Disabled**.

**NOTE:** The LOM option can be selected only if the card supports OS to iDRAC pass-through capability. Else, this option is grayed-out.

3. If you select **LOM** as the pass-through configuration, and if the server is connected using dedicated mode, enter the IPv4 address of the operating system.

**NOTE:** If the server is connected in shared LOM mode, then the **OS IP Address** field is disabled.

4. If you select **USB NIC** as the pass-through configuration, enter the IP address of the USB NIC.

The default value is 169.254.1.1. However, if this IP address conflicts with an IP address of other interfaces of the host system or the local network, you must change it. Do not enter 169.254.0.3 and 169.254.0.4 IPs. These IPs are reserved for the USB NIC port on the front panel when Type-C USB cable is used.

**NOTE:** If IPv6 is preferred, the default address is fde1:53ba:e9a0:de11::1. If needed, this address can be modified in the idrac.OS-BMC.UsbNicULA setting. If IPv6 is not wanted on the USB-NIC, it can be disabled by changing the address to ":::"

5. Click **Back**, click **Finish**, and then click **Yes**.  
The details are saved.

## Enabling or disabling an operating system to iDRAC Pass-through using web interface

### About this task

To enable operating system to iDRAC Pass-through using Web interface:

### Steps

1. Go to **iDRAC Settings > Connectivity > Network > OS to iDRAC Pass-through**.  
The **OS to iDRAC Pass-through** page is displayed.
2. Change the State to **Enabled**.
3. Select any of the following options for Pass-through Mode:
  - **LOM**—The operating system to iDRAC pass-through link between the iDRAC and the host operating system is established through the LOM or OCP.
  - **USB NIC**—The operating system to iDRAC pass-through link between the iDRAC and the host operating system is established through the internal USB bus.

**NOTE:** If you set the pass-through mode to LOM, ensure that:

  - The operating system and iDRAC are on the same subnet.
  - NIC selection in Network Settings is set to LOM.
4. If the server is connected in shared LOM mode, then the **OS IP Address** field is disabled.

**NOTE:** If the VLAN is enabled on the iDRAC, the LOM-Passthrough functions only in shared LOM mode with VLAN taggings that are configured on the host.

**NOTE:**

  - When Pass-through mode is set to LOM, it is not possible to launch iDRAC from the host operating system after cold boot.
  - The LOM Pass-through is removed using the Dedicated mode feature.
5. If you select **USB NIC** as the pass-through configuration, enter the IP address of the USB NIC.  
The default value is 169.254.1.1. It is recommended to use the default IP address. However, if this IP address conflicts with an IP address of other interfaces of the host system or the local network, you must change it.  
Do not enter 169.254.0.3 and 169.254.0.4 IPs. These IPs are reserved for the USB NIC port on the front panel when Type-C USB cable is used.

**NOTE:** If IPv6 is preferred, the default address is fde1:53ba:e9a0:de11::1. If needed, this address can be modified in the idrac.OS-BMC.UsbNicULA setting. If IPv6 is not wanted on the USB-NIC, it can be disabled by changing the address to ":::"

**NOTE:** When you modify the static IP address of the USB NIC, it automatically adjusts the DHCP address range to align with the new static IP. For instance, if you set the static IP to 169.254.1.1, the DHCP address updates to 169.254.1.2.  
This change is compatible with the network manager, Wicked, which accepts the new DHCP address.
6. Click **Apply**.
7. Click **Test Network Configuration** to check if the IP is accessible and the link is established between the iDRAC and the host operating system.

# Obtaining certificates

The following table lists the types of certificates based on the login type.

**Table 14. Types of certificate based on login type**

| Login Type                  | Certificate Type       | How to Obtain?                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Active Directory user login | Trusted CA certificate | This certificate is issued by a CA.<br> <b>NOTE:</b> SHA-2 certificates are also supported.                                                                                                                                                                                                                                                                  |
| Local User login            | SSL Certificate        | Generate a CSR and get it signed from a trusted CA<br> <b>NOTE:</b> iDRAC ships with a default self-signed SSL server certificate. The iDRAC web server, Virtual Media, and Virtual Console use this certificate.<br> <b>NOTE:</b> SHA-2 certificates are also supported. |

## SSL server certificates

iDRAC includes a web server that is configured to use the industry-standard SSL security protocol to transfer encrypted data over a network. An SSL encryption option is provided to disable weak ciphers. Built upon asymmetric encryption technology, SSL is widely accepted for providing authenticated and encrypted communication between clients and servers to prevent eavesdropping across a network.

An SSL-enabled system can perform the following tasks:

- Authenticate itself to an SSL-enabled client
- Allow the two systems to establish an encrypted connection

 **NOTE:** If SSL encryption is set to 256-bit or higher and 168-bit or higher, the cryptography settings for your virtual machine environment (JVM, IcedTea) may require installing the Unlimited Strength Java Cryptography Extension Policy Files to permit usage of iDRAC plugins such as vConsole with this level of encryption. For information about installing the policy files, see the documentation for Java.

iDRAC Web server has a Dell self-signed unique SSL digital certificate by default. You can replace the default SSL certificate with a certificate signed by a well-known Certificate Authority (CA). A Certificate Authority is a business entity that is recognized in the Information Technology industry for meeting high standards of reliable screening, identification, and other important security criteria. Examples of CAs include Thawte and VeriSign. To initiate the process of obtaining a CA-signed certificate, use either iDRAC Web interface or RACADM interface to generate a Certificate Signing Request (CSR) with your company's information. Then, submit the generated CSR to a CA such as VeriSign or Thawte. The CA can be a root CA or an intermediate CA. After you receive the CA-signed SSL certificate, upload this to iDRAC.

You can upload CA certificate from iDRAC UI via **iDRAC Settings > Services > Web Server > SSL/TLS Certificate Signing Request**. You can also see certificate details from other interfaces.

For each iDRAC to be trusted by the management station, that iDRAC's SSL certificate must be placed in the management station's certificate store. Once the SSL certificate is installed on the management stations, supported browsers can access iDRAC without certificate warnings.

You can also upload a custom signing certificate to sign the SSL certificate, rather than relying on the default signing certificate for this function. By importing one custom signing certificate into all management stations, all the iDRACs using the custom signing certificate are trusted. If a custom signing certificate is uploaded when a custom SSL certificate is already in-use, then the custom SSL certificate is disabled and a one-time auto-generated SSL certificate, signed with the custom signing certificate, is used. You can download the custom signing certificate (without the private key). You can also delete an existing custom signing certificate. After deleting the custom signing certificate, iDRAC resets and auto-generates a new self-signed SSL certificate. If a self-signed certificate is regenerated, then the trust must be re-established between that iDRAC and the management workstation. Auto-generated SSL certificates are self-signed and have an expiration date of seven years and one day and a start date of one day in the past (for different time zone settings on management stations and the iDRAC).

The iDRAC Web server SSL certificate supports the asterisk character (\*) as part of the left-most component of the Common Name when generating a Certificate Signing Request (CSR). For example, \*.qa.com, or \*.company.qa.com. This is called a wildcard certificate. If a wildcard CSR is generated outside of iDRAC, you can have a signed single wildcard SSL certificate that you can upload for multiple iDRACs and all the iDRACs are trusted by the supported browsers. While connecting to iDRAC Web interface using a supported browser that supports a wildcard certificate, the iDRAC is trusted by the browser. While launching viewers, the iDRACs are trusted by the viewer clients.

You can enable the Certificate Expiry Notification feature and also can configure the Notification Interval and Notification Frequency. iDRAC provides notification about certificate expiry.

 **NOTE:** The default self-signed certificates are automatically updated on iDRAC reboot. Hence, self signed certificates are not considered for the certificate expiry.

You can enable Certificate Expiry Notification and the notification interval from - **iDRAC Settings > Services > Web Server > Settings**. Also, on iDRAC login page, you can see the security warning at the bottom of the page about certificate expiry.

## Generating a new certificate signing request

A CSR is a digital request to a Certificate Authority (CA) for a SSL server certificate. SSL server certificates allow clients of the server to trust the identity of the server and to negotiate an encrypted session with the server.

After the CA receives a CSR, they review and verify the information the CSR contains. If the applicant meets the CA's security standards, the CA issues a digitally-signed SSL server certificate that uniquely identifies the applicant's server when it establishes SSL connections with browsers running on management stations.

After the CA approves the CSR and issues the SSL server certificate, it can be uploaded to iDRAC. The information used to generate the CSR, stored on the iDRAC firmware, must match the information contained in the SSL server certificate, that is, the certificate must have been generated using the CSR created by iDRAC.

## Generating CSR using RACADM

To generate a CSR using RACADM, use the `set` command with the objects in the `iDRAC.Security` group, and then use the `sslcsrgen` command to generate the CSR.

For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

## Generating CSR using web interface

### About this task

To generate a new CSR:

 **NOTE:** Each new CSR overwrites any previous CSR data that are stored in the firmware. The information in the CSR must match the information in the SSL server certificate. Else, iDRAC does not accept the certificate.

### Steps

1. In the iDRAC Web interface, go to **iDRAC Settings > Services > Web Server > SSL certificate**, select **Generate Certificate Signing Request (CSR)** and click **Next**.  
The **Generate a New Certificate Signing Request** page is displayed.
2. Enter a value for each CSR attribute.  
For more information, see **iDRAC Online Help**.
3. Click **Generate**.  
A new CSR is generated. Save it to the management station.

## Automatic Certificate Enrollment

In iDRAC, the Automatic Certificate Enrollment (ACE) feature enables you to automatically install and renew the certificates that are used by the web server. When this feature is enabled, the existing web server certificate is replaced by a new certificate. Enter the Certificate Signing Request (CSR) details before you enable ACE.

 **NOTE:**

- ACE is a licensed feature and requires a Datacenter license.
- Valid Network Device Enrollment Service (NDES) setup is required to issue the server certificate.

iDRAC time needs to be synchronized with the NDES/Certificate Authority.

**NOTE:** If the time is not synchronized, iDRAC may receive invalid or expired certificates during the enrollment and renewal process.

The following are the ACE configuration parameters:

- Enable and Disable.
- SCEP server URL/ACEM (Automatic Certificate Management Environment)
- Challenge password

**NOTE:** For more information about these parameters, see **iDRAC Online Help**.

The following are the available status for ACE:

- Enrolled—ACE is enabled. The certificate is monitored, and a new certificate can be issued on expiry.
- Enrolling—Intermediate state after ACE is enabled.
- Error—Problem encountered with NDES server.
- None—Default.

**NOTE:** When you enable ACE, the web server is restarted and all the existing web sessions are logged out.

**NOTE:** To view the success and failure messages, see the lifecycle logs.

## Uploading server certificate

After generating a CSR, you can upload the signed SSL server certificate to the iDRAC firmware. iDRAC must be reset to apply the certificate. iDRAC accepts only X509, Base 64 encoded Web server certificates. SHA-2 certificates are also supported.

**CAUTION:** During reset, iDRAC is not available for a few minutes.

## Uploading server certificate using RACADM

### About this task

To upload the SSL server certificate, use the `sslcertupload` command. For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

If the CSR is generated outside of iDRAC with a private key available, then to upload the certificate to iDRAC:

### Steps

1. Send the CSR to a well-known root CA. CA signs the CSR and the CSR becomes a valid certificate.
2. Upload the private key using the remote `racadm sslkeyupload` command.
3. Upload the signed certificate to iDRAC using the remote `racadm sslcertupload` command.  
The new certificate is uploaded iDRAC. A message is displayed asking you to reset iDRAC.
4. Run the `racadm racreset` command to reset iDRAC.  
iDRAC resets and the new certificate is applied. The iDRAC is not available for a few minutes during the reset.

**NOTE:** You must reset iDRAC to apply the new certificate. Until iDRAC is reset, the existing certificate is active.

## Uploading server certificate using web interface

### About this task

To upload the SSL server certificate:

### Steps

1. In the iDRAC Web interface, go to **iDRAC Settings > Services > Web Server > SSL/TLS Certificate Signing Request**, select **Upload Server Certificate** and click **Next**.  
The **Certificate Upload** page is displayed.
  2. Under **File Path**, click **Browse** and select the certificate on the management station.
  3. Click **Apply**.  
The SSL server certificate is uploaded to iDRAC.
  4. A pop-up message is displayed asking you to reset iDRAC immediately or at a later time. Click **Reset iDRAC** or **Reset iDRAC Later** as required.  
iDRAC resets and the new certificate is applied. The iDRAC is not available for a few minutes during the reset.
-  **NOTE:** You must reset iDRAC to apply the new certificate. Until iDRAC is reset, the existing certificate is active.

## Viewing server certificate

You can view the SSL server certificate that is currently being used in iDRAC.

### Viewing server certificate using RACADM

To view the SSL server certificate, use the `sslcertview` command.

For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

### Viewing server certificate using web interface

In the iDRAC Web interface, go to **iDRAC Settings > Services > Web Server > SSL certificate**. The **SSL** page displays the SSL server certificate that is currently in use at the top of the page.

## Uploading custom signing certificate

You can upload a custom signing certificate to sign the SSL certificate. SHA-2 certificates are also supported.

### Uploading custom signing certificate using web interface

#### About this task

To upload the custom signing certificate using iDRAC web interface:

#### Steps

1. Go to **iDRAC Settings > Services > Web Server > SSL/TLS Custom Signing Certificate**.  
The **SSL** page is displayed.
  2. Under **SSL/TLS Custom Signing Certificate**, click **Upload Signing Certificate**.  
The **Upload Custom SSL Certificate Signing Certificate** page is displayed.
  3. Click **Choose File** and select the custom SSL certificate signing certificate file.  
Only Public-Key Cryptography Standards #12 (PKCS #12) compliant certificate is supported.
  4. If the certificate is password protected, in the **PKCS#12 Password** field, enter the password.
  5. Click **Apply**.  
The certificate is uploaded to iDRAC.
  6. A pop-up message is displayed asking you to reset iDRAC immediately or at a later time. Click **Reset iDRAC** or **Reset iDRAC Later** as required.  
After iDRAC resets, the new certificate is applied. The iDRAC is not available for a few minutes during the reset.
-  **NOTE:** You must reset iDRAC to apply the new certificate. Until iDRAC is reset, the existing certificate is active.

## Uploading custom SSL certificate signing certificate using RACADM

To upload the custom SSL certificate signing certificate using RACADM, use the `sslcertupload` command, and then use the `racreset` command to reset iDRAC.

For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

## Downloading custom SSL certificate signing certificate

You can download the custom signing certificate using iDRAC Web interface or RACADM.

## Downloading custom SSL certificate signing certificate using RACADM

To download the custom SSL certificate signing certificate, use the `sslcertdownload` subcommand. For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

## Downloading custom signing certificate

### About this task

To download the custom signing certificate using iDRAC Web interface:

### Steps

1. Go to **iDRAC Settings > Connectivity > SSL**.  
The **SSL** page is displayed.
2. Under **Custom SSL Certificate Signing Certificate**, select **Download Custom SSL Certificate Signing Certificate** and click **Next**.  
A pop-up message is displayed that allows you to save the custom signing certificate to a location of your choice.

## Downloading custom SSL certificate signing certificate

You can download the custom signing certificate using iDRAC Web interface or RACADM.

## Downloading custom SSL certificate signing certificate using RACADM

To download the custom SSL certificate signing certificate, use the `sslcertdownload` subcommand. For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

## Downloading custom signing certificate

### About this task

To download the custom signing certificate using iDRAC Web interface:

### Steps

1. Go to **iDRAC Settings > Connectivity > SSL**.  
The **SSL** page is displayed.
2. Under **Custom SSL Certificate Signing Certificate**, select **Download Custom SSL Certificate Signing Certificate** and click **Next**.  
A pop-up message is displayed that allows you to save the custom signing certificate to a location of your choice.



# Configuring multiple iDRACs using RACADM

## About this task

You can configure one or more iDRACs with identical properties using RACADM. When you query a specific iDRAC using its group ID and object ID, RACADM creates a configuration file from the retrieved information. Import the file to other iDRACs to identically configure them.

### NOTE:

- The configuration file contains information that is applicable for the particular server. The information is organized under various object groups.
- Some configuration files contain unique iDRAC information, such as the static IP address, that you must modify before you import the file into other iDRACs.

You can also use the System Configuration Profile (SCP) to configure multiple iDRACs using RACADM. SCP file contains the component configuration information. You can use this file to apply the configuration for BIOS, iDRAC, RAID, and NIC by importing the file into a target system. For more information, see **XML Configuration Workflow** white paper available at [Dell Manuals](#) page.

To configure multiple iDRACs using the configuration file:

## Steps

1. Query the target iDRAC that contains the required configuration using the following command:

```
racadm get -f <file_name>.xml -t xml -c iDRAC.Embedded.1
```

The command requests the iDRAC configuration and generates the configuration file.

 **NOTE:** Redirecting the iDRAC configuration to a file using `get -f` is only supported with the local and remote RACADM interfaces.

 **NOTE:** The generated configuration file does not contain user passwords.

The `get` command displays all configuration properties in a group (specified by group name and index) and all configuration properties for a user.

2. Modify the configuration file using a text editor, if required.

 **NOTE:** It is recommended that you edit this file with a simple text editor. The RACADM utility uses an ASCII text parser. Any formatting confuses the parser, which may corrupt the RACADM database.

3. On the target iDRAC, use the following command to modify the settings:

```
racadm set -f <file_name>.xml -t xml
```

This loads the information into the other iDRAC. You can use `set` command to synchronize the user and password database with Server Administrator.

4. Reset the target iDRAC using the command: `racadm racreset`

# Disabling access to modify iDRAC configuration settings on host system

## About this task

You can disable access to modify the iDRAC configuration settings through Local RACADM or iDRAC Settings utility. However, you can view these configuration settings. To do this:

## Steps

1. In iDRAC Web interface, go to **iDRAC Settings > Services > Local Configurations**.

2. Select one or both of the following:

- **Disable the iDRAC Local Configuration using iDRAC Settings** — Disables access to modify the configuration settings in iDRAC Settings utility.
- **Disable the iDRAC Local Configuration using RACADM** — Disables access to modify the configuration settings in Local RACADM.

3. Click **Apply**.



**NOTE:** If access is disabled, you cannot use Server Administrator or IPMITool to perform iDRAC configurations. However, you can use IPMI Over LAN.

# Delegated Authorization using OAuth 2.0

The Delegated Authorization feature allows a user or console to access iDRAC API using OAuth 2.0 JSON Web Tokens (JWT) that the user or console first obtains from an Authorization Server. Once an OAuth JWT has been retrieved, the user or console may use it to invoke iDRAC API. This circumvents the need for specifying a username and password to access the API. For iDRAC API to accept the tokens, ensure to update the mandatory claims such as **aud (audience)**, **iss (issuer)**, **roles**, **exp (expiry)**, and **sub (subject)** during token creation in the authorization server.

 **NOTE:** This feature is available with DataCenter or Enterprise license. You must have Configure iDRAC or Configure Users privilege to use this feature.

iDRAC supports configuration of up to 2 Authorization Servers. The configuration requires a user to specify the following Authorization Server details:

- **Name**—A string to identify the Authorization Server on the iDRAC.
- **Metadata URL**—The OpenID Connect compliant URL as advertised by the server.
- **HTTPS certificate**—The server public key the iDRAC should use to communicate with the server.
- **Offline Key**—The JWK set document for the Authorization Server.
- **Offline Issuer**—The issuer string as used in tokens issued by the Authorization Server.

For Online configuration:

- When configuring an Authorization Server, the iDRAC administrator needs to ensure that the iDRAC has online network access to the Authorization Server.
- If iDRAC cannot access the Authorization Server, the configuration fails and a subsequent attempt to access the iDRAC API fails even though a valid token is presented.

For offline configuration:

- iDRAC does not need to communicate with the Auth server, but instead it is configured with the metadata details that it has downloaded offline. When configured offline, iDRAC has the public portion of the signing keys and can validate the token without a network connection to the Auth server.

# Viewing iDRAC and managed system information

You can view iDRAC and managed system health and properties, hardware and firmware inventory, sensor health, storage devices, network devices, and view and terminate user sessions.

## Topics:

- [Viewing managed system health and properties](#)
- [Configuring Asset Tracking](#)
- [Viewing system inventory](#)
- [Viewing system components](#)
- [Monitoring performance index of CPU, memory, and input output modules](#)
- [Reading Firmware and Hardware inventories](#)
- [Performing and checking system/ component configuration status](#)
- [Performing and checking firmware update status](#)
- [Idle Server Detection](#)
- [GPU \(Accelerators\) Management](#)
- [Checking the system for Fresh Air compliance](#)
- [Viewing historical temperature data](#)
- [Configuring warning threshold for inlet temperature](#)
- [Viewing network interfaces available on host OS](#)
- [Viewing or terminating iDRAC sessions](#)

## Viewing managed system health and properties

When you log in to the iDRAC web interface, the **System Summary** page allows you to view the managed system's health, basic iDRAC information, preview the virtual console, add and view work notes, and quickly launch tasks such as power on or off, power cycle, view logs, update and rollback firmware, switch on or switch off the front panel LED, and reset iDRAC.

To access the **System Summary** page, go to **System > Overview > Summary**. The **System Summary** page is displayed. For more information, see the **iDRAC Online Help**.

You can also view the basic system summary information using the iDRAC Settings utility. To do this, in iDRAC Settings utility, go to **System Summary**. The **iDRAC Settings System Summary** page is displayed. For more information, see the **iDRAC Settings Utility Online Help**.

## Configuring Asset Tracking

### About this task

The Asset Tracking feature in iDRAC provides you the ability to configure various attributes that are related to your server. This includes information such as acquisition, warranty, service, and so on.

 **NOTE:** Asset Tracking in iDRAC is similar to the Asset Tag feature in OpenManage Server Administrator. However, the attribute information has to be entered separately in both these tools for them to report the relevant Asset data.

To configure Asset Tracking:

### Steps

1. In the iDRAC interface, go to **Configuration > Asset Tracking**.
2. Click **Add Custom Assets** to add any additional attributes which are not specified by default on this page.

3. Enter all the relevant information of your server asset and click **Apply**.
4. To view the Asset Tracking Report, go to **System > Details > Asset Tracking**.

## Viewing system inventory

You can view information about the hardware and firmware components that are installed on the managed system. To view the system inventory in the iDRAC web interface, go to **System > Inventory**. For information about the displayed properties, see **iDRAC Online Help**.

The **Hardware Inventory** section displays the information for the following components available on the managed system:

- iDRAC
- OEM
- RAID controller
- Batteries
- CPUs
- GPU
- DIMMs
- HDDs
- Backplanes
- Network Interface Cards (integrated and embedded)
- Video card
- Power Supply Units (PSUs)
- Fans
- Fibre Channel HBAs
- USB
- NVMe PCIe SSD devices

**i NOTE:** In hardware Inventory for any GPU, entries BuildDate, GPUGUID, and OEMInfo are supported and populated for NVIDIA devices only. OEMInfo data is populated only if there is any data given by the NVIDIA device for that field.

**i NOTE:**

Channel cards, also known as industry standard devices, adhere to industry standard specifications for both hardware and firmware, making them compatible with various server manufacturers. The following lists the HII attributes of the Channel card for which the values are invalid or not displayed in the **Hardware Inventory**:

- LastSystemInventoryTime
- LastUpdateTime
- FCoEOffloadMode
- ScsiOffloadMode
- NicMode
- MaxBandwidth
- MinBandwidth
- Virtual iSCSI MAC Address
- Permanent iSCSI MAC Address
- Virtual FIP MAC Address
- Permanent FIP MAC Address
- Last Update Time
- Last System Inventory Time
- WWN
- VirtWWN
- WWPN
- VirtWWPN
- Controller BIOS Version
- EFI Version
- FCoE WWNN
- OS Driver Version
- iSCSI OS Driver Version

- FCOE OS Driver Version
- FC OS Driver Version
- RDMA OS Driver Version
- Protocol
- Media Type
- NIC Mode
- FCoE Offload Mode
- iSCSI Offload Mode
- SNAPI Support
- SNAPI State
- VPI Support
- Update Lockdown
- Update Lockdown State
- Max Bandwidth
- Min Bandwidth
- LAN Driver Version
- iSCSI Driver Version
- FCoE Driver Version
- FC Driver Version
- RDMA Driver Version
- CableLength
- FamilyVersion
- TransceiverPartNumber
- TransceiverSerialNumber
- TransceiverVendorName
- Revision
- NodeGUID
- PermanentPortGUID
- VPI support
- PrimaryStatus
- NumberOfPorts
- NumberPCIEFunctionsEnabled
- NumberPCIEFunctionsSupported
- MediaType
- VirtNodeGUID
- VirtPortGUID
- SlotLength
- SlotType
- PCIDeviceID
- PCISubDeviceID
- PCISubVendorID
- PCIVendorID
- CurrentMACAddress
- PermanentMACAddress
- PCI Device ID
- Link Speed
- Family Firmware Version

**NOTE:** Staged inventory is updated only after host boot. Inventories such as PCIe Slots and PCIe devices under hardware inventory are part of the staged inventory that is generated during the Host boot (after CSIOR). Even Hot plug or removal of drives does not change the inventory data unless the host is rebooted.

In the UI, **Network Device > Partition State** page, PCI Device ID, Minimum Bandwidth, and Maximum Bandwidth values are empty.

Under **Network Device > Settings and Capabilities** page, Supported Boot Protocol value is empty.

The **Firmware Inventory** section displays the firmware version for the following components:

- BIOS

- Lifecycle Controller
- iDRAC
- Operating system driver pack
- FPGA
- PERC controllers
- Physical disks
- Power supply
- NIC
- Fibre Channel
- Backplane
- Enclosure
- PCIe SSDs
- TPM
- iSM

**NOTE:** Components that lack roll-back capability using iDRAC do not display their Release Date in the software inventory. When a component is updated from the HOST operating system, it may not populate rollback contents and Release Date details.

**NOTE:** After performing an in-band operating system firmware update on an external enclosure (JBOD), cold reboot (power cycle) the server to update the enclosure information and system inventory in iDRAC.

**NOTE:** Firmware inventory may take long or may not show up sometimes. Use the `racadm` command `getremoteservicestatus` before running firmware inventory.

**NOTE:**

- The software inventory displays only the last 4 bytes of the firmware version and the Release date information. For example, if the firmware version is FLVDL06, the firmware inventory displays DL06.
- For SATA drives, the firmware version shows four characters always. If any SATA drive has a firmware version more than four characters, the software inventory displays the last four characters of the firmware version and the storage page and hardware inventory displays the full version.
- When you are viewing software inventory using Redfish interface, the Release date information is displayed only for components that support rollback.

**NOTE:**

- The firmware version of the GPU bundle is displayed as 00.00.00.00:
  - Until a firmware update is performed using DUP.
  - After the system, erase is performed on the system.
- If there is a communication failure between the GPU baseboard and iDRAC, the GPU bundle version may not be displayed in the firmware inventory. To resolve the communication issue, perform the AC cycle. The firmware version is displayed as 00.00.00.00 when communication is restored.
- If any device (Example: TPM) is in the OFF state, then the software inventory displays the version as **Not Available** or **0**. And if the application is not installed, then it shows the version as **Not Installed**.
- The system displays the default initial system date and time as the **Installed date/time** in the **System Inventory** page until a new device firmware version is installed using the DUP. Also, BIOS and iDRAC date/ time should be synchronized for components whose inventory details are obtained from BIOS (example: BIOS, TPM).
- The installation date does not change if the updated version is the same as the installed version.
- Sometimes the **LastUpdateTime** field of any component in iDRAC hardware inventory is shown as future/past date. This can happen when either BIOS or HOST time is set to the wrong date. To fix this issue, correct the BIOS or HOST date.

When you replace any hardware component or update the firmware versions, enable and run the **Collect System Inventory on Reboot** (CSIOR) option to collect the system inventory on reboot. After a few minutes, log in to iDRAC, and go to the **System Inventory** page to view the details. It may take up to 5 minutes for the information to be available depending on the hardware installed on the server.

**NOTE:** CSIOR option is enabled by default.

 **NOTE:** Configuration changes and firmware updates that are made within the operating system may not reflect properly in the inventory until you perform a server restart.

Click **Export** to export the hardware inventory in an XML format and save it to a location of your choice.

# Viewing system components

The following components in iDRAC UI help you to monitor the health of a managed system:

- **Batteries**—Provides information about the batteries on the system board CMOS and storage RAID on motherboard (ROMB).
-  **NOTE:** The storage ROMB battery settings are available only if the system has a ROMB with a battery.
- **CPU**—Indicates the health and state of the CPUs in a managed system. It also reports automatic processor throttling and predictive failure.
- **Memory**—Indicates the health and state of the Dual In-line memory modules (DIMMs) present in a managed system.
- **Intrusion**—Provides information about the chassis.
- **Power** (available only for rack servers)—Provides the status and the rating of the PSUs. **Zone** is displayed only for specific server models. The chassis is divided into two zones, zone **1** and zone **2**. Zone **1** is displayed for PSUs 1 and 2. Zone **2** is displayed for PSUs 3, 4, 5, 6, 7 and 8.

 **CAUTION:** If PSU 1 or PSU 2 fail, replace the failed PSU to prevent the system from shutting down. This is applicable for zone PSU configurations.

 **NOTE:** If there is a redundancy failure, only the PSU number is displayed in the error message. It is recommended to replace the PSU. This is applicable for zone PSU configurations.

 **NOTE:** If there is only one power supply in the system, the power supply redundancy is set to **Disabled**.

- **Voltages**—Indicates the status and reading of the voltage sensors on various system components.
- **Cooling**—Provides details about fans and hardware temperatures. The four types of fans are Bundled Dual Rotor (Fan1A, Fan1B), Bundled Single Rotor (Fan1A, Fan1B), Non-Bundled Dual Rotor (Fan1, Fan2), and Non-Bundled Single Rotor (Fan1, Fan2).
- **Accelerator**—Provides the details of the GPUs and Processing Accelerators.
- **PCIe Slots**—Provides details for all PCIe devices including PCIeSSD (NVMe) devices.
-  **NOTE:** PCIe slots containing embedded devices display **Populate** as No and **PCIe Type** as N/A. Examples of such embedded devices include M.2 BOSS, M.2 Interposer SSD, and OCP NIC.
- **Network Devices**—Provides details for all network devices including DPUs.
- **Fan Board**—Provides the details of the cooling controllers. The fans are monitored and controlled by cooling controllers. The number of fan controllers can vary based on the server model.
- **Front Panel**—Provides the system ID and details about the health of the system.

The following table lists the system components that can be monitored:

 **NOTE:** The **System Overview** page displays data only for sensors present on your system.

**Table 15. System components monitored from the iDRAC UI**

| System components | Navigation path in iDRAC      |
|-------------------|-------------------------------|
| Batteries         | System > Overview > Batteries |
| Cooling           | System > Overview > Cooling   |
| CPU               | System > Overview > CPU       |
| Memory            | System > Overview > Memory    |
| Intrusion         | System > Overview > Intrusion |



**Table 15. System components monitored from the iDRAC UI (continued)**

| System components | Navigation path in iDRAC            |
|-------------------|-------------------------------------|
| Power             | System > Overview > Power           |
| Removable Media   | System > Overview > Removable Media |
| Voltages          | System > Overview > Voltages        |
| Network Devices   | System > Overview > Network Devices |
| Accelerators      | System > Overview > Accelerators    |
| PCIe Slots        | System > Overview > PCIe Slots      |
| Front Panel       | System > Overview > Front Panel     |
| Fan Board         | System > Overview > Fan Board       |

Use the `racadm getsensorinfo` command to get the details of each of these components.

 **NOTE:** For current information of supported properties and their values, see the **iDRAC Online Help**.

## Monitoring performance index of CPU, memory, and input output modules

In Dell PowerEdge servers, Intel ME supports Compute Usage Per Second (CUPS) functionality. The CUPS functionality provides real-time monitoring of CPU, memory, and I/O utilization and system-level utilization index for the system. Intel ME allows out-of-band (OOB) performance monitoring and does not consume CPU resources. The Intel ME has a system CUPS sensor that provides computation, memory, and I/O resource utilization values as a CUPS Index. iDRAC monitors this CUPS index for the overall system utilization and also monitors the instantaneous utilization index of the CPU, Memory, and I/O.

The CPU and chipset have dedicated Resource monitoring Counters (RMC). The data from these RMCs is queried to obtain utilization information of system resources. The data from RMCs is aggregated by the node manager to measure the cumulative utilization of each of these system resources that is read from iDRAC using existing intercommunication mechanisms to provide data through out-of-band management interfaces.

The Intel sensor representation of performance parameters and index values is for a complete physical system. Therefore, the performance data representation on the interfaces is for the complete physical system, even if the system is virtualized and has multiple virtual hosts.

To display the performance parameters, the supported sensors must be present in the server.

The four system utilization parameters are:

- **CPU Utilization** — Data from RMCs for each CPU core is aggregated to provide cumulative utilization of all the cores in the system. This utilization is based on time that is spent in active and inactive states. A sample of RMC is taken every six seconds. To view the current CPU utilization, check the CPU utilization that is displayed in the host operating system.
- **Memory Utilization** — RMCs measure memory traffic occurring at each memory channel or memory controller instance. Data from these RMCs is aggregated to measure the cumulative memory traffic across all the memory channels on the system. This is a measure of memory bandwidth consumption and not the amount of memory utilization. iDRAC aggregates it for one minute, so it may or may not match the memory utilization that other operating system tools, such as **top** in Linux, show. Memory bandwidth utilization that the iDRAC shows is an indication of whether workload is memory intensive or not.
- **I/O Utilization** — There is one RMC per root port in the PCI Express Root Complex to measure PCI Express traffic emanating from or directed to that root port and the lower segment. Data from these RMCs is aggregated for measuring PCI express traffic for all PCI Express segments emanating from the package. This is a measure of I/O bandwidth utilization for the system.
- **System Level CUPS Index** — The CUPS index is calculated by aggregating CPU, Memory, and I/O index considering a predefined load factor of each system resource. The load factor depends on the nature of the workload on the system. CUPS Index represents the measurement of the compute headroom available on the server. If the system has a large CUPS Index, then there is limited headroom to place more workloads on that system. As the resource consumption decreases, the system's CUPS index decreases. A low CUPS index indicates that there is a large compute headroom and the server can receive new workloads and the server is in a lower power state to reduce power consumption. Workload monitoring can then be applied throughout the data center to provide a high-level and holistic view of the data center's workload, providing a dynamic data center solution.

**NOTE:** The CPU, memory, and I/O utilization indexes are aggregated over one minute. Therefore, if there are any instantaneous spikes in these indexes, they may be suppressed. They are an indication of workload patterns not the amount of resource utilization.

The IPMI, SEL, and SNMP traps are generated if the thresholds of the utilization indexes are reached and the sensor events are enabled. The sensor event flags are disabled by default. It can be enabled using the standard IPMI interface.

The required privileges are:

- Login privilege is required to monitor performance data.
- Configure privilege is required for setting warning thresholds and reset historical peaks.
- Login privilege and Enterprise license are required to read historical static data.

## Monitoring performance index for of CPU, memory, and input output modules using RACADM

Use the **SystemPerfStatistics** sub command to monitor performance index for CPU, memory, and I/O modules. For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

## Monitoring performance index of CPU, memory, and input output modules using web interface

To monitor the performance index of CPU, memory, and I/O modules, in the iDRAC web interface, go to **System > Performance**.

- **System Performance** section — Displays the current reading and the warning reading for CPU, Memory and I/O utilization index, and system level CUPS index in a graphical view.
- **System Performance Historical Data** section:
  - Provides the statistics for CPU, memory, IO utilization, and the system level CUPS index. If the host system is powered off, then the graph displays the power off line below 0 percent.
  - You can reset the peak utilization for a particular sensor. Click **Reset Historical Peak**. You must have Configure privilege to reset the peak value.
- **Performance Metrics** section:
  - Displays status and present reading
  - Displays or specifies the warning threshold utilization limit. You must have server configure privilege to set the threshold values.

For information about the displayed properties, see the **iDRAC Online Help**.

## Reading Firmware and Hardware inventories

**NOTE:** Ensure that you give few seconds of interval while using the command `getremoteservicesstatus` with a timeout of 5 minutes.

1. Use `getremoteservicesstatus method/URI/command` to check whether Lifecycle Controller (LC) status is ready. Ensure that system is **ON** at least once and **Collect System Inventory On Restart (CSIOR)** had run at least once to get the proper details. Based on the requirement for some components like storage and network, you may also need to check if system is in **Out of POST** and **Real Time (RT)** status.
2. Maximum timeout for LC to be ready should be 5 minutes. Ensure that system is in the following status for LC status to be ready:
  - Out of POST
  - Job is not already running
  - Not in LC GUI
  - Host struck in POST
3. Once LC status is ready, use the `getinventory method/ URI/ command`.

# Performing and checking system/ component configuration status

**NOTE:** Ensure that you give few seconds of interval while using the command `getremoteservicesstatus` with a timeout of 5 minutes.

1. Check for firmware inventory (follow the procedure mentioned above).
2. To avoid possible failures later, ensure that the required component is present in the system.
3. After initial checks, use `getremoteservicesstatus` method/URI/command to check if LC status is ready. Based on the requirement for some components like storage and network, other status may need to be checked, for instance, whether system is **Out of POST** and **Real Time (RT)** status.
4. Once LC is ready, use the system/ component configurations and create the job.
5. Create reboot job or reboot the host if the configuration requires a host reboot. Specific configuration setting might require a cold reboot.
6. The caller must check for job status to be completed - **Success/ Failure**. View LifeCycle Log events, job queue status and check config results for more details on the failures.
7. Only after **Collect System Inventory On Restart (CSIOR)** in host if required is completed successfully, the job marks as completed if not failed. This is applicable if host reboot is required.
8. Once the job is completed, wait for 30 s, and then use `getremoteservicesstatus` method/URI/command to check if LC status is ready, with other required status and then read the expected values.

# Performing and checking firmware update status

**NOTE:** Make sure to wait a few seconds between each use of the `getremoteservicesstatus` command and set a timeout of 5 minutes.

1. Check for firmware inventory (follow the procedure mentioned above).
2. To avoid potential issues, verify that the component you want to update is present in the system. Then, ensure that you have selected the correct and supported Dell Update Package (DUP) for the upload
3. After initial checks, use `getremoteservicesstatus` method/URI/command to check if LC status is ready.
4. After LC is ready, use the `firmwareupdate` method/ URI/ command and pass the correct DUP to start the update.
5. If the update requires a host reboot, create a reboot job or reboot the host. Power, OSM, and PERC updates require a cold reboot.
6. Check the job status **Success/ Failure**. View LifeCycle Log events and job queue status, and check for configuration results for more details on the failures.
7. The job will only be marked as completed after the **Collect System Inventory On Restart (CSIOR)** process in the host, if required, is successfully finished. If CSIOR fails, the job is marked as failed, even if there are multiple catalog updates. Therefore, it is recommended that the caller either does not set its own timeout or sets a timeout longer than the CSIOR timeout.
8. If the update takes more than 6 hours (meaning the job module does not receive a status from the update module for 6 hours), the job may timeout and fail.
9. The update timeouts are determined based on the device team's recommendations, which are read at run-time.
10. After the job is marked as completed, if the inventory does not reflect the newly applied changes, wait for 30 seconds, and then check the inventory again

# Idle Server Detection

iDRAC provides out-of-band performance monitoring index of server components like CPU, memory, and I/O.

The history data of the server level CUPS index is used to monitor whether the server is utilized or running idle for long time. If the server is underutilized below certain threshold for a defined span of interval (in hours), then it will be reported as idle server.

This feature is only supported on Intel platforms with CUPS ability. AMD and Intel platforms without CUPS capability do not support this feature.

**NOTE:**

- This feature requires Datacenter license.

- To read the configurations of Idle Server Configuration parameters, you need Login privilege and to modify the parameters you need iDRAC Configure privilege.

To view or modify the parameters, navigate to **Configuration > System Settings**.

Idle server detection is reported based on following parameters:

- Idle Server Threshold (%) - This is set to 20% by default and can be configured from 0 to 50%. The reset operation sets the threshold to 20%.
- Idle Server Scan Interval (in hours) - This is the time period over which the hourly samples are collected to determine the idle server. This is set to 240 hours by default and can be configured from 1 to 9000 hours. The reset operation sets the interval to 240 hours.
- Server Utilization Percentile (%) - The utilization percentile value can be set to 80 to 100%. The default value is 80%. If the 80% of the hourly samples falls below utilization threshold, then it is considered as idle server.

## Modifying idle Server Detection parameters using RACADM

```
racadm get system.idleServerDetection
```

## Modifying idle Server Detection parameters using Redfish

```
https://<iDRAC IP>/redfish/v1/Managers/iDRAC.Embedded.1/Oem/Dell/DellAttributes/  
System.Embedded.1
```

# GPU (Accelerators) Management

Dell PowerEdge servers are shipped with Graphics Processing Unit (GPU). GPU management enables you to view the various GPUs connected to the system and also monitor power, temperature, and thermal information of the GPUs.

The following are the GPU properties and the license details:

**Table 16. GPU properties and the license details**

| GPU Properties    | License      |
|-------------------|--------------|
| <b>Inventory:</b> |              |
| Board Part Number | All Licenses |
| OEM Info          | All Licenses |
| Serial Number     | All Licenses |
| Marketing Name    | All Licenses |
| GPU Part Number   | All Licenses |
| Build Date.       | All Licenses |
| Firmware Version  | All Licenses |
| GPU GUID          | All Licenses |
| PCI vendorid      | All Licenses |
| PCI deviceid      | All Licenses |
| PCI Subvendorid   | All Licenses |
| PCI Subdeviceid   | All Licenses |
| GPU Status        | All Licenses |
| GPU Health        | All Licenses |

**Table 16. GPU properties and the license details (continued)**

| GPU Properties                       | License      |
|--------------------------------------|--------------|
| <b>Thermal Metrics:</b>              |              |
| Primary GPU Temperature              | All Licenses |
| Secondary GPU Temperature            | All Licenses |
| Board Temperature                    | All Licenses |
| Memory Temperature                   | All Licenses |
| GPU Target Temperature               | Datacenter   |
| Minimum GPU HW Slowdown Temperature  | Datacenter   |
| GPU Shutdown Temperature             | Datacenter   |
| Maximum Memory Operating Temperature | Datacenter   |
| Maximum GPU Operating Temperature    | Datacenter   |
| Thermal Alert State                  | Datacenter   |
| Power Brake State                    | Datacenter   |
| <b>Power Metrics:</b>                |              |
| Power Consumption                    | All Licenses |
| Power Supply Status                  | Datacenter   |
| Board Power Supply Status            | Datacenter   |
| Current Power Cap Limit              | Datacenter   |
| <b>Utilization:</b>                  |              |
| GPU                                  | Datacenter   |
| Memory                               | Datacenter   |
| <b>Clocks:</b>                       |              |
| Graphics                             | Datacenter   |
| Memory                               | Datacenter   |

**NOTE:**

- GPU properties are not listed for Embedded GPU cards, and the Status is marked as **Unknown**.
- The operating temperature may be different for AMD-based systems.
- The number of GPU entries per PCIe slot that is displayed in the host may differ from that in the iDRAC.
- When a manual AC power cycle is required after performing any component or bundled firmware updates for GPUs or Power Distribution Board (PDB) FPGAs, SUP0545 event in Lifecycle(LC) logs is displayed. After this event, ensure to perform a manual AC or virtual AC power cycle to avoid any unexpected behavior in the server.
- After a GPU firmware update that includes component firmware updates or bundled firmware updates, ensure to perform an AC power cycle or virtual AC power cycle to complete the update. Doing so avoids any unexpected behavior in iDRAC related to GPUs.
- In Persistent Mode, during warm reboot, the GPU Power capping limit values may not be accurate.
- GPU Power Capping feature is not available in Non-A2 GPU configurations.

GPU has to be in ready state before the command fetches the data. GPUStatus field in Inventory shows the availability of the GPU and whether the GPU device is responding or not. If the GPU status is ready, GPUStatus shows **OK**, otherwise the status shows **Unavailable**.

The GPU offers multiple health parameters that can be pulled through the SMBPB interface of the NVIDIA controllers. This feature is limited only to NVIDIA cards. The following are the health parameters that are retrieved from the GPU device:

- Power

- Temperature
- Thermal

**NOTE:** This feature is only limited to NVIDIA cards. This information is not available for any other GPU that the server may support. The interval for polling the GPU cards over the PBI is 5 seconds.

With warm reboot and Persistent mode is disabled, we can see the following behavior:

- Power Consumption shows as N/A.
- Power Cap limit shows with older inventory Limit Values.

The host system must have the NVIDIA GPU driver that is installed and running for various GPU features to be available. Some of the GPU features that are available are power consumption, current power cap limit, GPU power capping and limiting, GPU target temperature, minimum GPU slowdown temperature, GPU shutdown temperature, maximum memory operating temperature, and maximum GPU operating temperature, GPU utilization, and so on. These values are shown as **N/A** if the NVIDIA GPU driver is not installed. GPU features that are dependent on the driver that is loaded and running are not limited to this list.

In Linux, when the card is unused, the driver down-trains the card and unloads to save power. In such cases, the power consumption, current power cap limit, GPU power capping and limiting, GPU target temperature, minimum GPU slowdown temperature, GPU shutdown temperature, max memory operating temperature, max memory operating temperature, max GPU operating temperature, GPU utilization, and other features are not available. Persistent mode should be enabled for the device to avoid unload. You can use `nvidia-smi` tool to enable this using the command `nvidia-smi -pm 1`.

You can generate GPU reports using Telemetry. For more information about telemetry features, see [Telemetry Streaming](#).

**NOTE:** In Racadm, You may see dummy GPU entries with empty values. This may happen if the device is not ready to respond when iDRAC queries the GPU device for the information. Perform an iDRAC `racrest` operation to resolve this issue.

## Monitoring Processing Accelerators

Accelerator devices with PCIe class Processing Accelerators need real-time temperature and sensor monitoring because it generates significant heat when in use.

Perform the following steps to get the inventory information of **Processing Accelerators**:

1. Power off the server.
2. Install the accelerators on the riser card.
3. Power on the server.
4. Wait until POST is complete.
5. Log in to the iDRAC UI.
6. Go to **System > Overview > Accelerators**. You can see both GPU and Processing Accelerators sections.
7. Expand the specific accelerator to see the following sensor information:
  - Power consumption
  - Temperature details

**NOTE:** Logical temperature sensors are not displayed in the iDRAC interfaces. Only physical temperature sensors are displayed.

**NOTE:** You must have iDRAC login privilege to access the accelerators' information.

**NOTE:** Power consumption sensors are available only for the supported accelerators and are available only with Datacenter license.

**NOTE:** iDRAC interfaces may not display the information of power and thermal sensors that are dependent on the host operating system (operating system). In this case, install the GPU drivers (ROCm package) in the host operating system.

**NOTE:**

- It is recommended to perform A100 GPU CEC firmware update before updating the Accelerators firmware.
- Do not perform GPU CEC and Accelerators firmware update simultaneously to avoid failure of updates. Perform an AC or virtual AC power cycle after the firmware update failures. Doing so avoids further failures of a single update that is caused by a previous update failure.
- The firmware update for the HGX A100 8-GPU Baseboard FPGA may take between 60 and 90 minutes to complete.

- HGX A100 8-GPU Baseboard FPGA and CEC DUP updates must not be triggered simultaneously. It is recommended to follow these steps:
  1. Update the CEC firmware.
  2. Perform a virtual AC or a manual AC cycle.
  3. Update the FPGA firmware.
  4. Perform another virtual AC or manual AC cycle.
- To update the PDB FPGA from the operating system, start a cold reboot. After the update, a virtual AC cycle is performed.

**NOTE:** Occasionally, the accelerators send 0 values for power consumption. Therefore, PLDM also uses 0 values and displays the same in the UI. Although, the values are automatically corrected in subsequent readings.

**NOTE:** PCIe Devices are dependent on the device drivers and firmware to respond to iDRAC requests. These devices log LC message HWC9053 (communication that is lost with the device) when the required drivers and firmware are not loaded or when the server is in the Pre-Operating System environment (UEFI shell and Lifecycle Controller page).

## Checking the system for Fresh Air compliance

### About this task

Fresh Air cooling directly uses outside air to cool systems in the data center. Fresh Air compliant systems can operate above its normal ambient operating range (temperatures up to 113 °F (45 °C)).

**NOTE:** Some servers or certain configurations of a server may not be Fresh Air compliant. See the specific server manual for details related to Fresh Air compliance or contact Dell for more details.

To check the system for Fresh Air compliance:

### Steps

1. In the iDRAC Web interface, go to **System > Overview > Cooling > Temperature overview**. The **Temperature overview** page is displayed.
2. See the **Fresh Air** section that indicates whether the server is fresh air compliant or not.

## Viewing historical temperature data

You can monitor the percentage of time the system has operated at ambient temperature that is greater than the normally supported fresh air temperature threshold. The system board temperature sensor reading is collected over a period of time to monitor the temperature. The data collection starts when the system is first powered on after it is shipped from the factory. The data is collected and displayed for the duration when the system is powered on. You can track and store the monitored temperature for the last seven years.

**NOTE:** You can track the temperature history even for systems that are not Fresh-Air compliant. However, the threshold limits and fresh air related warnings generated are based on fresh air supported limits. The limits are 42°C for warning and 47°C for critical. These values correspond to 40°C and 45°C fresh air limits with 2°C margin for accuracy.

Two fixed temperature bands are tracked that are associated to fresh air limits:

- Warning band — Consists of the duration a system has operated above the temperature sensor warning threshold (42°C). The system can operate in the warning band for 10% of the time for 12 months.
- Critical band — Consists of the duration a system has operated above the temperature sensor critical threshold (47°C). The system can operate in the critical band for 1% of the time for 12 months which also increments time in the warning band.

The collected data is represented in a graphical format to track the 10% and 1% levels. The logged temperature data can be cleared only before shipping from the factory.

An event is generated if the system continues to operate above the normally supported temperature threshold for a specified operational time. If the average temperature over the specified operational time is greater than or equal to the warning level (> = 8%) or the critical level (> = 0.8%), an event is logged in the Lifecycle Log and the corresponding SNMP trap is generated. The events are:

- Warning event when the temperature was greater than the warning threshold for duration of 8% or more in the last 12 months.
- Critical event when the temperature was greater than the warning threshold for duration of 10% or more in the last 12 months.
- Warning event when the temperature was greater than the critical threshold for duration of 0.8% or more in the last 12 months.
- Critical event when the temperature was greater than the critical threshold for duration of 1% or more in the last 12 months.

You can also configure iDRAC to generate additional events. For more information, see the [Setting alert recurrence event](#) section.

## Viewing historical temperature data using RACADM

To view historical data using RACADM, use the `inlettemphistory` command.

For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

**NOTE:** You may find mismatch in the inlet temperature values in iDRAC UI and RACADM. To compare the data between these interfaces, please note the following:

- **iDRAC UI** options from drop-down and their output:
  - Last Day - Hourly data displayed for last 24 hours.
  - Last Month - Daily data displayed for last 30 days.
  - Last Year - Monthly data displayed for last 12 months.
- **RACADM** output: Displays exact values for hour, day, month, and year for respective RACADM commands.

## Viewing historical temperature data using iDRAC web interface

### About this task

To view historical temperature data:

### Steps

1. In the iDRAC Web interface, go to **System > Overview > Cooling > Temperature overview**. The **Temperature overview** page is displayed.
2. See the **System Board Temperature Historical Data** section that provides a graphical display of the stored temperature (average and peak values) for the last day, last 30 days, and last year.

For more information, see the **iDRAC Online Help**.

**NOTE:** After an iDRAC firmware update or iDRAC reset, some temperature data may not be displayed in the graph.

**NOTE:** WX3200 AMD GPU card currently doesnot support I2C interface for temperature sensors. Hence, temperature readings will not be available for this card from iDRAC interfaces.

## Configuring warning threshold for inlet temperature

You can modify the minimum and maximum warning threshold values for the system board inlet temperature sensor. If a reset to the default action is performed, the temperature thresholds are set to the default values. You must have Configure user privilege to set the warning threshold values for the inlet temperature sensor.

**NOTE:** The difference between **Upper Warning** and **Reading** of **System Board Inlet Temp** must be more than 2 degrees to avoid any health status warnings.

**NOTE:** Each time that you reboot the system or update the iDRAC firmware, LC logs that provide information about the Threshold limit set for the inlet temperature sensor are displayed.



## Configuring warning threshold for inlet temperature using web interface

### About this task

To configure warning threshold for inlet temperature:

### Steps

1. In the iDRAC Web interface, go to **System > Overview > Cooling > Temperature overview**. The **Temperature overview** page is displayed.
2. In the **Temperature Probes** section, for the **System Board Inlet Temp**, enter the minimum and maximum values for the **Warning Threshold** in Centigrade or Fahrenheit. If you enter the value in centigrade, the system automatically calculates and displays the Fahrenheit value. Similarly, if you enter Fahrenheit, the value for Centigrade is displayed.
3. Click **Apply**.

The values are configured.

**NOTE:** Changes to default thresholds are not reflected in the historical data chart since the chart limits are for fresh air limit values only. Warnings for exceeding the custom thresholds are different from warning associated to exceeding fresh air thresholds.

## Viewing network interfaces available on host OS

You can view information about all the network interfaces that are available on the host operating system such as the IP addresses that are assigned to the server. The iDRAC Service Module provides this information to iDRAC. The OS IP address information includes the IPv4 and IPv6 addresses, MAC address, Subnet mask or prefix length, the FQDD of the network device, network interface name, network interface description, network interface status, network interface type (Ethernet, tunnel, loopback, and so on.), Gateway address, DNS server address, and DHCP server address.

**NOTE:** This feature is available with iDRAC Express and iDRAC Enterprise/Datacenter licenses.

To view the OS information, make sure that:

- You have Login privilege.
- iDRAC Service Module is installed and running on the host operating system.
- OS Information option is enabled in the **iDRAC Settings > Overview > iDRAC Service Module** page.

iDRAC can display the IPv4 and IPv6 addresses for all the interfaces configured on the Host OS.

Depending on how the Host OS detects the DHCP server, the corresponding IPv4 or IPv6 DHCP server address may not be displayed.

## Viewing network interfaces available on host OS using RACADM

Use the `gethostnetworkinterfaces` command to view the network interfaces available on the host operating systems using RACADM. For more information, see the *iDRAC RACADM CLI Guide*.

## Viewing network interfaces available on the host operating system using web interface

### About this task

To view the network interfaces available on the host operating system using Web interface:

### Steps

1. Go to **System > Host OS > Network Interfaces**. The **Network Interfaces** page displays all the network interfaces that are available on the host operating system.

2. To view the list of network interfaces associated with a network device, from the **Network Device FQDD** drop-down menu, select a network device and click **Apply**.  
The operating system IP details are displayed in the **Host OS Network Interfaces** section.
3. From the **Device FQDD** column, click the network device link.  
The corresponding device page is displayed from the **Hardware > Network Devices** section, where you can view the device details. For information about the properties, see the **iDRAC Online Help**.
4. Click the  icon to display more details.

Similarly, you can view the host operating system network interface information that is associated with a network device from the **Hardware > Network Devices** page. Click **View Host OS Network Interfaces**.

 **NOTE:** For the ESXi host operating system in the iDRAC Service Module v2.3.0 or later, the **Description** column in the **Additional Details** list is displayed in the following format:

```
<List-of-Uplinks-Configured-on-the-vSwitch>/<Port-Group>/<Interface-name>
```

## Viewing or terminating iDRAC sessions

You can view the number of users currently logged in to iDRAC and terminate the user sessions.

### Terminating iDRAC sessions using RACADM

You must have administrator privileges to terminate iDRAC sessions using RACADM.

To view the current user sessions, use the `getssninfo` command.

To terminate a user session, use the `closesn` command.

For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

### Terminating iDRAC sessions using web interface

#### About this task

The users who do not have administrative privileges must have Configure iDRAC privilege to terminate iDRAC sessions using iDRAC Web interface.

To view and terminate the iDRAC sessions:

#### Steps

1. In the iDRAC Web interface, go to **iDRAC Settings > users > Sessions**.  
The **Sessions** page displays the session ID, username, IP address, and session type. For more information about these properties, see the **iDRAC Online Help**.
2. To terminate the session, under the **Terminate** column, click the Trashcan icon for a session.

# Setting up iDRAC communication

You can communicate with iDRAC using any of the following modes:

- iDRAC Web Interface
- Serial connection using DB9 cable (RAC serial or IPMI serial)
- IPMI Serial Over LAN
- IPMI Over LAN
- Remote RACADM
- Local RACADM

**NOTE:** To ensure that Local RACADM import or export commands work properly, ensure that the USB mass-storage host is enabled in the operating system. For information about enabling USB storage host, see the documentation for your operating system.

The following table provides an overview of the supported protocols, supported commands, and pre-requisites:

**Table 17. Communication modes — summary**

| Mode of Communication                    | Supported Protocol                                  | Supported Commands | Pre-requisite                                                   |
|------------------------------------------|-----------------------------------------------------|--------------------|-----------------------------------------------------------------|
| <b>iDRAC Web Interface</b>               | Internet Protocol (https)                           | N/A                | Web Server                                                      |
| <b>Serial using Null modem DB9 cable</b> | Serial Protocol                                     | RACADM and IPMI    | Part of iDRAC firmware and RAC Serial or IPMI Serial is enabled |
| <b>IPMI Serial Over LAN</b>              | Intelligent Platform Management Bus protocol<br>SSH | IPMI               | IPMITool is installed and IPMI Serial Over LAN is enabled       |
| <b>IPMI over LAN</b>                     | Intelligent Platform Management Bus protocol        | IPMI               | IPMITool is installed and IPMI Settings is enabled              |
| <b>Remote RACADM</b>                     | https                                               | Remote RACADM      | Remote RACADM is installed and enabled                          |
| <b>Firmware RACADM</b>                   | SSH                                                 | Firmware RACADM    | Firmware RACADM is installed and enabled                        |
| <b>Local RACADM</b>                      | IPMI                                                | Local RACADM       | Local RACADM is installed                                       |

## Topics:

- [Communicating with iDRAC through serial connection using DB9 cable](#)
- [Switching between RAC serial and serial console while using DB9 cable](#)
- [Communicating with iDRAC using IPMI SOL](#)
- [Communicating with iDRAC using IPMI over LAN](#)
- [Enabling or disabling remote RACADM](#)
- [Disabling local RACADM](#)
- [Configuring Linux for serial console during boot in RHEL](#)
- [Configuring a serial terminal in RHEL](#)
- [Supported SSH cryptography schemes](#)

# Communicating with iDRAC through serial connection using DB9 cable

## About this task

You can use any of the following communication methods to perform systems management tasks through serial connection to rack and tower servers:

- RAC Serial
- IPMI Serial—Direct Connect Basic mode and Direct Connect Terminal mode.

 **NOTE:** Only USB-to-DB9 serial adapters with FTDI controllers are supported.

 **NOTE:** When USB DB9 is connected to the system, the **Baud Rate (iDRAC Settings > Connectivity > Serial)** is set automatically depending on whether **RAC Serial** is enabled or disabled. If **RAC Serial** is enabled, the **RAC Serial > Baud Rate** is set for USB DB9. If **RAC Serial** is disabled (IPMI is enabled), the **IPMI Serial > Baud Rate** is set for USB DB9.

To establish the serial connection:

## Steps

1. Configure the BIOS to enable serial connection.
2. Connect the Null Modem DB9 cable from the management station's serial port to the managed system's external serial connector.

 **NOTE:** A server power cycle is required from vConsole or UI for any change in Baud-rate.

 **NOTE:** If iDRAC serial connection authentication is disabled, then iDRAC racreset is required for any change in BAUD-rate.
3. Make sure that the management station's terminal emulation software is configured for serial connection using any of the following:
  - Linux Minicom in an Xterm
  - Hilgraeve's HyperTerminal Private Edition (version 6.3)

Based on where the managed system is in its boot process, you can see either the POST screen or the operating system screen. This is based on the configuration: SAC for Windows and Linux text mode screens for Linux.

4. Enable RAC serial or IPMI serial connections in iDRAC.

## Configuring BIOS for serial connection

### About this task

To configure BIOS for Serial Connection:

 **NOTE:** This is applicable only for iDRAC on rack and tower servers.

### Steps

1. Turn on or restart the system.
2. Press F2.
3. Go to **System BIOS Settings > Serial Communication**.
4. Select **External Serial Connector** to **Remote Access device**.
5. Click **Back**, click **Finish**, and then click **Yes**.
6. Press Esc to exit **System Setup**.

## Enabling RAC serial connection

After configuring serial connection in BIOS, enable RAC serial in iDRAC.

 **NOTE:** This is applicable only for iDRAC on rack and tower servers.

## Enabling RAC serial connection using RACADM

To enable RAC serial connection using RACADM, use the `set` command with the object in the `iDRAC.Serial` group.

## Enabling RAC serial connection using web interface

### About this task

To enable RAC serial connection:

### Steps

1. In the iDRAC Web interface, go to **iDRAC Settings > Network > Serial**.  
The **Serial** page is displayed.
2. Under **RAC Serial**, select **Enabled** and specify the values for the attributes.
3. Click **Apply**.  
The RAC serial settings are configured.

## Enabling IPMI serial connection basic and terminal modes

To enable IPMI serial routing of BIOS to iDRAC, configure IPMI Serial in any of the following modes in iDRAC:

 **NOTE:** This is applicable only for iDRAC on rack and tower servers.

- IPMI basic mode — Supports a binary interface for program access, such as the IPMI shell (ipmish) that is included with the Baseboard Management Utility (BMU). For example, to print the System Event Log using ipmish using IPMI Basic mode, run the following command: `ipmish -com 1 -baud 57600 -flow cts -u <username> -p <password> sel get`

 **NOTE:** The default iDRAC username and password are provided on the system badge.

- IPMI terminal mode—Supports ASCII commands that are sent from a serial terminal. This mode supports a limited number of commands (including power control) and raw IPMI commands that are typed as hexadecimal ASCII characters. It allows you to view the operating system boot sequences up to BIOS, when you log in to iDRAC through SSH. You need to log out from the IPMI terminal using `[sys pwd -x]`. The following are the examples for IPMI Terminal mode commands.
  - `[sys tmode]`
  - `[sys pwd -u root calvin]`
  - `[sys health query -v]`
  - `[18 00 01]`
  - `[sys pwd -x]`

## Enabling serial connection using web interface

### About this task

Make sure to disable the RAC serial interface to enable IPMI Serial.

To configure IPMI Serial settings:

### Steps

1. In the iDRAC Web interface, go to **iDRAC Settings > Connectivity > Serial**.
2. Under **IPMI Serial**, specify the values for the attributes. For information about the options, see the **iDRAC Online Help**.
3. Click **Apply**.

## Enabling serial connection IPMI mode using RACADM

To configure the IPMI mode, disable the RAC serial interface and then enable the IPMI mode.

```
racadm set iDRAC.Serial.Enable 0
racadm set iDRAC.IPMISerial.ConnectionMode <n>
```

n=0 — Terminal Mode  
n=1 — Basic Mode

## Enabling serial connection IPMI serial settings using RACADM

### Steps

1. Change the IPMI serial-connection mode to the appropriate setting using the command.

```
racadm set iDRAC.Serial.Enable 0
```

2. Set the IPMI Serial baud rate using the command.

```
racadm set iDRAC.IPMISerial.BaudRate <baud_rate>
```

| Parameter   | Allowed values (in bps)         |
|-------------|---------------------------------|
| <baud_rate> | 9600, 19200, 57600, and 115200. |

3. Enable the IPMI serial hardware flow control using the command.

```
racadm set iDRAC.IPMISerial.FlowContro 1
```

4. Set the IPMI serial channel minimum privilege level using the command.

```
racadm set iDRAC.IPMISerial.ChanPrivLimit <level>
```

| Parameter   | Privilege level |
|-------------|-----------------|
| <level> = 2 | User            |
| <level> = 3 | Operator        |
| <level> = 4 | Administrator   |

5. Ensure that the serial MUX (external serial connector) is set correctly to the remote access device in the BIOS Setup program to configure BIOS for serial connection.

### Example

For more information about these properties, see the IPMI 2.0 specification.

## Additional settings for ipmi serial terminal mode

This section provides additional configuration settings for IPMI serial terminal mode.

## Configuring additional settings for IPMI serial terminal mode using RACADM

To configure the Terminal Mode settings, use the `set` command with the objects in the `idrac.ipmiserial` group.  
For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

## Configuring additional settings for IPMI serial terminal mode using web interface

### About this task

To set the Terminal Mode settings:

### Steps

1. In the iDRAC Web interface, go to **iDRAC Settings > Connectivity > Serial**.  
The **Serial** page is displayed.
2. Enable IPMI serial.
3. Click **Terminal Mode Settings**.  
The **Terminal Mode Settings** page is displayed.
4. Specify the following values:
  - Line editing
  - Delete control
  - Echo Control
  - Handshaking control
  - New line sequence
  - Input new line sequences

For information about the options, see the **iDRAC Online Help**.
5. Click **Apply**.  
The terminal mode settings are configured.
6. Make sure that the serial MUX (external serial connector) is set correctly to the remote access device in the BIOS Setup program to configure BIOS for serial connection.

## Switching between RAC serial and serial console while using DB9 cable

iDRAC supports Escape key sequences that allow switching between RAC Serial Interface communication and Serial Console on rack and tower servers.

### Switching from RAC serial to serial console

To switch to Serial Console Mode when in RAC Serial Interface Communication Mode, press Esc+Shift, Q.

When in terminal mode, to switch the connection to the Serial Console mode, press Esc+Shift, Q.

To go back to the terminal mode use, when connected in Serial Console mode, press Esc+Shift, 9.

### Switching from serial console to RAC serial

To switch to RAC Serial Interface communication mode when in Serial Console Mode, press Esc+Shift, 9.

The key sequence directs you to the `iDRAC Login` prompt (if the iDRAC is set to RAC Serial mode) or to the Serial Connection mode where terminal commands can be issued if iDRAC is set to IPMI Serial Direct Connect Terminal Mode.

## Communicating with iDRAC using IPMI SOL

### About this task

IPMI Serial Over LAN (SOL) allows a managed system's text-based console serial data to be redirected over iDRAC's dedicated or shared out-of-band ethernet management network. Using SOL you can:

- Remotely access operating systems with no time-out.

- Diagnose host systems on Emergency Management Services (EMS) or Special Administrator Console (SAC) for Windows or Linux shell.
- View the progress of a servers during POST and reconfigure the BIOS setup program.

To setup the SOL communication mode:

### Steps

1. Configure BIOS for serial connection.
2. Configure iDRAC to Use SOL.
3. Enable a supported protocol (SSH, IPMITool).

## Configuring BIOS for serial connection

### About this task

 **NOTE:** This is applicable only for iDRAC on rack and tower servers.

### Steps

1. Turn on or restart the system.
2. Press F2.
3. Go to **System BIOS Settings > Serial Communication**.
4. Specify the following values:
  - Serial Communication — On With Console Redirection
  - Serial Port Address — COM2.

 **NOTE:** You can set the **serial communication** field to **On with serial redirection via com1** if **serial device2** in the **serial port address** field is also set to com1.

- External serial connector — Serial device 2
  - Failsafe Baud Rate — 115200
  - Remote Terminal Type — VT100/VT220
  - Redirection After Boot — Enabled
5. Click **Back** and then click **Finish**.
  6. Click **Yes** to save the changes.
  7. Press <Esc> to exit **System Setup**.

 **NOTE:** BIOS sends the screen serial data in 25 x 80 format. The SSH window that is used to invoke the `console com2` command must be set to 25 x 80. Then, the redirected screen appears correctly.

 **NOTE:** If the boot loader or operating system provides serial redirection such as GRUB or Linux, then the BIOS **Redirection After Boot** setting must be disabled. This is to avoid potential race condition of multiple components accessing the serial port.

## Configuring iDRAC to use SOL

You can specify the SOL settings in iDRAC using Web interface, RACADM, or iDRAC Settings utility.

### Configuring iDRAC to use SOL using RACADM

#### About this task

To configure IPMI Serial over LAN (SOL):



## Steps

1. Enable IPMI Serial over LAN using the command.

```
racadm set iDRAC.IPMISol.Enable 1
```

2. Update the IPMI SOL minimum privilege level using the command.

```
racadm set iDRAC.IPMISol.MinPrivilege <level>
```

| Parameter   | Privilege level |
|-------------|-----------------|
| <level> = 2 | User            |
| <level> = 3 | Operator        |
| <level> = 4 | Administrator   |

**NOTE:** To activate IPMI SOL, you must have the minimum privilege defined in IMPI SOL. For more information, see the IPMI 2.0 specification.

3. Update the IPMI SOL baud rate using the command.

```
racadm set iDRAC.IPMISol.BaudRate <baud_rate>
```

**NOTE:** To redirect the serial console over LAN, make sure that the SOL baud rate is identical to the managed system's baud rate.

| Parameter   | Allowed values (in bps)         |
|-------------|---------------------------------|
| <baud_rate> | 9600, 19200, 57600, and 115200. |

4. Enable SOL for each user using the command.

```
racadm set iDRAC.Users.<id>.SolEnable 2
```

| Parameter | Description           |
|-----------|-----------------------|
| <id>      | Unique ID of the user |

## Example

**NOTE:** To redirect the serial console over LAN, ensure that the SOL baud rate is identical to the baud rate of the managed system.

## Configuring iDRAC to use SOL using iDRAC web interface

### About this task

To configure IPMI Serial over LAN (SOL):

## Steps

1. In the iDRAC Web interface, go to **iDRAC Settings > Connectivity > Serial Over LAN**. The **Serial over LAN** page is displayed.
2. Enable SOL, specify the values, and click **Apply**. The IPMI SOL settings are configured.
3. To set the character accumulate interval and the character send threshold, select **Advanced Settings**. The **Serial Over LAN Advanced Settings** page is displayed.
4. Specify the values for the attributes and click **Apply**.

The IPMI SOL advanced settings are configured. These values help to improve the performance.  
For information about the options, see the **iDRAC Online Help**.

## Enabling supported protocol

The supported protocols are IPMI and SSH.

## Enabling supported protocol using RACADM

To enable the SSH, run the following command.

SSH

```
racadm set iDRAC.SSH.Enable 1
```

To change the SSH port, run the following command:

```
racadm set iDRAC.SSH.Port <port number>
```

You can use tools such as:

- IPMITool for using IPMI protocol
- Putty/OpenSSH for using SSH protocol

## Enabling supported protocol using web interface

To enable SSH, go to **iDRAC Settings > Services** and select **Enabled** for SSH.

To enable IPMI, go to **iDRAC Settings > Connectivity** and select **IPMI Settings**. Make sure that the **Encryption Key** value is all zeroes or press the backspace key to clear and change the value to NULL characters.

## SOL using IPMI protocol

### About this task

The IPMI-based SOL utility and IPMITool use RMCP+ delivered using UDP datagrams to port 623. The RMCP+ provides improved authentication, data integrity checks, encryption, and the ability to carry multiple types of payloads while using IPMI 2.0. For more information, see <http://ipmitool.sourceforge.net/manpage.html>.

The RMCP+ uses a 40-character hexadecimal string (characters 0-9, a-f, and A-F) encryption key for authentication. The default value is a string of 40 zeros.

An RMCP+ connection to iDRAC must be encrypted using the encryption key (Key Generator Key). You can configure the encryption key using the iDRAC web interface or iDRAC Settings utility.

To start SOL session using IPMITool from a management station:

 **NOTE:** If required, you can change the default SOL time-out at **iDRAC Settings > Services**.

### Steps

1. Install IPMITool from the **Dell Systems Management Tools and Documentation** DVD.  
For installation instructions, see the **Software Quick Installation Guide**.
2. At the command prompt (Windows or Linux), run the following command to start SOL from iDRAC:

```
ipmitool -H <iDRAC-ip-address> -I lanplus -U <login name> -P <login password> sol  
activate
```

This command connected the management station to the managed system's serial port.

3. To quit a SOL session from IPMITool, press ~ and then . (period).

 **NOTE:** If a SOL session does not terminate, reset iDRAC and allow up to two minutes to complete booting.

### Example

-  **NOTE:** IPMI SOL session may terminate while copying large input text from a client running Windows OS to a host running Linux OS. To avoid the session from getting terminated abruptly, convert any large text to a UNIX-based line ending.
-  **NOTE:** If a SOL session created using RACADM tool exists, starting another SOL session using IPMI tool will not show any notification or error about the existing sessions.
-  **NOTE:** Due to windows OS settings, SOL session connected through ssh and IPMI tool may go to blank screen after booting. Disconnect and Re-connect the SOL Session again to get back SAC prompt.

## SOL using SSH

### About this task

Secure Shell (SSH) is a network protocol used to perform command line communications to iDRAC. You can parse remote RACADM commands through this interface.

SSH has improved security. iDRAC supports only SSH version 2 with password authentication, and is enabled by default. iDRAC supports up to two to four SSH sessions at a time.

-  **NOTE:** While establishing an SSH connection, a security message is displayed `Further Authentication required` even though 2FA is disabled.

Use open-source programs such as PuTTY or OpenSSH that support SSH on a management station to connect to iDRAC.

-  **NOTE:** Run `OpenSSH` from a VT100 or ANSI terminal emulator on Windows. Running `OpenSSH` at the Windows command prompt does not result in full functionality (that is, some keys do not respond and no graphics are displayed).

Before using SSH to communicate with iDRAC, complete the following steps:

### Steps

1. Configure BIOS to enable Serial Console.
2. Configure SOL in iDRAC.
3. Enable SSH using iDRAC Web interface or RACADM.

#### SSH (port 22) client > WAN connection > iDRAC

The IPMI-based SOL that uses SSH protocol eliminates the need for an additional utility because the serial-to-network translation happens within iDRAC. The SSH console that you use must be able to interpret and respond to the data arriving from the serial port of the managed system. The serial port usually attaches to a shell that emulates an ANSI- or VT100/VT220-terminal. The serial console is automatically redirected to the SSH.

## Using SOL from OpenSSH on Linux

### About this task

To start SOL from OpenSSH on a Linux management station:

-  **NOTE:** If required, you can change the default SSH session time-out at **iDRAC Settings > Services**.

### Steps

1. Start a shell.
2. Connect to iDRAC using the following command: `ssh <iDRAC-ip-address> -l <login name>`
3. Enter one of the following commands at the command prompt to start SOL:
  - `connect com2`
  - `console com2`

This connects iDRAC to the managed system's SOL port. Once a SOL session is established, iDRAC command line console is not available. Follow the escape sequence correctly to open the iDRAC command line console. The escape sequence is

also printed on the screen as soon as a SOL session is connected. When the managed system is off, it takes sometime to establish the SOL session.

**NOTE:** You can use console com1 or console com2 to start SOL. Reboot the server to establish the connection.

To view history of the SOL interface, enable Serial Data Capture. It writes all serial data received from the host to iDRAC memory in a 512 KB rolling window. This requires Datacenter license.

4. Quit the SOL session to close an active SOL session.

## Using SOL from PuTTY on Windows

### About this task

**NOTE:** If required, you can change the default SSH time-out at **iDRAC Settings > Services**.

To start IPMI SOL from PuTTY on a Windows management station:

### Steps

1. Run the following command to connect to iDRAC

```
putty.exe [-ssh] <login name>@<iDRAC-ip-address> <port number>
```

**NOTE:** The port number is optional. It is required only when the port number is reassigned.

2. Run the command `console com2` or `connect com2` to start SOL and boot the managed system.

A SOL session from the management station to the managed system using the SSH protocol is opened. To access the iDRAC command-line console, follow the ESC key sequence. Putty and SOL connection behavior:

- While accessing the managed system through putty during POST, if the Function keys and keypad option on putty is set to:
  - VT100+ — F2 passes, but F12 cannot pass.
  - ESC[n~ — F12 passes, but F2 cannot pass.
- In Windows, if the Emergency Management System (EMS) console is opened immediately after a host reboot, the Special Admin Console (SAC) terminal may get corrupted. Quit the SOL session, close the terminal, open another terminal, and start the SOL session using the same command.

**NOTE:** Due to windows OS settings, SOL session connected through ssh and IPMI tool may go to blank screen after booting. Disconnect and Re-connect the SOL Session again to get back SAC prompt.

## Disconnecting SOL session in iDRAC command line console

### About this task

The commands to disconnect a SOL session are based on the utility. You can exit the utility only when a SOL session is completely terminated.

To disconnect a SOL session, terminate the SOL session from the iDRAC command line console.

### Steps

To quit SOL redirection, press Enter, Esc, T.  
The SOL session closes.

### Example

If a SOL session is not terminated completely in the utility, other SOL sessions may not be available. To resolve this, terminate the command line console in the Web interface under **iDRAC Settings > Connectivity > Serial Over LAN**.

# Communicating with iDRAC using IPMI over LAN

You must configure IPMI over LAN for iDRAC to enable or disable IPMI commands over LAN channels to any external systems. If IPMI over LAN is not configured, then external systems cannot communicate with the iDRAC server using IPMI commands.

 **NOTE:** IPMI also supports IPv6 address protocol for Linux-based operating systems.

## Configuring IPMI over LAN using iDRAC settings utility

### About this task

To configure IPMI over LAN:

### Steps

1. In the **iDRAC Settings Utility**, go to **Network**.  
The **iDRAC Settings Network** page is displayed.
2. For **IPMI Settings**, specify the values.  
For information about the options, see the **iDRAC Settings Utility Online Help**.
3. Click **Back**, click **Finish**, and then click **Yes**.  
The IPMI over LAN settings are configured.

## Configuring IPMI over LAN using RACADM

### Steps

1. Enable IPMI over LAN.

```
racadm set iDRAC.IPMILan.Enable 1
```

 **NOTE:** This setting determines the IPMI commands that are performed using IPMI over LAN interface. For more information, see the IPMI 2.0 specifications at **intel.com**.

2. Update the IPMI channel privileges.

```
racadm set iDRAC.IPMILan.PrivLimit <level>
```

| Parameter   | Privilege level |
|-------------|-----------------|
| <level> = 2 | User            |
| <level> = 3 | Operator        |
| <level> = 4 | Administrator   |

3. Set the IPMI LAN channel encryption key, if required.

```
racadm set iDRAC.IPMILan.EncryptionKey <key>
```

| Parameter | Description                                                |
|-----------|------------------------------------------------------------|
| <key>     | 20-character encryption key in a valid hexadecimal format. |

### Example

 **NOTE:** The iDRAC IPMI supports the RMCP+ protocol. For more information, see the IPMI 2.0 specifications at **intel.com**.

## Configuring IPMI over LAN using web interface

### About this task

To configure IPMI over LAN:

### Steps

1. In the iDRAC Web interface, go to **iDRAC Settings > Connectivity**.  
The **Network** page is displayed.
2. Under **IPMI Settings**, specify the values for the attributes and click **Apply**.  
For information about the options, see the **iDRAC Online Help**.

The IPMI over LAN settings are configured.

## Enabling or disabling remote RACADM

You can enable or disable remote RACADM using the iDRAC Web interface or RACADM. You can run up to five remote RACADM sessions in parallel.

 **NOTE:** Remote RACADM is enabled by default.

## Enabling or disabling remote RACADM using RACADM

 **NOTE:** It is recommended to run these commands using local RACADM or firmware RACADM.

To disable remote RACADM:

- To disable remote RACADM:

```
racadm set iDRAC.Racadm.Enable 0
```

- To enable remote RACADM:

```
racadm set iDRAC.Racadm.Enable 1
```

## Enabling or disabling remote RACADM using web interface

### Steps

1. In iDRAC Web interface, go to **iDRAC Settings > Services**.
2. Under **Remote RACADM**, select the desired option and click **Apply**.  
The remote RACADM is enabled or disabled based on the selection.

## Disabling local RACADM

The local RACADM is enabled by default. To disable, see [Disabling access to modify iDRAC configuration settings on host system](#).

# Configuring Linux for serial console during boot in RHEL

## About this task

The following steps are specific to the Linux GRand Unified Bootloader (GRUB). Similar changes are required if a different boot loader is used.

**NOTE:** When you configure the client VT100 emulation window, set the window or application that is displaying the redirected Virtual Console to 25 rows x 80 columns to make sure the correct text displays. Else, some text screens may be garbled.

Edit the **/etc/grub.conf** file as follows:

## Steps

1. Locate the General Setting sections in the file and add the following:

```
serial --unit=1 --speed=57600 terminal --timeout=10 serial
```

2. Append two options to the kernel line:

```
kernel ..... console=ttyS1,115200n8r console=tty1
```

3. Disable GRUB's graphical interface and use the text-based interface. Else, the GRUB screen is not displayed in RAC Virtual Console. To disable the graphical interface, comment-out the line starting with **splashimage**.

The following example provides a sample **/etc/grub.conf** file that shows the changes that are described in this procedure.

```
# grub.conf generated by anaconda
# Note that you do not have to rerun grub after making changes to this file
# NOTICE: You do not have a /boot partition. This means that all
# kernel and initrd paths are relative to /, e.g.
# root (hd0,0)
# kernel /boot/vmlinuz-version ro root=/dev/sda1
# initrd /boot/initrd-version.img
#boot=/dev/sda
default=0
timeout=10
#splashimage=(hd0,2)/grub/splash.xpm.gz

serial --unit=1 --speed=57600
terminal --timeout=10 serial

title Red Hat Linux Advanced Server (2.4.9-e.3smp) root (hd0,0)
kernel /boot/vmlinuz-2.4.9-e.3smp ro root=/dev/sda1 hda=ide-scsi console=ttyS0
console=ttyS1,115200n8r
initrd /boot/initrd-2.4.9-e.3smp.img
title Red Hat Linux Advanced Server-up (2.4.9-e.3) root (hd0,00)
kernel /boot/vmlinuz-2.4.9-e.3 ro root=/dev/sda1 s
initrd /boot/initrd-2.4.9-e.3.im
```

4. To enable multiple GRUB options to start Virtual Console sessions through the RAC serial connection, add the following line to all options:

```
console=ttyS1,115200n8r console=tty1
```

The example shows **console=ttyS1,57600** added to the first option.

**NOTE:** If the boot loader or operating system provides serial redirection such as GRUB or Linux, then the BIOS **Redirection After Boot** setting must be disabled. This is to avoid the potential race condition of multiple components accessing the serial port.

## Enabling login to the virtual console after boot

In the file `/etc/inittab`, add a new line to configure `agetty` on the COM2 serial port:

```
co:2345:respawn:/sbin/agetty -h -L 57600 ttyS1 ansi
```

The following example shows a sample file with the new line.

```
#inittab This file describes how the INIT process should set up
#the system in a certain run-level.
#Author:Miquel van Smoorenburg
#Modified for RHS Linux by Marc Ewing and Donnie Barnes
#Default runlevel. The runlevels used by RHS are:
#0 - halt (Do NOT set initdefault to this)
#1 - Single user mode
#2 - Multiuser, without NFS (The same as 3, if you do not have #networking)
#3 - Full multiuser mode
#4 - unused
#5 - X11
#6 - reboot (Do NOT set initdefault to this)
id:3:initdefault:
#System initialization.
si::sysinit:/etc/rc.d/rc.sysinit
l0:0:wait:/etc/rc.d/rc 0
l1:1:wait:/etc/rc.d/rc 1
l2:2:wait:/etc/rc.d/rc 2
l3:3:wait:/etc/rc.d/rc 3
l4:4:wait:/etc/rc.d/rc 4
l5:5:wait:/etc/rc.d/rc 5
l6:6:wait:/etc/rc.d/rc 6
#Things to run in every runlevel.
ud::once:/sbin/update
ud::once:/sbin/update
#Trap CTRL-ALT-DELETE
ca::ctrlaltdel:/sbin/shutdown -t3 -r now
#When our UPS tells us power has failed, assume we have a few
#minutes of power left. Schedule a shutdown for 2 minutes from now.
#This does, of course, assume you have power installed and your
#UPS is connected and working correctly.
pf::powerfail:/sbin/shutdown -f -h +2 "Power Failure; System Shutting Down"
#If power was restored before the shutdown kicked in, cancel it.
pr:12345:powerokwait:/sbin/shutdown -c "Power Restored; Shutdown Cancelled"

#Run gettys in standard runlevels
co:2345:respawn:/sbin/agetty -h -L 57600 ttyS1 ansi
1:2345:respawn:/sbin/mingetty tty1
2:2345:respawn:/sbin/mingetty tty2
3:2345:respawn:/sbin/mingetty tty3
4:2345:respawn:/sbin/mingetty tty4
5:2345:respawn:/sbin/mingetty tty5
6:2345:respawn:/sbin/mingetty tty6

#Run xdm in runlevel 5
#xdm is now a separate service
x:5:respawn:/etc/X11/prefdm -nodaemon
```

In the file `/etc/securetty`, add a new line with the name of the serial tty for COM2:

```
ttyS1
```

The following example shows a sample file with the new line.

 **NOTE:** Use the Break Key Sequence (`~B`) to perform the Linux **Magic SysRq** key commands on the serial console using IPMI Tool.

```
vc/1
vc/2
vc/3
vc/4
```



```
vc/5
vc/6
vc/7
vc/8
vc/9
vc/10
vc/11
tty1
tty2
tty3
tty4
tty5
tty6
tty7
tty8
tty9
tty10
tty11
ttyS1
```

## Configuring a serial terminal in RHEL

### About this task

To configure a serial terminal in RHEL:

### Steps

1. Add, or update the following lines to `/etc/default/grub`:

```
GRUB_CMDLINE_LINUX_DEFAULT="console=tty0 console=ttyS0,115200n8"
```

```
GRUB_TERMINAL="console serial"
```

```
GRUB_SERIAL_COMMAND="serial --speed=115200 --unit=0 --word=8 --parity=no --stop=1"
```

`GRUB_CMDLINE_LINUX_DEFAULT` applies this configuration only to the default menu entry, use `GRUB_CMDLINE_LINUX` to apply it to all the menu entries.

Each line should only appear once within the `/etc/default/grub`. If the line exists, then modify it to avoid another copy. Therefore, only one `GRUB_CMDLINE_LINUX_DEFAULT` line is allowed.

2. Rebuild the `/boot/grub2/grub.cfg` configuration file by running the `grub2-mkconfig -o` command as follows:

- On BIOS-based systems:

```
~]# grub2-mkconfig -o /boot/grub2/grub.cfg
```

- On UEFI-based systems:

```
~]# grub2-mkconfig -o /boot/efi/EFI/redhat/grub.cfg
```

### Next steps

For more information, see the RHEL System Administrator's Guide at [redhat.com](https://www.redhat.com).

# Controlling GRUB from serial console

## About this task

You can configure GRUB to use the serial console instead of the VGA console. This allows you to interrupt the boot process and choose a different kernel or add kernel parameters, for example, to boot into single user mode.

## Steps

To configure GRUB to use serial console, comment out the splash image and add the `serial` and `terminal` options to `grub.conf`:

```
[root@localhost ~]# cat /boot/grub/grub.conf

# grub.conf generated by anaconda

#

# Note that you do not have to rerun grub after making changes to this file

# NOTICE:  You have a /boot partition.  This means that

#           all kernel and initrd paths are relative to /boot/, eg.

#           root (hd0,0)

#           kernel /vmlinuz-version ro root=/dev/hda2

#           initrd /initrd-version.img

#boot=/dev/hda

default=0

timeout=10

#splashimage=(hd0,0)/grub/splash.xpm.gz

serial --unit=0 --speed=1152001
```

 **NOTE:** Restart the system for the settings to take effect.

# Supported SSH cryptography schemes

To communicate with iDRAC using SSH protocol, it supports multiple cryptography schemes listed in the following table.

Table 18. SSH cryptography schemes

| Scheme Type             | Algorithms                                                                                             |
|-------------------------|--------------------------------------------------------------------------------------------------------|
| Asymmetric Cryptography |                                                                                                        |
| Public key              | <ul style="list-style-type: none"><li>curve25519-sha256</li><li>curve25519-sha256@libssh.org</li></ul> |

**Table 18. SSH cryptography schemes (continued)**

| Scheme Type                   | Algorithms                                                                                                                                                                                                                                                                                |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | <ul style="list-style-type: none"> <li>ssh-rsa</li> <li>ecdsa-sha2-nistp256</li> <li>diffie-hellman-group16-sha512</li> <li>diffie-hellman-group18-sha512</li> <li>diffie-hellman-group14-sha256</li> </ul>                                                                               |
| <b>Symmetric Cryptography</b> |                                                                                                                                                                                                                                                                                           |
| Key Exchange                  | <ul style="list-style-type: none"> <li>rsa-sha2-512</li> <li>rsa-sha2-256</li> <li>ssh-rsa</li> <li>ecdsa-sha2-nistp256</li> <li>ssh-ed25519</li> <li>ecdh-sha2-nistp256</li> <li>ecdh-sha2-nistp384</li> <li>ecdh-sha2-nistp521</li> <li>diffie-hellman-group-exchange-sha256</li> </ul> |
| Encryption                    | <ul style="list-style-type: none"> <li>chacha20-poly1305@openssh.com</li> <li>aes128-ctr</li> <li>aes192-ctr</li> <li>aes256-ctr</li> <li>aes128-gcm@openssh.com</li> <li>aes256-gcm@openssh.com</li> </ul>                                                                               |
| MAC                           | <ul style="list-style-type: none"> <li>umac-64@openssh.com</li> <li>umac-128-etm@openssh.com</li> <li>hmac-sha2-256-etm@openssh.com</li> <li>hmac-sha2-512-etm@openssh.com</li> <li>umac-128@openssh.com</li> <li>hmac-sha2-256</li> <li>hmac-sha2-512</li> </ul>                         |
| Compression                   | None                                                                                                                                                                                                                                                                                      |

 **NOTE:** If you enable OpenSSH 7.0 or later, DSA public key support is disabled. To ensure better security for iDRAC, Dell recommends not enabling DSA public key support.

## Using public key authentication for SSH

iDRAC supports the Public Key Authentication (PKA) over SSH. This is a licensed feature. When the PKA over SSH is set up and used correctly, you must enter the user name while logging into iDRAC. This is useful for setting up automated scripts that perform various functions. The uploaded keys must be in RFC 4716 or OpenSSH format. Else, you must convert the keys into that format.

In any scenario, a pair of private and public key must be generated on the management station. The public key is uploaded to iDRAC local user and private key is used by the SSH client to establish the trust relationship between the management station and iDRAC.

You can generate the public or private key pair using:

- **PuTTY Key Generator** application for clients running Windows
- **ssh-keygen** CLI for clients running Linux.

 **CAUTION:** This privilege is normally reserved for users who are members of the Administrator user group on iDRAC. However, users in the 'Custom' user group can be assigned this privilege. A user with this privilege can modify any user's configuration. This includes creation or deletion of any user, SSH Key management for users, and so on. For these reasons, assign this privilege carefully.

 **CAUTION:** The capability to upload, view, and/ or delete SSH keys is based on the 'Configure Users' user privilege. This privilege allows user(s) to configure another user's SSH key. You should grant this privilege carefully.

## Generating public keys for Linux

To use the **ssh-keygen** application to create the basic key, open a terminal window and at the shell prompt, enter `ssh-keygen -t rsa -b 2048 -C testing`

Where:

- `-t` is **rsa**.
- `-b` specifies the bit encryption size between 2048 and 4096.
- `-C` allows modifying the public key comment and is optional.

 **NOTE:** The options are case-sensitive.

Follow the instructions. After the command is performed, upload the public file.

 **CAUTION:** Keys that are generated from the Linux management station using `ssh-keygen` are in the non-4716 format. Convert the keys into the 4716 format using `ssh-keygen -e -f /root/.ssh/id_rsa.pub > std_rsa.pub`. Do not change the permissions of the key file. The conversion must be done using default permissions.

 **NOTE:** iDRAC does not support ssh-agent forward of keys.

## Generating public keys for Windows

### About this task

To use the **PuTTY Key Generator** application to create the basic key:

### Steps

1. Start the application and select RSA for the key type.
2. Enter the number of bits for the key. The number of bits must be between 2048 and 4096 bits.
3. Click **Generate** and move the mouse in the window as directed.  
The keys are generated.
4. You can modify the key comment field.
5. Enter a passphrase to secure the key.
6. Save the public and private key.

## Uploading SSH keys

You can upload up to four public keys **per user** to use over an SSH interface. Before adding the public keys, make sure that you view the keys if they are set up, so that a key is not accidentally overwritten.

When adding new public keys, make sure that the existing keys are not at the index where the new key is added. iDRAC does not perform checks to make sure previous key(s) are deleted before a new key(s) are added. When a new key is added, it is usable if the SSH interface is enabled.

## Uploading SSH keys using web interface

### About this task

To upload the SSH keys:

### Steps

1. In the iDRAC Web interface, go to **iDRAC Settings > Users > Local Users**.  
The **Local Users** page is displayed.
2. In the **User ID** column, click a user ID number.  
The **Users Main Menu** page is displayed.
3. Under **SSH Key Configurations**, select **Upload SSH Key(s)** and click **Next**.  
The **Upload SSH Key(s)** page is displayed.
4. Upload the SSH keys in one of the following ways:
  - Upload the key file.
  - Copy the contents of the key file into the text boxFor more information, see iDRAC Online Help.
5. Click **Apply**.

## Uploading SSH keys using RACADM

To upload the SSH keys, run the following command:

 **NOTE:** You cannot upload and copy a key at the same time.

- For local RACADM: `racadm sshpkauth -i <2 to 16> -k <1 to 4> -f <filename>`
- From remote RACADM using or SSH: `racadm sshpkauth -i <2 to 16> -k <1 to 4> -t <key-text>`

For example, to upload a valid key to iDRAC User ID 2 in the first key space using a file, run the following command:

```
$ racadm sshpkauth -i 2 -k 1 -f pkkey.key
```

 **NOTE:** The `-f` option is not supported on ssh/serial RACADM.

## Deleting SSH keys

Before deleting the public keys, make sure that you view the keys if they are set up, so that a key is not accidentally deleted.

## Deleting SSH keys using RACADM

To delete the SSH key(s), run the following commands:

- Specific key — `racadm sshpkauth -i <2 to 16> -d -k <1 to 4>`
- All keys — `racadm sshpkauth -i <2 to 16> -d -k all`

## Deleting SSH keys using web interface

### About this task

To delete the SSH key(s):

### Steps

1. In Web interface, go to **iDRAC Settings > Users**.  
The **Local Users** page is displayed.
2. In the **ID** column, select a user ID number, click **Edit**.  
The **Edit User** page is displayed.
3. Under **SSH Key Configurations**, select a SSH Key and click **Edit**.  
The **SSH Key** page displays the **Edit From** details.
4. Select **Remove** for the key(s) you want to delete, and click **Apply**.  
The selected key(s) is deleted.

## Viewing SSH keys

You can view the keys that are uploaded to iDRAC.

### Viewing SSH keys using web interface

#### About this task

To view the SSH keys:

#### Steps

1. In Web interface, go to **iDRAC Settings > Users**.  
The **Local Users** page is displayed.
2. In the **User ID** column, click a user ID number.  
The **Users Main Menu** page is displayed.
3. Under **SSH Key Configurations**, select **View/Remove SSH Key(s)** and click **Next**.  
The **View/Remove SSH Key(s)** page is displayed with the key details.

## User roles and user accounts

You can create user roles with specific privileges using iDRAC to manage your system and maintain system security. By default, iDRAC is configured with a local administrator role. The default iDRAC username and password are provided with the system badge. For more information, see the documentation for your server.

As an administrator, you can create user roles with the associated privileges. You can create user accounts and assign the newly created roles or the existing roles **Administrator**, **Operator**, **ReadOnly** in iDRAC. You can set up local users or use directory services such as Microsoft Active Directory or LDAP to set up user accounts. Using a directory service provides a central location for managing authorized user accounts.

### Topics:

- [iDRAC user roles and privileges](#)
- [Recommended characters in usernames and passwords](#)
- [Creating user roles](#)
- [Configuring local users](#)
- [Configuring Active Directory users](#)
- [Configuring generic LDAP users](#)
- [Testing LDAP directory service settings](#)

## iDRAC user roles and privileges

The default iDRAC roles are **Administrator**, **Operator**, and **ReadOnly**. These roles have specific user privileges.

The following table lists the default iDRAC role names:

**Table 19. iDRAC roles**

| Roles         | Privileges                                                                                                                                                              |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Administrator | Log in to iDRAC, Configure iDRAC, Configure Users, Clear Logs, Configure System, Access Virtual Console, Access Virtual Media, Test Alerts, and Execute Debug Commands. |
| Operator      | Log in to iDRAC, Configure iDRAC, Clear Logs, Configure System, Access Virtual Media, Test Alerts, and Execute Debug Commands.                                          |
| ReadOnly      | Log in to iDRAC.                                                                                                                                                        |

The following table describes the user privileges:

**Table 20. iDRAC user privileges**

| Privileges                                                                                                                                                                                                  | Description                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Log in to iDRAC                                                                                                                                                                                             | Enables the users to log in to iDRAC. This is the default privilege across all interfaces (UI, RACADM, and IPMI ).                                                                                                |
| Configure iDRAC.                                                                                                                                                                                            | Enables the users to configure iDRAC. With this privilege, a user can also configure power management, virtual console, virtual media, licenses, system settings, storage devices, BIOS settings, SCP, and so on. |
|  <b>NOTE:</b> The administrator role overrides all the privileges from the other components such as BIOS setup password. |                                                                                                                                                                                                                   |
| Configure Users.                                                                                                                                                                                            | Enables the users to create user accounts.                                                                                                                                                                        |
| Clear Logs                                                                                                                                                                                                  | Enables the users to clear only the System Event Logs (SELs).                                                                                                                                                     |
| Configure System.                                                                                                                                                                                           | Enables the users to power-cycle the host system.                                                                                                                                                                 |

**Table 20. iDRAC user privileges (continued)**

| Privileges              | Description                                                                                   |
|-------------------------|-----------------------------------------------------------------------------------------------|
| Access Virtual Console  | Enables the users to run Virtual Console.                                                     |
| Access Virtual Media    | Enables the users to run and use Virtual Media.                                               |
| Test Alerts             | Enables the users to test email alerts, SNMP traps, and other configured alert notifications. |
| Execute Debug Commands. | Enables the users to run diagnostic commands.                                                 |

## Recommended characters in usernames and passwords

This section provides details about the recommended characters while creating and using usernames and passwords.

**NOTE:** The password must include one uppercase and one lower case letter, one number, and a special character.

Use the following characters while creating usernames and passwords:

**Table 21. Recommended characters for usernames**

| Characters                                                                                                                                                                        | Length |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <ul style="list-style-type: none"> <li>0-9</li> <li>A-Z</li> <li>a-z</li> <li>-, !, #, \$, %, &amp;, (, ), *, ;, ?, [, \, ], ^, _ , ` , { ,   , }, ~, +, &lt;, =, &gt;</li> </ul> | 1–16   |

**Table 22. Recommended characters for passwords**

| Characters                                                                                                                                                                                            | iDRAC10 versions     | Length |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|--------|
| <ul style="list-style-type: none"> <li>0-9</li> <li>A-Z</li> <li>a-z</li> <li>!, -, ., ", #, \$, %, &amp;, ( ), *, +, ,, /, :, ;, ?, @, [, \, ], ^, _ , ` , { ,   , }, ~, +, &lt;, =, &gt;</li> </ul> | 1.10.17.00 and later | 1–127  |

### **NOTE:**

- You may be able to create usernames and passwords that include other characters. However, to ensure compatibility with all interfaces, Dell Technologies recommends using only the characters that are listed here.
- The network-share type determines the characters that are allowed in usernames and passwords for network shares. iDRAC supports valid characters for network share credentials as defined by the share type, except <, >, and , (comma).
- To improve security, use complex passwords that have eight or more characters and include lowercase alphabets, uppercase alphabets, numbers, and special characters. It is also recommended to regularly change the passwords, if possible.

## Creating user roles

You can create user roles with the required privileges so that you can delegate the tasks to the users efficiently. Only users with the Administrator role can create user roles.

### Steps

- Go to **iDRAC Settings > Users > Local Users > User Roles**.
- Click **+Add**.  
The **Add New Role** dialog box is displayed.



3. Select the **User ID**.
4. Enter the **User Role Name**.
5. Select the **User Privileges**.
6. Click **Save**.  
The user role is listed in the **User Roles** list.

## Configuring local users

You can configure up to 32 local users in iDRAC with specific user roles. Before you create an iDRAC user, verify if any current users exist. You can set user names, passwords, and roles with the privileges for these users. The user names and passwords can be changed using any of the iDRAC secured interfaces (that is, web interface, RACADM or Redfish).

### Create local users using iDRAC UI

#### About this task

You can add users and assign specific roles to users based on the tasks that are assigned to them.

 **NOTE:** If the user role assigned to you has the **Configure Users** privilege, only then you can create users.

#### Steps

1. In the iDRAC UI, go to **iDRAC Settings > Local Users > Overview**.  
The local users are listed.
2. Click **+Add**.  
The **Add New User** dialog box is displayed.
3. Select the **ID**.
4. Enter the **User Name**, **Password**, and **Confirm Password** fields.
5. Select the **User Role**.  
The corresponding **User Privileges** are displayed.
6. Click **Save**.  
The user name is displayed in the list.

### Configuring local users using RACADM

 **NOTE:** You must be logged in as user **root** to perform RACADM commands on a remote Linux system.

You can configure single or multiple iDRAC users using RACADM.

To configure multiple iDRAC users with identical configuration settings, follow these procedures:

- Use the RACADM examples in this section as a guide to create a batch file of RACADM commands, and then perform the batch file on each managed system.
- Create the iDRAC configuration file and perform the `racadm set` command on each managed system using the same configuration file.

If you are configuring a new iDRAC or if you have used the `racadm racresetcfg` command, then check for the default iDRAC user name and password on the system badge. The `racadm racresetcfg` command resets the iDRAC to the default values.

 **NOTE:** If SEKM is enabled on the server, then disable SEKM using the `racadm sekm disable` command before using this command. This can avoid any storage devices being locked out which are secured by iDRAC, if SEKM settings are erased from iDRAC by performing this command.

 **NOTE:** Users can be enabled and disabled over time. As a result, a user may have a different index number on each iDRAC.

To verify if a user exists, type the following command once for each index (1–16):

```
racadm get iDRAC.Users.<index>.UserName
```

Several parameters and object IDs are displayed with their current values. The key field is `iDRAC.Users.UserName=`. If a user name is displayed after `=`, that index number is taken.

**NOTE:** You can use

```
racadm get -f <myfile.cfg>
```

and view or edit the

```
myfile.cfg
```

file, which includes all iDRAC configuration parameters.

To enable SNMP v3 authentication for a user, use **SNMPv3AuthenticationType**, **SNMPv3Enable**, **SNMPv3PrivacyType** objects. For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

If you use the Server Configuration Profile file to configure users, use the **AuthenticationProtocol**, **ProtocolEnable**, and **PrivacyProtocol** attributes to enable SNMPv3 authentication.

## Adding iDRAC user using RACADM

### Steps

1. Set the index and user name.

```
racadm set idrac.users.<index>.username <user_name>
```

| Parameter   | Description              |
|-------------|--------------------------|
| <index>     | Unique index of the user |
| <user_name> | User name                |

2. Set the password.

```
racadm set idrac.users.<index>.password <password>
```

3. Set the user privileges.

For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

4. Enable the user.

```
racadm set idrac.users.<index>.enable 1
```

### Example

To verify, use the following command:

```
racadm get idrac.users.<index>
```

For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

## Enabling iDRAC user with permissions

### About this task

To enable a user with specific administrative permissions (role-based authority):

## Steps

1. Locate an available user index.

```
racadm get iDRAC.Users <index>
```

2. Type the following commands with the new user name and password.

```
racadm set iDRAC.Users.<index>.Privilege <user privilege bit mask value>
```

 **NOTE:** The default privilege value is 0, which indicates the user has no privileges enabled. For a list of valid bit-mask values for specific user privileges, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

## Configuring Active Directory users

If your company uses the Microsoft Active Directory software, you can configure the software to provide access to iDRAC, allowing you to add and control iDRAC user privileges to your existing users in your directory service. This is a licensed feature.

You can configure user authentication through Active Directory to log in to the iDRAC. You can also provide role-based authority, which enables an administrator to configure specific privileges for each user.

 **NOTE:** By default STARTLS on port 389 is configured. The connection protocol can be reconfigured to LDAPS on port 636 using Redfish or the RACADM command `racadm set iDRAC.ActiveDirectory.Connection LDAPS`.

## Prerequisites for using Active Directory authentication for iDRAC

To use the Active Directory authentication feature of iDRAC, make sure that you have:

- Deployed an Active Directory infrastructure. See the Microsoft website for more information.
- Integrated PKI into the Active Directory infrastructure. iDRAC uses the standard Public Key Infrastructure (PKI) mechanism to authenticate securely into the Active Directory. See the Microsoft website for more information.
- Enabled the Secure Socket Layer (SSL) on all domain controllers that iDRAC connects to for authenticating to all the domain controllers.

## Enabling SSL on domain controller

### About this task

When iDRAC authenticates users with an Active Directory domain controller, it starts an SSL session with the domain controller. At this time, the domain controller must publish a certificate signed by the Certificate Authority (CA)—the root certificate of which is also uploaded into iDRAC. For iDRAC to authenticate to **any** domain controller—whether it is the root or the child domain controller—that domain controller must have an SSL-enabled certificate signed by the domain's CA.

If you are using Microsoft Enterprise Root CA to **automatically** assign all your domain controllers to an SSL certificate, you must:

### Steps

1. Install the SSL certificate on each domain controller.
2. Export the Domain Controller Root CA Certificate to iDRAC.
3. Import iDRAC Firmware SSL Certificate.

## Installing SSL certificate for each domain controller

### About this task

To install the SSL certificate for each controller:

### Steps

1. Click **Start > Administrative Tools > Domain Security Policy**.
2. Expand the **Public Key Policies** folder, right-click **Automatic Certificate Request Settings** and click **Automatic Certificate Request**.  
The **Automatic Certificate Request Setup Wizard** is displayed.
3. Click **Next** and select **Domain Controller**.
4. Click **Next** and click **Finish**. The SSL certificate is installed.

## Exporting domain controller root CA certificate to iDRAC

### About this task

To export the domain controller root CA certificate to iDRAC:

### Steps

1. Locate the domain controller that is running the Microsoft Enterprise CA service.
2. Click **Start > Run**.
3. Enter `mmc` and click **OK**.
4. In the **Console 1** (MMC) window, click **File (or Console )** and select **Add/Remove Snap-in**.
5. In the **Add/Remove Snap-In** window, click **Add**.
6. In the **Standalone Snap-In** window, select **Certificates** and click **Add**.
7. Select **Computer** and click **Next**.
8. Select **Local Computer**, click **Finish**, and click **OK**.
9. In the **Console 1** window, go to **Certificates Personal Certificates** folder.
10. Locate and right-click the root CA certificate, select **All Tasks**, and click **Export....**
11. In the **Certificate Export Wizard**, click **Next**, and select **No do not export the private key**.
12. Click **Next** and select **Base-64 encoded X.509 (.cer)** as the format.
13. Click **Next** and save the certificate to a directory on your system.
14. Upload the certificate you saved in step 13 to iDRAC.

## Importing iDRAC firmware SSL certificate

### About this task

iDRAC SSL certificate is the identical certificate that is used for iDRAC web server. All iDRAC controllers are shipped with a default self-signed certificate.

If the Active Directory Server is set to authenticate the client during an SSL session initialization phase, you need to upload an iDRAC Server certificate to the Active Directory Domain controller. This additional step is not required if the Active Directory does not perform a client authentication during an SSL session's initialization phase.

 **NOTE:** If the iDRAC firmware SSL certificate is CA-signed and the certificate of that CA is already in the domain controller's Trusted Root Certificate Authority list, do not perform the steps in this section.

To import iDRAC firmware SSL certificate to all domain controller trusted certificate lists:

### Steps

1. Download iDRAC SSL certificate using the following RACADM command:  
`racadm sslcertdownload -t 1 -f <RAC SSL certificate>`
2. On the domain controller, open an **MMC Console** window and select **Certificates > Trusted Root Certification Authorities**.
3. Right-click **Certificates**, select **All Tasks** and click **Import**.
4. Click **Next** and browse to the SSL certificate file.
5. Install iDRAC SSL Certificate in each domain controller's **Trusted Root Certification Authority**.

If you have installed your own certificate, make sure that the CA signing your certificate is in the **Trusted Root Certification Authority** list. If the Authority is not in the list, you must install it on all your domain controllers.

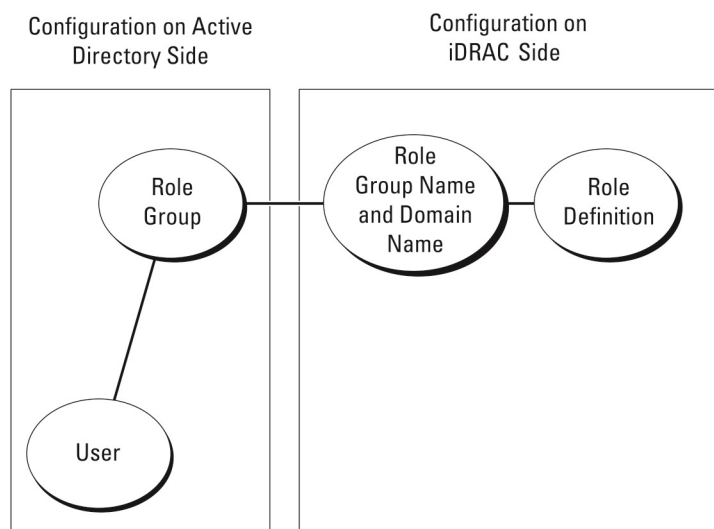
6. Click **Next** and select whether you want Windows to automatically select the certificate store based on the type of certificate, or browse to a store of your choice.
7. Click **Finish** and click **OK**. The iDRAC firmware SSL certificate is imported to all domain controller trusted certificate lists.

## Supported Active Directory authentication mechanisms

You can use Active Directory to define iDRAC user access, which uses only the default Microsoft Active Directory group objects.

## Microsoft Active Directory integration overview

As shown in the following figure, Active Directory integration requires configuration on both Active Directory and iDRAC.



**Figure 1. Configuration of iDRAC with Active Directory**

In Active Directory, a standard group object is used as a role group. A user who has iDRAC access is a member of the role group. To provide this user access to a specific iDRAC, the role group name and its domain name must be configured on the specific iDRAC. The role and the privilege level are defined on each iDRAC and not in the Active Directory. You can configure up to 15 role groups in each iDRAC. Table reference no shows the default role group privileges.

**Table 23. Default role group privileges**

| Role Groups  | Default Privilege Level | Permissions Granted                                                                                                                                                                     | Bit Mask    |
|--------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Role Group 1 | None                    | Log in to iDRAC, Configure iDRAC, Configure Users, Clear Logs, perform Server Control Commands, Access Virtual Console, Access Virtual Media, Test Alerts, perform Diagnostic Commands. | 0x000001ff  |
| Role Group 2 | None                    | Log in to iDRAC, Configure iDRAC, perform Server Control Commands, Access Virtual Console, Access Virtual Media, Test Alerts, perform Diagnostic Commands.                              | 0x0000001f3 |
| Role Group 3 | None                    | Log in to iDRAC                                                                                                                                                                         | 0x00000001  |

**Table 23. Default role group privileges (continued)**

| Role Groups  | Default Privilege Level | Permissions Granted     | Bit Mask   |
|--------------|-------------------------|-------------------------|------------|
| Role Group 4 | None                    | No assigned permissions | 0x00000000 |
| Role Group 5 | None                    | No assigned permissions | 0x00000000 |

 **NOTE:** The Bit Mask values are used only when setting Active Directory permissions with RACADM.

## Single domain versus multiple domain scenarios

If all the login users and role groups, including the nested groups, are in the same domain, then only the domain controllers' addresses must be configured on iDRAC. In this single domain scenario, any group type is supported.

If all the login users and role groups, or any of the nested groups, are from multiple domains, then Global Catalog server addresses must be configured on iDRAC. In this multiple domain scenario, all the role groups and nested groups, if any, must be a Universal Group type.

## Configuring Active Directory

### Prerequisites

Before configuring the Active Directory, ensure that:

- You have the iDRAC Enterprise or Datacenter license.
- The configuration is performed on a server that is used as the Domain Controller.
- The date, time, and time zone on the server are correct.
- The iDRAC network settings are configured, or in the iDRAC web interface go to **iDRAC Settings > Connectivity > Network > Common Settings** to configure the network settings.

### About this task

To configure iDRAC for an Active Directory login access:

### Steps

1. On an Active Directory server (domain controller), open the Active Directory users and Snap-in in the system.
2. Create the iDRAC groups and users.
3. Configure the group name, domain name, and the role privileges on iDRAC using the iDRAC web interface or RACADM.

## Configuring Active Directory using RACADM

### Steps

1. Use the following commands:

```
racadm set iDRAC.ActiveDirectory.Enable 1
racadm set iDRAC.ADGroup.Name <common name of the role group>
racadm set iDRAC.ADGroup.Domain <fully qualified domain name>
racadm set iDRAC.ADGroup.Privilege <Bit-mask value for specific RoleGroup permissions>
racadm set iDRAC.ActiveDirectory.DomainController1 <fully qualified domain name or IP
address of the domain controller>
racadm set iDRAC.ActiveDirectory.DomainController2 <fully qualified domain name or IP
address of the domain controller>
racadm set iDRAC.ActiveDirectory.DomainController3 <fully qualified domain name or IP
address of the domain controller>
racadm set iDRAC.ActiveDirectory.GlobalCatalog1 <fully qualified domain name or IP
address of the domain controller>
racadm set iDRAC.ActiveDirectory.GlobalCatalog2 <fully qualified domain name or IP
address of the domain controller>
racadm set iDRAC.ActiveDirectory.GlobalCatalog3 <fully qualified domain name or IP
address of the domain controller>
```

- Enter the Fully Qualified Domain Name (FQDN) of the domain controller, not the FQDN of the domain. For example, enter `servername.dell.com` instead of `dell.com`.
- For bit-mask values for specific Role Group permissions, see [Default role group privileges](#).
- Provide at least one of the three domain controller addresses. iDRAC attempts to connect to each of the configured addresses one-by-one until it makes a successful connection. These are the addresses of the domain controllers where the user accounts and the role groups are located.
- The Global Catalog server is only required for Active Directory when the user accounts and role groups are in different domains. In multiple domain cases, only the Universal Group can be used.
- If certificate validation is enabled, the FQDN or IP address that you specify in this field must match the Subject or Subject Alternative Name field of your domain controller certificate.
- To disable the certificate validation during an SSL handshake, use the following command:

```
racadm set iDRAC.ActiveDirectory.CertValidationEnable 0
```

In this case, no Certificate Authority (CA) certificate must be uploaded.

- To enforce the certificate validation during an SSL handshake (optional), use the following command:

```
racadm set iDRAC.ActiveDirectory.CertValidationEnable 1
```

In this case, you must upload the CA certificate using the following command:

```
racadm sslcertupload -t 0x2 -f <ADS root CA certificate>
```

**NOTE:** If certificate validation is enabled, specify the Domain Controller Server addresses and the Global Catalog FQDN. Ensure that DNS is configured correctly under **Overview > iDRAC Settings > Network**.

Using the following RACADM command may be optional.

```
racadm sslcertdownload -t 1 -f <RAC SSL certificate>
```

2. If DHCP is enabled on iDRAC and you want to use the DNS provided by the DHCP server, enter the following command:

```
racadm set iDRAC.IPv4.DNSFromDHCP 1
```

3. If DHCP is disabled on iDRAC or you want manually enter the DNS IP address, enter the following RACADM command:

```
racadm set iDRAC.IPv4.DNSFromDHCP 0
racadm set iDRAC.IPv4.DNSFromDHCP.DNS1 <primary DNS IP address>
racadm set iDRAC.IPv4.DNSFromDHCP.DNS2 <secondary DNS IP address>
```

4. If you want to configure a list of user domains so that you must only enter the user name when logging in to the web interface, use the following command:

```
racadm set iDRAC.UserDomain.<index>.Name <fully qualified domain name or IP Address of the domain controller>
```

You can configure up to 40 user domains with index numbers between 1 and 40.

## Configuring Active Directory using iDRAC web interface

### About this task

**NOTE:** For information about the various fields, see the **iDRAC Online Help**.

### Steps

1. In the iDRAC web interface, go to **iDRAC Settings > Users > Directory Services**. The **Directory Services** page is displayed.
2. Select **Microsoft Active Directory**, and then click **Edit**. The **Active Directory Configuration and Management-Certificate Settings** page is displayed.

3. Optionally, enable **Certificate Validation** and click **Choose file** and select the CA-signed digital certificate that is used during initiation of either TLS or SSL connections when communicating with the Active Directory (AD) server.
4. Click **Upload**.
5. Click **Next**.

The **Active Directory Configuration and Management Step-Common Settings** page is displayed.

6. Enable **Active Directory** and add the **User Domain Name**.
7. If certificate validation is enabled, specify the **Domain Controller Server Addresses**.

 **NOTE:** Ensure that DNS is configured correctly under **iDRAC Settings > Network**.

8. Enable **Lookup Global Catalog Servers with DNS** and specify the Global Catalog Server addresses.
9. Click a **Role Group** to configure the control authorization policy for users under the standard schema mode. **The Standard Schema Role Groups** dialog box is displayed.
10. Enter the **Group Name** and **Group Domain**, select **Group Role** and the privileges, and then click **Save**.
11. Click **Save**. The Active Directory settings are configured.

## Adding iDRAC users and privileges to Active Directory

Using the Dell-extended Active Directory Users and Computers Snap-in, you can add iDRAC users and privileges by creating device, association, and privilege objects. To add each object, perform the following:

- Create an iDRAC device Object
- Create a Privilege Object
- Create an Association Object
- Add objects to an Association Object

### Creating iDRAC device object

#### About this task

To create iDRAC device object:

#### Steps

1. In the MMC **Console Root** window, right-click a container.
2. Select **New > Dell Remote Management Object Advanced**.  
The **New Object** window is displayed.
3. Enter a name for the new object. The name must be identical to iDRAC name that you enter while configuring Active Directory properties using iDRAC Web interface.
4. Select iDRAC **Device Object** and click OK.

### Creating privilege object

#### About this task

To create a privilege object:

 **NOTE:** You must create a privilege object in the same domain as the related association object.

#### Steps

1. In the **Console Root** (MMC) window, right-click a container.
2. Select **New > Dell Remote Management Object Advanced**.  
The **New Object** window is displayed.
3. Enter a name for the new object.
4. Select **Privilege Object** and click OK.
5. Right-click the privilege object that you created, and select **Properties**.



6. Click the **Remote Management Privileges** tab and assign the privileges for the user or group.

## Creating association object

### About this task

To create association object:

 **NOTE:** iDRAC association object is derived from the group and its scope is set to Domain Local.

### Steps

1. In the **Console Root** (MMC) window, right-click a container.
2. Select **New > Dell Remote Management Object Advanced**.  
This **New Object** window is displayed.
3. Enter a name for the new object and select **Association Object**.
4. Select the scope for the **Association Object** and click OK.
5. Provide access privileges to the authenticated users for accessing the created association objects.

## Providing user access privileges for association objects

### About this task

To provide access privileges to the authenticated users for accessing the created association objects:

### Steps

1. Go to **Administrative Tools > ADSI Edit**. The **ADSI Edit** window is displayed.
2. In the right-pane, navigate to the created association object, right-click and select **Properties**.
3. In the **Security** tab, click **Add**.
4. Type `Authenticated Users`, click **Check Names**, and click **OK**. The authenticated users is added to the list of **Groups and user names**.
5. Click **OK**.

## Adding objects to association object

Using the **Association Object Properties** window, you can associate users or user groups, privilege objects, and iDRAC devices or iDRAC device groups.

You can add groups of users and iDRAC devices.

## Adding privileges

### About this task

To add privileges:

Click the **Privilege Object** tab to add the privilege object to the association that defines the user's or user group's privileges when authenticating to an iDRAC device. Only one privilege object can be added to an Association Object.

### Steps

1. Select the **Privileges Object** tab and click **Add**.
2. Enter the privilege object name and click **OK**.
3. Click the **Privilege Object** tab to add the privilege object to the association that defines the user's or user group's privileges when authenticating to an iDRAC device. Only one privilege object can be added to an Association Object.

## Adding users or user groups

### About this task

To add users or user groups:

### Steps

1. Right-click the **Association Object** and select **Properties**.
2. Select the **Users** tab and click **Add**.
3. Enter the user or user group name and click **OK**.

## Adding iDRAC devices or iDRAC device groups

### About this task

To add iDRAC devices or iDRAC device groups:

### Steps

1. Select the **Products** tab and click **Add**.
2. Enter iDRAC devices or iDRAC device group name and click **OK**.
3. In the **Properties** window, click **Apply** and click **OK**.
4. Click the **Products** tab to add one iDRAC device connected to the network that is available for the defined users or user groups. You can add multiple iDRAC devices to an Association Object.

## Testing Active Directory settings

You can test the Active Directory settings to verify whether your configuration is correct, or to diagnose the problem with a failed Active Directory log in.

## Testing Active Directory settings using iDRAC web interface

### About this task

To test the Active Directory settings:

### Steps

1. In iDRAC Web Interface, go to **iDRAC Settings > Users > Directory Services > Microsoft Active Directory**, click **Test**. The **Test Active Directory Settings** page is displayed.
2. Click **Test**.
3. Enter a test user's name (for example, **username@domain.com**) and password and click **Start Test**. A detailed test results and the test log displays.

If there is a failure in any step, examine the details in the test log to identify the problem and a possible solution.

 **NOTE:** When testing Active Directory settings with Enable Certificate Validation checked, iDRAC requires that the Active Directory server be identified by the FQDN and not an IP address. If the Active Directory server is identified by an IP address, certificate validation fails because iDRAC is not able to communicate with the Active Directory server.

## Configuring generic LDAP users

iDRAC provides a generic solution to support Lightweight Directory Access Protocol (LDAP)-based authentication. This feature does not require any schema extension on your directory services.

To make iDRAC LDAP implementation generic, the commonality between different directory services is used to group users and then map the user-group relationship. The directory service-specific action is the schema. For example, they may have different attribute names for the group, user, and the link between the user and the group. These actions can be configured in iDRAC.

**NOTE:** By default STARTLS on port 389 is configured. The connection protocol can be reconfigured to LDAPS using Redfish or the RACADM command `racadm set idrac.ldap.connection LDAPS`. The port can be reconfigured using Redfish or the RACADM command `racadm set idrac.ldap.port 636`.

## Configuring generic LDAP directory service using RACADM

To configure the LDAP directory service, use the objects in the `iDRAC.LDAP` and `iDRAC.LDAPRole` groups.

For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

## Configuring generic LDAP directory service using iDRAC web-based interface

### About this task

To configure the generic LDAP directory service using Web interface:

**NOTE:** For information about the various fields, see the [iDRAC Online Help](#).

### Steps

1. In the iDRAC Web interface, go to **iDRAC Settings > Users > Directory Services > Generic LDAP Directory Service**, click **Edit**.  
The **Generic LDAP Configuration and Management Step 1 of 3** page displays the current generic LDAP settings.
2. Optionally, enable certificate validation and upload the digital certificate used during initiation of SSL connections when communicating with a generic LDAP server.
3. Click **Next**.  
The **Generic LDAP Configuration and Management Step 2 of 3** page is displayed.
4. Enable generic LDAP authentication and specify the location information about generic LDAP servers and user accounts.  
**NOTE:** If certificate validation is enabled, specify the LDAP Server's FQDN and make sure that DNS is configured correctly under **iDRAC Settings > Network**.  
**NOTE:** In this release, nested group is not supported. The firmware searches for the direct member of the group to match the user DN. Also, only single domain is supported. Cross domain is not supported.
5. Click **Next**.  
The **Generic LDAP Configuration and Management Step 3a of 3** page is displayed.
6. Click **Role Group**.  
The **Generic LDAP Configuration and Management Step 3b of 3** page is displayed.
7. Specify the group distinguished name, the privileges associated with the group, and click **Apply**.  
**NOTE:** If you are using Novell eDirectory and if you have used these characters—#(hash), "(double quotes), :(semi colon), > (greater than), , (comma), or <(lesser than)—for the Group DN name, they must be escaped.

The role group settings are saved. The **Generic LDAP Configuration and Management Step 3a of 3** page displays the role group settings.

8. If you want to configure additional role groups, repeat steps 7 and 8.
9. Click **Finish**. The generic LDAP directory service is configured.

## Testing LDAP directory service settings

You can test the LDAP directory service settings to verify whether your configuration is correct, or to diagnose the problem with a failed LDAP log in.

# Testing LDAP directory service settings using iDRAC web interface

## About this task

To test the LDAP directory service settings:

## Steps

1. In iDRAC Web Interface, go to **iDRAC Settings > Users > Directory Services > Generic LDAP Directory Service**. The **Generic LDAP Configuration and Management** page displays the current generic LDAP settings.
2. Click **Test**.
3. Enter the user name and password of a directory user that is chosen to test the LDAP settings. The format depends on the **Attribute of User Login** is used and the user name entered must match the value of the chosen attribute.
  -  **NOTE:** When testing LDAP settings with **Enable Certificate Validation** checked, iDRAC requires that the LDAP server be identified by the FQDN and not an IP address. If the LDAP server is identified by an IP address, certificate validation fails because iDRAC is not able to communicate with the LDAP server.
  -  **NOTE:** When generic LDAP is enabled, iDRAC first tries to login the user as a directory user. If it fails, local user lookup is enabled.

The test results and the test log are displayed.

# System Configuration Lockdown mode

System Configuration Lockdown mode helps in preventing unintended changes after a system is provisioned. Lockdown mode is applicable to both configuration and firmware updates. When the system is locked down, any attempt to change the system configuration is blocked. If any attempts are made to change the critical system settings, an error message is displayed. Enabling System lockdown mode blocks the firmware update of third-party I/O cards using the vendor tools.

System Lockdown mode is only available for Enterprise licensed customers.

**NOTE:** Enhanced Lockdown for NICs only includes firmware lockdown to prevent firmware updates. Configuration (x-UEFI) lockdown is not supported.

**NOTE:** After the System Lockdown mode is enabled, you cannot change any configuration settings. System settings fields are disabled.

Lockdown mode can be enabled or disabled using the following interfaces:

- iDRAC web interface
- RACADM
- System Configuration Profile (SCP)
- Redfish
- Using F2 during POST and selecting iDRAC Settings
- Factory System Erase

**NOTE:** To enable Lockdown mode, you must have an iDRAC Enterprise or Datacenter license, and Control and Configure system privileges.

**NOTE:** You may be able to access vMedia while system is in the Lockdown mode but configuring remote file share is not enabled.

**NOTE:** The interfaces such as OMSA, SysCfgr, and USC can only check the settings but cannot modify the configurations.

**NOTE:** When the Lockdown mode is enabled, you cannot configure any alert settings. However, you can trigger a test email.

The following table lists the functional and nonfunctional features, interfaces, and utilities that are affected by Lockdown mode:

**NOTE:** Changing the boot order using iDRAC is not supported when Lockdown mode is enabled. However, a boot-control option is available in the vConsole menu, which has no effect when iDRAC is in the Lockdown mode.

**Table 24. Items affected by Lockdown mode**

| Disabled                                                                                                                                                                                                                                                                                                                                                          | Remains functional                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Deleting Licenses</li> <li>• DUP updates</li> <li>• SCP import</li> <li>• Reset to defaults</li> <li>• IPMI</li> <li>• DRAC/LC</li> <li>• DTK-Syscfg</li> <li>• Redfish</li> <li>• OpenManage Essentials</li> <li>• BIOS (F2 settings become read-only)</li> <li>• Select network cards.</li> <li>• iLKM/SEKM</li> </ul> | <ul style="list-style-type: none"> <li>• Power Operations - Power ON/OFF, Reset</li> <li>• Power cap setting</li> <li>• Power priority</li> <li>• Identify devices (Chassis or PERC)</li> <li>• Part replacement, Easy Restore, and System Board replacement</li> <li>• Running diagnostics</li> <li>• Modular operations (FlexAddress or Remote-Assigned Address)</li> <li>• All vendor tools that have direct access to the device (excludes selected NICs)</li> <li>• License export</li> <li>• PERC               <ul style="list-style-type: none"> <li>◦ PERC CLI</li> <li>◦ DTK-RAIDCFG</li> <li>◦ F2</li> </ul> </li> <li>• All Vendor tools that have direct access to the device.</li> </ul> |

**Table 24. Items affected by Lockdown mode**

| Disabled | Remains functional                                                                                                                                                                                                                                                                                                |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | <ul style="list-style-type: none"> <li>• NVMe <ul style="list-style-type: none"> <li>◦ DTK-RAIDCFG</li> <li>◦ F2</li> </ul> </li> <li>• BOSS-N1</li> <li>• ISM settings (operating system BMC enable, watchdog ping, operating system name, operating system version)</li> <li>• Deleting User Session</li> </ul> |

 **NOTE:** When lockdown mode is enabled, the OpenID Connect login option is not displayed in the iDRAC login page.

# Configuring iDRAC to send alerts

## About this task

You can set alerts and actions for certain events that occur on the managed system. An event occurs when the status of a system component is greater than the pre-defined condition. If an event matches an event filter and you have configured this filter to generate an alert (e-mail, SNMP trap, IPMI alert, remote system logs, Redfish event, or WS events), then an alert is sent to one or more configured destinations. If the same event filter is also configured to perform an action (such as reboot, power cycle, or power off the system), the action is performed. You can set only one action for each event.

To configure iDRAC to send alerts:

## Steps

1. Enable alerts.
2. Optionally, you can filter the alerts based on category or severity.
3. Configure the e-mail alert, IPMI alert, SNMP trap, remote system log, Redfish event, operating system log, and/or WS-event settings.
4. Enable event alerts and actions such as:
  - Send an email alert, IPMI alert, SNMP traps, remote system logs, Redfish event, operating system log, or WS events to configured destinations.
  - Perform a reboot, power off, or power cycle the managed system.

## Next steps

 **NOTE:** For any update requiring iDRAC reset/ reboot or in case iDRAC is rebooted, it is recommended to check if iDRAC is fully ready by waiting for few seconds of interval with maximum timeout of 5 minutes before using any other command.

## Topics:

- [Enabling or disabling alerts](#)
- [Setting event alerts](#)
- [Setting alert recurrence event](#)
- [Setting event actions](#)
- [Configuring email alert, SNMP trap, or IPMI trap settings](#)
- [Configuring Redfish Eventing](#)
- [Configuring Remote System Logging](#)
- [Alerts message IDs](#)
- [Liquid cooling leak detection](#)

## Enabling or disabling alerts

For sending an alert to configured destinations or to perform an event action, you must enable the global alerting option. This property overrides individual alerting or event actions that is set.

## Enabling or disabling alerts using web interface

### About this task

To enable or disable generating alerts:

### Steps

1. In the iDRAC web interface, go to **Configuration > System Settings > Alert Configuration**.

The **Alerts** page is displayed.

2. Under **Alerts** section:
  - Select **Enable** to enable alert generation or perform an event action.
  - Select **Disable** to disable alert generation or disable an event action.
3. Click **Apply** to save the setting.

## Quick Alert Configuration

### About this task

To configure alerts in bulk:

### Steps

1. Go to **Quick Alert Configuration** under **Alert Configuration** page.
2. Under **Quick Alert Configuration** section:
  - Select the alert category.
  - Select the issue severity notification.
  - Select the location where you would like to receive these notifications.
3. Click **Apply** to save the settings.  
All the alerts that are configured are displayed in total under **Alerts Configuration Summary**.

### Next steps

-  **NOTE:** You must select at least one category, one severity, and one destination type to apply the configuration.
-  **NOTE:** After the alerts are configured using the **Quick Alerts** tab, and when you go to the **Alert Configuration** tab, the configured alerts are not enabled in the respective alert categories. To enable the respective alert categories:
1. Select one of the other category tabs (**System Health**, **Audit**, **Updates**, and **Configuration**) in the **Alert Configuration** page.
  2. Revert to the category that was originally used for alert configuration.

## Enabling or disabling alerts using RACADM

Use the following command:

```
racadm set iDRAC.IPMILan.AlertEnable <n>
```

n=0 — Disabled

n=1 — Enabled

## Enabling or disabling alerts using iDRAC settings utility

### About this task

To enable or disable generating alerts or event actions:

### Steps

1. In the iDRAC Settings utility, go to **Alerts**.  
The **iDRAC Settings Alerts** page is displayed.
2. Under **Platform Events**, select **Enabled** to enable alert generation or event action. Else, select **Disabled**. For more information about the options, see **iDRAC Settings Utility Online Help**.
3. Click **Back**, click **Finish**, and then click **Yes**.  
The alert settings are configured.



# Setting event alerts

You can set event alerts such as e-mail alerts, IPMI alerts, SNMP traps, remote system logs, operating system logs, and WS events to be sent to configured destinations.

## Setting event alerts using web interface

### About this task

To set an event alert using the web interface:

### Steps

1. Make sure that you have configured the e-mail alert, IPMI alert, SNMP trap settings, and/or remote system log settings.
2. In iDRAC Web interface, go to **Configuration > System Settings > Alerts and Remote System Log Configuration**.
3. Under **Category**, select one or all of the following alerts for the required events:
  - Email
  - SNMP Trap
  - IPMI Alert
  - Remote System Log
  - WS Eventing
  - OS Log
  - Redfish Event
4. Select **Action**.  
The setting is saved.
5. Optionally, you can send a test event. In the **Message ID to Test Event** field, enter the message ID to test if the alert is generated and click **Test**. For more information about the event and error messages generated by the system firmware and agents that monitor system components, see the **Event and Error Message Reference Guide** at [iDRACmanuals](#)

## Setting event alerts using RACADM

To set an event alert, use the **eventfilters** command. For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

## Setting alert recurrence event

You can configure iDRAC to generate additional events at specific intervals if the system continues to operate at a temperature which is greater than the inlet temperature threshold limit. The default interval is 30 days. The valid range is 0 to 366 days. A value of '0' indicates no event recurrence.

 **NOTE:** You must have Configure iDRAC privilege to set the alert recurrence value.

## Setting alert recurrence events using RACADM

To set the alert recurrence event using RACADM, use the **eventfilters** command. For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

## Setting alert recurrence events using iDRAC web interface

### About this task

To set the alert recurrence value:

### Steps

1. In iDRAC Web interface, go to **Configuration > System Settings > Alert Recurrence**.
2. In the **Recurrence** column, enter the alert frequency value for the required category, alert, and severity type(s).  
For more information, see the **iDRAC Online help**.
3. Click **Apply**.  
The alert recurrence settings are saved.

## Setting event actions

You can set event actions such as perform a reboot, power cycle, power off, or perform no action on the system.

### Setting event actions using web interface

#### About this task

To set an event action:

#### Steps

1. In iDRAC Web interface, go to **Configuration > System Settings > Alert and Remote System Log Configuration**.
2. From the **Actions** drop-down menu, for each event select an action:
  - Reboot
  - Power Cycle
  - Power Off
  - No Action
3. Click **Apply**.  
The setting is saved.

### Setting event actions using RACADM

To configure an event action, use the `eventfilters` command. For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

## Configuring email alert, SNMP trap, or IPMI trap settings

The management station uses Simple Network Management Protocol (SNMP) and Intelligent Platform Management Interface (IPMI) traps to receive data from iDRAC. For systems with large number of nodes, it may not be efficient for a management station to poll each iDRAC for every condition that may occur. For example, event traps can help a management station with load balancing between nodes or by issuing an alert if an authentication failure occurs. SNMP v1, v2, and v3 formats are supported.

You can configure the IPv4 and IPv6 alert destinations, email settings, and SMTP server settings, and test these settings. You can also specify the SNMP v3 user to whom you want to send the SNMP traps.

Before configuring the email, SNMP, or IPMI trap settings, make sure that:

- You have Configure RAC permission.
- You have configured the event filters.

### Configuring IP alert destinations

You can configure the IPv6 or IPv4 addresses to receive the IPMI alerts or SNMP traps.

For information about the iDRAC MIBs required to monitor the servers using SNMP, see the *Dell OpenManage SNMP Reference Guide* available on the [OpenManage manuals](#) page..

## Configuring IP alert destinations using web interface

### About this task

To configure alert destination settings using Web interface:

#### Steps

1. In iDRAC Web interface, go to **Configuration > System Settings > SNMP and E-mail Settings**.
2. Select the **State** option to enable an alert destination (IPv4 address, IPv6 address, or Fully Qualified Domain Name (FQDN)) to receive the traps.  
You can specify up to eight destination addresses. For more information about the options, see the **iDRAC Online Help**.
3. Select the SNMP v3 user to whom you want to send the SNMP trap.
4. Enter the iDRAC SNMP community string (applicable only for SNMPv1 and v2) and the SNMP alert port number.  
For more information about the options, see the **iDRAC Online Help**.  
**NOTE:** The Community String value indicates the community string to use in a Simple Network Management Protocol (SNMP) alert trap sent from iDRAC. Make sure that the destination community string is the same as the iDRAC community string. The default value is Public.
5. To test whether the IP address is receiving the IPMI or SNMP traps, click **Send** under **Test IPMI Trap** and **Test SNMP Trap** respectively.
6. Click **Apply**.  
The alert destinations are configured.
7. In the **SNMP Trap Format** section, select the protocol version to be used to send the traps on the trap destination(s) — **SNMP v1**, **SNMP v2**, or **SNMP v3** and click **Apply**.  
**NOTE:** The **SNMP Trap Format** option applies only for SNMP Traps and not for IPMI Traps. IPMI Traps are always sent in SNMP v1 format and is not based on the configured **SNMP Trap Format** option.

The SNMP trap format is configured.

## Configuring IP alert destinations using RACADM

### About this task

To configure the trap alert settings:

#### Steps

1. To enable traps:

```
racadm set idrac.SNMP.Alert.<index>.Enable <n>
```

| Parameter | Description                                        |
|-----------|----------------------------------------------------|
| <index>   | Destination index. Allowed values are 1 through 8. |
| <n>=0     | Disable the trap                                   |
| <n>=1     | Enable the trap                                    |

2. To configure the trap destination address:

```
racadm set idrac.SNMP.Alert.<index>.DestAddr <Address>
```

| Parameter | Description                                        |
|-----------|----------------------------------------------------|
| <index>   | Destination index. Allowed values are 1 through 8. |
| <Address> | A valid IPv4, IPv6, or FQDN address                |

3. Configure the SNMP community name string:

```
racadm set idrac.ipmilan.communityname <community_name>
```

| Parameter        | Description              |
|------------------|--------------------------|
| <community_name> | The SNMP Community Name. |

4. To configure SNMP destination:

- Set the SNMP trap destination for SNMPv3:

```
racadm set idrac.SNMP.Alert.<index>.DestAddr <IP address>
```

- Set SNMPv3 users for trap destinations:

```
racadm set idrac.SNMP.Alert.<index>.SNMPv3Username <user_name>
```

- Enable SNMPv3 for a user:

```
racadm set idrac.users.<index>.SNMPv3Enable Enabled
```

5. To test the trap, if required:

```
racadm testtrap -i <index>
```

### Example

For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

## Configuring IP alert destinations using iDRAC settings utility

### About this task

You can configure alert destinations (IPv4, IPv6, or FQDN) using the iDRAC Settings utility. To do this:

### Steps

1. In the **iDRAC Settings utility**, go to **Alerts**.  
The **iDRAC Settings Alerts** page is displayed.
2. Under **Trap Settings**, enable the IP address(es) to receive the traps and enter the IPv4, IPv6, or FQDN destination address(es). You can specify up to eight addresses.
3. Enter the community string name.  
For information about the options, see the **iDRAC Settings Utility Online Help**.
4. Click **Back**, click **Finish**, and then click **Yes**.  
The alert destinations are configured.

## Configuring email alert settings

You can configure the sender email address and receiver (destination) email address to receive the email alerts. Also, configure the SMTP server address settings.

**NOTE:** Email alerts support both IPv4 and IPv6 addresses. The iDRAC DNS Domain Name must be specified when using IPv6.

**NOTE:** If you are using an external SMTP server, ensure that iDRAC can communicate with that server. If the server is unreachable, the error RAC0225 is displayed while trying to send a test mail.

## Configuring email alert settings using web interface

### About this task

To configure the email alert settings using Web interface:

### Steps

1. In the iDRAC Web interface, go to **Configuration > System Settings > SMTP (E-mail) Configuration**.
2. Type a valid email address.
3. Click **Send** under **Test Email** to test the configured email alert settings.
4. Click **Apply**.
5. For SMTP (Email) Server Settings provide the following details:
  - SMTP (Email) Server IP Address or FQDN/DNS Name
  - Custom Sender Address—This field has the following options:
    - **Default** —Address field is not editable.
    - **Custom**—You can enter the email ID from which you can receive the email alerts.
  - Custom Message Subject Prefix—This field has the following options:
    - **Default**—Default message is not editable.
    - **Custom**—You can choose the message to appear in the **Subject** line of the email.
  - SMTP Port Number—The connection can be encrypted and emails can be sent over secure ports:
    - **No Encryption**—Port 25 (default)
    - **SSL**— Port 465
  - Connection Encryption—When you do not have an email server in your premises, you can use cloud-based email servers or SMTP Relays. To configure a cloud email server, you can set this feature to any of the following values from the drop-down:
    - **None**—No encryption on the connection to the SMTP server. It is the default value.
    - **SSL**—Runs SMTP protocol over SSL



#### NOTE:

- This is a licensed feature and is not available in the iDRAC Core License.
- You must have Configure iDRAC privilege to use this feature.

- Authentication
- Username

For Server settings, the port usage depends on `connectionencryptiontype` and this can be configured only using RACADM.

6. Click **Apply**. For more information about the options, see the **iDRAC Online Help**.

## Configuring email alert settings using RACADM

### Steps

1. To enable email alert:

```
racadm set iDRAC.EmailAlert.Enable.[index] [n]
```

| Parameter    | Description                                              |
|--------------|----------------------------------------------------------|
| <b>index</b> | Email destination index. Allowed values are 1 through 4. |
| <b>n=0</b>   | Disables email alerts.                                   |
| <b>n=1</b>   | Enables email alerts.                                    |

2. To configure email settings:

```
racadm set iDRAC.EmailAlert.Address.[index] [email-address]
```

| Parameter            | Description                                                        |
|----------------------|--------------------------------------------------------------------|
| <b>index</b>         | Email destination index. Allowed values are 1 through 4.           |
| <b>email-address</b> | Destination email address that receives the platform event alerts. |

3. To configure sender email settings:

```
racadm set iDRAC.RemoteHosts.[index] [email-address]
```

| Parameter            | Description                                                |
|----------------------|------------------------------------------------------------|
| <b>index</b>         | Sender Email index.                                        |
| <b>email-address</b> | Sender email address that sends the platform event alerts. |

4. To configure a custom message:

```
racadm set iDRAC.EmailAlert.CustomMsg.[index] [custom-message]
```

| Parameter             | Description                                              |
|-----------------------|----------------------------------------------------------|
| <b>index</b>          | Email destination index. Allowed values are 1 through 4. |
| <b>custom-message</b> | Custom message                                           |

5. To test the configured email alert, if required:

```
racadm testemail -i [index]
```

| Parameter    | Description                                                           |
|--------------|-----------------------------------------------------------------------|
| <b>index</b> | Email destination index to be tested. Allowed values are 1 through 4. |

For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

## Configuring SMTP email server address settings

You must configure the SMTP server address for email alerts to be sent to specified destinations.

### Configuring SMTP email server address settings using iDRAC web interface

#### About this task

To configure the SMTP server address:

#### Steps

1. In iDRAC Web interface, go to **Configuration > System Settings > Alert Configuration > SNMP (E-mail Configuration)**.
2. Enter the valid IP address or fully qualified domain name (FQDN) of the SMTP server to be used in the configuration.
3. Select the **Enable Authentication** option and then provide the user name and password (of a user who has access to SMTP server).
4. Enter the SMTP port number.  
For more information about the fields, see the **iDRAC Online Help**.
5. Click **Apply**.

The SMTP settings are configured.

## Configuring SMTP email server address settings using RACADM

To configure the SMTP email server:

```
racadm set iDRAC.RemoteHosts.SMTPServerIPAddress <SMTP E-mail Server IP Address>
```

## Configuring Redfish Eventing

The Redfish eventing protocol is used for a client service (subscriber) to register interest (subscription) with a server (event source) for receiving messages containing the Redfish events (notifications or event messages). Clients interested in receiving the Redfish eventing messages can subscribe with iDRAC and receive Lifecycle Controller job related events.

## Configuring Remote System Logging

You can send lifecycle logs to a remote system. Before doing this, ensure that:

- There is network connectivity between iDRAC and the remote system.
- The remote system and iDRAC are on the same network.

 **NOTE:** This feature is available with iDRAC Enterprise and Datacenter licenses.

Remote Syslog identity certificate can be generated within the company's internal certificate signing server setup. TLS based Remote Syslog servers and clients use the same CA certificate in the configuration settings, which is obtained from a CA server. iDRAC provides a user interface to upload this CA certificate and add it to its configuration file and restart the Remote Syslog service.

## Configuring remote system logging using web interface

### About this task

To configure the remote syslog server settings:

### Steps

1. In the iDRAC Web interface, go to **Configuration > System Settings > Alert Configuration > Remote Syslog > Settings**.
2. The following settings are available. Select the required setting:
  - **Basic Settings**—For legacy solutions
  - **Secure Settings**—For new Implementation (Encrypt Remote Syslog traffic with TLS). For
  - **None**—For disabling Remote Syslog alerts

For information about the field values of these options, see the **iDRAC Online Help**.

3. Click **Apply**.  
The settings are saved. All logs written to the lifecycle log are also simultaneously written to configured remote server(s).

## Configuring remote system logging using RACADM

To configure the remote system-logging settings, use the `set` command with the objects in the `iDRAC.SysLog` group.

For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

## Alerts message IDs

The following table provides the list of message IDs that are displayed for the alerts.

**Table 25. Alert message IDs**

| Message ID | Description          |
|------------|----------------------|
| AMP        | Amperage             |
| ASR        | Auto Sys Reset       |
| BAT        | Battery Event        |
| BIOS       | BIOS Management      |
| BOOT       | BOOT Control         |
| CBL        | Cable                |
| CPU        | Processor            |
| CPUA       | Proc Absent          |
| CTL        | Storage Contr        |
| DH         | Cert Mgmt            |
| DIS        | Auto-Discovery       |
| ENC        | Storage Encloser     |
| FAN        | Fan Event            |
| FSD        | Debug                |
| HWC        | Hardware Config      |
| IPA        | DRAC IP Change       |
| ITR        | Intrusion            |
| JCP        | Job Control          |
| LC         | Lifecycle Controller |
| LIC        | Licensing            |
| LNK        | Link Status          |
| LOG        | Log event            |
| MEM        | Memory               |
| NDR        | NIC OS Driver        |
| NIC        | NIC Config           |
| OSD        | OS Deployment        |
| OSE        | OS Event             |
| PCI        | PCI Device           |
| PDR        | Physical Disk        |
| PR         | Part Exchange        |
| PST        | BIOS POST            |
| PSU        | Power Supply         |
| PSUA       | PSU Absent           |
| PWR        | Power Usage          |
| RAC        | RAC Event            |
| RDU        | Redundancy           |
| RED        | FW Download          |



**Table 25. Alert message IDs (continued)**

| Message ID | Description     |
|------------|-----------------|
| RFM        | FlexAddress SD  |
| RSI        | Remote Service  |
| SEC        | Security Event  |
| SEL        | Sys Event Log   |
| SSD        | PCIe SSD        |
| STOR       | Storage         |
| SUP        | FW Update Job   |
| SWC        | Software Config |
| SWU        | Software Change |
| SYS        | System Info     |
| TMP        | Temperature     |
| TST        | Test Alert      |
| UEFI       | UEFI Event      |
| USR        | User Tracking   |
| VDR        | Virtual Disk    |
| VLT        | Voltage         |
| VME        | Virtual Media   |
| VRM        | Virtual Console |
| WRK        | Work Note       |

## Liquid cooling leak detection

iDRAC detects liquid cooling leaks in the CPU and GPU through critical, warning, and information signals received from the respective OEM IPMI sensors. The leaks are notified as System Event Logs (SELs), LC logs, WS events, Emails, and SNMP traps based on the configured alert settings. iDRAC performs appropriate actions according to the alert configuration settings.

By default, servers are configured to perform an automatic **Ungraceful Power Off** when any leak is detected. If you want to change the default configuration (**Ungraceful Power Off**) to other options, see [Configuring leak detection](#).

If iDRAC detects a leak while accessing the pre-boot environment (BIOS or boot manager), or during the booting process, the server may trigger an immediate shutdown.

## Configuring leak detection

**Liquid Cooling System Default Action Ungraceful Power Off** is the alert group that is applicable to liquid cooling leak alerts and actions. By default the alert group is configured for ungraceful **Power Off** action. If there is a leak in the system, ungraceful power off is initiated. You can configure the options according to your requirements.

### About this task

#### Steps

1. Go to **Configuration > System Settings > Alert Configuration > Alerts > Alert Configuration > Liquid Cooling System Default Action UnGraceful Power Off**.
2. Click **+**.  
The **Severity** and **SNMP trap** check boxes are selected. In the **Actions** list, **Power Off** is selected by default.

3. If you want to change the default **Power Off** action, select **No action** or **Graceful Power Off** from the **Action** list.

 **NOTE:** If **No action** is selected, the system does not power off when there is a liquid leak.

 **NOTE:** If the server fails to perform a **Graceful Power Off** within 15 minutes, perform a force power off.

 **CAUTION:** If iDRAC is rebooting and the Graceful Power Off is also in progress simultaneously, the system takes more than 20 minutes to power off. It is recommended to force power off the server and not wait for the Graceful Power Off operation to complete.

4. If you want to include more notifications, select **Email**, **SNMP Trap**, **IPMI Alert**, **Remote System Log**, **WS Event**, **OS Log**, and **Redfish Event** check boxes.
5. To change the power action for small leaks, modify the **Action** for Warning severity (yellow ).
6. To change the action for large leaks, modify the **Action** for Critical severity (red X).

## Managing logs

iDRAC provides Lifecycle logs that contains events that are related to system, storage devices, network devices, firmware updates, configuration changes, license messages, and so on. However, the system events are also available as a separate log called System Event Log (SEL). The lifecycle log is accessible through iDRAC Web interface, and RACADM.

When the size of the lifecycle log reaches 800 KB, the logs are compressed and archived. You can only view the nonarchived log entries, and apply filters and comments to nonarchived logs. To view the archived logs, you must export the entire lifecycle log to a location on your system.

### Topics:

- [Viewing System Event Log](#)
- [Viewing Lifecycle log](#)
- [Viewing Lifecycle log using RACADM](#)
- [Exporting Lifecycle Controller logs](#)
- [Prevent Lifecycle Log Overflow](#)
- [Adding work notes](#)
- [Viewing Lifecycle log](#)

## Viewing System Event Log

When a system event occurs on a managed system, it is recorded in the System Event Log (SEL). The same SEL entry is also available in the LC log.

 **NOTE:** SEL and LC logs may have mismatch in timestamp when iDRAC is rebooting.

## Viewing System Event Log using RACADM

To view the SEL:

```
racadm getsel <options>
```

If no arguments are specified, the entire log is displayed.

To display the number of SEL entries: `racadm getsel -i`

To clear the SEL entries: `racadm clrsel`

For more information, see [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

## Viewing System Event Log using web interface

To view the SEL, in iDRAC Web interface, go to **Maintenance > System Event Log**.

The **System Event Log** page displays a system health indicator, a time stamp, and a description for each event logged. For more information, see the **iDRAC Online Help**.

Click **Save As** to save the **SEL** to a location of your choice.

 **NOTE:** If you are using Internet Explorer and if there is a problem when saving, download the Cumulative Security Update for Internet Explorer. You can download it from the Microsoft Support website at [support.microsoft.com](http://support.microsoft.com).

To clear the logs, click **Clear Log**.

 **NOTE:** **Clear Log** only appears if you have Clear Logs permission.

After the SEL is cleared, an entry is logged in the Lifecycle Controller log. The log entry includes the user name and the IP address from where the SEL was cleared.

## Viewing System Event Log using iDRAC settings utility

### About this task

You can view the total number of records in the System Event Log (SEL) using the iDRAC Settings Utility and clear the logs. To do this:

### Steps

1. In the iDRAC Settings Utility, go to **System Event Log**.  
The **iDRAC Settings.System Event Log** displays the **Total Number of Records**.
2. To clear the records, select **Yes**. Else, select **No**.
3. To view the system events, click **Display System Event Log**.
4. Click **Back**, click **Finish**, and then click **Yes**.

## Viewing Lifecycle log

Lifecycle Controller logs provide the history of changes that are related to components installed on a managed system. You can also add work notes to each log entry.

The following events and activities are logged:

- All
- SystemHealth—System m Health category represents all the alerts that are related to hardware within the system chassis.
- Storage — Storage Health category represents alerts that are related to the storage subsystem.
- Updates—Update category represents alerts that are generated due to firmware/Driver upgrades/downgrades.
- Audit — Audit category represents the audit log.
- Configuration — Configuration category represents alerts that are related to hardware, firmware, and software configuration changes.
- Work Notes

When you log in to or log out of iDRAC using any of the following interfaces, the log-in, log-out, or login failure events are recorded in the Lifecycle logs:

- SSH
- Web interface
- RACADM
- Redfish
- IPMI over LAN
- Serial
- Virtual console
- Virtual media

You can view, and filter logs based on the category and severity level. You can also export and add a work note to a log event.

 **NOTE:** Lifecycle logs for Personality Mode change are generated only during the warm boot of the host.

If you initiate configuration jobs using RACADM CLI or iDRAC web interface, the Lifecycle log contains information about the user, interface used, and the IP address of the system from which you initiate the job.

 **NOTE:** When an event occurs multiple times, a single event log is displayed in the LC logs. An additional log (LOG007) is also displayed indicating the number of times this event has occurred. By default, duplicate event logs are disabled in iDRAC. If you want all the events to be displayed in the LC logs, run the RACADM command `set idrac.logging.LCDuplicateEventEnable enabled`.

## Viewing Lifecycle log using web interface

To view the Lifecycle Logs, click **Maintenance > Lifecycle Log**. The **Lifecycle Log** page is displayed. For more information about the options, see the **iDRAC Online Help**.

### Filtering Lifecycle logs

#### About this task

You can filter logs based on category, severity, keyword, or date range.

To filter the lifecycle logs:

#### Steps

1. In the **Lifecycle Log** page, under the **Log Filter** section, do any or all of the following:
  - Select the **Log Type** from the drop-down list.
  - Select the severity level from the **Severity** drop-down list.
  - Enter a keyword.
  - Specify the date range.
2. Click **Apply**.  
The filtered log entries are displayed in **Log Results**.

### Adding comments to Lifecycle logs

#### About this task

To add comments to the Lifecycle logs:

#### Steps

1. In the **Lifecycle Log** page, click the + icon for the required log entry.  
The Message ID details are displayed.
2. Enter the comments for the log entry in the **Comment** box.  
The comments are displayed in the **Comment** box.

## Viewing Lifecycle log using RACADM

To view Lifecycle logs, use the `lcllog` command.

For more information, see the *iDRAC RACADM CLI Guide*.

## Exporting Lifecycle Controller logs

You can export the entire Lifecycle Controller log (active and archived entries) in a single zipped XML file to a network share or to the local system. The zipped XML file extension is `.xml.gz`. The file entries are ordered sequentially based on their sequence numbers, ordered from the lowest sequence number to the highest.

### Exporting Lifecycle Controller logs using RACADM

To export the Lifecycle Controller logs, use the `lcllog export` command.

For more information, see the *iDRAC RACADM CLI Guide*.

## Exporting Lifecycle Controller logs using web interface

### About this task

To export the Lifecycle Controller logs using the Web interface:

### Steps

1. In the **Lifecycle Log** page, click **Export**.
2. Select any of the following options:
  - **Network**—Export the Lifecycle Controller logs to a shared location on the network.
  - **Local**—Export the Lifecycle Controller logs to a location on the local system.

 **NOTE:** While specifying the network share settings, it is recommended to avoid special characters for user name and password or percent encode the special characters.

For information about the fields, see the **iDRAC Online Help**.
3. Click **Export** to export the log to the specified location.

## Prevent Lifecycle Log Overflow

iDRAC supports the ability to prevent Lifecycle Log overflow from consoles due to high frequency of logins from consoles.

- The USR0030/USR0032 events are logged in the lifecycle log for every successful login/logout respectively.
  - These events can be aggregated into a new single log, based on an attribute setting.
  - A new USR0036 log shall be logged in the lifecycle log containing an aggregation of the login-logout events that have occurred within a duration of time that is specified by attribute `LCLoggingAggregationTimeout`.
-  **NOTE:**
- By default, the feature attribute `LCLogAggregation` is disabled.
  - By default, the timeout is set to 60 minutes and is applicable only if `LCLogAggregation` is Enabled.
- USR0030 and USR0032 shall not be logged in the lifecycle log, but continues to be sent individual alerts if the corresponding alerts are enabled (SNMP/Email/Redfish Event/WS-Event).

## Adding work notes

### About this task

Each user who logs in to iDRAC can add work notes and this is stored in the lifecycle log as an event. You must have iDRAC logs privilege to add work notes. A maximum of 255 characters are supported for each new work note.

 **NOTE:** You cannot delete a work note.

To add a work note:

### Steps

1. In the iDRAC Web interface, go to **Dashboard > Notes > add note**.  
The **Work Notes** page is displayed.
2. Under **Work Notes**, enter the text in the blank text box.

 **NOTE:** It is recommended not to use too many special characters.
3. Click **Save**.  
The work note is added to the log. For more information, see the **iDRAC Online Help**.

# Viewing Lifecycle log

Lifecycle Controller logs provide the history of changes that are related to components installed on a managed system. You can also add work notes to each log entry.

The following events and activities are logged:

- All
- SystemHealth—System m Health category represents all the alerts that are related to hardware within the system chassis.
- Storage — Storage Health category represents alerts that are related to the storage subsystem.
- Updates—Update category represents alerts that are generated due to firmware/Driver upgrades/downgrades.
- Audit — Audit category represents the audit log.
- Configuration — Configuration category represents alerts that are related to hardware, firmware, and software configuration changes.
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- Redfish
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You can view, and filter logs based on the category and severity level. You can also export and add a work note to a log event.

 **NOTE:** Lifecycle logs for Personality Mode change are generated only during the warm boot of the host.

If you initiate configuration jobs using RACADM CLI or iDRAC web interface, the Lifecycle log contains information about the user, interface used, and the IP address of the system from which you initiate the job.

 **NOTE:** When an event occurs multiple times, a single event log is displayed in the LC logs. An additional log (LOG007) is also displayed indicating the number of times this event has occurred. By default, duplicate event logs are disabled in iDRAC. If you want all the events to be displayed in the LC logs, run the RACADM command `set idrac.logging.LCDuplicateEventEnable enabled`.

## Viewing Lifecycle log using web interface

To view the Lifecycle Logs, click **Maintenance > Lifecycle Log**. The **Lifecycle Log** page is displayed. For more information about the options, see the **iDRAC Online Help**.

# Monitoring and managing power in iDRAC

You can use iDRAC to monitor and manage the power requirements of the managed system. This helps to protect the system from power outages by appropriately distributing and regulating the power consumption on the system.

The key features are:

- **Power Monitoring** —View the power status, history of power measurements, the current averages, peaks, and so on, for the managed system.
- **Power Capping** —View and set the power cap for the managed system, including displaying the minimum and maximum potential power consumption. This is a licensed feature.
- **Power Control** —Enables you to remotely perform power control operations (such as, power on, power off, system reset, power cycle, and graceful shutdown) on the managed system.
- **Power Supply Options**—Configure the power supply options such as redundancy policy, cold redundancy, and PSU rotation.
- **AC Power Recovery Options**—Configure the power recovery options so that the system is recovered based on your requirement.

## Topics:

- [Monitoring power](#)
- [Setting warning threshold for power consumption](#)
- [Configuring power supply options](#)
- [Enabling or disabling power button](#)
- [Multi-Vector Cooling](#)

## Monitoring power

iDRAC monitors the power consumption in the system continuously and displays the following power values:

- Power consumption warning and critical thresholds.
- Cumulative power, peak power, and peak amperage values.
- Power consumption over the last hour, last day, or last week.
- Average, minimum, and maximum power consumption.
- Historical peak values and peak timestamps.
- Peak headroom and instantaneous headroom values.

**NOTE:** The histogram for the system power consumption trend (hourly, daily, weekly) is maintained only while iDRAC is running. If iDRAC is restarted, the existing power consumption data is lost and the histogram is restarted.

**NOTE:** After an iDRAC firmware update or reset, the power consumption graph will be wiped or reset.

## Monitoring performance index of CPU, memory, and input output modules using web interface

To monitor the performance index of CPU, memory, and I/O modules, in the iDRAC web interface, go to **System > Performance**.

- **System Performance** section — Displays the current reading and the warning reading for CPU, Memory and I/O utilization index, and system level CUPS index in a graphical view.
- **System Performance Historical Data** section:
  - Provides the statistics for CPU, memory, IO utilization, and the system level CUPS index. If the host system is powered off, then the graph displays the power off line below 0 percent.
  - You can reset the peak utilization for a particular sensor. Click **Reset Historical Peak**. You must have Configure privilege to reset the peak value.



- **Performance Metrics** section:
  - Displays status and present reading
  - Displays or specifies the warning threshold utilization limit. You must have server configure privilege to set the threshold values.

For information about the displayed properties, see the [iDRAC Online Help](#).

## Monitoring performance index for of CPU, memory, and input output modules using RACADM

Use the **SystemPerfStatistics** sub command to monitor performance index for CPU, memory, and I/O modules. For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

## Setting warning threshold for power consumption

You can set the warning threshold value for the power consumption sensor in the system. The warning/critical power threshold may change after the system is power-cycled based on PSU capacity and redundancy policy. However, the warning threshold must not exceed the critical threshold even if the PSU capacity or the redundancy policy is changed.

If a **Reset** action is performed, the power thresholds are set to default.

You must have Configure User privilege to set the **Warning Threshold** value for the power consumption sensor.

 **NOTE:** The **Warning Threshold** value is reset to the default value after performing a `racreset` or an iDRAC update.

## Setting warning threshold for power consumption

### Steps

1. In the iDRAC Web interface, go to **System > Overview > Present Power Reading and Thresholds**.
2. In the **Present Power Reading and Thresholds** section, click **Edit Warning Threshold**.  
The **Edit Warning Threshold** dialog box is displayed.
3. In the **Warning Threshold** field, enter the threshold value in **Watts** or **BTU/hr**.  
The values must be lower than the **Failure Threshold** values. The values are rounded off to the nearest value that is divisible by 14. If you enter **Watts**, the system automatically calculates and displays the **BTU/hr** value. Similarly, if you enter BTU/hr, the value for **Watts** is displayed.
4. Click **Save**. The values are configured.

## Configuring power supply options

You can configure the power supply options such as power control, AC power recovery, redundancy policy, cold redundancy policy, and power cap policy.

 **NOTE:** For PSUs, individual zone-level messages for PSU redundancy loss and recovery are not logged. Instead, a single entry (RDU0012 for loss, RDU0011 for recovery) is recorded for both zones when PSU redundancy fails and recovers.

## Power control operations

iDRAC enables you to remotely perform a power on, power off, reset, graceful shutdown , Non-Masking Interrupt (NMI), or power cycle using UI or RACADM.

Server power-control operations that are initiated from iDRAC are independent of the power-button behavior that is configured in the BIOS. You can use the `PushPowerButton` function to gracefully shut down the system, or power it on, even if the BIOS is configured to do nothing when the physical power button is pressed.

## Performing power control operations in the Web Interface

### Steps

1. Go to **Configuration > Power Management > Power Control**. The **Power Control** options are displayed.
2. Select the required power operation:
  - Power Off System
  - Non-Masking Interrupt (NMI)
  - Graceful Shutdown
  - Reset System (warm boot)
  - Power Cycle System (cold boot)
3. Click **Apply**. For more information, see the **iDRAC Online Help**.

## Performing power control operations using RACADM

To perform power actions, use the **serveraction** command.

For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

## Configuring AC Power Recovery

You can configure the expected power state after a power loss and power recovery of the server.

### Steps

1. Go to **Configuration > Power Management > AC Power Recovery**
2. Select the expected state of the server in the **AC Power Recovery** field after power recovery. The default option is **Last**.  
 **NOTE:** Select **Last** if you want to restore the server to its state before the power loss occurred.
3. Select the **AC Power Recovery Delay** option depending on when you want to power on the server.
4. If you have selected **User Defined** as the **AC Power Recovery Delay**, enter the **User Defined Delay (120s to 600s)** delay in seconds (between 120 and 600 seconds).

## Power capping

You can view the power threshold limits that cover the range of AC and DC power consumption. These limits reflect the power usage of a system under heavy workload in the data center. Power Capping is a licensed feature.

## Viewing and configuring power cap policy

When power cap policy is enabled, it enforces a user-defined power limits on the system. If power-capping is not enabled, the default hardware power-protection policy is used. This power-protection policy is independent of the user-defined policy. The system performance is dynamically adjusted to maintain power consumption close to the specified threshold.

Actual power consumption depends on the workload. It may momentarily exceed the threshold until performance adjustments are completed. For example, consider a system that has a minimum and maximum Potential Power Consumption values of 500 W and 700 W respectively. You can specify a Power Budget Threshold to reduce consumption to 525 W. When this power budget is configured, the performance of the system is dynamically adjusted to maintain power consumption of 525 W or less.

If you set a very low power cap or if the ambient temperature is unusually high, power consumption may temporarily exceed the power-cap while the system is powering up or being reset.

If the power cap value is set lower than the minimum recommended threshold, iDRAC may not be able maintain the requested power cap.

You can specify the value in Watts, BTU/hr, or as a percentage of the recommended maximum power limit.

When setting the power cap threshold in BTU/hr, the conversion to Watts is rounded off to the nearest integer. When the power cap threshold are read from the system, the Watts to BTU/hr conversion is also rounded off. Because of the rounding off, the actual values may slightly differ.

 **NOTE:** Setting a power cap limit to a value below the recommended range may cause varied performance including increased boot time.

## Configuring power cap policy using web interface

### About this task

To view and configure the power policies:

#### Steps

1. In iDRAC Web interface, go to **Configuration > Power Management > Power Cap Policy**.  
The current power policy limit is displayed under the **Power Cap Limits** section.
2. Select **Enable** under **Power Cap**.
3. Under **Power Cap Limits** section, enter the power limit within recommended range in Watts and BTU/hr or the maximum % of recommended system limit.
4. Click **Apply** to apply the values.

## Configuring the power cap policy using RACADM

To view and configure the current power cap values, use the following objects with the `set` command:

- `system.serverpwr.PowerCapSetting`
- `system.serverpwr.PowerCapValue`

For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

## Configuring power cap policy using iDRAC settings utility

### About this task

To view and configure power policies:

#### Steps

1. In iDRAC Settings utility, go to **Power Configuration**.

 **NOTE:** The **Power Configuration** link is available only if the server power supply unit supports power monitoring.

The **iDRAC Settings Power Configuration** page is displayed.

2. Select **Enabled** to enable the **Power Cap Policy** Else, select **Disabled**.
3. Use the recommended settings, or under **User Defined Power Cap Policy**, enter the required limits.  
For more information about the options, see the **iDRAC Settings Utility Online Help**.
4. Click **Back**, click **Finish**, and then click **Yes**.  
The power cap values are configured.

## Redundancy policies

You can select the redundancy policies in iDRAC based on your requirement.

If you want the backup PSU to remain inactive until the primary PSU fails, enable the **Cold Redundancy** policy. The backup PSU becomes the cold redundant PSU. If the primary PSU fails or cannot handle the load, the cold redundant PSU becomes the primary PSU to perform the power supply tasks. The Cold Redundancy policy optimizes the input power consumption of the enabled PSUs without jeopardizing the redundancy policy.

If you want to switch between the cold redundant PSU and the active PSU periodically, enable the **Cold Redundancy PSU Rotation**.

 **NOTE:** If **Cold Redundancy** is enabled, the power consumption of the cold redundant PSU displays 0 **Amps** in the **Power Supply Unit Readings (System >Power )** page.

## Configuring redundancy policy in the Web Interface

You can enable **Cold Redundancy** and **Cold Redundancy PSU Rotation** policies for the PSUs.

### Steps

1. Go to **Configuration > Power Management > Power Configuration**.
2. Under **Power Redundancy Policy**, select **A/B Grid Redundant**.
3. If you want to enable cold redundancy and cold redundancy PSU rotation policies, select **True** in the **Cold Redundancy** and **Cold Redundancy PSU Rotation** boxes.
4. Click **Apply**.

## Configuring power supply options using RACADM

To configure the power supply options, use the following objects with the `get/set` command:

- `system.serverpwr.PSRedPolicy`

For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

## Configuring power supply options using iDRAC settings utility

### About this task

To configure the power supply options:

### Steps

1. In iDRAC Settings utility, go to **Power Configuration**.

 **NOTE:** The **Power Configuration** link is available only if the server power supply unit supports power monitoring.

The **iDRAC Settings Power Configuration** page is displayed.

2. Under **Power Supply Options**:
  - Enable or disable power supply redundancy.
  - Enable or disable Cold redundancy.
  - Enable or disable PSU rotation.
3. Click **Back**, click **Finish**, and then click **Yes**.  
The power supply options are configured.

## Enabling or disabling power button

### About this task

To enable or disable the power button on the managed system:

### Steps

1. In iDRAC Settings utility, go to **Front Panel Security**.  
The **iDRAC Settings Front Panel Security** page is displayed.
2. Select **Enabled** to enable the power button or **Disabled** to disable it.
3. Click **Back**, click **Finish**, and then click **Yes**.  
The settings are saved.

# Multi-Vector Cooling

Multi-Vector Cooling implements a multiprong approach to Thermal Controls in Dell Server Platforms. You can configure multivector cooling options through iDRAC web interface by navigating to **Configuration > System Settings > Hardware Settings > Cooling Configuration**. It includes (but not limited to):

- Large set of sensors (thermal, power, inventory so on) that allows accurate interpretation of real-time system thermal state at various locations within the server. It displays only a small subset of sensors that are relevant to users needs based on the configuration.
- Intelligent and adaptive closed loop control algorithm optimizes fan response to maintain component temperatures. It also conserves fan power, airflow consumption, and acoustics.
- Using fan zone mapping, cooling can be initiated for the components when it requires. Thus, it results in maximum performance without compromising the efficiency of power utilization.
- Accurate representation of slot by slot PCIe airflow in terms of LFM metric (Linear Feet per Minute - an accepted industry standard on how PCIe card airflow requirement is specified). Display of this metric in various iDRAC interfaces allows user to:
  - Know the maximum LFM capability of each slot within the server.
  - Know what approach is being taken for PCIe cooling for each slot (airflow that is controlled, temperature that is controlled).
  - Know the minimum LFM being delivered to a slot, if the card is a third-party Card (user-defined custom card).
  - Dial-in custom minimum LFM value for the third-party Card allowing more accurate definition of the card cooling needs for which the user is better aware of through their custom card specification.
- Displays real-time system airflow metric (CFM, cubic feet per minute) in various iDRAC interfaces to the user to enable data center airflow balancing based on aggregation of per server CFM consumption.
- Allows custom thermal settings like Thermal Profiles (Maximum Performance vs. Maximum Performance per Watt, Sound Cap), custom fan speed options (minimum fan speed, fan speed offsets) and custom Exhaust Temperature settings.
  - Most of these settings allow additional cooling over the baseline cooling generated by thermal algorithms and do not allow fan speeds to go below system cooling requirements.

**NOTE:** One exception to the above statement is for fan speeds that are added for third-party PCIe cards. The thermal algorithm provision airflow for third-party cards may be more or less than the actual card cooling needs, and the customer may fine-tune the response for the card by entering the LFM corresponding to the third-party card.

- The Custom Exhaust Temperature option limits exhaust temperature to customer wanted settings.

**NOTE:** It is important to note that with certain configurations and workloads, it may not be physically possible to reduce exhaust below a wanted set point (for example Custom exhaust setting of 45C with a high inlet temp (for example 30C) and a loaded config {high system power consumption, low airflow}).

- The Sound Cap option limits the CPU power consumption and controls fan speed and acoustical ceiling. This is unique for acoustical deployments and may result in reduced system performance.
- System layout and design enables increased airflow capability (by allowing high power) and dense system configurations. It provides less system restrictions and increased feature density.
  - Streamlined airflow permits efficient airflow to fan power consumption ratio.
- Custom fans are designed for higher efficiency, better performance, longer life, and less vibration. It also delivers a better acoustic outcome.
  - The average life expectancy of a server fan varies according to the platform specification.
  - If a fan is hot removed or inserted, it may take up to 90 seconds for iDRAC interfaces to reflect the changes in the **Cooling** page (**System > Overview > Cooling > Fans**)
- Custom heat-sinks are designed for optimized component cooling at minimum (required) airflow yet support high-performance CPUs.

## iDRAC Direct Updates

iDRAC provides out of band ability to update the firmware of various components of a PowerEdge server. iDRAC direct update helps in eliminating staged jobs during updates. Only SEP (passive) backplanes are supported for direct updates.

iDRAC used to have staged updates to initiate firmware update of the components. From this release, Direct updates have been applied to PSU and Backplane. With the use of Direct Updates and Backplane can have quicker updates. In case of PSU, one reboot (for initializing the updates) is avoided and the update can happen in a single reboot.

With the Direct update feature in iDRAC, you can eliminate the first reboot to initiate the updates. The second reboot is controlled by the device itself, and iDRAC notifies the user if there is a need for a separate reset using job status.

 **NOTE:** For any update requiring iDRAC reset/ reboot or in case iDRAC is rebooted, it is recommended to check if iDRAC is fully ready by waiting for a few seconds of interval with a maximum timeout of 5 minutes before using any other command.

# Inventorying, monitoring, and configuring network devices

You can inventory, monitor, and configure the following network devices:

- Network Interface Cards (NICs)
- Converged Network Adapters (CNAs)
- LAN On Motherboards (LOMs)
- Open Compute Project (OCP) cards

Before you disable NPAR or an individual partition on CNA devices, ensure that you clear all I/O identity attributes (Example: IP address, virtual addresses, initiator, and storage targets) and partition-level attributes (Example: Bandwidth allocation). You can disable a partition either by changing the `VirtualizationMode` attribute setting to NPAR or by disabling all personalities on a partition.

Depending on the type of installed CNA device, the settings of partition attributes may not be retained from the last time the partition was active. Set all I/O identity attributes and partition-related attributes when enabling a partition. You can enable a partition by either changing the `VirtualizationMode` attribute setting to NPAR or by enabling a personality (Example: `NicMode`) on the partition.

## Topics:

- [Inventorying and monitoring FC HBA devices](#)
- [Inventorying and monitoring network devices](#)
- [Inventorying and monitoring SFP transceiver devices](#)
- [Telemetry Streaming](#)
- [Serial Data Capture](#)
- [Dynamic configuration of virtual addresses, initiator, and storage target settings](#)
- [SSD Wear Threshold](#)
- [Configuring persistence policy settings](#)

## Inventorying and monitoring FC HBA devices

You can remotely monitor the health and view the inventory of the Fibre Channel Host Bus Adapters (FC HBA) devices in the managed system. The Emulex and QLogic FC HBAs are supported. For each FC HBA device, you can view the following information for the ports:

- FC storage target information
- NVMe storage target information
- Port Properties
- Receive and Transmit Statistics

 **NOTE:** Emulex FC8 HBAs are not supported.

## Monitoring FC HBA devices using RACADM

To view the FC HBA device information using RACADM, use the `hwinventory` command.

For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

## Monitoring FC HBA devices using web interface

To view the FC HBA device information using Web interface, go to **System > Overview > Network Devices > Fibre Channel**. For more information about the displayed properties, see **iDRAC Online Help**.

The page name also displays the slot number where the FC HBA device is available and the type of FC HBA device.

## Inventorying and monitoring network devices

You can remotely monitor the health and view the inventory of the network devices in the managed system.

For each device, you can view the following information of the ports and enabled partitions:

- Link Status
- Properties
- Settings and Capabilities
- Receive and Transmit Statistics
- iSCSI, FCoE initiator, and target information: For Channel cards, these details are not displayed in the **Network Devices** page.

**NOTE:** In case of Embedded NIC device, BIOS representation of each LOM port is considered as individual NIC device so that the FQDD string is shown as **Embedded NIC 1 Port 1 Partition 1** and **Embedded NIC 2 Port 1 Partition 1**.

## Monitoring network devices using RACADM

To view information about network devices, use the `hwinventory` and `nicstatistics` commands.

For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

Additional properties may be displayed when using RACADM in addition to the properties displayed in the iDRAC web interface.

## Monitoring network devices using web interface

To view the information about the integrated NIC, network devices, FC devices, and ports, go to **System > Overview > Network Devices**. The **Network Devices** page is displayed. For more information about the displayed properties, see the **iDRAC Online Help**. For each network device, FC device, or network port present in the system, a corresponding tab is displayed.

**NOTE:** The **Switch Connection ID** and **Switch Port Connection ID** are not displayed for the Channel adapters.

**NOTE:** **Wake On LAN** Port property for the Network Devices in iDRAC UI may contain stale data as it is updated during CSIOR. See the RACADM output for the correct data of this property.

## Connection View

Manually checking and troubleshooting the servers' networking connections is unmanageable in a datacenter environment. iDRAC streamlines the job with the iDRAC Connection View. This feature allows you to remotely check and troubleshoot network connections from the same centralized GUI that you are using for deploying, updating, monitoring, and maintaining the servers. Connection View in iDRAC provide details of the physical mapping of switch ports to server's network ports and iDRAC dedicated port connections. All supported network cards are visible in Connection View, irrespective of the brand.

Instead of manually checking and troubleshooting the server's networking connections, you can view and manage network cable connections remotely.

Connection View provides the information of the switch ports that are connected to the server ports, and iDRAC dedicated port. The server network ports include those on PowerEdge LOM, OCP, Mezz cards, and PCIe add-in cards.

To view the network devices connection view, navigate to **System > Overview > Network Device > Connection View** to view the Connection View.

Also, you can click **iDRAC Settings > Connectivity > Network > Common Settings > Connection View** to enable or disable the connection view.



Connection View can be explored with the RACADM `SwitchConnection View` command.

|                                  |                                                                                                                       |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>Enabled</b>                   | Select <b>Enabled</b> to enable Connection View. By default the <b>Enabled</b> option is selected.                    |
| <b>State</b>                     | Displays <b>Enabled</b> if you enable the connection view option from the <b>Connection View</b> from iDRAC settings. |
| <b>Switch Connection ID</b>      | Displays the LLDP chassis ID of the switch through which the device port is connected.                                |
| <b>Switch Port Connection ID</b> | Displays the LLDP port ID of the switch port to which the device port is connected.                                   |

**NOTE:** Switch Connection ID and Switch Port Connection ID are available after the Connection View is enabled and the Link is connected. The associated network card needs to be compatible with the Connection View. Connection View feature is supported only on network cards with DELL NCSI OEM commands support. Only users with iDRAC Configure privilege can modify the Connection View settings.

iDRAC supports to send standard LLDP packets to external switches. This provides options to discover iDRACs on the network. iDRAC sends two types of LLDP packets to the outbound network:

- **Topology LLDP** - In this feature, the LLDP packet goes through all the supported server NIC ports so that an external switch can locate the originating server, OCP port[NIC FQDD], IOM location in the chassis, and so on. Topology LLDP is available as an option for all PowerEdge servers. The LLDP packets contain server network device connectivity information and are used by I/O modules and external switches to update their configuration.

**NOTE:** The Topology LLDP is not supported on 1 GbE controllers and select 10 GbE controllers (Intel X520, QLogic 578xx).

- **Discovery LLDP** - In this feature, the LLDP packet goes only through the active iDRAC NIC port in use (dedicated NIC or shared LOM), so an adjacent switch can locate the iDRAC connection port in the switch. Discovery LLDP is specific only to the active iDRAC network port and will not be seen in all the Network ports in the server. Discovery LLDP has some details of iDRAC such as IP address, MAC address, service tag and so on. The switch can automatically discover iDRAC devices that are connected to it and some data of iDRAC.

**NOTE:** If Virtual MAC Address is cleared on a port/partition, then Virtual MAC Address is the same as MAC Address.

To enable or disable the Topology LLDP, navigate to **iDRAC Settings > Connectivity > Network > Common Settings > Topology LLDP**.

To enable or disable the iDRAC Discovery LLDP, navigate to **iDRAC Settings > Connectivity > Network > Common Settings > iDrac Discovery LLDP**. By default, the Enabled option is selected.

LLDP packet that is originated from iDRAC can be viewed from the switch using the command: `show lldp neighbors`.

## Refresh Connection View

Use **Refresh Connection View** to get the latest information of Switch Connection ID and Switch Port Connection ID.

**NOTE:** If iDRAC has switch connection and switch port connection information for server network port or iDRAC network port. If the switch connection and switch port connection information are not refreshed for five minutes, then the switch connection and switch port connection information is shown as stale (last known good data) data for all user interfaces.

## Connection View Possible Values

|                         |                                                                                                                                                                                           |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Feature Disabled</b> | Connection view feature is disabled, to view the connection view data enable the feature.                                                                                                 |
| <b>No Link</b>          | Indicates that the link that is associated with the network controller port is down.                                                                                                      |
| <b>Not Available</b>    | LLDP is not enabled on the switch. Check whether LLDP is enabled on the switch port.                                                                                                      |
| <b>Not Supported</b>    | Network controller does not support the Connection view feature.                                                                                                                          |
| <b>Stale Data</b>       | Last known good data, either the Network controller port link is down or the system is powered off. Use the refresh option to refresh the connection view details to get the latest data. |

**Valid Data**

Displays the Valid Switch Connection ID and the Switch Port Connection ID information.

## Inventorying and monitoring SFP transceiver devices

You can remotely monitor the health and view the inventory of SFP transceiver devices that are connected to the system. Following are the supported transceivers:

- SFP
- SFP+
- SFP28
- SFP-DD
- QSFP
- QSFP+
- QSFP28
- QSFP-DD
- Base-T modules
- AOC and DAC cables
- RJ-45 Base-T connected with Ethernet
- Fiber channel
- IB adapter ports

The most useful transceiver information is Serial number and Part number from transceiver EPROM. These would allow to verify the remotely installed transceivers, when troubleshooting connectivity issues. For each SFP Transceiver device, you can view the following information for the ports:

- Vendor Name
- Part Number
- Revision
- Serial Number
- Device Identifier
- Interface Type

## Monitoring SFP Transceiver devices using web interface

To view the SFP Transceiver device information using Web interface, go to **System > Overview > Network Devices** and click on particular device. For more information about the displayed properties, see **iDRAC Online Help**.

The page name also displays the slot number where the transceiver device is available under Port statistics.

Monitoring data for SFP devices is only available for active SFPs. Following are the information displayed:

- TX Output Power
- TX Bias Current
- RX Input Power
- Vcc Voltage
- Temperature

## Monitoring SFP Transceiver devices using RACADM

To view the SFP Transceiver device information using RACADM, use the `networktransceiverstatistics` command.

For more information, see the *Integrated Dell Remote Access Controller RACADM CLI Guide*.

## Telemetry Streaming

Telemetry enables users to collect and stream real-time device metrics, events, and data logs from a PowerEdge server to a subscribed external client or server application. Using Telemetry, you can set the type and frequency of reports that needs to be generated.

**NOTE:** The feature is supported on all the platforms, and it requires an iDRAC Datacenter license.

Telemetry is a one-to-many solution for collecting and streaming the live system data from one or more PowerEdge servers (iDRAC) to a centralized 'Remote Server Monitoring, Analysis, and Alerting service'. The feature also supports on-demand data collection of the data.

The Telemetry data includes metrics/inventory and logs/events. The data can be streamed (pushed out) or collected (pulled) from iDRAC to or by remote consumers like Redfish client and Remote Syslog Server. The telemetry data is also provided to the iDRAC SupportAssist data collector on demand. The data collection and report is based on predefined Redfish telemetry metrics, trigger, and report definitions. The telemetry streaming settings can be configured using iDRAC web interface, RACADM, Redfish, and Server Configuration Profile (SCP).

To enable Telemetry, go to **Configuration > System Settings > Telemetry Screening**, and select **Enabled** in the **Telemetry Data Stream** list. Data streaming is automatic until Telemetry is enabled.

The following table describes the metric reports that can be generated using telemetry:

**Table 26. Metric report**

| Type           | Metric Group | Inventory | Sensor | Statistics | Configuration | Metrics |
|----------------|--------------|-----------|--------|------------|---------------|---------|
| I/O Devices    | NICs         | No        | Yes    | Yes        | No            | No      |
|                | FC HBAs      | No        | Yes    | Yes        | No            | No      |
| Server Devices | CPUs         | No        | Yes    | No         | No            | Yes     |
|                | Memory       | No        | Yes    | No         | No            | Yes     |
|                | Fans         | No        | Yes    | No         | No            | No      |
|                | PSUs         | No        | No     | No         | No            | Yes     |
|                | Sensors      | No        | Yes    | No         | No            | No      |
| Environmental  | Thermal      | No        | Yes    | No         | No            | Yes     |
|                | Power        | No        | No     | Yes        | No            | Yes     |
|                | Performance  | No        | No     | Yes        | No            | No      |
| Accelerators   | GPUs         | No        | No     | Yes        | No            | Yes     |

To know more about the field descriptions of Telemetry section, see **iDRAC Online Help**.

**NOTE:**

- When an SAS/SATA backplane is connected to onboard SATA controller then it is expected that the backplane may not show as Enclosure in the system and may not be shown in hardware inventory as well.
- StorageDiskSMARTDATA is only supported on SSD drives with SAS/SATA bus protocol and behind the BOSS controller.
- StorageSensor data is reported only for the drives in Ready/Online/Non-RAID mode and not behind the BOSS controller.
- NVMeSMARTData is only supported for direct-attached and BOSS N-1 NVMe drives.
- GPGPUStatistics data is only available in specific GPGPU models that support ECC memory capability.
- PSUMetrics is not available on modular platforms.
- Fan Power and PCIe Power Metrics may be displayed as 0 for some platforms.
- CUPS report has been renamed to SystemUsage and it is supported on both INTEL and AMD platforms.

#### Telemetry Workflow:

1. Install Datacenter license, if not installed already.
2. Configure global Telemetry settings including Enabling the telemetry and Rsyslog server network address and port using RACADM, Redfish, SCP, or iDRAC UI.
3. Configure the following Telemetry report streaming parameters on the required device report or log using either RACADM or Redfish interface:
  - EnableTelemetry
  - ReportInterval
  - ReportTriggers

 **NOTE:** Enable iDRAC Alerts and Redfish events for the specific hardware for which you need telemetry reports.

4. Redfish client makes a subscription request to the Redfish EventService on iDRAC.
5. iDRAC generates and pushes the metric report or log/event data to the subscribed client when the predefined trigger conditions are met.

#### Feature Constraints:

1. For security reasons, iDRAC supports only HTTPS-based communication to the client.
2. For stability reasons, iDRAC supports up to eight subscriptions.
3. Deletion of subscriptions is supported through Redfish interface only, even for the manual deletion by the Admin.

#### Behavior of Telemetry feature:

- iDRAC generates and pushes (HTTP POST) the Metric Report or log/event data to all the subscribed clients to the destination specified in the subscription when the predefined trigger conditions are met. The clients receive new data only upon successful subscription creation.
- The metric data includes the timestamp in ISO format, UTC time (ends in 'Z'), at the time of data collection from the source.
- Clients can terminate a subscription by sending an HTTP DELETE message to the URI of the subscription resource through the Redfish interface.
- If the subscription is deleted either by iDRAC or the client, then iDRAC does not send (HTTP POST) reports. If the number of delivery errors exceeds predefined thresholds, then iDRAC may delete a subscription.
- If a user has Admin privilege, they can delete the subscriptions but only through Redfish interface.
- The client is notified about the termination of a subscription by iDRAC by sending the 'Subscription terminated' event as the last message.
- Subscriptions are persistent and can remain even after iDRAC restarts. However, you can delete the subscriptions by performing `racresetcfg` or `racadm systemerase idrac` operations.
- User interfaces like RACADM, Redfish, SCP, and iDRAC display the current status of the client subscriptions.
- TelemetryService readiness can be checked using a new attribute `TelemetryServiceStatus` added under `GetRemoteServiceAPIStatus` API call. This attribute is added to the existing list of `LTStatus`, `RTStatus`, `ServerStatus`, and `Status`.

## Metric Report Definition

A Metric Report Definition provides a means to define the set of metrics that must be in a telemetry report and how the report must be generated and streamed.

iDRAC telemetry streaming provides metrics that can provide data on server status with no performance impact on the main server. These metrics include various system parameters such as CPU usage, memory usage, power consumption, temperature readings, fan speed, and more.

 **NOTE:** Telemetry metric report attributes are removed from the Attributes Registry. You cannot enable metric reports in the Attribute Registry by using RACADM. Use Redfish or iDRAC UI to enable metric reports.

 **NOTE:** CPU Registers and SerialLog reports are decommissioned in 17G. You can no longer view the reports under `/redfish/v1/TelemetryService/MetricReportDefinitions/`.

## Import and edit the Metric Report Definition

If you want to customize a Metric Report Definition for your specific needs, import the Metric Report Definition and edit the properties.

#### Steps

1. Go to **Configuration > System Settings > Telemetry Configuration > Metric Report Definition**
2. Select the **Location Type**.
3. Click **Choose File** and select the file.
4. Click **Import**.  
The metrics report file is imported. The report is displayed in the **Telemetry reports** list.
5. If you want to edit a specific metrics report, click **Actions > Edit Report Properties** for that specific report.  
The **Report Settings** dialog box is displayed.

6. Edit the report settings and click **Save**.

## Export Metric Report Definition

Export the metric report definition if you want to compare the performance of servers or to use the metric report as a template for other servers.

### Steps

1. Go to **Configuration > System Settings > Telemetry Configuration > Metric Report Definition**.
2. Select the **Location Type**.
3. Select the **Metric Report Definition**.
4. Click **Save**.  
The metrics report file is saved.

## Triggers

Telemetry Triggers define a set of conditions. Based on these conditions, the associated Metric Reports are generated and streamed. The conditions can include a system event or a user-defined condition, such as a metric value crossing a threshold limit or reaching equal to a discrete value.

Triggers can be configured to monitor a wide range of conditions, such as hardware failures, changes in system performance, or other significant events. iDRAC sends the associated Metric Report using one of the configured streaming methods. These methods are Server-Sent Events (SSE) or Post to subscription.

## Export triggers

Export the triggers if you want to compare the triggers of servers or to use the trigger as a template for other servers.

### Steps

1. Go to **Configuration > System Settings > Telemetry Configuration > Triggers**.
2. Select the **Location Type**.
3. Select the trigger from the **File Name** list.
4. Click **Export**.  
The triggers file is displayed in the **Triggers** list.

## Import triggers

If you want to customize a trigger for your specific needs, import the trigger.

### Steps

1. Go to **Configuration > System Settings > Telemetry Configuration > Triggers**.
2. Select the **Location Type**.
3. Click **Choose File** and select the file.
4. Click **Import**.  
The trigger file is imported. The trigger is displayed in the **Triggers** list.

## Serial Data Capture

iDRAC allows you to capture console redirection serial for later retrieval with the use of the Serial Data Capture feature. This feature requires an iDRAC Datacenter license.

The purpose of the Serial Data Capture feature is to capture the system serial data and store it so that the customer can later retrieve it for debugging purpose.

You can enable or disable a serial data capture using RACADM, Redfish, and iDRAC interfaces. When this attribute is enabled, iDRAC captures serial traffic received on Host Serial Device2 irrespective of serial Mux mode settings.

To enable or disable Serial Data Capture using iDRAC UI, go to **Maintenance > Diagnostics > Serial Data Logs** page, and select the check box to enable or disable.

**NOTE:**

- This attribute is persistent over iDRAC reboot.
- Firmware reset to default disables this feature.
- While Serial Data capture is enabled, the buffer keeps getting appended with recent data. If the user disables Serial capture and enables it again, iDRAC starts appending from last update.

The System serial data capture starts when the user enables the serial data capture flag from any of the interfaces. If serial data capture is enabled after the system has booted, you have to reboot the system, so BIOS can see the new setting (console redirection Enabled requested by iDRAC) to get the serial data. iDRAC will start the data capture continuously and stores to the shared memory with a limit of 512 KB. This buffer is circular.

**NOTE:**

- For this feature to be functional, one must have Login privilege and System control privilege.
- This feature requires an iDRAC Datacenter license.

## Dynamic configuration of virtual addresses, initiator, and storage target settings

You can dynamically view and configure the virtual address, initiator and storage target settings, and apply a persistence policy. It allows the application to apply the settings based on power state changes (that is, operating system restart, warm reset, cold reset, or AC cycle) and also based on persistence policy setting for that power state. This provides more flexibility in deployments that need rapid re-configuration of system workloads to another system.

The virtual addresses are:

- Virtual MAC Address
- Virtual iSCSI MAC Address
- Virtual FIP MAC Address
- Virtual WWN
- Virtual WWPN

**NOTE:** When you clear the persistence policy, all the virtual addresses are reset to the default permanent address set at the factory.

**NOTE:** Some cards with the virtual FIP, virtual WWN, and virtual WWPN MAC attributes, the virtual WWN and virtual WWPN MAC attributes are automatically configured when you configure virtual FIP.

Using the IO Identity feature, you can:

- View and configure the virtual addresses for network and fibre channel devices (for example, NIC, CNA, FC HBA).
- Configure the initiator (for iSCSI and FCoE) and storage target settings (for iSCSI, FCoE, and FC).
- Specify persistence or clearance of the configured values over a system AC power loss, cold, and warm system resets.

The values configured for virtual addresses, initiator and storage targets may change based on the way the main power is handled during system reset and whether the NIC, CNA, or FC HBA device has auxiliary power. The persistence of IO identity settings can be achieved based on the policy setting made using iDRAC.

Only if the I/O identity feature is enabled, the persistence policies take effect. Each time the system resets or powers on, the values are persisted or cleared based on the policy settings.

**NOTE:** After the values are cleared, you cannot re-apply the values before running the configuration job.

## View I/O Identity Optimization support in the web interface

Under the **Port Properties** of the specific NIC card (**System > Network Devices**), if **Persistence Policy Capability** displays **Capable** for the specific NIC card, the card supports IO Identity Optimization.

## Virtual or Remote assigned Address and Persistence Policy behavior when iDRAC is set to Remote-Assigned Address mode or Console mode

The following table describes the Virtual Address Management (VAM) configuration and Persistence Policy behavior, and the dependencies.

**Table 27. Virtual/Remote-Assigned Address and Persistence Policy behavior**

| Remote assigned Address Feature State in OME Modular | Mode set in iDRAC          | IO Identity Feature State in iDRAC | SCP                                                            | Persistence Policy                                         | Clear Persistence Policy—Virtual Address           |
|------------------------------------------------------|----------------------------|------------------------------------|----------------------------------------------------------------|------------------------------------------------------------|----------------------------------------------------|
| Remote-Assigned Address enabled                      | RemoteAssignedAddress Mode | Enabled                            | Virtual address management (VAM) configured                    | Configured VAM persists                                    | Set to Remote assigned Address                     |
| Remote-Assigned Address enabled                      | RemoteAssignedAddress Mode | Enabled                            | VAM not configured                                             | Set to Remote assigned Address                             | No persistence—Is set to Remote assigned Address   |
| Remote-Assigned Address enabled                      | RemoteAssignedAddress Mode | Disabled                           | Configured using the path provided in Lifecycle Controller     | Set to Remote assigned Address for that cycle              | No persistence—Is set to Remote assigned Address   |
| Remote-Assigned Address enabled                      | RemoteAssignedAddress Mode | Disabled                           | VAM not configured                                             | Set to Remote assigned Address                             | Set to Remote assigned Address                     |
| Remote-Assigned Address disabled                     | RemoteAssignedAddress Mode | Enabled                            | VAM configured                                                 | Configured VAM persists                                    | Persistence only—clear is not possible             |
| Remote-Assigned Address disabled                     | RemoteAssignedAddress Mode | Enabled                            | VAM not configured                                             | Set to hardware MAC address                                | No persistence supported. Depends on card behavior |
| Remote-Assigned Address disabled                     | RemoteAssignedAddress Mode | Disabled                           | Configured using the path provided in the Lifecycle Controller | Lifecycle Controller configuration persists for that cycle | No persistence supported. Depends on card behavior |
| Remote-Assigned Address disabled                     | RemoteAssignedAddress Mode | Disabled                           | VAM not configured                                             | Set to hardware MAC address                                | Set to hardware MAC address                        |
| Remote-Assigned Address enabled                      | Console Mode               | Enabled                            | VAM configured                                                 | Configured VAM persists                                    | Both persistence and clear must work               |
| Remote-Assigned Address enabled                      | Console Mode               | Enabled                            | VAM not configured                                             | Set to hardware MAC address                                | Set to hardware MAC address                        |
| Remote-Assigned Address enabled                      | Console Mode               | Disabled                           | Configured using the path provided in the Lifecycle Controller | Lifecycle Controller configuration persists for that cycle | No persistence supported. Depends on card behavior |
| Remote-Assigned Address disabled                     | Console Mode               | Enabled                            | VAM configured                                                 | Configured VAM persists                                    | Both persistence and clear must work               |
| Remote-Assigned Address disabled                     | Console Mode               | Enabled                            | VAM not configured                                             | Set to hardware MAC address                                | Set to hardware MAC address                        |
| Remote-Assigned Address disabled                     | Console Mode               | Disabled                           | Configured using the path provided                             | Lifecycle Controller configuration                         | No persistence supported.                          |

**Table 27. Virtual/Remote-Assigned Address and Persistence Policy behavior (continued)**

| Remote assigned Address Feature State in OME Modular | Mode set in iDRAC | IO Identity Feature State in iDRAC | SCP                         | Persistence Policy          | Clear Persistence Policy—Virtual Address |
|------------------------------------------------------|-------------------|------------------------------------|-----------------------------|-----------------------------|------------------------------------------|
|                                                      |                   |                                    | in the Lifecycle Controller | persists for that cycle     | Depends on card behavior                 |
| Remote-Assigned Address enabled                      | Console Mode      | Disabled                           | VAM not configured          | Set to hardware MAC address | Set to hardware MAC address              |

**NOTE:**

- Part replacement configuration for Partition capable cards works fine when VirtualizationMode (the attribute to enable number of partitions) is the same as the replaced card and the NIC card present in the server.
- Part replacement configuration does not trigger when replaced card VirtualizationMode (number of partitions) are not matching with the NIC card present in the server.
- During the Part Replacement window before CSIOR, the Lifecycle Controller restores the NIC configuration. It involves a cold boot followed by warm boot. After both are rebooted, NIC has the same firmware which is installed during restore process.
- Persistence policy applies for every reboot based on policy. Here on cold boot, virtual identities are not applied due to firmware version mismatch and persistence data being deleted.
- The persistence policy feature checks the PCI IDs and firmware version of the current and previous NIC of the same vendor which is replaced. In case these fields are not matched, virtual identities are not applied and persistence data (virtual identities) are also deleted from iDRAC.
- For part replacement, the vendor should maintain the same PCI ids and firmware version or you must perform VAM job/ template deployment.

## System behavior for FlexAddress and IO Identity

**Table 28. System behavior for FlexAddress and I/O Identity**

| Type                                       | FlexAddress Feature State in CMC | IO Identity Feature State in iDRAC | Availability of Remote Agent VA for the Reboot Cycle | VA Programming Source        | Reboot Cycle VA Persistence Behavior |
|--------------------------------------------|----------------------------------|------------------------------------|------------------------------------------------------|------------------------------|--------------------------------------|
| Server with FA-equivalent Persistence      | Enabled                          | Disabled                           | NA                                                   | FlexAddress from CMC         | Per FlexAddress spec                 |
|                                            | N/A, Enabled, or Disabled        | Enabled                            | Yes - New or Persisted                               | Remote Agent Virtual Address | Per FlexAddress spec                 |
|                                            |                                  |                                    | No                                                   | Virtual Address Cleared      |                                      |
|                                            | Disabled                         | Disabled                           | NA                                                   | NA                           | NA                                   |
| Server with VAM Persistence Policy Feature | Enabled                          | Disabled                           | NA                                                   | FlexAddress from CMC         | Per FlexAddress spec                 |
|                                            | Enabled                          | Enabled                            | Yes — New or Persisted                               | Remote Agent Virtual Address | Per Remote Agent Policy Setting      |
|                                            |                                  |                                    | No                                                   | FlexAddress from CMC         | Per FlexAddress spec                 |
|                                            | Disabled                         | Enabled                            | Yes — New or Persisted                               | Remote Agent Virtual Address | Per Remote Agent Policy Setting      |
|                                            |                                  |                                    | No                                                   | Virtual Address Cleared      |                                      |
|                                            | Disabled                         | Disabled                           | NA                                                   | NA                           | NA                                   |



## Enabling or disabling IO Identity Optimization

Normally, after the system boots, the devices are configured and then after a reboot the devices are initialized. You can enable the I/O Identity Optimization feature to achieve boot optimization. If it is enabled, it sets the virtual address, initiator, and storage target attributes after the device is reset and before it is initialized, thus eliminating a second BIOS restart. The device configuration and boot operation occur in a single system start and is optimized for boot time performance.

Before enabling I/O identity optimization, make sure that:

- You have the Login, Configure, and System Control privileges.
- BIOS, iDRAC, and network cards are updated to the latest firmware.

After enabling I/O Identity Optimization feature, export the Server Configuration Profile file from iDRAC, modify the required I/O Identity attributes in the SCP file, and import the file back to iDRAC.

 **NOTE:** I/O Identity attributes should only be set using SCP to make them persistent across reboots. Using other methods to set them is not persistent.

For the list of I/O Identity Optimization attributes that you can modify in the SCP file, see the **NIC Profile** document available at [Dell Support](#) page.

 **NOTE:** Do not modify non- I/O identity Optimization attributes.

## Enabling or disabling IO Identity Optimization using web interface

### About this task

To enable or disable I/O Identity Optimization:

### Steps

1. In the iDRAC Web interface, go to **Configuration > System Settings > Hardware Settings > I/O Identity Optimization**.  
The **I/O Identity Optimization** page is displayed.
2. Click the **I/O Identity Optimization** tab, select the **Enable** option to enable this feature. To disable, clear this option.
3. Click **Apply** to apply the setting.

## Enabling or disabling IO Identity Optimization using RACADM

To enable I/O Identity Optimization, use the command:

```
racadm set idrac.ioidopt.IOIDOptEnable Enabled
```

After enabling this feature, you must restart the system for the settings to take effect.

To disable I/O Identity Optimization, use the command:

```
racadm set idrac.ioidopt.IOIDOptEnable Disabled
```

To view the I/O Identity Optimization setting, use the command:

```
racadm get iDRAC.IOIDOpt
```

## SSD Wear Threshold

iDRAC provides you with the ability to configure thresholds of Remaining Rated Write Endurance for all SSDs and Available Spare of NVMe PCIe SSDs.

When SSD Remaining Rated Write Endurance and NVMe PCIe SSD Available Spare values are less than the threshold, then iDRAC logs this event in the LC log and depending on the alert type selection, iDRAC also performs Email alert, SNMP Trap, IPMI Alert, Logging in Remote Syslog, WS Eventing, and operating system log.

iDRAC alerts the user when the SSD Remaining Rated Write Endurance goes below the set threshold, so that the system admin can take a backup of the SSD or replace it.

For only NVMe PCIe SSDs, iDRAC displays **Available Spare** and provide a threshold to warn. **Available Spare** is not available for SSDs which are connected behind PERC and HBA. **Capable Speed** is not available for drives (SAS, SATA) that are connected behind HBA465e controllers.

## Configuring SSD Wear Threshold alert features using web interface

### About this task

To configure Remaining Rated Write Endurance and Available Spare Alert Threshold using web interface:

### Steps

1. In the iDRAC Web interface, go to **Configuration > System Settings > Hardware Settings > SSD Wear Thresholds**. The **SSD Wear Thresholds** page is displayed.
2. **Remaining Rated Write Endurance** — You can set the value between 1-99%. The default value is 10%. Alert type for this feature is **SSD Wear Write Endurance** and security alert is **Warning** as a result of threshold event..
3. **Available Spare Alert Threshold** — You can set the value between 1-99%. The default value is 10%. Alert type for this feature is **SSD Wear Available Spare** and security alert is **Warning** as a result of threshold event.

## Configuring SSD Wear Threshold alert features using RACADM

To configure Remaining Rated Write Endurance, use the command:

```
racadm set System.Storage.RemainingRatedWriteEnduranceAlertThreshold n
```

, where n= 1 to 99%.

To configure Available Spare Alert Threshold, use the command:

```
racadm set System.Storage.AvailableSpareAlertThreshold n
```

, where n= 1 to 99%.

## Configuring persistence policy settings

Using IO identity, you can configure policies specifying the system reset and power cycle behaviors that determine the persistence or clearance of the virtual address, initiator, and storage target settings. Each individual persistence policy attribute applies to all ports and partitions of all applicable devices in the system. The device behavior changes between auxiliary powered devices and non-auxiliary powered devices.

 **NOTE:** The **Persistence Policy** feature may not work when set to default, if the **VirtualAddressManagement** attribute is set to **FlexAddress** on iDRAC, ensure that you set the **VirtualAddressManagement** attribute to **Console** mode in iDRAC.

You can configure the following persistence policies:

- Virtual Address: Auxiliary powered devices
- Virtual Address: Non-Auxiliary powered devices
- Initiator
- Storage target

Before applying the persistence policy, make sure to:

- Inventory the network hardware at least once, that is, enabled Collect System Inventory On Restart.
- Enable I/O Identity Optimization.

Events are logged to the Lifecycle Controller log when:

- I/O Identity Optimization is enabled or disabled.
- Persistence policy is changed.

- Virtual address, initiator and target values are set based on the policy. A single log entry is logged for the configured devices and the values that are set for those devices when the policy is applied.

Event actions are enabled for SNMP or email notifications. Logs are also included in the remote syslogs.

## Default values for persistence policy

**Table 29. Default values for persistence policy**

| Persistence Policy                             | AC Power Loss | Cold Boot    | Warm Boot |
|------------------------------------------------|---------------|--------------|-----------|
| Virtual Address: Auxiliary Powered Devices     | Not selected  | Selected     | Selected  |
| Virtual Address: Non-Auxiliary Powered Devices | Not selected  | Not selected | Selected  |
| Initiator                                      | Selected      | Selected     | Selected  |
| Storage Target                                 | Selected      | Selected     | Selected  |

**NOTE:** When a persistent policy is disabled and when you perform the action to lose the virtual address, re-enabling the persistent policy does not retrieve the virtual address. You must set the virtual address again after you enable the persistent policy.

**NOTE:** If there is a persistence policy in effect and the virtual addresses, initiator, or storage targets are set on a CNA-device partition, do not reset or clear the values configured for virtual addresses, initiator, and storage targets before changing the VirtualizationMode or the personality of the partition. The action is performed automatically when you disable the persistence policy. You can also use a configuration job to explicitly set the virtual address attributes to 0s and the initiator and storage targets values as defined in [iSCSI initiator and storage target default values](#).

## Configuring persistence policy settings using iDRAC web interface

### About this task

To configure the persistence policy:

### Steps

1. In the iDRAC Web interface, go to **Configuration > System Settings > Hardware Settings > I/O Identity Optimization**.
2. Click **I/O Identity Optimization** tab.
3. In the **Persistence Policy** section, select one or more of the following for each persistence policy:
  - **Warm Reset** - The virtual address or target settings persist when warm reset condition occurs.
  - **Cold Reset** - The virtual address or target settings persist when cold reset conditions occur.
  - **AC Power Loss** - The virtual address or target settings persist when AC power loss conditions occur.
4. Click **Apply**.  
The persistence policies are configured.

## Configuring persistence policy settings using RACADM

To set persistence policy, use the following racadm object with the **set** sub command:

- For virtual addresses, use **iDRAC.IOIDOpt.VirtualAddressPersistencePolicyAuxPwr** and **iDRAC.IOIDOpt.VirtualAddressPersistencePolicyNonAuxPwr** objects
- For initiator, use **iDRAC.IOIDOpt.InitiatorPersistencePolicy** object
- For storage targets, use **iDRAC.IOIDOpt.StorageTargetPersistencePolicy** object

## iSCSI initiator and storage target default values

The following tables provide the list of default values for iSCSI initiator and storage targets when the persistence policies are cleared.

**Table 30. iSCSI initiator —default values**

| iSCSI Initiator            | Default Values in IPv4 mode | Default Values in IPv6 mode |
|----------------------------|-----------------------------|-----------------------------|
| IscsiInitiatorIpAddr       | 0.0.0.0                     | ::                          |
| IscsiInitiatorIpv4Addr     | 0.0.0.0                     | 0.0.0.0                     |
| IscsiInitiatorIpv6Addr     | ::                          | ::                          |
| IscsiInitiatorSubnet       | 0.0.0.0                     | 0.0.0.0                     |
| IscsiInitiatorSubnetPrefix | 0                           | 0                           |
| IscsiInitiatorGateway      | 0.0.0.0                     | ::                          |
| IscsiInitiatorIpv4Gateway  | 0.0.0.0                     | 0.0.0.0                     |
| IscsiInitiatorIpv6Gateway  | ::                          | ::                          |
| IscsiInitiatorPrimDns      | 0.0.0.0                     | ::                          |
| IscsiInitiatorIpv4PrimDns  | 0.0.0.0                     | 0.0.0.0                     |
| IscsiInitiatorIpv6PrimDns  | ::                          | ::                          |
| IscsiInitiatorSecDns       | 0.0.0.0                     | ::                          |
| IscsiInitiatorIpv4SecDns   | 0.0.0.0                     | 0.0.0.0                     |
| IscsiInitiatorIpv6SecDns   | ::                          | ::                          |
| IscsiInitiatorName         | Value Cleared               | Value Cleared               |
| IscsiInitiatorChapId       | Value Cleared               | Value Cleared               |
| IscsiInitiatorChapPwd      | Value Cleared               | Value Cleared               |
| IPVer                      | Ipv4                        | Ipv6                        |

**Table 31. iSCSI storage target attributes — default values**

| iSCSI Storage Target Attributes | Default Values in IPv4 mode | Default Values in IPv6 mode |
|---------------------------------|-----------------------------|-----------------------------|
| ConnectFirstTgt                 | Disabled                    | Disabled                    |
| FirstTgtIpAddress               | 0.0.0.0                     | ::                          |
| FirstTgtTcpPort                 | 3260                        | 3260                        |
| FirstTgtBootLun                 | 0                           | 0                           |
| FirstTgtIscsiName               | Value Cleared               | Value Cleared               |
| FirstTgtChapId                  | Value Cleared               | Value Cleared               |
| FirstTgtChapPwd                 | Value Cleared               | Value Cleared               |
| FirstTgtIpVer                   | Ipv4                        | NA                          |
| ConnectSecondTgt                | Disabled                    | Disabled                    |
| SecondTgtIpAddress              | 0.0.0.0                     | ::                          |
| SecondTgtTcpPort                | 3260                        | 3260                        |
| SecondTgtBootLun                | 0                           | 0                           |
| SecondTgtIscsiName              | Value Cleared               | Value Cleared               |

**Table 31. iSCSI storage target attributes — default values (continued)**

| <b>iSCSI Storage Target Attributes</b> | <b>Default Values in IPv4 mode</b> | <b>Default Values in IPv6 mode</b> |
|----------------------------------------|------------------------------------|------------------------------------|
| SecondTgtChapId                        | Value Cleared                      | Value Cleared                      |
| SecondTgtChapPwd                       | Value Cleared                      | Value Cleared                      |
| SecondTgtIpVer                         | Ipv4                               | NA                                 |

## Managing storage devices

iDRAC supports PERC 12 and BOSS N1 controllers. BOSS controllers support only RAID 0 and RAID 1.

iDRAC has expanded its agent-free management to include direct configuration of the PERC controllers. It enables you to remotely configure the storage components that are attached to your system at run-time. These components include RAID and non-RAID controllers and the channels, ports, enclosures, and disks attached to them.

The complete storage subsystem discovery, topology, health monitoring, and configuration are accomplished in the Comprehensive Embedded Management (CEM) framework by interfacing with the internal and external PERC controllers through the MCTP protocol over I2C interface.

Using iDRAC, you can perform most of the functions that are available in OpenManage Storage Management including real-time (no reboot) configuration commands (for example, create virtual disk). You can completely configure RAID before installing the operating system.

You can configure and manage the controller functions without accessing the BIOS. These functions include configuring virtual disks and applying RAID levels and hot spares for data protection. You can initiate many other controller functions such as rebuild and troubleshooting. You can protect your data by configuring data-redundancy or assigning hot spares.

### NOTE:

- Any Foreign Virtual Disks that are detected behind BOSS controllers should be cleared either from BIOS HII or by using reset controller operation.
- For any update requiring iDRAC reset or reboot or in case iDRAC is rebooted, it is recommended to check if iDRAC is fully ready by waiting for a few seconds of interval with a maximum timeout of 5 minutes before using any other command.
- For PERC12, Online Capacity Expansion (OCE) with drive addition is possible only on a virtual disk with full size. OCE with drive addition is not performed on sliced virtual disks.
- To avoid any unpredictable error, it is recommended not to perform any storage-related operation when a storage job is in progress.
- Storage operations must be initiated only after the host has completed the boot process, and after all storage devices have been fully initialized and enumerated by the system.

 **NOTE:** If the Volume Management Device (VMD) BIOS setting (with ID module support) is enabled on a PowerEdge Server, avoid configuring CPU-attached NVMe drives to prevent unpredictable behavior.

The storage devices are:

- Controllers—Most operating systems do not read and write data directly from the disks, but instead send read and write instructions to a controller. The controller is the hardware in your system that interacts directly with the disks to write and retrieve data. A controller has connectors (channels or ports) which are attached to one or more physical disks or an enclosure containing physical disks. RAID controllers can span the boundaries of the disks to create an extended amount of storage space or a virtual disk using the capacity of more than one disk. Controllers also perform other tasks, such as initiating rebuilds, initializing disks, and more. To complete their tasks, controllers require special software known as firmware and drivers. In order to function properly, the controller must have the minimum required version of the firmware and the drivers installed. Different controllers have different characteristics in the way that they read and write data and perform tasks. It is helpful to understand these features to most efficiently manage the storage.

 **NOTE:** For BOSS Controllers, the complete VD information may not be available when both PDs are plugged-out and plugged-in back.

 **NOTE:** Hardware caches erase fails on a configured external PERC12 controller. Run the reset configuration operation before performing Hardware cache erase.

- Physical disks or physical devices—Reside within an enclosure or are attached to the controller. On a RAID controller, physical disks or devices are used to create virtual disks. The physical devices connected to the controller must have the latest firmware. For the latest supported firmware, contact your service provider.

- **Virtual disk**—It is storage that is created by a RAID controller from one or more physical disks. Although a virtual disk may be created from several physical disks, it is viewed by the operating system as a single disk. Depending on the RAID level used, the virtual disk may retain redundant data if there is a disk failure or have particular performance attributes. Virtual disks can only be created on a RAID controller.
- **Enclosure**—It is attached to the system externally while the backplane and its physical disks are internal. If the enclosures are connected in multipath configuration, ensure that the following port combinations are used to connect to the controllers:
  - Port 0 and Port 2
  - Port 1 and Port 3
- **Backplane**—It is similar to an enclosure. In a backplane, the controller connector and hard drives are attached to the enclosure, but it does not have the management features (temperature probes, alarms, and so on) associated with external enclosures. Physical disks can be contained in an enclosure or attached to the backplane of a system.

**NOTE:** A maximum configuration setup with around 192 hard drives may take up to minimum 30 minutes to complete the inventory.

**NOTE:** When a system has an extensive array of storage components, such as 240 Virtual Disks and 60 Drives, it is expected that certain pending storage operations or Discard Pending operations may encounter failures that are accompanied by the RAC0508 error.

**NOTE:** When one or more backplanes are connected to an expander, then enclosure position is displayed as **Unknown**.

**NOTE:** In some platforms, hot removal or insertion of SLED is not supported and you may see some unexpected errors. It needs to be powered off before hot removal or insertion.

In addition to managing the hard drives contained in the enclosure, you can monitor the status of the fans, power supply, and temperature probes in an enclosure. You can hot-plug enclosures. Hot-plugging is defined as adding of a component to a system while the operating system is still running. The status for the drives that are hot removed is indicated as **Removed** and the drive information is available within the hardware and firmware inventories until the next host reboot.

**NOTE:** If you perform drive firmware update on hot-plugged drives, the PR36 log may get missed in the LifeCycle logs even though the update is successful. To avoid this, perform a host reboot before the firmware update.

**NOTE:** PR36 is not logged for Battery Backup Units (BBUs) and Data processing Units (DPUs) after firmware updates.

Storage events from PERC are mapped to SNMP traps as applicable. Any changes to the storage configurations are logged in the Lifecycle Log.

**NOTE:** After performing a server warm reboot, iDRAC may report PDR8 LC log for drives that are connected behind PERC controllers.

**NOTE:** In iDRAC, you can see backplane and enclosures that are associated with the PERC controller. This enclosure reports 16 slots (even though the system does not support that many drives).

In systems where drives are cabled directly to the RAID controller there is an entry that is created for each possible drive connection to PERC. PERC supports up to 16 cable-connected drives so you see that 16 slots are reported.

**Table 32. PERC capability**

| PERC Capability | CEM configuration Capable Controller                                                                                                                                                                                                              |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Real-time       | If there are pending or scheduled jobs for any controller, then the jobs have to be cleared or you must wait for the jobs to be completed before applying the configuration at run-time. A reboot is not required for Run-time or real-time jobs. |
| Staged          | N/A                                                                                                                                                                                                                                               |

## Topics:

- [Understanding RAID concepts](#)
- [Supported controllers](#)
- [Summary of supported features for storage devices](#)
- [Inventorying and monitoring storage devices](#)
- [Viewing storage device topology](#)
- [Managing physical disks](#)
- [Converting a physical disk to RAID or non-RAID mode](#)

- Erasing physical disks
- Erasing SED/ISE device data
- Rebuild Physical Disk
- Managing virtual disks
- RAID Configuration Features
- Managing PCIe SSDs
- Managing enclosures or backplanes
- Storage devices — apply operation scenarios
- Blinking or unblinking component LEDs
- Warm reboot

## Understanding RAID concepts

Storage Management uses the Redundant Array of Independent Disks (RAID) technology to provide Storage Management capability. Understanding Storage Management requires an understanding of RAID concepts, as well as some familiarity with how the RAID controllers and operating system view disk space on your system.

### What is RAID

RAID is a technology for managing the storage of data on the physical disks that reside or are attached to the system. A key aspect of RAID is the ability to span physical disks so that the combined storage capacity of multiple physical disks can be treated as a single, extended disk space. Another key aspect of RAID is the ability to maintain redundant data which can be used to restore data in the event of a disk failure. RAID uses different techniques, such as striping, mirroring, and parity, to store and reconstruct data. There are different RAID levels that use different methods for storing and reconstructing data. The RAID levels have different characteristics in terms of read/write performance, data protection, and storage capacity. Not all RAID levels maintain redundant data, which means for some RAID levels lost data cannot be restored. The RAID level you choose depends on whether your priority is performance, protection, or storage capacity.

**i NOTE:** The RAID Advisory Board (RAB) defines the specifications used to implement RAID. Although RAB defines the RAID levels, commercial implementation of RAID levels by different vendors may vary from the actual RAID specifications. An implementation of a particular vendor may affect the read and write performance and the degree of data redundancy.

### Hardware RAID

RAID can be implemented on hardware. A system using hardware RAID has a RAID controller that implements the RAID levels and processes data reads and writes to the hard disks.

### RAID concepts

RAID uses particular techniques for writing data to disks. These techniques enable RAID to provide data redundancy or better performance. These techniques include:

- **Mirroring**—Duplicating data from one physical disk to another physical disk. Mirroring provides data redundancy by maintaining two copies of the same data on different physical disks. If one of the disks in the mirror fails, the system can continue to operate using the unaffected disk. Both sides of the mirror contain the same data always. Either side of the mirror can act as the operational side. A mirrored RAID disk group is comparable in performance to a RAID 5 disk group in read operations but faster in write operations.
- **Striping**—Disk striping writes data across all physical disks in a virtual disk. Each stripe consists of consecutive virtual disk data addresses that are mapped in fixed-size units to each physical disk in the virtual disk using a sequential pattern. For example, if the virtual disk includes five physical disks, the stripe writes data to physical disks one through five without repeating any of the physical disks. The amount of space consumed by a stripe is the same on each physical disk. The portion of a stripe that resides on a physical disk is a stripe element. Striping by itself does not provide data redundancy. Striping in combination with parity does provide data redundancy.
- **Stripe size**—The total disk space consumed by a stripe not including a parity disk. For example, consider a stripe that contains 64KB of disk space and has 16KB of data residing on each disk in the stripe. In this case, the stripe size is 64KB and the stripe element size is 16KB.
- **Stripe element**—A stripe element is the portion of a stripe that resides on a single physical disk.



- **Stripe element size**—The amount of disk space consumed by a stripe element. For example, consider a stripe that contains 64KB of disk space and has 16KB of data residing on each disk in the stripe. In this case, the stripe element size is 16 KB and the stripe size is 64 KB.
- **Parity**—Parity refers to redundant data that is maintained using an algorithm with striping. When one of the striped disks fails, the data can be reconstructed from the parity information using the algorithm.
- **Span**—A span is a RAID technique that is used to combine storage space from groups of physical disks into a RAID 10, 50, or 60 virtual disk.

## RAID levels

Each RAID level uses some combination of mirroring, striping, and parity to provide data redundancy or improved read and write performance. For specific information on each RAID level, see [Choosing RAID levels](#).

## Organizing data storage for availability and performance

RAID provides different methods or RAID levels for organizing the disk storage. Some RAID levels maintain redundant data so that you can restore data after a disk failure. Different RAID levels also entail an increase or decrease in the I/O (read and write) performance of a system.

Maintaining redundant data requires the use of additional physical disks. The possibility of a disk failure increases with an increase in the number of disks. Since the differences in I/O performance and redundancy, one RAID level may be more appropriate than another based on the applications in the operating environment and the nature of the data being stored.

When choosing a RAID level, the following performance and cost considerations apply:

- **Availability or fault-tolerance** — Availability or fault-tolerance refers to the ability of a system to maintain operations and provide access to data even when one of its components has failed. In RAID volumes, availability or fault-tolerance is achieved by maintaining redundant data. Redundant data includes mirrors (duplicate data) and parity information (reconstructing data using an algorithm).
- **Performance** — Read and write performance can be increased or decreased depending on the RAID level you choose. Some RAID levels may be more appropriate for particular applications.
- **Cost efficiency** — Maintaining the redundant data or parity information associated with RAID volumes requires additional disk space. In situations where the data is temporary, easily reproduced, or non-essential, the increased cost of data redundancy may not be justified.
- **Mean Time Between Failure (MTBF)** — Using additional disks to maintain data redundancy also increases the chance of disk failure at any given moment. Although this option cannot be avoided in situations where redundant data is a requirement, it does have implications on the workload of the system support staff within your organization.
- **Volume** — Volume refers to a single disk non-RAID virtual disk. You can create volumes using external utilities like the O-ROM <Ctrl> <r>. Storage Management does not support the creation of volumes. However, you can view volumes and use drives from these volumes for creation of new virtual disks or Online Capacity Expansion (OCE) of existing virtual disks, provided free space is available.

## Choosing RAID levels

You can use RAID to control data storage on multiple disks. Each RAID level or concatenation has different performance and data protection characteristics.

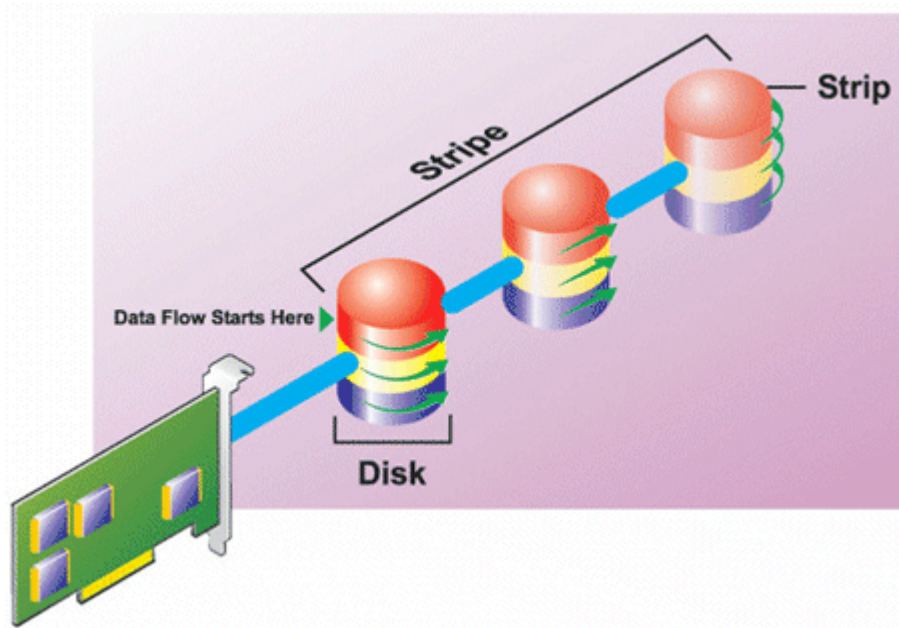
 **NOTE:** The H3xx PERC controllers do not support RAID levels 6 and 60.

The following topics provide specific information on how each RAID level store data as well as their performance and protection characteristics:

- [RAID level 0 \(striping\)](#)
- [RAID level 1 mirroring](#)
- [RAID level 5 \(striping with distributed parity\)](#)
- [RAID level 6 \(striping with additional distributed parity\)](#)
- [RAID level 50 \(striping over RAID 5 sets\)](#)
- [RAID level 60 \(striping over RAID 6 sets\)](#)
- [RAID level 10 \(striping over mirror sets\)](#)

## RAID level 0 - striping

RAID 0 uses data striping, which is writing data in equal-sized segments across the physical disks. RAID 0 does not provide data redundancy.

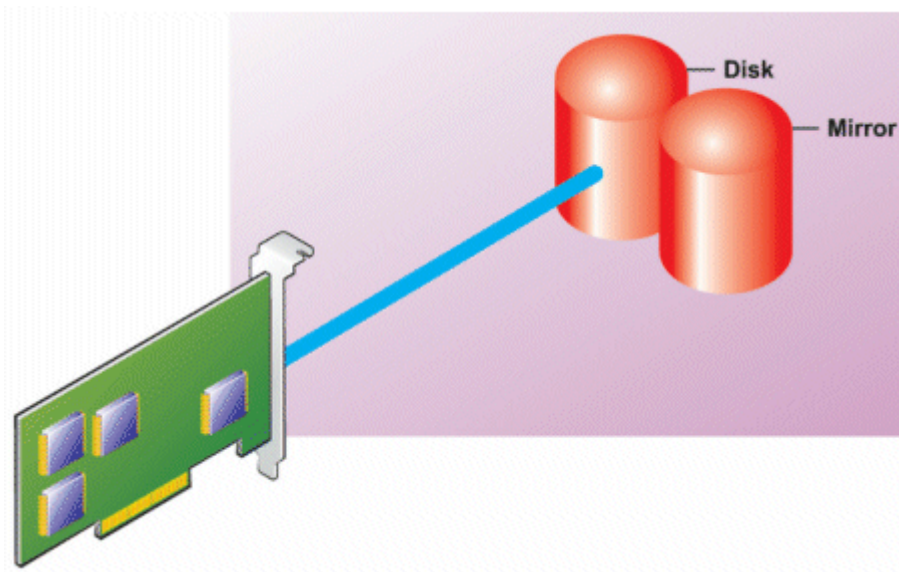


### RAID 0 characteristics:

- Groups **n** disks as one large virtual disk with a capacity of (smallest disk size) \* **n** disks.
- Data is stored to the disks alternately.
- No redundant data is stored. When a disk fails, the large virtual disk fails with no means of rebuilding the data.
- Better read and write performance.

## RAID level 1 - mirroring

RAID 1 is the simplest form of maintaining redundant data. In RAID 1, data is mirrored or duplicated on one or more physical disks. If a physical disk fails, data can be rebuilt using the data from the other side of the mirror.

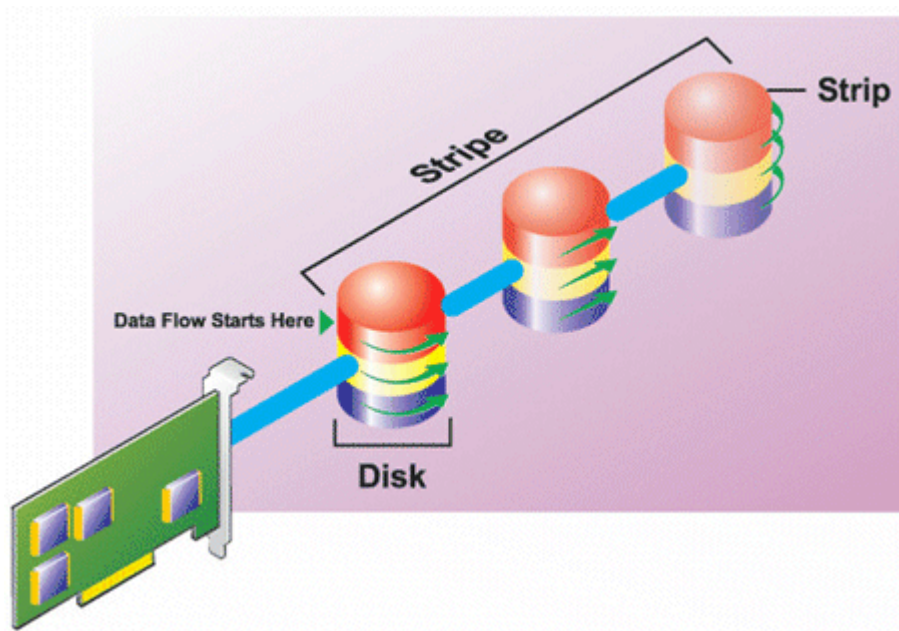


### RAID 1 characteristics:

- Groups  $n + n$  disks as one virtual disk with the capacity of  $n$  disks. The controllers currently supported by Storage Management allow the selection of two disks when creating a RAID 1. Because these disks are mirrored, the total storage capacity is equal to one disk.
- Data is replicated on both the disks.
- When a disk fails, the virtual disk still works. The data is read from the mirror of the failed disk.
- Better read performance, but slightly slower write performance.
- Redundancy for protection of data.
- RAID 1 is more expensive in terms of disk space since twice the number of disks are used than required to store the data without redundancy.

## RAID level 5 or striping with distributed parity

RAID 5 provides data redundancy by using data striping in combination with parity information. Rather than dedicating a physical disk to parity, the parity information is striped across all physical disks in the disk group.

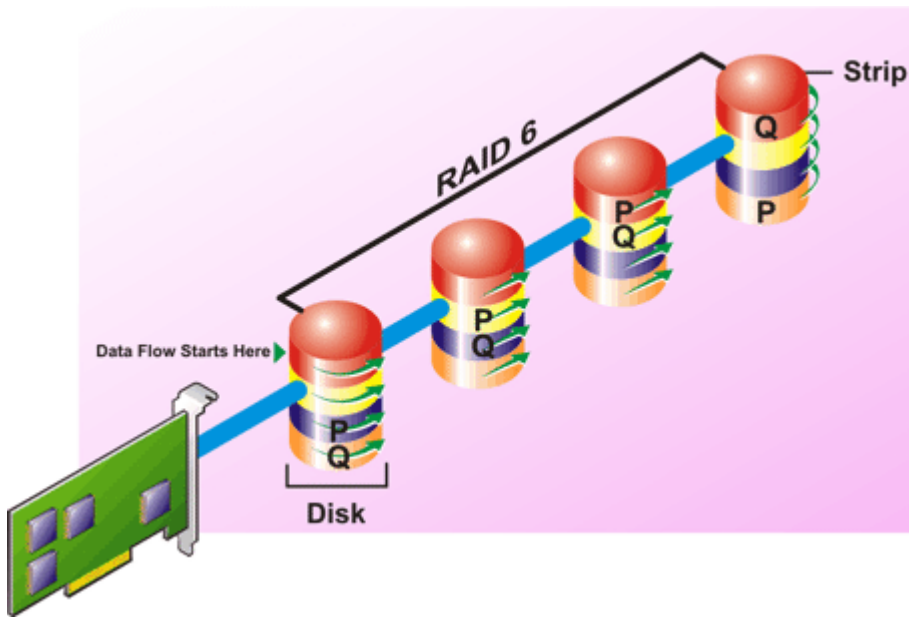


### RAID 5 characteristics:

- Groups  $n$  disks as one large virtual disk with a capacity of  $(n-1)$  disks.
- Redundant information (parity) is alternately stored on all disks.
- When a disk fails, the virtual disk still works, but it is operating in a degraded state. The data is reconstructed from the surviving disks.
- Better read performance, but slower write performance.
- Redundancy for protection of data.

## RAID level 6-striping with additional distributed parity

RAID 6 provides data redundancy by using data striping in combination with parity information. Similar to RAID 5, the parity is distributed within each stripe. RAID 6, however, uses an additional physical disk to maintain parity, such that each stripe in the disk group maintains two disk blocks with parity information. The additional parity provides data protection in the event of two disk failures. In the following image, the two sets of parity information are identified as **P** and **Q**.



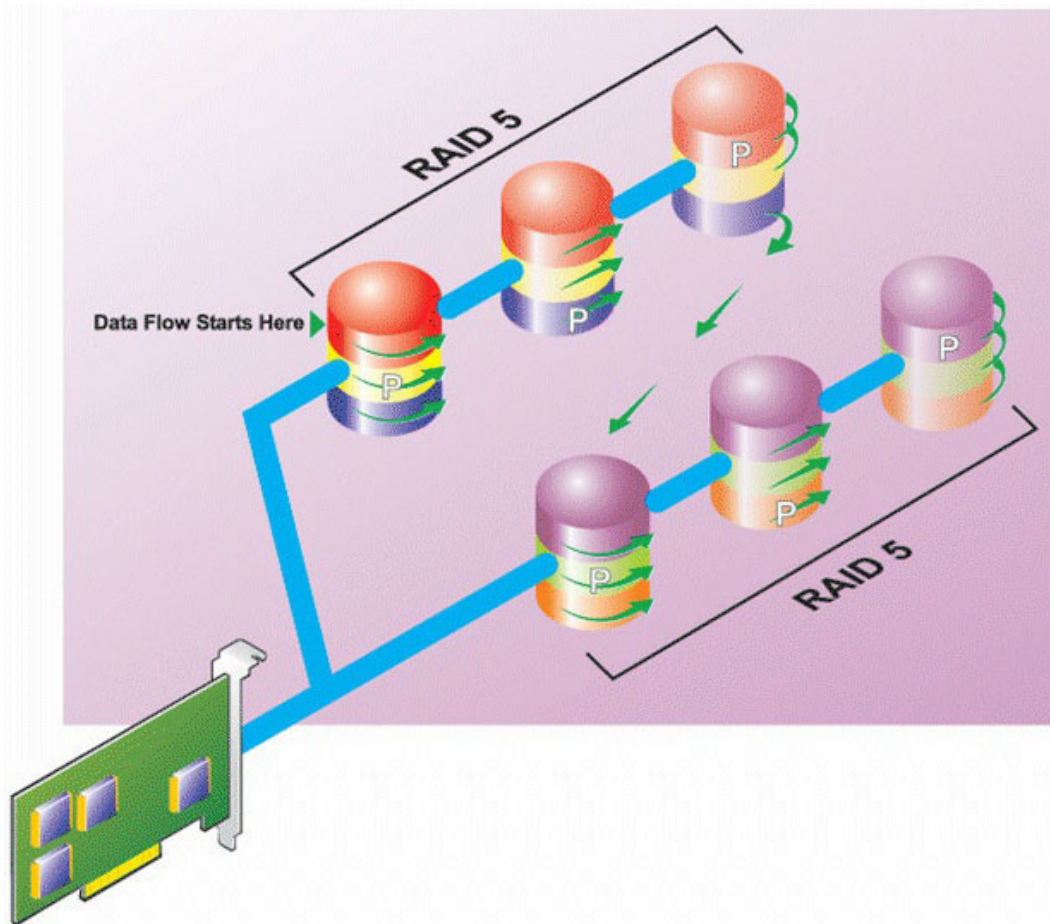
#### RAID 6 characteristics:

- Groups **n** disks as one large virtual disk with a capacity of **(n-2)** disks.
- Redundant information (parity) is alternately stored on all disks.
- The virtual disk remains functional with up to two disk failures. The data is reconstructed from the surviving disks.
- Better read performance, but slower write performance.
- Increased redundancy for protection of data.
- Two disks per span are required for parity. RAID 6 is more expensive in terms of disk space.

### RAID level 50 - striping over RAID 5 sets

RAID 50 is striping over more than one span of physical disks. For example, a RAID 5 disk group that is implemented with three physical disks and then continues on with a disk group of three more physical disks would be a RAID 50.

It is possible to implement RAID 50 even when the hardware does not directly support it. In this case, you can implement more than one RAID 5 virtual disks and then convert the RAID 5 disks to dynamic disks. You can then create a dynamic volume that is spanned across all RAID 5 virtual disks.



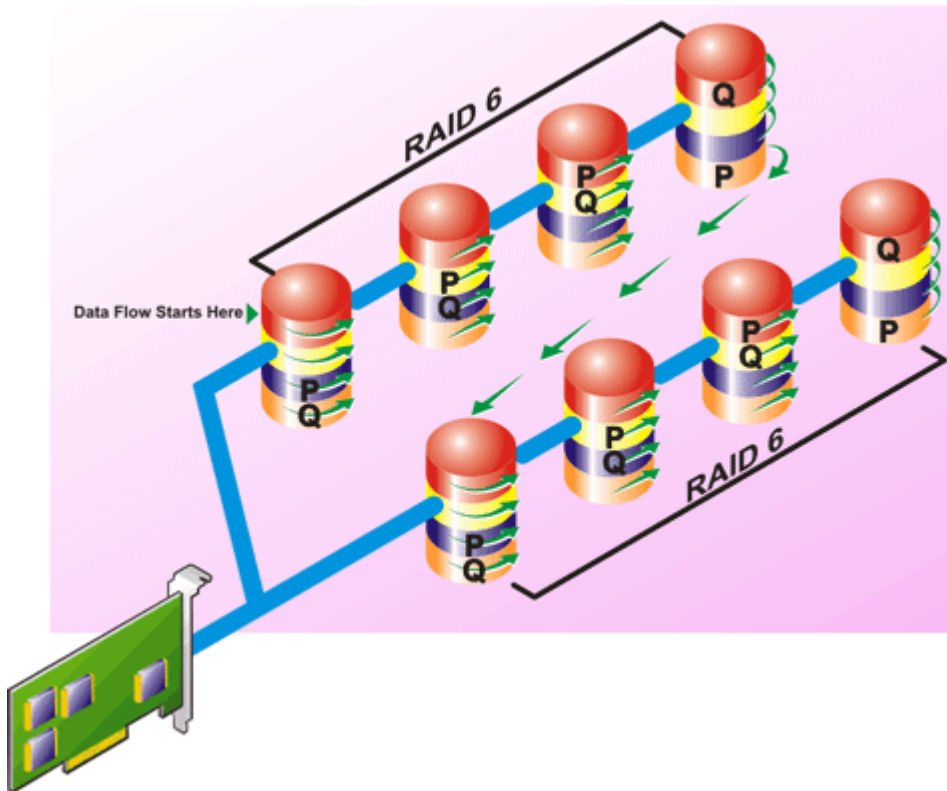
#### RAID 50 characteristics:

- Groups  $n \times s$  disks as one large virtual disk with a capacity of  $s \times (n-1)$  disks, where  $s$  is the number of spans and  $n$  is the number of disks within each span.
- Redundant information (parity) is alternately stored on all disks of each RAID 5 span.
- Better read performance, but slower write performance.
- Requires as much parity information as standard RAID 5.
- Data is striped across all spans. RAID 50 is more expensive in terms of disk space.

### RAID level 60 - striping over RAID 6 sets

RAID 60 is striping over more than one span of physical disks that are configured as a RAID 6. For example, a RAID 6 disk group that is implemented with four physical disks and then continues on with a disk group of four more physical disks would be a RAID 60.



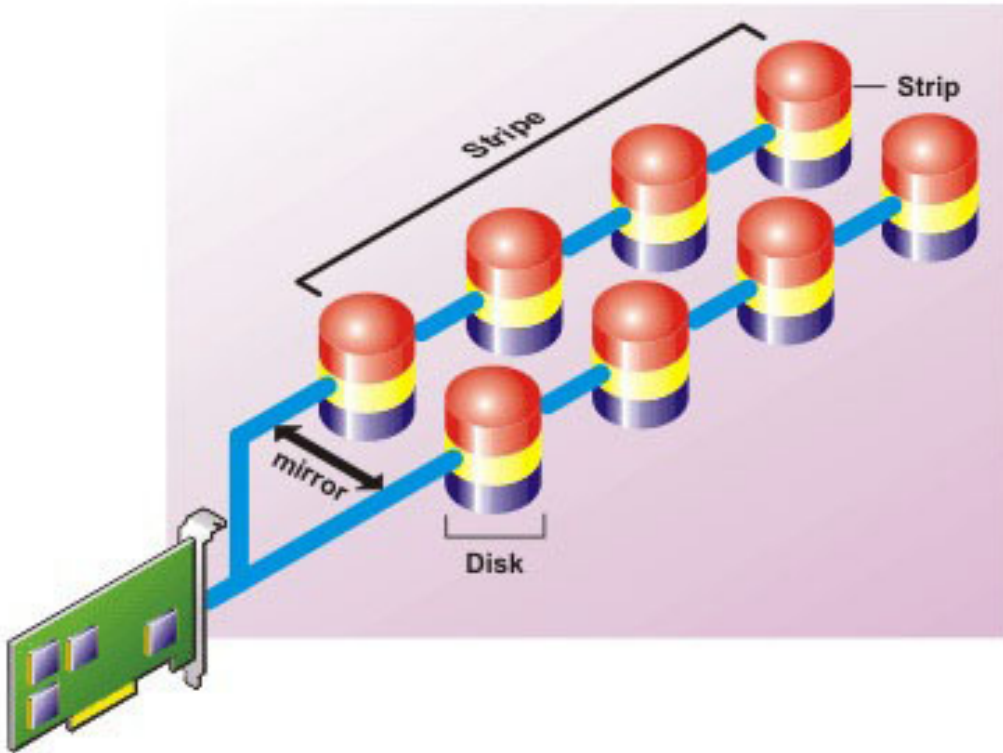


#### RAID 60 characteristics:

- Groups  $n \times s$  disks as one large virtual disk with a capacity of  $s \times (n-2)$  disks, where  $s$  is the number of spans and  $n$  is the number of disks within each span.
- Redundant information (parity) is alternately stored on all disks of each RAID 6 span.
- Better read performance, but slower write performance.
- Increased redundancy provides greater data protection than a RAID 50.
- Requires proportionally as much parity information as RAID 6.
- Two disks per span are required for parity. RAID 60 is more expensive in terms of disk space.

## RAID level 10 - striped with mirrors

The RAB considers RAID level 10 to be an implementation of RAID level 1. RAID 10 combines mirrored physical disks (RAID 1) with data striping (RAID 0). With RAID 10, data is striped across multiple physical disks. The striped disk group is then mirrored onto another set of physical disks. RAID 10 can be considered a **mirror of stripes**.



#### RAID 10 characteristics:

- Groups **n** disks as one large virtual disk with a capacity of  $(n/2)$  disks, where **n** is an even integer.
- Mirror images of the data are striped across sets of physical disks. This level provides redundancy through mirroring.
- When a disk fails, the virtual disk still works. The data is read from the surviving mirrored disk.
- Improved read performance and write performance.
- Redundancy for protection of data.

## Comparing RAID level performance

The following table compares the performance characteristics associated with the more common RAID levels. This table provides general guidelines for choosing a RAID level. Evaluate your specific environment requirements before choosing a RAID level.

**Table 33. RAID level performance comparison**

| RAID Level | Data Redundancy | Read Performance                                          | Write Performance                  | Rebuild Performance | Minimum Disks Required         | Suggested Uses                                            |
|------------|-----------------|-----------------------------------------------------------|------------------------------------|---------------------|--------------------------------|-----------------------------------------------------------|
| RAID 0     | None            | Very Good                                                 | Very Good                          | N/A                 | N                              | Noncritical data.                                         |
| RAID 1     | Excellent       | Very Good                                                 | Good                               | Good                | 2N (N = 1)                     | Small databases, database logs, and critical information. |
| RAID 5     | Good            | Sequential reads: good.<br>Transactional reads: Very good | Fair, unless using writeback cache | Fair                | N + 1 (N = at least two disks) | Databases and other read intensive transactional uses.    |
| RAID 10    | Excellent       | Very Good                                                 | Fair                               | Good                | 2N x X                         | Data intensive environments (large records).              |

**Table 33. RAID level performance comparison (continued)**

| RAID Level | Data Redundancy | Read Performance                                          | Write Performance                  | Rebuild Performance | Minimum Disks Required                 | Suggested Uses                                                               |
|------------|-----------------|-----------------------------------------------------------|------------------------------------|---------------------|----------------------------------------|------------------------------------------------------------------------------|
| RAID 50    | Good            | Very Good                                                 | Fair                               | Fair                | $N + 2$ ( $N =$ at least 4)            | Medium sized transactional or data intensive uses.                           |
| RAID 6     | Excellent       | Sequential reads: good.<br>Transactional reads: Very good | Fair, unless using writeback cache | Poor                | $N + 2$ ( $N =$ at least two disks)    | Critical information. Databases and other read intensive transactional uses. |
| RAID 60    | Excellent       | Very Good                                                 | Fair                               | Poor                | $X \times (N + 2)$ ( $N =$ at least 2) | Critical information. Medium sized transactional or data intensive uses.     |

$N$  = Number of physical disks and  $X$  = Number of RAID sets

## Supported controllers

### Supported RAID controllers

The iDRAC interfaces support the following PERC12 controllers:

- PERC H965i Front
- PERC H965i Adapter
- PERC H365i Adapter
- PERC H365i Front

The iDRAC interfaces support the following BOSS controller:

- BOSS-N1

The iDRAC interfaces support the following PERC13 controllers:

- PERC H975i Front

## Summary of supported features for storage devices

The following tables provide the features that are supported by the storage devices through iDRAC.

**Table 34. Supported features for storage controllers - PERC 12**

| Features                                                 | H965i Front and H965i Adapter                 | H365i Front and H365i Adapter |
|----------------------------------------------------------|-----------------------------------------------|-------------------------------|
| Assign or unassign a physical disk as a global hot spare | Real-time                                     | Real-time                     |
| Convert to RAID/Non RAID                                 | Real-time (converts drive to non-RAID volume) | Real-time                     |
| Rebuild                                                  | Real-time                                     | Not applicable                |
| Cancel Rebuild                                           | Real-time                                     | Not applicable                |
| Create virtual disks                                     | Real-time                                     | Real-time                     |



**Table 34. Supported features for storage controllers - PERC 12 (continued)**

| <b>Features</b>                                               | <b>H965i Front and H965i Adapter</b> | <b>H365i Front and H365i Adapter</b> |
|---------------------------------------------------------------|--------------------------------------|--------------------------------------|
| Rename virtual disks                                          | Real-time                            | Real-time                            |
| Edit virtual disks cache policies                             | Real-time                            | Not applicable                       |
| Check virtual disk consistency                                | Real-time                            | Not applicable                       |
| Cancel check consistency                                      | Not applicable                       | From HII                             |
| Initialize virtual disks                                      | Real-time                            | Not applicable                       |
| Cancel initialization                                         | Real-time                            | Not applicable                       |
| Encrypt virtual disks                                         | Real-time                            | Not applicable                       |
| Assign or unassign dedicated hot spare                        | Real-time                            | Not applicable                       |
| Delete virtual disks                                          | Real-time                            | Real-time                            |
| Cancel Background Initialization                              | Real-time                            | Real-time                            |
| Online Capacity Expansion (OCE)                               | Not applicable                       | Not applicable                       |
| RAID Level Migration (RLM)                                    | Not applicable                       | Not applicable                       |
| Discard Preserved Cache                                       | Real-time                            | Real-time                            |
| Set Patrol Read Mode                                          | Not applicable                       | Only from HII                        |
| Manual Patrol Read Mode                                       | Real-time                            | Not applicable                       |
| Patrol Read Unconfigured Areas                                | Real-time                            | Not applicable                       |
| Check Consistency Mode                                        | Not applicable                       | Not applicable                       |
| Copyback Mode                                                 | Not applicable                       | Not applicable                       |
| Load Balance Mode                                             | Not applicable                       | Real-time                            |
| Check Consistency Rate                                        | Real-time                            | Not applicable                       |
| Boot VD                                                       | Not applicable                       | Not applicable                       |
| Change PD State                                               | Not applicable                       | Real-time                            |
| Rebuild Rate                                                  | Real-time                            | Not applicable                       |
| BGI Rate                                                      | Real-time                            | Not applicable                       |
| Reconstruct Rate                                              | Real-time                            | Not applicable                       |
| Import foreign configuration                                  | Real-time                            | Real-time                            |
| Auto-import foreign configuration                             | Not applicable                       | Not applicable                       |
| Clear foreign configuration                                   | Real-time                            | Real-time                            |
| Reset controller configuration                                | Real-time                            | Real-time                            |
| Create or change security keys                                | Real-time                            | Real-time                            |
| Secure Enterprise Key Manger                                  | Real-time                            | Real-time                            |
| Inventory and remotely monitor the health of PCIe SSD devices | Not applicable                       | Not applicable                       |
| Prepare the PCIe SSD to be removed                            | Not applicable                       | Not applicable                       |
| Securely erase the data for PCIe SSD                          | Not applicable                       | Real-time                            |

**Table 34. Supported features for storage controllers - PERC 12 (continued)**

| Features                                 | H965i Front and H965i Adapter | H365i Front and H365i Adapter |
|------------------------------------------|-------------------------------|-------------------------------|
| Configure Backplane mode (split/unified) | Real-time                     | Not applicable                |
| Blink or unblink component LEDs          | Real-time                     | Real-time                     |
| Switch controller mode                   | Not applicable                | Not applicable                |
| T10PI support for Virtual Disks          | Not applicable                | Not applicable                |

**Table 35. Supported features for storage controllers - PERC 13**

| Features                                                 | H975i Front                                   |
|----------------------------------------------------------|-----------------------------------------------|
| Assign or unassign a physical disk as a global hot spare | Real-time                                     |
| Convert to RAID or Non RAID                              | Real-time (converts drive to non-RAID volume) |
| Rebuild                                                  | Real-time                                     |
| Cancel Rebuild                                           | Real-time                                     |
| Create virtual disks                                     | Real-time                                     |
| Rename virtual disks                                     | Real-time                                     |
| Edit virtual disks cache policies                        | Real-time                                     |
| Check virtual disk consistency                           | Real-time                                     |
| Cancel check consistency                                 | Not applicable                                |
| Initialize virtual disks                                 | Real-time                                     |
| Cancel initialization                                    | Real-time                                     |
| Encrypt virtual disks                                    | Real-time                                     |
| Assign or unassign dedicated hot spare                   | Real-time                                     |
| Delete virtual disks                                     | Real-time                                     |
| Cancel Background Initialization                         | Real-time                                     |
| Online Capacity Expansion (OCE)                          | Not applicable                                |
| RAID Level Migration (RLM)                               | Not applicable                                |
| Discard Preserved Cache                                  | Real-time                                     |
| Set Patrol Read Mode                                     | Not applicable                                |
| Manual Patrol Read Mode                                  | Real-time                                     |
| Patrol Read Unconfigured Areas                           | Real-time                                     |
| Check Consistency Mode                                   | Not applicable                                |
| Copyback Mode                                            | Not applicable                                |
| Load Balance Mode                                        | Not applicable                                |
| Check Consistency Rate                                   | Real-time                                     |
| Boot VD                                                  | Not applicable                                |
| Change PD State                                          | Not applicable                                |
| Rebuild Rate                                             | Real-time                                     |
| BGI Rate                                                 | Real-time                                     |
| Reconstruct Rate                                         | Real-time                                     |

**Table 35. Supported features for storage controllers - PERC 13 (continued)**

| Features                                                      | H975i Front    |
|---------------------------------------------------------------|----------------|
| Import foreign configuration                                  | Real-time      |
| Auto-import foreign configuration                             | Not applicable |
| Clear foreign configuration                                   | Real-time      |
| Reset controller configuration                                | Real-time      |
| Create or change security keys                                | Real-time      |
| Secure Enterprise Key Manger                                  | Real-time      |
| Inventory and remotely monitor the health of PCIe SSD devices | Not applicable |
| Prepare the PCIe SSD to be removed                            | Not applicable |
| Securely erase the data for PCIe SSD                          | Not applicable |
| Configure Backplane mode (split/unified)                      | Real-time      |
| Blink or unblink component LEDs                               | Real-time      |
| Switch controller mode                                        | Not applicable |
| T10PI support for Virtual Disks                               | Not applicable |

**Table 36. Supported features for storage devices**

| Feature                                                       | BOSS-N1           |
|---------------------------------------------------------------|-------------------|
| Create Virtual Disks                                          | Staged            |
| Reset Controller Configuration                                | Staged            |
| Fast Initialization                                           | Staged            |
| Delete Virtual Disks                                          | Staged            |
| Full Initialization                                           | Not applicable    |
| Inventory and remotely monitor the health of PCIe SSD devices | Not applicable    |
| Prepare the PCIe SSD to be removed                            | Not applicable    |
| Securely erase the data for PCIe SSD                          | Not applicable    |
| Blink or unblink component LEDs                               | Real-time         |
| Hot plugging of drives                                        | Real-time         |
| SEKM                                                          | Staged            |
| Supported RAID levels                                         | RAID 0 and RAID 1 |

## Inventorying and monitoring storage devices

You can remotely monitor the health and view the inventory of the following Comprehensive Embedded Management (CEM) enabled storage devices in the managed system using iDRAC web interface:

- RAID controllers, non-RAID controllers, BOSS controllers, and PCIe extenders
- Enclosures that include Enclosure Management Modules (EMMs), power supply, fan probe, and temperature probe.
- Physical disks
- Virtual disks
- Batteries



- On a system with more virtual disks, hardware inventory may show empty physical drive data for some of the virtual drives.
- If you connect SAS cables or PSU to the EMM module or reconnect the EMM module during runtime on maximum configuration of PowerVault MD 2412, PowerVault MD 2424, or PowerVault MD 2460, the values and states of enclosure components such as temperature probe, fan, EMM, and PSU may not be updated accurately. Therefore, it is recommended to perform `racreset` after connecting the SAS cable to the EMM module or after reconnecting the EMM module.
- For external enclosures connected to PERC or HBA, if you perform hot plug or remove any of the enclosure components such as SAS cables, PSU, or EMM, the iDRAC storage inventory data may not be updated or incorrect Lifecycle Log entries may be generated. It is recommended to perform iDRAC reset after any such operation.
- The display of the Physical Disks inventory is delayed when EMM is connected to HBA465e. To expedite the retrieval of the Physical Disks inventory, run the `racreset` command.
- For external JBODs, after you perform EMM hot plug, some of the enclosure drives may not appear under storage enclosure. This is a known limitation. Perform `racreset` to recover the configuration.
- Alerts and SNMP traps are generated for storage events. The events are logged in the Lifecycle Log.
- If you try to delete completed jobs from a job queue when a job is in progress, then job which is in progress may fail. Hence, it is recommended to wait for the ongoing job to complete before deleting the job.
- For an accurate inventory of BOSS controllers, ensure that Collect System Inventory On Reboot Operation (CSIOR) is completed. CSIOR is enabled by default.
- Physical disks in a system with multiple backplanes may be listed under a different backplane. Use the blink function to identify the disks.
- FQDD of certain Backplanes may not be the same in Software Inventory and Hardware Inventory.
- The lifecycle log for the PERC controller is not available when the past PERC controller events are being processed and this does not affect the functionality. Past event processing can vary depending on the configuration.
- During hot removal of M.2 drive for BOSS N1 controller, iDRAC dashboard health status turns amber, but server front/back health indicator LED stays blue.

**NOTE:** You can see SRV015 error in two cases- when the device does not support TSR collection like AHCI controllers or if the device supports TSR collection but not yet inventoried on sideband yet.

## Monitoring storage devices using web interface

To view the storage device information using web interface:

- Go to **Storage > Overview > Summary** to view the summary of the storage components and the recently logged events. This page is automatically refreshed every 30 seconds.
- Go to **Storage > Overview > Controllers** to view the RAID controller information. The **Controllers** page is displayed.
- Go to **Storage > Overview > Physical Disks** to view physical disk information. The **Physical Disks** page is displayed.
- Go to **Storage > Overview > Virtual Disks** to view virtual disk information. The **Virtual Disks** page is displayed.
- Go to **Storage > Overview > Enclosures** to view the enclosure information. The **Enclosures** page is displayed.

**NOTE:** If there are odd number of slots in the server, an empty slot row is added in the **Summary of Slots** list in the **Enclosure** page.

**NOTE:** For the latest information on supported properties and their values, see the **iDRAC Online Help**.

You can also use filters to view specific device information.

- NOTE:**
- The storage hardware list is not displayed in case the system does not have storage devices with CEM support.
  - Behavior of non-Dell certified or third-party NVMe devices may not be consistent in iDRAC.
  - If the NVMe SSDs in the backplane slots support NVMe-MI commands and the I2C connection to backplane slots are fine, iDRAC discovers these NVMe SSDs and reports them in the interfaces irrespective of the PCI connections to the respective backplane slots.

**NOTE:**

**Table 37. GUI and other interfaces support**

| Type | Web GUI Support              | Other Interfaces support         |
|------|------------------------------|----------------------------------|
| SATA | Not available                | Inventory and RAID configuration |
| NVMe | Physical disk inventory only | Inventory and RAID configuration |

For more information about the displayed properties and to use the filter options, see the iDRAC Online Help.

## Monitoring storage devices using RACADM

To view the storage device information, use the `storage` command.

For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

## Monitoring backplane using iDRAC settings utility

In the iDRAC Settings utility, go to **System Summary**. The **iDRAC Settings.System Summary** page is displayed. The **Backplane Inventory** section displays the backplane information. For information about the fields, see the **iDRAC Settings Utility Online Help**.

## Viewing storage device topology

You can view the hierarchical physical containment view of the key storage components, that is, a list of controllers, enclosures connected to the controller and a link to the physical disk contained in each enclosure. The physical disks attached directly to the controller are also displayed.

To view the storage device topology, go to **Storage > Overview**. The **Overview** page displays the hierarchical representation of the storage components in the system. The available options are:

- Controllers
- Physical Disks
- Virtual Disks
- Enclosures

Click the links to view the respective component details.

## Managing physical disks

You can perform the following for physical disks:

- View physical disk properties.
- Assign or unassign a physical disk as a global hot spare.
- Convert to RAID capable disk.
- Convert to non-RAID disk.
- Blink or unblink the LED.
- Rebuild physical disk
- Cancel rebuild physical disk
- Cryptographic erase

**NOTE:** If any of the SEKM secured drives that are directly attached to the server or behind a controller are not seen or accessible by the operating system, then it is recommended to review the lifecycle logs and ensure that all the secured drives are unlocked. Otherwise take the recommended actions mentioned in lifecycle logs.

## Assigning or unassigning dedicated hot spares

A dedicated hot spare is an unused backup disk that is assigned to a virtual disk. When a physical disk in the virtual disk fails, the hot spare is activated to replace the failed physical disk without interrupting the system or requiring your intervention.

You must have Login and Server Control privilege to run this operation.

You can assign only 4K drives as hot spare to 4K virtual disks.

If you have assigned a physical disk as a dedicated hot spare in Add to Pending Operation mode, the pending operation is created but a job is not created. Then, if you try to unassign the dedicated hot spare, the assign dedicated hot spare pending operation is cleared.

If you have unassigned a physical disk as a dedicated hot spare in Add to Pending Operation mode, the pending operation is created but a job is not created. Then, if you try to assign the dedicated hot spare, the unassign dedicated hot spare pending operation is cleared.

**NOTE:** While the log export operation is in progress, you cannot view information about dedicated hot spares on the **Manage Virtual Disks** page. After the log export operation is complete, reload or refresh the **Manage Virtual Disks** page to view the information.

## Assigning or unassigning global hot spare using web interface

### About this task

To assign or unassign a global hot spare for a physical disk drive:

### Steps

1. In the iDRAC web interface, go to **Storage > Overview > Physical Disk**.
2. All the physical disks are displayed.
3. To assign as a global hot spare, from the drop-down menus in the **Action** column, select **Assign Global Hotspare** for one or more physical disks.
4. To unassign a hot spare, from the drop-down menus in the **Action** column, select **Unassign Hotspare** for one or more physical disks.
5. Click **Apply Now**.  
Depending on your requirement, you can also choose to apply **At Next Reboot** or **At Scheduled Time**. Based on the selected operation mode, the settings are applied.

## Assigning or unassigning global hot spare using RACADM

Use the `storage` command and specify the type as global hot spare.

For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

## Converting a physical disk to RAID or non-RAID mode

Converting a physical disk to RAID mode enables the disk for all RAID operations. When a disk is in a non-RAID mode, the disk is exposed to the operating system unlike unconfigured good disks and is used in a direct pass-through mode.

You can convert the physical disk drives to RAID or non-RAID mode by:

- Using iDRAC interfaces such as iDRAC web interface, RACADM, or Redfish
- Pressing <Ctrl+R> while restarting the server and selecting the required controller.

**NOTE:** If the physical drives connected to a PERC controller are in non-RAID mode, the size of the disk displayed in the iDRAC interfaces, such as iDRAC GUI, RACADM, and Redfish may be slightly less than the actual size of the disk. However, you can use the full capacity of the disk to deploy operating systems.

# Converting physical disks to RAID capable or non-RAID mode using the iDRAC web interface

## About this task

To convert the physical disks to RAID mode or non-RAID mode, perform the following steps:

## Steps

1. In the iDRAC web interface, click **Storage > Overview > Physical Disks**.
2. Click **Filter options**. Two options are displayed - **Clear All Filters** and **Advanced Filter**. Click **Advanced Filter** option. An elaborated list is displayed that allows you to configure different parameters.
3. From the **Group By** drop-down menu, select an enclosure or virtual disks. The parameters associated with the enclosure or the VD are displayed.
4. Click **Apply**, once you select all the desired parameters. For more information about the fields, see the **iDRAC Online Help**. The settings are applied based on the option selected in the operation mode.

# Converting hard drives to RAID capable or non-RAID mode using RACADM

Depending on whether you want to convert to RAID or Non-RAID mode, use the following RACADM commands:

- To convert to RAID mode, use the `racadm storage converttoraid` command.
- To convert to Non-RAID mode, use the `racadm storage converttononraid` command.

For more information about the commands, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

## Erasing physical disks

The System Erase feature allows you to erase the contents of the physical drives. This feature is accessible using RACADM. Physical drives on the server are grouped into two categories.

- Secure erase drives—Includes drives that provide cryptographic erase such as ISE, SED SAS, and SATA drives, and PCIe SSDs.

 **NOTE:** ISE drives follow the NIST SP 800-88r1 standard and are NIST purge compliant. This means that all **old data** is irretrievable upon erasure.

- Overwrite erase drives—Includes all drives that do not support cryptographic erase.

 **NOTE:** The system erase option only applies to drives within the server. iDRAC is not able to erase drives in an external enclosure such as a JBOD.

The RACADM SystemErase subcommand includes options for the following categories:

- The **SecureErasePD** option cryptographically erases all the secure erase drives.
- The **OverwritePD** option overwrites data on all drives.

 **NOTE:** Cryptographic Erase of BOSS physical disks can be done by the `SystemErase` method and it is supported from RACADM.

Before performing SystemErase, use the following command to check the erase capability of all physical disks for a server:

```
# racadm storage get pdisks -o -p SystemEraseCapability
```

 **NOTE:** If SEKM is enabled on the server, then disable SEKM using the `racadm sekm disable` command before using this command. This can avoid any storage devices being locked out which are secured by iDRAC, if SEKM settings are erased from iDRAC by performing this command.

To erase ISE and SED drives, use this command:

```
# racadm systemerase -secureerasepd
```

To erase overwrite erase drives, use the following command:

```
# racadm systemerase -overwritepd
```

**NOTE:** RACADM SystemErase removes all the virtual disks from the physical disks that are erased by the above commands.

**NOTE:** RACADM SystemErase causes the server to restart in order to perform the erase operations.

**NOTE:** Individual PCIe SSD or SED devices can be erased using the iDRAC UI or RACADM. For more information, see the [Erasing PCIe SSD device data](#) and [Erasing SED device data using UI](#) sections.

## Erasing SED/ISE device data

**NOTE:** This operation is not supported when the supported device is a part of a Virtual Disk. The target-supported device must be removed from the virtual disk prior to performing device erase.

Cryptographic Erase permanently erases all data present on the disk. Performing a Cryptographic Erase on an SED/ISE overwrites all blocks and results in permanent loss of all data on the supported devices. During Cryptographic Erase, the host is unable to access the supported device. SED/ISE device erase can be performed either in real time (drives behind RAID Controller and PSID erase) or be applied after a system reboot.

If the system reboots or experiences a power loss during cryptographic erase, the operation is canceled. You must reboot the system and restart the process.

Before erasing SED/ISE device data, ensure that:

- Lifecycle Controller is enabled.
- You have Server Control and Login privileges.
- Selected supported drive is not part of a virtual disk.

**NOTE:**

- Erasing SED/ISE can be performed either as a real time (drives behind RAID Controller or PSID erase) or as a staged operation.
- After the drive is erased, it may still be displayed as active within the operating system due to data caching. If this occurs, reboot the operating system and the erased drive will no longer be displayed or report any data.
- Reboot the server before starting the operation. If the operation continues to fail, ensure that CSIOR is enabled and that the NVMe disks qualified by Dell Technologies.
- Cryptographic erase can also be performed using PSID.

## Erasing SED device data using RACADM

To securely erase an SED device:

```
racadm storage cryptographicerase:<SED FQDD>
```

To create the target job after performing the `cryptographicerase psid` command:

```
racadm jobqueue create <SED FQDD> -s TIME_NOW -realtime
```

To create the target staged job after performing the `cryptographicerase` command:(non-PSID)

```
racadm jobqueue create <SED FQDD> -s TIME_NOW -e <start_time>
```

To query the job ID returned:

```
racadm jobqueue view -i <job ID>
```

To perform cryptographic erase:

```
racadm storage cryptographicerase:<SED FQDD> -psid<PSID>
```



For more information, see the *iDRAC10 RACADM CLI Guide*.

## Erasing SED/ISE device data using web interface

### About this task

To erase the data on the supported device:

### Steps

1. In the iDRAC Web interface, go to **Storage > Overview > Physical Disks**.  
The **Physical Disk** page is displayed.
2. From the **Controller** drop-down menu, select the controller to view the associated devices.
3. From the drop-down menus, select **Cryptographic Erase** for one or more SED/ISEs.  
If you have selected **Cryptographic Erase** and you want to view the other options in the drop-down menu, then select **Action** and then click the drop-down menu to view the other options.
4. From the **Apply Operation Mode** drop-down menu, select one of the following options:
  - **Apply Now** — Select this option to apply the actions immediately with no system reboot required.
  - **At Next Reboot** — Select this option to apply the actions during the next system reboot.
  - **At Scheduled Time** — Select this option to apply the actions at a scheduled day and time:
    - **Start Time** and **End Time** — Click the calendar icons and select the days. From the drop-down menus, select the time. The action is applied between the start time and end time.
    - From the drop-down menu, select the type of reboot:
      - No Reboot (Manually Reboot System)
      - Graceful Shutdown
      - Force Shutdown
      - Power Cycle System (cold boot)
5. Click **Apply**.  
If the job is not created, a message indicating that the job creation was not successful is displayed. Also, the message ID and the recommended response action is displayed.  
If the job is created successfully, a message indicating that the job ID is created for the selected controller is displayed. Click **Job Queue** to view the progress of the job in the Job Queue page.  
If pending operation is not created, an error message is displayed. If pending operation is successful and job creation is not successful, then an error message is displayed.

## Rebuild Physical Disk

Rebuild Physical Disk is the ability to reconstruct the contents of a failed disk. This is true only when auto rebuild option is set to false. If there is a redundant virtual disk, the rebuild operation can reconstruct the contents of a failed physical disk. A rebuild can take place during normal operation, but it degrades performance.

Cancel Rebuild can be used to cancel a rebuild that is in progress. If you cancel a rebuild, the virtual disk remains in a degraded state. The failure of an additional physical disk can cause the virtual disk to fail and may result in data loss. It is recommended to perform a rebuild on the failed physical disk at the earliest.

In case, you cancel the rebuild of a physical disk that is assigned as a hot spare, reinitiate the rebuild on the same physical disk in order to restore the data. Canceling the rebuild of a physical disk and then assigning another physical disk as a hot spare does not cause the newly assigned hot spare to rebuild the data.

## Managing virtual disks

You can perform the following operations for the virtual disks:

- Create
- Delete
- Edit policies

- Initialize
- Check consistency
- Cancel check consistency
- Encrypt virtual disks
- Assign or unassign dedicated hot spares
- Blink and unblink virtual disk
- Cancel background initialization
- Online capacity expansion
- RAID level migration

**NOTE:** You can manage and monitor 240 virtual disks using iDRAC interface. To create VDs, use either Device Setup (F2) or PERCCLI command line tool.

## Creating virtual disks

To implement RAID functions, you must create a virtual disk. A virtual disk refers to storage created by a RAID controller from one or more physical disks. Although a virtual disk may be created from several physical disks, it is seen by the operating system as a single disk.

Before creating a virtual disk, you should be familiar with the information in [Considerations Before Creating Virtual Disks](#).

You can create a Virtual Disk using the Physical Disks that are attached to the PERC controller. To create a Virtual Disk, you must have the Server Control user privilege. You can create a maximum of 64 virtual drives and a maximum of 16 virtual drives in the same drive group.

You cannot create a virtual disk if:

- Physical disk drives are not available for virtual disk creation. Install additional physical disk drives.
- The maximum number of virtual disks that can be created on the controller has been reached. You must delete at least one virtual disk and then create a new virtual disk.
- The maximum number of virtual disks that are supported by a drive group has been reached. You must delete one virtual disk from the selected group and then create a new virtual disk.
- A job is running or scheduled on the selected controller. You must wait for this job to complete, or you can delete the job before attempting a new operation. You can view and manage the status of the scheduled job in the Job Queue page.
- Physical disk is in non-RAID mode. You must convert to RAID mode using iDRAC interfaces such as iDRAC web interface, RACADM, Redfish, or <CTRL+R>.

**NOTE:** If you create a virtual disk in Add to Pending Operation mode and a job is not created, and then if you delete the Virtual disk, then the create pending operation for the virtual disk is cleared.

**NOTE:** BOSS controller allows you to create virtual disk only of size equal to the full size of the M.2 physical storage media. Ensure that you set the virtual disk size to zero when using the Server Configuration Profile to create a BOSS virtual disk. For other interfaces such as RACADM and Redfish, the virtual disk size should not be specified.

**NOTE:** Creating virtual disks is not allowed on drives that are already secured.

## Considerations before creating virtual disks

Before creating virtual disks, consider the following:

- Virtual disk names not stored on controller—The names of the virtual disks that you create are not stored on the controller. This means that if you reboot using a different operating system, the new operating system may rename the virtual disk using its own naming conventions.
- Disk grouping is a logical grouping of disks attached to a RAID controller on which one or more virtual disks are created, such that all virtual disks in the disk group use all the physical disks in the disk group. The current implementation supports the blocking of mixed disk groups during the creation of logical devices.
- Physical disks are bound to disk groups. Therefore, there is no RAID level mixing on one disk group.
- There are limitations on the number of physical disks that can be in the virtual disk. These limitations depend on the controller. When creating a virtual disk, controllers support some stripes and spans (methods for combining the storage on physical disks). Because the number of total stripes and spans is limited, the number of physical disks that can be used is also limited. The limitations on stripes and spans affect the RAID levels as follows:
  - The maximum number of spans affects RAID 10, RAID 50, and RAID 60.
  - The maximum number of stripes affects RAID 0, RAID 5, RAID 50, RAID 6, and RAID 60.

- The number of physical disks in a mirror is always 2. This affects RAID 1 and RAID 10.

**NOTE:** RAID 1 and RAID 0 are only supported for BOSS controllers.

- Cannot create virtual disks on PCIe SSDs. But PERC 11 and later controllers support creating virtual disks using PCIe SSDs.

**NOTE:** Some actions can result in the boot target ID not being reset to ffff when no VD or EPD-PT is configured.

## Creating virtual disks using RACADM

Use the `racadm storage createvd` command.

For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

**NOTE:** Disk slicing or configuring partial VDs is not supported using RACADM on the drives managed by S140 controller.

## Creating virtual disks using web interface

### About this task

To create virtual disk:

### Steps

1. In the iDRAC Web interface, go to **Storage > Overview > Virtual DisksAdvanced Filter**.
2. In the **Virtual Disk** section, do the following:
  - a. From the **Controller** drop-down menu, select the controller for which you want to create the virtual disk.
  - b. From the **Layout** drop-down menu, select the RAID level for the Virtual Disk.  
Only those RAID levels supported by the controller appear in the drop-down menu and it is based on the RAID levels are available based on the total number of physical disks available.
  - c. Select the **Media Type, Stripe Size, Read Policy, Write Policy, Disk Cache Policy**.  
Only those values supported by the controller appear in the drop-down menus for these properties.
  - d. In the **Capacity** field, enter the size of the virtual disk.  
The maximum size is displayed and then updated as disks are selected.
  - e. The **Span Count** field is displayed based on the selected physical disks (step 3). You cannot set this value. It is automatically calculated after selecting disks for multi-raid level. **Span Count** field is applicable to RAID 10, RAID 50, and RAID 60. If you have selected RAID 10 and if the controller supports uneven RAID 10, then the span count value is not displayed. The controller automatically sets the appropriate value. For RAID 50 and RAID 60, this field is not displayed when minimum number of disks are used to create RAID. It can be changed if more disks are used.
3. In the **Select Physical Disks** section, select the number of physical disks.  
For more information about the fields, see the **iDRAC Online Help**.
4. From the **Apply Operation Mode** drop-down menu, select when you want to apply the settings.
5. Click **Create Virtual Disk**.  
Based on the selected **Apply Operation Mode**, the settings are applied.

**NOTE:** You can use alphanumeric characters, dashes, and underscores in the disk name.

## Editing virtual disk cache policies

You can change the read, write, or disk cache policy of a virtual disk.

**NOTE:** Some of the controllers do not support all read or write policies. Therefore, when a policy is applied, an error message is displayed.

The read policies indicate whether the controller must read sequential sectors of the virtual disk searching for data:

- **Adaptive Read Ahead** — The controller initiates read ahead only if the two most recent reads requests accessed sequential sectors of the disk. If subsequent read requests access random sectors of the disk, the controller reverts to no read ahead

policy. The controller continues to evaluate whether read requests are accessing sequential sectors of the disk, and initiates read ahead if necessary.

- **Read Ahead** — The controller reads sequential sectors of the virtual disk when seeking data. Read ahead policy may improve system performance if the data is written to the sequential sectors of the virtual disk.
- **No Read Ahead** — Selecting no read ahead policy indicates that the controller should not use read ahead policy.

The write policies specify if the controller sends a write-request completion signal when the data is in the cache or after it has been written to the disk.

- **Write Through** — The controller sends a write-request completion signal only after the data is written to the disk. Write-through caching provides better data security than write-back caching, since the system assumes that the data is available only after it has been safely written to the disk.
- **Write Back** — The controller sends a write-request completion signal as soon as the data is in the controller cache but has not yet been written to disk. Write back caching may provide improved performance since subsequent read requests can retrieve data quickly from the cache then from the disk. However, data loss may occur in the event of a system failure which prevents that data from being written on a disk. Other applications may also experience problems when actions assume that the data is available on the disk.
- **Force Write Back** — The write cache is enabled regardless of whether the controller has a battery. If the controller does not have a battery and force write-back caching is used, data loss may occur in the event of a power failure.

The Disk Cache policy applies to readings on a specific virtual disk. These settings do not affect the read-ahead policy.

#### **NOTE:**

- Controller non-volatile cache and battery backup of controller cache affects the read-policy or the write policy that a controller can support. All PERCs do not have battery and cache.
- Read ahead and write back requires cache. Therefore, if the controller does not have cache, it does not allow you to set the policy value.
  - Similarly, if the PERC has cache but not battery and the policy is set that requires accessing cache, then data loss may occur if base of power off. So few PERCs may not allow that policy.
  - Therefore, depending upon the PERC, the policy value is set.

## Deleting virtual disks

Deleting a virtual disk destroys all information including file systems and volumes on the virtual disk and removes the virtual disk from the controller's configuration. When deleting virtual disks, all assigned global hot spares may be automatically unassigned when the last virtual disk associated with the controller is deleted. When deleting the last virtual disk of a disk group, all assigned dedicated hot spares automatically become global hot spares.

If you delete all the VDs for a global hot spare, then the global hot spare is automatically deleted.

You must have the Login and Server Control privilege to perform delete virtual disks.

When this operation is allowed, you can delete a boot virtual drive. It is done from the sideband and the independent of the operating system. Hence, a warning message appears before you delete the virtual drive.

If you delete a virtual disk, and then immediately create a new virtual disk with all the same characteristics as the one that was deleted, the controller recognizes the data as if the first virtual disk were never deleted. In this situation, if you do not want the old data after re-creating a new virtual disk, reinitialize the virtual disk.

-  **NOTE:** Reset configuration and delete Virtual Disk operations cannot be stacked with a maximum 240 virtual disk creation operations. This results in the failure of the operation. These two operations can be run as separate jobs with a minimum 2-minute gap.

## Checking virtual disk consistency

This operation verifies the accuracy of the redundant (parity) information. This task only applies to redundant virtual disks. When necessary, the check consistency task rebuilds the redundant data. If the virtual drive has a degraded status, running a check consistency may be able to return the virtual drive to ready status. You can perform a consistency check using the web interface or RACADM.

You can also cancel the check consistency operation. The cancel check consistency is a real-time operation.

You must have Login and Server Control privilege to check consistency of virtual disks.

 **NOTE:** Consistency check is not supported when the drives are set up in RAID0 mode.

 **NOTE:** If you perform Cancel Consistency operation when there is no consistency check operation is in progress, then the pending operation in GUI is shown as Cancel BGI instead of Cancel Consistency check.

## Initializing virtual disks

Initializing virtual disks erases the all the data on the disk but does not change the virtual disk configuration. You must initialize a virtual disk that is configured before it is used.

 **NOTE:** Do not initialize virtual disks when attempting to recreate an existing configuration.

You can perform a fast initialization, a full Initialization, or cancel the initialization operation.

 **NOTE:** The cancel initialization is a real-time operation. You can cancel the initialization using only the iDRAC Web interface and not RACADM.

### Fast initialization

The fast initialize operation initializes all physical disks included in the virtual disk. It updates the metadata on the physical disks so that all disk space is available for future write operations. The initialize task can be completed quickly because the existing information on the physical disks is not erased, although future write operations overwrite any information that remains on the physical disks.

Fast initialization only deletes the boot sector and stripe information. Perform a fast initialize only if you are constrained for time or the hard drives are new or unused. Fast Initialization takes less time to complete (usually 30-60 seconds).

 **CAUTION:** Performing a fast initialize causes existing data to be inaccessible.

The fast initialize task does not write zeroes to the disk blocks on the physical disks. It is because the Fast Initialize task does not perform a write operation, it causes less degradation to the disk.

A fast initialization on a virtual disk overwrites the first and last 8 MB of the virtual disk, clearing any boot records or partition information. The operation takes only 2-3 seconds to complete and is recommended when you are recreating virtual disks.

A background initialization starts five minutes after the Fast Initialization is completed.

### Full or slow initialization

The full initialization (also called slow initialize) operation initializes all physical disks included in the virtual disk. It updates the metadata on the physical disks and erases all existing data and file systems. You can perform a full initialization after creating the virtual disk. In comparison with the fast initialize operation, you may want to use the full initialize if you have trouble with a physical disk or suspect that it has bad disk blocks. The full initialize operation remaps bad blocks and writes zeroes to all disk blocks.

If full initialization of a virtual disk is performed, background initialization is not required. During full initialization, the host is not able to access the virtual disk. If the system reboots during a full initialization, the operation terminates and a background initialization process starts on the virtual disk.

It is always recommended to do a full initialization on drives that previously contained data. Full initialization can take up to 1-2 minutes per GB. The speed of initialization depends on the controller model, speed of hard drives, and the firmware version.

The full initialize task initializes one physical disk at a time.

 **NOTE:** Full initialize is supported only in real-time. Only few controllers support full initialization.

## Encrypting virtual disks

When encryption is disabled on a controller (that is, the security key is deleted), manually enable encryption for virtual disks created using SED drives. If the virtual disk is created after encryption is enabled on a controller, the virtual disk is automatically encrypted. It is automatically configured as an encrypted virtual disk unless the enabled encryption option is disabled during the virtual disk creation.

You must have Login and Server Control privilege to manage the encryption keys.

**NOTE:** Though encryption is enabled in the controllers, user needs to manually enable encryption on the VD if VD is created from iDRAC.

## Assigning or unassigning dedicated hot spares

A dedicated hot spare is an unused backup disk that is assigned to a virtual disk. When a physical disk in the virtual disk fails, the hot spare is activated to replace the failed physical disk without interrupting the system or requiring your intervention.

You must have Login and Server Control privilege to run this operation.

You can assign only 4K drives as hot spare to 4K virtual disks.

If you have assigned a physical disk as a dedicated hot spare in Add to Pending Operation mode, the pending operation is created but a job is not created. Then, if you try to unassign the dedicated hot spare, the assign dedicated hot spare pending operation is cleared.

If you have unassigned a physical disk as a dedicated hot spare in Add to Pending Operation mode, the pending operation is created but a job is not created. Then, if you try to assign the dedicated hot spare, the unassign dedicated hot spare pending operation is cleared.

**NOTE:** While the log export operation is in progress, you cannot view information about dedicated hot spares on the **Manage Virtual Disks** page. After the log export operation is complete, reload or refresh the **Manage Virtual Disks** page to view the information.

## Rename VD

To change the name of a Virtual Disk, the user must have System Control privilege. The virtual disk name can contain only alphanumeric characters, dashes, and underscores. The maximum length of the name depends on the individual controller. In most cases, the maximum length is 15 characters. Every time a virtual disk is renamed, an LC Log gets created.

## Edit Disk capacity

Online Capacity Expansion (OCE) allows you to increase the storage capacity of selected RAID levels while the system remains online. The controller redistributes the data on the array (called Reconfiguration), placing new space available at the end of each RAID array.

Online Capacity Expansion (OCE) can be achieved in two ways:

- If free space is available on the smallest physical drive on the virtual disks group after starting LBA of Virtual disks, the virtual disk's capacity can be expanded within that free space. This option allows you to enter the new increased virtual disk size. If disk group in a virtual disk has space available only before starting LBA, then Edit Disk Capacity in same disk group is not permitted even though there is Available Space on a physical drive.
- A virtual disk's capacity can also be expanded by adding additional compatible physical disks to the existing virtual disk group. This option does not allow you to enter the new increased virtual disk size. New increased virtual disk size is calculated and displayed to the user based on the used disk space of existing physical disk group on a particular virtual disk, existing raid level of the virtual disk and the number of new drives added to the virtual disk.

Capacity Expansion allows user to specify the final VD size. Internally final VD size is conveyed to PERC in percentage (this percentage is the space user would like to use from empty space left in the array for the local disk to expand). Because of this percentage logic final VD size after reconfiguration completes may be different from what user provided for scenario where user is not giving maximum VD size possible as the final VD size (percentage turns out to be less than 100%). User does not see difference in this entered VD size and final VD size after reconfiguration, if maximum possible VD size is entered by user.

## Raid-Level Migration

Changing the RAID level of a virtual disk is called RAID-Level Migration (RLM). iDRAC provides an option to increase the VD size using RLM. In a way, RLM allows migrating the RAID level of a virtual disk which in turn may increase the size of virtual disks.

RAID level migration is the process of converting a VD from one RAID level to another. When you migrate a VD to a different RAID level, the user data on it is redistributed to the format of the new configuration.

This configuration is supported by both staged and realtime.

The following table describes possible reconfigurable VD layouts while reconfiguring (RLM) a VD with addition of disks and without addition of disks.

**Table 38. Possible VD Layouts**

| Source VD Layout | Possible target VD layout with disk addition | Possible target VD layout without disk addition |
|------------------|----------------------------------------------|-------------------------------------------------|
| R0 (single disk) | R1                                           | NA                                              |
| R0               | R5/R6                                        | NA                                              |
| R1               | R0/R5/R6                                     | R0                                              |
| R5               | R0/R6                                        | R0                                              |
| R6               | R0/R5                                        | R0/R5                                           |

## Permitted operations when OCE or RLM is going on

The following operations are allowed when OCE/RLM is going on:

**Table 39. Permitted operations**

| From Controller End behind which a VD is going through OCE/RLM | From VD End (which is going through OCE/RLM) | From any other Ready State Physical Disk on the same controller | From any other VD (which is not going through OCE/RLM) End on the same controller |
|----------------------------------------------------------------|----------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Reset Configuration                                            | Delete                                       | Blink                                                           | Delete                                                                            |
| Export Log                                                     | Blink                                        | Unblink                                                         | Blink                                                                             |
| Set Patrol Read Mode                                           | Unblink                                      | Assign Global Hot Spare                                         | Unblink                                                                           |
| Start Patrol Read                                              | N/A                                          | Convert to non-RAID Disks                                       | Rename                                                                            |
| Change Controller Properties                                   | N/A                                          | N/A                                                             | Change Policy                                                                     |
| Manage Physical Disk Power                                     | N/A                                          | N/A                                                             | Slow Initialize                                                                   |
| Convert to RAID Capable Disks                                  | N/A                                          | N/A                                                             | Fast Initialize                                                                   |
| Convert to Non-RAID Disks                                      | N/A                                          | N/A                                                             | Replace Member Disk                                                               |
| Change Controller Mode                                         | N/A                                          | N/A                                                             | N/A                                                                               |

## Cancel Initialization

This feature is the ability to cancel the background initialization on a virtual disk. On PERC controllers, the background initialization of redundant virtual disk starts automatically after a virtual disk is created. The background initialization of redundant virtual disk prepares the virtual disk for parity information and improves write performance. However, some processes such as creating a virtual disk cannot be run while the background initialization is in progress. Cancel Initialization provides the ability to cancel the background initialization manually. Once cancelled, the background initialization automatically restarts within 0 to 5 minutes.

 **NOTE:** Background initialization is not applicable for RAID 0 virtual disks.

## OCE and RLM Restrictions or Limitations

The following are the common limitations for OCE and RLM:

- OCE and RLM is restricted to the scenario where the disk group contains only one VD.
- OCE is not supported on RAID 50 and RAID 60. RLM is not supported on RAID 10, RAID 50, and RAID 60.
- If the controller already contains the maximum number of virtual disks, you cannot perform a RAID-level migration or capacity expansion on any virtual disk.

- The controller changes the write-cache policy of all virtual disks undergoing an RLM or OCE to write-through until RLM/OCE is complete.
- Reconfiguring Virtual Disks typically impacts disk performance until the reconfiguration operation is complete.
- The total number of physical disks in a disk group cannot exceed 32.
- If any background operation (such as BGL, rebuild, copyback or patrol read) is already running on the corresponding VD or PD, then reconfiguration (OCE or RLM) is not allowed at that time.
- Disk migration during reconfiguration (OCE or RLM) on drives associated with a virtual disk (VD) will cause the reconfiguration process to fail.
- Any new drive added for OCE/RLM becomes part of the VD after reconstruction completes. But the state for those new drive changes to Online just after reconstruction starts.

## Managing virtual disks using RACADM

Use the following commands to manage virtual disks:

- To delete virtual disk:

```
racadm storage deletevd:<VD FQDD>
```

- To initialize virtual disk:

```
racadm storage init:<VD FQDD> -speed {fast|full}
```

- To check consistency of virtual disks (not supported on RAID0):

```
racadm storage ccheck:<vdisk fqdd>
```

To cancel the consistency check:

```
racadm storage cancelcheck: <vdisk fqdd>
```

- To encrypt virtual disks:

```
racadm storage encryptvd:<VD FQDD>
```

- To assign or unassign dedicated hot spares:

```
racadm storage hotspare:<Physical Disk FQDD> -assign <option> -type dhs -vdkey: <FQDD of VD>
```

**<option>=yes**

Assign hot spare

**<Option>=no**

Unassign hot spare

## Managing virtual disks using web interface

### Steps

1. In the iDRAC web interface, go to **Storage > Overview > Virtual Disks**.
2. From the **Virtual Disks** list, select the controller for which you want to manage the virtual disks.
3. From **Action** drop-down menu, select an action.

When you select an action, an additional **Action** window displayed. Select or enter the value.

- **Rename**
- **Delete**
- **Edit Cache Policy**—You can change the cache policy for the following options:
  - **Read Policy**—Following values are available for selection:
    - **Adaptive Read Ahead**—Indicates that for the given volume, the control uses the Read-Ahead cache policy if the two most recent disks access occurred in sequential sectors. If the read requests are random, the controller returns to No Read Ahead mode.
    - **No Read Ahead**— Indicates that for the given volume, no read ahead policy is used.



- **Read Ahead**—Indicates that for the given volume, the controller reads sequentially ahead of the requested data and stores the additional data in cache memory, anticipating a data requirement. This speeds up sequential data reads, but there is less improvement when accessing random data.
- **Write Policy**—Change the write cache policy to one of the following options:
  - **Write Through**—Indicates that for the given volume, the controller sends a data transfer completion signal to the host system when the disk subsystem has received all the data in a transaction.
  - **Write Back**—Indicates that for the given volume, the controller sends a data transfer completion signal to the host system when the controller cache has received all the data in a transaction. The controller then writes the cached data to the storage device in the background.
  - **Force Write Back**—When using force write-back caching, the write cache is enabled regardless of whether the controller has a battery. If the controller does not have a battery and force write-back caching is used, data loss may occur in the event of a power failure.
- **Disk Cache Policy**—Change the disk cache policy to one of the following options:
  - **Default**—Indicates that the disk is using its default write cache mode. For SATA disks, this is enabled and for SAS disks this is disabled.
  - **Enabled**—Indicates that the disk's write cache is enabled. This increases performance and the probability of data loss if there is power loss.
  - **Disabled**—Indicates that the disk's write cache is disabled. This decreases performance and the probability of data loss.
- **Edit Disk Capacity**—You can add the physical disks to the selected virtual disk in this window. This window also shows the current capacity and new capacity of the virtual disk after adding the physical disks.
- **RAID Level Migration**—Displays the Disk Name, Current RAID Level, and size of the virtual disk. Allows you to select a New RAID Level. User may have to add additional drives to existing Virtual disks to migrate to the new raid level. This feature is not applicable on RAID 10, 50 and 60.
- **Initialize: Fast** —Updates the metadata on the physical disks so that all the disk space is available for future write operations. The initialize option can be completed quickly because existing information about the physical disks is not erased, although future write operations overwrites any information that remains on the physical disks.
- **Initialize: Full**—All existing data and file systems are erased.

 **NOTE:** The **Initialize: Full** option is not applicable for PERC H330 controllers.

- **Check Consistency**—To check the consistency of a virtual disk, select **Check Consistency** from the corresponding drop-down menu.

 **NOTE:** Consistency check is not supported on drives set up in RAID 0 mode.

For more information about these options, see the **iDRAC Online Help**.

4. Click **Apply Now** to apply the changes immediately. Click **At Next Reboot** to apply the changes after next reboot. Click **At Scheduled Time** to apply the changes at a particular time, and **Discard All Pending** to discard the changes.

Based on the selected operation mode, the settings are applied.

## RAID Configuration Features

Following table lists some of the RAID configuration features which are available in RACADM:

 **CAUTION:** Forcing a physical disk to go online or offline may result in data loss.

**Table 40. RAID Configuration Features**

| Feature      | RACADM Command                                        | Description                                                                                                                                                                                                                                                                                                                                                                    |
|--------------|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Force Online | <pre>racadm storage forceonline:&lt;PD FQDD&gt;</pre> | A power failure, corrupted data, or some other reason may lead to a physical disk going offline. You can use this feature to force a physical disk back into an online state when all other options have been exhausted. Once the command is run, the controller places the drive back into online state and restore its membership within the virtual disk. This happens only |

**Table 40. RAID Configuration Features (continued)**

| Feature                                                                                                                                          | RACADM Command                                                                                           | Description                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                  |                                                                                                          | if the controller can read from the drive and can write into its metadata.                                                                                                                                                                                                                                                                                 |
| <b>NOTE:</b> Data recovery is only possible if a limited portion of the disk is damaged. Force Online feature cannot fix an already failed disk. |                                                                                                          |                                                                                                                                                                                                                                                                                                                                                            |
| Force Offline                                                                                                                                    | <pre>racadm storage forceoffline:&lt;PD FQDD&gt;</pre>                                                   | This feature removes a drive from a virtual disk configuration so that it goes offline, resulting in a degraded VD configuration. It is helpful if a drive is likely to fail in near future or is reporting a SMART failure but is still online. It can be also used if you would like to utilize a drive which is part of an existing RAID configuration. |
| Replace Physical Disk                                                                                                                            | <pre>racadm storage replacephysicaldisk:&lt;Source PD FQDD &gt; -dstpd &lt;Destination PD FQDD&gt;</pre> | Allows you to copy data from a physical disk which is a member of a VD, to another physical disk. The source disk should be in online state, while the destination disk should be in ready state and of a similar size and type to replace the source.                                                                                                     |
| Virtual Disk as boot device                                                                                                                      | <pre>racadm storage setbootvd:&lt;controller FQDD&gt; -vd &lt;VirtualDisk FQDD&gt;</pre>                 | A virtual disk can be configured as a boot device using this feature. This enables fault tolerance when a VD with redundancy is selected as the boot device, and also has the operating system installed on it.                                                                                                                                            |
| Unlock Foreign Configuration                                                                                                                     | <pre>racadm storage unlock:&lt;Controller FQDD&gt; -key &lt;Key id&gt; -passwd &lt;passphrase&gt;</pre>  | This feature is used to authenticate locked drives which have a different source controller encryption than the destination. Once unlocked, the drive can be successfully migrated from one controller to another.                                                                                                                                         |

## Managing controllers

You can perform the following for controllers:

- Configure controller properties
- Import or auto import foreign configuration
- Clear foreign configuration
- Reset controller configuration
- Create, change, or delete security keys
- Discard preserved cache

## Configuring controller properties

You can configure the following properties for the controller:

- Patrol read mode (auto or manual)
- Start or stop patrol read if patrol read mode is manual
- Patrol read unconfigured areas
- Check consistency mode
- Copyback mode
- Load balance mode

- Check consistency rate
- Rebuild rate
- BGI rate
- Reconstruct rate
- Enhanced auto import foreign configuration
- Create or change security keys
- Encryption mode (iDRAC Local Key Management and Secure Enterprise key Manager)

You must have Login and Server Control privilege to configure the controller properties.

## Patrol read mode considerations

Patrol read identifies disk errors to avoid disk failures, data loss, or corruption. It runs automatically once a week on SAS and SATA HDDs.

The Patrol Read does not run on a physical disk in the following circumstances:

- The physical disk is an SSD.
- The physical disk is not included in a virtual disk or assigned as a hot spare.
- The physical disk is included in a virtual disk that is undergoing one of the following:
  - A rebuild
  - A reconfiguration or reconstruction
  - A background initialization
  - A check consistency

In addition, the Patrol Read operation suspends during heavy I/O activity and resumes when the I/O is complete.

**i NOTE:** For more information on how often the Patrol Read operation runs when in auto mode, see the respective controller documentation.

**i NOTE:** Patrol read mode operations such as **Start** and **Stop** are not supported if there are no virtual disks available in the controller. Though you can invoke the operations successfully using the iDRAC interfaces, the operations fail when the associated job is started.

## Load balance

The Load Balance property provides the ability to automatically use both controller ports or connectors connected to the same enclosure to route I/O requests. This property is available only on SAS controllers.

## Bgi rate

On PERC controllers, background initialization of a redundant virtual disk begins automatically within 0 to 5 minutes after the virtual disk is created. The background initialization of a redundant virtual disk prepares the virtual disk to maintain redundant data and improves write performance. For example, after the background initialization of a RAID 5 virtual disk completes, the parity information has been initialized. After the background initialization of a RAID 1 virtual disk completes, the physical disks are mirrored.

The background initialization process helps the controller identify and correct problems that may occur with the redundant data later. In this regard, the background initialization process is similar to a check consistency. The background initialization should be allowed to run to completion. If cancelled, the background initialization automatically restarts within 0 to 5 minutes. Some processes such as read and write operations are possible while the background initialization is running. Other processes, such as creating a virtual disk, cannot be run concurrently with a background initialization. These processes cause the background initialization to cancel.

The background initialization rate, configurable between 0% and 100%, represents the percentage of the system resources dedicated to running the background initialization task. At 0%, the background initialization has the lowest priority for the controller, takes the most time to complete, and is the setting with the least impact to system performance. A background initialization rate of 0% does not mean that the background initialization is stopped or paused. At 100%, the background initialization is the highest priority for the controller. The background initialization time is minimized and is the setting with the most impact to system performance.

## Check consistency

The Check Consistency task verifies the accuracy of the redundant (parity) information. This task only applies to redundant virtual disks. When necessary, the Check Consistency task rebuilds the redundant data. If the virtual disk is in a Failed Redundancy state, running a check consistency may be able to return the virtual disk to a Ready state.

The check consistency rate, configurable between 0% and 100%, represents the percentage of the system resources dedicated to running the check consistency task. At 0%, the check consistency has the lowest priority for the controller, takes the most time to complete, and is the setting with the least impact to system performance. A check consistency rate of 0% does not mean that the check consistency is stopped or paused. At 100%, the check consistency is the highest priority for the controller. The check consistency time is minimized and is the setting with the most impact to system performance.

## Configuring controller properties using RACADM

- To set Patrol Read Mode:

```
racadm set storage.controller.<index>.PatrolReadMode {Automatic | Manual | Disabled}
```

- If Patrol read mode is set to manual, use the following commands to start and stop Patrol read Mode:

```
racadm storage patrolread:<Controller FQDD> -state {start|stop}
```

**NOTE:** Patrol read mode operations such as Start and Stop are not supported if there are no virtual disks available in the controller. Though you can invoke the operations successfully using the iDRAC interface, the operations fail when the associated job is started.

**NOTE:** The storage attributes PatrolReadMode and PersistHotspare that are assigned to HBA12 and PERC12 controllers are immutable. If you attempt to modify these attributes, you may encounter errors. Though the job is created and run, the original values remain the same.

- To specify the Check Consistency Mode, use **Storage.Controller.CheckConsistencyMode** object.
- To enable or disable the Copyback Mode, use **Storage.Controller.CopybackMode** object.
- To enable or disable the Load Balance Mode, use **Storage.Controller.PossibleloadBalancedMode** object.
- To specify the percentage of the system's resources dedicated to perform a check consistency on a redundant virtual disk, use **Storage.Controller.CheckConsistencyRate** object.
- To specify the percentage of the controller's resources dedicated to rebuild a failed disk, use **Storage.Controller.RebuildRate** object.
- To specify the percentage of the controller's resources dedicated to perform the background initialization (BGI) of a virtual disk after it is created, use **Storage.Controller.BackgroundInitializationRate** object.
- To specify the percentage of the controller's resources dedicated to reconstruct a disk group after adding a physical disk or changing the RAID level of a virtual disk residing on the disk group, use **Storage.Controller.ReconstructRate** object.
- To enable or disable the enhanced auto import of foreign configuration for the controller, use **Storage.Controller.EnhancedAutoImportForeignConfig** object.
- To create, modify, or delete security key to encrypt virtual drives:

```
racadm storage createsecuritykey:<Controller FQDD> -key <Key id> -passwd <passphrase>
racadm storage modifysecuritykey:<Controller FQDD> -key <key id> -oldpasswd <old
passphrase> -newpasswd <new passphrase>
racadm storage deletesecuritykey:<Controller FQDD>
```

## Configuring controller properties using web interface

### Steps

1. In the iDRAC web interface, go to **Storage > Overview > Controllers**. The **Setup Controllers** page is displayed.
2. In the **Controller** section, select the controller that you want to configure.
3. Specify the required information for the various properties.

The **Current Value** column displays the existing values for each property. You can modify this value by selecting the option from the **Action** drop-down menu for each property.

For information about the fields, see the **iDRAC Online Help**.

4. From the **Apply Operation Mode**, select when you want to apply the settings.
5. Click **Apply**.

Based on the selected operation mode, the settings are applied.

## Security Protocol and Data Model (SPDM)

SPDM Protocol is used for establishing the security capabilities and authenticity between hardware components. SPDM allows message exchange between iDRAC and end devices such as storage controllers (PERC12), FC, and CPU. This includes hardware identity certificates. You can enable SPDM using Redfish or RACADM.

**Table 41. SPDM Feature Licensing**

| Feature                                            | License      |
|----------------------------------------------------|--------------|
| Inventory- Detecting SPDM capable devices          | Unlicensed   |
| Collecting Hardware Identity of devices            | Enterprise   |
| Collecting measurements of devices                 | Enterprise   |
| Establishing Trust on device certificate using SCV | SCV License  |
| Encrypted communication channel                    | SEKM License |

When a device is SPDM capable, the SCV data that is collected contains SPDM Hardware identity certificates in addition to the existing fields. Firmware identity certificates are not in the SCV certificate.

 **NOTE:** You may notice the NIC port number missing in the downloaded SPDM HW certificate file name.

 **NOTE:** Export SPDM Certificate job failure can be seen in the Job queue when you perform frequent reboots.

### Enable SPDM using Redfish

The SPDM feature in iDRAC enables you to manage and monitor the security status of various components.

#### Steps

1. Perform a PATCH request for the following URI using the payload {"Attributes": {"SPDM.1.Enable": "Enabled"}}:  
`/redfish/v1/Managers/iDRAC.Embedded.1/Oem/Dell/DellAttributes/iDRAC.Embedded.1`
2. Perform `racreset` to ensure that the changes are reflected.

### Enable SPDM using RACADM

The SPDM feature in iDRAC enables you to manage and monitor the security status of various components.

#### Steps

1. Run the following command to enable SPDM :  
`racadm set idrac.SPDM.enable enabled`
2. Run the following commands to ensure that the changes are reflected:  
`racadm racreset`  
`racadm get idrac.SPDM`

## Importing or auto importing foreign configuration

A foreign configuration is data residing on physical disks that have been moved from one controller to another. Virtual disks residing on physical disks that have been moved are considered to be a foreign configuration.

You can import foreign configurations so that virtual disks are not lost after moving physical disks. A foreign configuration can be imported only if it contains a virtual disk that is in either Ready or Degraded state or a hot spare that is dedicated to a virtual disk which can be imported or is already present.

All the virtual disk data must be present, but if the virtual disk is using a redundant RAID level, the additional redundant data is not required.

For example, if the foreign configuration contains only one side of a mirror in a RAID 1 virtual disk, then the virtual disk is in a Degraded state and can be imported. If the foreign configuration contains only one physical disk that was originally configured as a RAID 5 using three physical disks, then the RAID 5 virtual disk is in a Failed state and cannot be imported.

In addition to virtual disks, a foreign configuration may consist of a physical disk that was assigned as a hot spare on one controller and then moved to another controller. The Import Foreign Configuration task imports the new physical disk as a hot spare. If the physical disk was set as a dedicated hot spare on the previous controller, but the virtual disk to which the hot spare was assigned is no longer present in the foreign configuration, then the physical disk is imported as a global hot spare.

The Import Foreign Configuration task is only displayed when the controller has detected a foreign configuration. You can also identify whether a physical disk contains a foreign configuration (virtual disk or hot spare) by checking the physical disk state. If the physical disk state is Foreign, then the physical disk contains all or some portion of a virtual disk or has a hot spare assignment.

**NOTE:** As part of Foreign Configuration Import, if a configuration is incomplete, it is not imported. However, the job will not fail. Job status shows as **Completed successfully**. You have to check the PD state to know whether the VD is imported or not.

**NOTE:** The task of importing foreign configuration imports all virtual disks residing on physical disks that have been added to the controller. If more than one foreign virtual disk is present, all the configurations are imported.

PERC9 controller provides support for auto import of foreign configuration without requiring user interactions. The auto import can be enabled or disabled. If enabled, the PERC controller can auto import any foreign configuration detected without manual intervention. If disabled, the PERC does not auto import any foreign configuration.

You must have Login and Server Control privilege to import foreign configurations.

This task is not supported on PERC hardware controllers running on HBA mode.

**NOTE:** It is not recommended to remove an external enclosure cable while the operating system is running on the system. Removing the cable can result in a foreign configuration when the connection is re-established.

You can manage foreign configurations in the following cases:

- All the physical disks in a configuration are removed and re-inserted.
- Some of the physical disks in a configuration are removed and re-inserted.
- All the physical disks in a virtual disk are removed, but at different times, and then re-inserted.
- The physical disks in a non-redundant virtual disk are removed.

The following constraints apply to the physical disks that are considered for import:

- The drive state of a physical disk can change from the time the foreign configuration is scanned to when the actual import occurs. The foreign import occurs only on drives that are in the Unconfigured Good state.
- Drives in the failed or offline state cannot be imported.
- The firmware does not allow you to import or clear the foreign configurations if more than eight foreign drives are present.

## Importing foreign configuration using RACADM

To import foreign configuration:

```
racadm storage importconfig:<Controller FQDD>
```

For more information, see the **iDRAC RACADM Command Line Reference Guide** available at [dell.com/idracmanuals](http://dell.com/idracmanuals).

## Importing foreign configuration using web interface

### About this task

 **NOTE:** If there is an incomplete foreign disk configuration in the system, then the state of one or more existing online virtual disks is also displayed as foreign.

 **NOTE:** Importing foreign configuration for BOSS controller is not supported.

To import foreign configuration:

### Steps

1. In the iDRAC web interface, go to **Storage > Overview > Controllers**.
2. From the **Controller** options, select the controller you want to import the foreign configuration to.
3. Click **Import** under the **Foreign Configuration** and then click **Apply**.

## Clearing foreign configuration

After moving a physical disk from one controller to another, you may find that the physical disk contains all or some portion of a virtual disk (foreign configuration). You can identify whether a previously used physical disk contains a foreign configuration (virtual disk) by checking the physical disk state. If the physical disk state is Foreign, then the physical disk contains all or some portion of a virtual disk. You can clear or erase the virtual disk information from the newly attached physical disks.

The Clear Foreign Configuration operation permanently erases all data residing on the physical disks that are added to the controller. If more than one foreign virtual disk is present, all the configurations are erased. You may prefer to import the virtual disk rather than destroy the data. An initialization must be performed to remove foreign data. If you have an incomplete foreign configuration which cannot be imported, you can use the Clearing Foreign Configuration option to erase the foreign data on the physical disks.

## Clearing foreign configuration using web interface

### About this task

To clear the foreign configuration:

### Steps

1. In the iDRAC web interface, go to **Storage > Overview > Controllers**.  
The **Controller Configuration** page is displayed.
2. From the **Controller** options, select the controller for which you want to clear the foreign configuration.

 **NOTE:** To clear foreign configuration on BOSS controllers, click **Reset Configuration**.

3. Click **Clear Configuration**.
4. Click **Apply**  
Based on the selected operation mode, the virtual disks residing on the physical disk is erased.

## Clearing foreign configuration using RACADM

To clear foreign configuration:

```
racadm storage clearconfig:<Controller FQDD>
```

For more information, see the **iDRAC RACADM Command Line Reference Guide** available at [dell.com/idracmanuals](http://dell.com/idracmanuals).

## Resetting controller configuration

You can reset the configuration for a controller. This operation deletes virtual disk drives and unassigns all hot spares on the controller. It does not erase any data other than removing the disks from the configuration. Reset configuration also does not remove any foreign configurations. Reset configuration does not erase any data. You may recreate the exact same configuration without an initialize operation which may result in the data being recovered. You must have server control privilege.

 **NOTE:** Resetting the controller configuration does not remove a foreign configuration. To remove a foreign configuration, perform clear configuration operation.

## Resetting controller configuration using web interface

### About this task

To reset the controller configuration:

### Steps

1. In the iDRAC Web interface, go to **Storage > Overview > Controllers**.
2. From the **Actions**, select **Reset Configuration** for one or more controllers.
3. For each controller, from the **Apply Operation Mode** drop-down menu, select when you want to apply the settings.
4. Click **Apply**.

Based on the selected operation mode, the settings are applied.

## Resetting controller configuration using RACADM

To reset the controller configuration:

```
racadm storage resetconfig:<Controller FQDD>
```

For more information, see the **iDRAC RACADM Command Line Reference Guide** available at [dell.com/idracmanuals](https://dell.com/idracmanuals).

## HBA adapter operations

Dell PowerEdge servers must have an operating system installed and the appropriate device driver to be loaded in order for Dell HBAs to operate. Following POST, the HBA ports will be disabled. The HBA device driver is responsible for resetting the HBA and enabling its ports connected to storage devices. Without an operating system, the driver will not be loaded, and there is no guarantee that iDRAC will be able to display storage devices connected to Dell HBAs.

The non-RAID controllers are the HBAs that do not have RAID capabilities. They do not support virtual disks.

You can perform the following for non-RAID controllers:

- View controller, physical disks, and enclosure properties as applicable for the non-RAID controller. Also, view EMM, fan, power supply unit, and temperature probe properties associated with the enclosure. The properties are displayed based on the type of controller.
- View software and hardware inventory information.
- Update firmware for enclosures behind the HBA controller (staged)
- Monitor the polling or polling frequency for physical disk SMART trip status when there is change detected
- Monitor the physical disks hot plug or hot removal status
- Blink or unblink LEDs

 **NOTE:**

- Even though LED is not available for tape drive, blink/unblink option can be successful.

 **NOTE:**

- Enable Collect System Inventory On Reboot (CSIOR) operation before inventorying or monitoring the non-RAID controllers.

 **NOTE:**

Detection of failed drives behind SAS HBA controllers is not supported.



## Monitoring predictive failure analysis on drives

Storage management supports Self Monitoring Analysis and Reporting Technology (SMART) on physical disks that are SMART-enabled.

SMART performs predictive failure analysis on each disk and sends alerts if a disk failure is predicted. The controllers check physical disks for failure predictions and, if found, pass this information to iDRAC. iDRAC immediately logs an alert.

## Controller operations in non-RAID mode or HBA mode

If the controller is in non-RAID mode (HBA mode), then:

- Virtual disks or hot spares are not available.
- Security state of the controller is disabled.
- All physical disks are in non-RAID mode.

You can perform the following operations if the controller is in non-RAID mode:

- Blink/unblink the physical disk.
- Configure all properties including the following:
  - Load balanced mode
  - Check consistency mode
  - Patrol read mode
  - Copyback mode
  - Controller boot mode
  - Enhanced auto import foreign configuration
  - Rebuild rate
  - Check consistency rate
  - Reconstruct rate
  - BGI rate
  - Enclosure or backplane mode
  - Patrol read unconfigured areas
- View all properties that are applicable to a RAID controller expect for virtual disks.
- Clear foreign configuration

 **NOTE:** If an operation is not supported in non-RAID mode, an error message is displayed.

You cannot monitor the enclosure temperature probes, fans, and power supplies when the controller is in non-RAID mode.

## Running RAID configuration jobs on multiple storage controllers

While performing operations on more than two storage controllers from any supported iDRAC interface, make sure to:

- Run the jobs on each controller individually. Wait for each job to complete before starting the configuration and job creation on the next controller.
- Schedule multiple jobs to run at a later time using the scheduling options.

## Manage Preserved cache

The Managed Preserved Cache feature is a controller option which provides the user an option to discard the controller cache data. In the write-back policy, data is written to the cache before being written to the physical disk. If the virtual disk goes offline or is deleted for any reason, the data in the cache gets deleted.

The PREC Controller preserves the data written on the preserved or dirty cache in an event of power failure or cable disconnect until you recover the virtual disk or clear the cache.

The status of the controller is affected by the preserved cache. The controller status is displayed as degraded if the controller has preserved cache. Discard the preserved cache is possible only if all of the following conditions are met:

- The controller does not have any foreign configuration.
- The controller does not have any offline or missing virtual disks.

- Cables to any virtual disk are not disconnected.

## Managing PCIe SSDs

Peripheral Component Interconnect Express (PCIe) solid-state device (SSD) is a high-performance storage device that is designed for solutions requiring low latency, high Input Output Operations per Second (IOPS), and enterprise class storage reliability and serviceability. The PCIe SSD is designed based on Single Level Cell (SLC) and Multi-Level Cell (MLC) NAND flash technology with a high-speed PCIe 2.0, PCIe 3.0, or PCIe 4.0 compliant interface.

Using iDRAC interfaces, you can view and configure NVMe PCIe SSDs.

The key features of PCIe SSD are:

- Hot plug capability
- High-performance device

You can perform the following operations for PCIe SSDs:

- Inventory and remotely monitor the health of PCIe SSDs in the server
- Prepare to remove the PCIe SSD
- Securely erase the data
- Blink or unblink the device LED (Identify the device)

You can perform the following operations for HHHL SSDs:

- Inventory and real-time monitoring of the HHHL SSD in the server
- Failed card reporting and logging in iDRAC and OMSS
- Securely erasing the data and removing the card
- TTY logs reporting

You can perform the following operations for SSDs:

- Drive status reporting such as Online, Failed, and Offline

**i NOTE:** When NVMe devices are controlled behind SW RAID, prepare to remove and cryptographic erase operations are not supported, blink, and unblink are supported.

## Inventorying and monitoring PCIe SSDs

The following inventory and monitoring information is available for PCIe SSDs:

- PCIe SSD backplane

**i NOTE:** If the system has a dedicated PCIe backplane, two FQDDs are displayed. One FQDD is for regular drives and the other is for SSDs. If the backplane is shared (universal), only one FQDD is displayed. In case the SSDs are directly attached, the controller FQDD reports as CPU.1, indicating that the SSD is directly attached to the CPU.

- Software inventory includes only the firmware version for the PCIe SSD.

## Inventorying and monitoring PCIe SSDs using web interface

To inventory and monitor PCIe SSD devices, in the iDRAC web interface, go to **Storage > Overview > Physical Disks**. The **Properties** page is displayed. For PCIe SSDs, the **Name** column displays **PCIe SSD**. Expand to view the properties.

## Inventorying and monitoring PCIe SSDs using RACADM

To view all PCIe SSD drives:

```
racadm storage get pdisks
```

To view PCIe SSD backplane information:

```
racadm storage get enclosures
```

 **NOTE:** For all the mentioned commands, PERC devices are also displayed.

For more information, see the **iDRAC RACADM Command Line Reference Guide** available at [dell.com/idracmanuals](http://dell.com/idracmanuals)

## Preparing to remove PCIe SSD

 **NOTE:** This operation is not supported when an NVMe device is behind PERC.

PCIe SSDs support orderly hot swap allowing you to add or remove a device without halting or rebooting the system in which the devices are installed. To prevent data loss, you must use the Prepare to Remove operation before physically removing a device.

Orderly hot swap is supported only when PCIe SSDs are installed in a supported system running a supported operating system. To ensure that you have the correct configuration for your PCIe SSD, see the system-specific owner's manual.

The Prepare to Remove operation is not supported for PCIe SSDs on the VMware vSphere (ESXi) systems.

The Prepare to Remove operation can be performed in real-time using iDRAC Service Module.

The Prepare to Remove operation stops any background activity and any ongoing I/O activity so that the device can be removed safely. It causes the status LEDs on the device to blink. You can safely remove the device from the system under the following conditions after you initiate the Prepare to Remove operation:

- The PCIe SSD is blinking the safe to remove LED pattern (blinks amber).
- The PCIe SSD is no longer accessible by the system.

Before preparing the PCIe SSD for removal, ensure that:

- iDRAC Service Module is installed.
- Lifecycle Controller is enabled.
- You have Server Control and Login privileges.

## Preparing to remove PCIe SSD using web interface

### About this task

To prepare the PCIe SSD for removal:

### Steps

1. In the iDRAC Web interface, go to **Storage > Overview > Physical Disks**.  
The **Setup Physical Disk** page is displayed.
2. From the drop-down menus, select **Prepare to Remove** for one or more PCIe SSDs.  
If you have selected **Prepare to Remove** and you want to view the other options in the drop-down menu, then select **Action** and then click the drop-down menu to view the other options.  
 **NOTE:** Ensure that iSM is installed and running to perform the `preparetoremove` operation.
3. From the **Apply Operation Mode** drop-down menu, select **Apply Now** to apply the actions immediately.  
If there are jobs to be completed, then this option is grayed-out.  
 **NOTE:** For PCIe SSD devices, only the **Apply Now** option is available. This operation is not supported in staged mode.
4. Click **Apply**.  
If the job is not created, a message indicating that the job creation was not successful is displayed. Also, the message ID and the recommended response action is displayed.  
If the job is created successfully, a message indicating that the job ID is created for the selected controller is displayed. Click **Job Queue** to view the progress of the job in the **Job Queue** page.  
If a pending operation is not created, an error message is displayed. If pending operation is successful and job creation is not successful, then an error message is displayed.

## Preparing to remove PCIe SSD using RACADM

To prepare the PCIeSSD drive for removal:

```
racadm storage preparetoremove:<PCIeSSD FQDD>
```

To create the target job after executing preparetoremove command:

```
racadm jobqueue create <PCIe SSD FQDD> -s TIME_NOW --realtime
```

To query the job ID returned:

```
racadm jobqueue view -i <job ID>
```

For more information, see the *RACADM Reference CLI Guide*.

## Erasing PCIe SSD device data

Cryptographic Erase permanently erases all data present on the disk. Performing a Cryptographic Erase on a PCIe SSD overwrites all blocks and results in permanent loss of all data on the PCIe SSD. During Cryptographic Erase, the host is unable to access the PCIe SSD. The changes are applied after system reboot.

If the system reboots or experiences a power loss during cryptographic erase, the operation is canceled. You must reboot the system and restart the process.

Before erasing PCIe SSD device data, ensure that:

- Lifecycle Controller is enabled.
- You have Server Control and Login privileges.

### NOTE:

- Erasing PCIe SSDs can only be performed as a staged operation.
- After the drive is erased, it displays in the operating system as online but it is not initialized. You must initialize and format the drive before using it again.
- After you hot-plug a PCIe SSD, it may take several seconds to appear on the web interface.

## Erasing PCIe SSD device data using web interface

### About this task

To erase the data on the PCIe SSD device:

### Steps

1. In the iDRAC Web interface, go to **Storage > Overview > Physical Disks**.  
The **Physical Disk** page is displayed.
2. From the **Controller** drop-down menu, select the controller to view the associated PCIe SSDs.
3. From the drop-down menus, select **Cryptographic Erase** for one or more PCIe SSDs.  
If you have selected **Cryptographic Erase** and you want to view the other options in the drop-down menu, then select **Action** and then click the drop-down menu to view the other options.
4. From the **Apply Operation Mode** drop-down menu, select one of the following options:
  - **At Next Reboot** — Select this option to apply the actions during the next system reboot.
  - **At Scheduled Time** — Select this option to apply the actions at a scheduled day and time:
    - **Start Time** and **End Time** — Click the calendar icons and select the days. From the drop-down menus, select the time. The action is applied between the start time and end time.
    - From the drop-down menu, select the type of reboot:
      - No Reboot (Manually Reboot System)
      - Graceful Shutdown
      - Force Shutdown
      - Power Cycle System (cold boot)

5. Click **Apply**.

If the job is not created, a message indicating that the job creation was not successful is displayed. Also, the message ID and the recommended response action is displayed.

If the job is created successfully, a message indicating that the job ID is created for the selected controller is displayed. Click **Job Queue** to view the progress of the job in the Job Queue page.

If pending operation is not created, an error message is displayed. If pending operation is successful and job creation is not successful, then an error message is displayed.

## Erasing PCIe SSD device data using RACADM

To securely erase a PCIe SSD device:

```
racadm storage secureerase:<PCIeSSD FQDD>
```

To create the target job after performing the `secureerase` command:

```
racadm jobqueue create <PCIe SSD FQDD> -s TIME_NOW -e <start_time>
```

To query the job ID returned:

```
racadm jobqueue view -i <job ID>
```

For more information, see the *iDRAC RACADM Command Line Reference Guide*

## Managing enclosures or backplanes

You can perform the following for enclosures or backplanes:

- View properties
- Configure universal mode or split mode
- View slot information (universal or shared)
- Set SGPIO mode
- Set Asset Tag
- Asset Name

## Configuring backplane mode

PowerEdge servers support a new internal storage topology, where two storage controllers (PERCs) can be connected to internal drives through a single expander. This configuration is used for high-performance mode with no failover or High Availability (HA) functionality. The expander splits the internal drive array between the two storage controllers. In this mode, virtual disk creation only displays the drives that are connected to a particular controller. There are no licensing requirements for this feature. This feature is supported only on a few systems.

The backplane supports the following modes:

- Unified mode—This is the default mode. The primary PERC controller has access to all the drives connected to the backplane even if a second PERC controller is installed.
- Split mode—One controller has access to the first 12 drives and the second controller has access to the last 12 drives. The drives connected to the first controller are numbered 0-11 while the drives connected to the second controller are numbered 12-23.
- Split mode 4:20—One controller has access to the first four drives and the second controller has access to the last 20 drives. The drives connected to the first controller are numbered 0-3 while the drives connected to the second controller are numbered 4-23.
- Split mode 8:16—One controller has access to the first eight drives and the second controller has access to the last 16 drives. The drives connected to the first controller are numbered 0-7 while the drives connected to the second controller are numbered 8-23.

- Split mode 16:8—One controller has access to the first 16 drives and the second controller has access to the last eight drives. The drives connected to the first controller are numbered 0-15 while the drives connected to the second controller are numbered 16-23.
- Split mode 20:4—One controller has access to the first 20 drives and the second controller has access to the last four drives. The drives connected to the first controller are numbered 0-19 while the drives connected to the second controller are numbered 20-23.
- Information Not Available—Controller information is not available.

iDRAC allows the split mode setting if the expander can support the configuration. Ensure that you enable this mode before installing the second controller. iDRAC performs a check for expander capability before allowing this mode to be configured and does not check whether the second PERC controller is present.

#### NOTE:

- Cable errors (or other errors) may appear if you put the backplane into split mode with only one PERC attached, or if you put the backplane into Unified Mode with two PERCs attached.
- When multiple backplanes are connected to a single PERC controller, the controller merges them into a single enclosure. Therefore, you see only one backplane in the Hardware Inventory or Storage page. However, Firmware Inventory displays the actual number of backplanes in the system, and querying these from the Redfish interface may result in an error.

To modify the setting, you must have Server Control privilege.

If any other RAID operations are in pending state or any RAID job is scheduled, you cannot change the backplane mode. Similarly, if this setting is pending, you cannot schedule other RAID jobs.

#### NOTE:

- Warning messages are displayed when the setting is being changed as there is a possibility of data loss.
- LC Wipe or iDRAC reset operations do not change the expander setting for this mode.
- This operation is supported only in real-time and not staged.
- You can change the backplane configuration multiple times.
- If the backplane is configured with the **Physical Slots** attribute as **0**, the iDRAC HII does not display the backplane details.
- The backplane splitting operation can cause data loss or foreign configuration if the drive association changes from one controller to another controller.
- During the backplane splitting operation, the RAID configuration may be impacted depending on the drive association.

Changes to these settings take effect only after a system power reset. Switching from Split mode to Unified mode triggers an error message on the next boot because the second controller cannot detect any drives. Also, the first controller identifies a foreign configuration. Ignoring this error results in the loss of existing virtual disks.

## Configuring enclosure using RACADM

### About this task

To configure the enclosure or backplane, use the `set` command with the objects in **BackplaneMode**.

For example, to set the BackplaneMode attribute to split mode:

### Steps

1. Run the following command to view the current backplane mode:

```
racadm get storage.enclosure.1.backplanecurrentmode
```

The output is:

```
BackplaneCurrentMode=UnifiedMode
```

2. Run the following command to view the requested mode:

```
racadm get storage.enclosure.1.backplanerequestedmode
```

The output is:

```
BackplaneRequestedMode=None
```

3. Run the following command to set the requested backplane mode to split mode:

```
racadm set storage.enclosure.1.backplanerequestedmode "splitmode"
```

The message is displayed indicating that the command is successful.

4. Run the following command to verify if the **backplanerequestedmode** attribute is set to split mode:

```
racadm get storage.enclosure.1.backplanerequestedmode
```

The output is:

```
BackplaneRequestedMode=None (Pending=SplitMode)
```

5. Run `storage get controllers` command and note down the controller instance ID.

6. Run the following command to create a job:

```
racadm jobqueue create <controller instance ID> -s TIME_NOW --realtime
```

A job ID is returned.

7. Run the following command to query the job status:

```
racadm jobqueue view -i JID_xxxxxxxx
```

where, `JID_xxxxxxxx` is the job ID from step 6.

The status is displayed as Pending.

Continue to query the job ID until you view the Completed status (this process may take up to three minutes).

8. Run the following command to view the `backplanerequestedmode` attribute value:

```
racadm get storage.enclosure.1.backplanerequestedmode
```

The output is:

```
BackplaneRequestedMode=SplitMode
```

9. Run the following command to cold reboot the server:

```
racadm serveraction powercycle
```

10. After the system completes POST and CSIOR, type the following command to verify the `backplanerequestedmode`:

```
racadm get storage.enclosure.1.backplanerequestedmode
```

The output is:

```
BackplaneRequestedMode=None
```

11. Run the following to verify if the backplane mode is set to split mode:

```
racadm get storage.enclosure.1.backplanecurrentmode
```

The output is:

```
BackplaneCurrentMode=SplitMode
```

12. Run the following command and verify that only 0–11 drives are displayed:

```
racadm storage get pdisks
```

For more information about the RACADM commands, see the **iDRAC RACADM Command Line Interface Reference Guide** available at [dell.com/idracmanuals](http://dell.com/idracmanuals).

## Configuring backplane mode using web interface

### About this task

To configure backplane mode using iDRAC web interface:

### Steps

1. In the iDRAC web interface, go to **Storage > Overview > Enclosures**.
2. From the **Enclosure** option, select the enclosure to configure.
3. From the **Action** drop-down menu, select **Edit Enclosure Mode**.  
The **Edit Enclosure Mode** page is displayed.
4. In the **Current Value** column, select the required enclosure mode for the backplane or enclosure. The options are:
  - Unified Mode
  - Split Mode
  - Split Mode 4:20
  - Split Mode 8:16
  - Split Mode 16:8
  - Split Mode 20:4
5. Click **Add to Pending Operations**.  
A job ID is created.
6. Click **Apply Now**.
7. Go to the **Job Queue** page and verify that it displays the status as Completed for the job.
8. Power cycle the system for the settings to take effect.

 **NOTE:** To avoid inventory issues, in case of any backplane cable connection changes, an additional iDRAC reboot and Host power cycle are required.

## Setting SGPIO mode

The storage controller can connect to the backplane in I2C mode (default setting for Dell backplanes) or Serial General Purpose Input/Output (SGPIO) mode. This connection is required for blinking LEDs on the drives. Dell PERC controllers and backplane support both these modes. To support certain channel adapters, the backplane mode must be changed SGPIO mode.

The SGPIO mode is only supported for passive backplanes. It is not supported for expander-based backplanes or passive backplanes in downstream mode. Backplane firmware provides information on capability, current state, and requested state.

After LC wipe operation or iDRAC reset to default, the SGPIO mode is reset to disabled state. It compares the iDRAC setting with the backplane setting. If the backplane is set to SGPIO mode, iDRAC changes its setting to match the backplane setting.

Server power cycle is required for any change in setting to take effect.

You must have Server Control privilege to modify this setting.

 **NOTE:** You cannot set the SGPIO mode using iDRAC Web interface.

## Setting SGPIO mode using RACADM

To configure the SGPIO mode, use the `set` command with the objects in the `SGPIOMode` group.



If it is set to disabled, it is I2C mode. If enabled, it is set to SGPIO mode.

For more information, see the **iDRAC RACADM Command Line Interface Reference Guide** available at [dell.com/idracmanuals](http://dell.com/idracmanuals).

## Set Enclosure Asset Name

Set Enclosure Asset Name allows the user to configure the Asset Name of a storage enclosure.

The user can change the Asset Name property of the enclosure to identify enclosures easily. These fields are checked for invalid values and an error is displayed if an invalid value is entered. These fields are part of the enclosure firmware; the data initially shown are the values saved in the firmware.

 **NOTE:** Asset Name has a character limit of 32 that includes the null character.

 **NOTE:** These operations are not supported on internal enclosures.

## Set Enclosure Asset Tag

Set Enclosure Asset Tag allows you to configure Asset Tag of a storage enclosure.

User can change the Asset Tag property of the enclosure to identify enclosures. These fields are checked for invalid values and an error is displayed if an invalid value is entered. These fields are part of the enclosure firmware; the data initially shown are the values saved in the firmware.

 **NOTE:** Asset Tag has a character limit of 10 that includes the null character.

 **NOTE:** These operations are not supported on internal enclosures.

## Choosing operation mode to apply settings

While creating and managing virtual disks, setting up physical disks, controllers, and enclosures or resetting controllers, before you apply the various settings, you must select the operation mode. That is, specify when you want to apply the settings:

- Immediately
- During the next system reboot
- At a scheduled time
- As a pending operation to be applied as a batch as part of a single job.

## Choosing operation mode using web interface

### About this task

To select the operation mode to apply the settings:

### Steps

1. You can select the operation mode on when you are on any of the following pages:
  - **Storage > Physical Disks** .
  - **Storage > Virtual Disks**
  - **Storage > Controllers**
  - **Storage > Enclosures**
2. Select one of the following from the **Apply Operation Mode** drop-down menu:
  - **At Next Reboot** — Select this option to apply the settings during the next system reboot.
  - **At Scheduled Time** — Select this option to apply the settings at a scheduled day and time:
    - **Start Time** and **End Time** — Click the calendar icons and select the days. From the drop-down menus, select the time. The settings are applied between the start time and end time.
    - From the drop-down menu, select the type of reboot:

- No Reboot (Manually Reboot System)
- Graceful Shutdown
- Force Shutdown
- Power Cycle System (cold boot)
- **Add to Pending Operations** — Select this option to create a pending operation to apply the settings. You can view all pending operations for a controller in the **Storage > Overview > Pending Operations** page.

**NOTE:**

- The **Add to Pending Operations** option is not applicable for the **Pending Operations** page and for PCIe SSDs in the **Physical Disks > Setup** page.
- Only the **Apply Now** option is available on the **Enclosure Setup** page.

3. Click **Apply**.  
Based on the operation mode selected, the settings are applied.

## Choosing operation mode using RACADM

To select the operation mode, use the `jobqueue` command.

For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

## Viewing and applying pending operations

You can view and commit all pending operations for the storage controller. All the settings are either applied at once, during the next reboot, or at a scheduled time based on the selected options. You can delete all the pending operations for a controller. You cannot delete individual pending operations.

Pending Operations are created on the selected components (controllers, enclosures, physical disks, and virtual disks).

Configuration jobs are created only on controllers. In PCIe SSD, a job is created for the PCIe SSD disk.

## Viewing, applying, or deleting pending operations using web interface

### Steps

1. In the iDRAC web interface, go to **Storage > Overview > Pending Operations**.  
The **Pending Operations** page is displayed.
  2. From the **Component** drop-down menu, select the controller for which you want to view, commit, or delete the pending operations.  
The list of pending operations is displayed for the selected controller.
- NOTE:**
- Pending operations are created for import foreign configuration, clear foreign configuration, security key operations, and encrypt virtual disks. But, they are not displayed in the **Pending Operations** page and in the Pending Operations pop-up message.
  - Jobs for PCIe SSD cannot be created from the **Pending Operations** page
3. To delete the pending operations for the selected controller, click **Delete All Pending Operations**.
  4. From the drop-down menu, select one of the following and click **Apply** to commit the pending operations:
    - **At Next Reboot** — Select this option to commit all the operations during the next system reboot.
    - **At Scheduled Time** — Select this option to commit the operations at a scheduled day and time.
      - **Start Time** and **End Time** — Click the calendar icons and select the days. From the drop-down menus, select the time. The action is applied between the start time and end time.
      - From the drop-down menu, select the type of reboot:
        - No Reboot (Manually Reboot System)
        - Graceful Shutdown
        - Force Shutdown
        - Power Cycle System (cold boot)

5. If the commit job is not created, a message indicating that the job creation was not successful is displayed. Also, the message ID and the recommended response action are displayed.
6. If the commit job is created successfully, a message indicating that the job ID is created for the selected controller is displayed. Click **Job Queue** to view the progress of the job in the **Job Queue** page.

If the clear foreign configuration, import foreign configuration, security key operations, or encrypt virtual disk operations are in pending state, and if these are the only operations pending, then you cannot create a job from the **Pending Operations** page. You must perform any other storage configuration operation or use RACADM to create the required configuration job on the required controller.

You cannot view or clear pending operations for PCIe SSDs in the **Pending Operations** page. Use the `racadm` command to clear the pending operations for PCIe SSDs.

## Viewing and applying pending operations using RACADM

To apply pending operations, use the `jobqueue` command.

For more information, see the **iDRAC RACADM Command Line Reference Guide** available at [dell.com/idracmanuals](http://dell.com/idracmanuals).

## Storage devices — apply operation scenarios

### Case 1: selected an apply operation (apply now, at next reboot, or at scheduled time) and there are no existing pending operations

If you have selected **Apply Now**, **At Next Reboot**, or **At Scheduled Time** and then clicked **Apply**, first the pending operation is created for the selected storage configuration operation.

- If the pending operation is successful and there are no prior existing pending operations, then the job is created. If the job is created successfully, a message indicating that the job ID is created for the selected device is displayed. Click **Job Queue** to view the progress of the job in the **Job Queue** page. If the job is not created, a message indicating that the job creation was not successful is displayed. Also, the message ID and the recommended response action are displayed.
- If the pending operation creation is unsuccessful and there are no prior existing pending operations, an error message with ID and recommended response action is displayed.

### Case 2: selected an apply operation (apply now, at next reboot, or at scheduled time) and there are existing pending operations

If you have selected **Apply Now**, **At Next Reboot**, or **At Scheduled Time** and then clicked **Apply**, first the pending operation is created for the selected storage configuration operation.

- If the pending operation is created successfully and if there are existing pending operations, then a message is displayed.
  - Click the **View Pending Operations** link to view the pending operations for the device.
  - Click **Create Job** to create job for the selected device. If the job is created successfully, a message indicating that the job ID is created for the selected device is displayed. Click **Job Queue** to view the progress of the job in the **Job Queue** page. If the job is not created, a message indicating that the job creation was not successful is displayed. Also, the message ID and the recommended response action is displayed.
  - Click **Cancel** to not create the job and remain on the page to perform more storage configuration operations.
- If the pending operation is not created successfully and if there are existing pending operations, then an error message is displayed.
  - Click **Pending Operations** to view the pending operations for the device.
  - Click **Create Job For Successful Operations** to create the job for the existing pending operations. If the job is created successfully, a message indicating that the job ID is created for the selected device is displayed. Click **Job Queue** to view the progress of the job in the **Job Queue** page. If the job is not created, a message indicating that the job creation was not successful is displayed. Also, the message ID and the recommended response action are displayed.
  - Click **Cancel** to not create the job and remain on the page to perform more storage configuration operations.

### Case 3: selected add to pending operations and there are no existing pending operations

If you have selected **Add to Pending Operations** and then clicked **Apply**, first the pending operation is created for the selected storage configuration operation.

- If the pending operation is created successfully and if there are no existing pending operations, then an information message is displayed:
  - Click **OK** to remain on the page to perform more storage configuration operations.

- Click **Pending Operations** to view the pending operations for the device. Until the job is created on the selected controller, these pending operations are not applied.
- If the pending operation is not created successfully and if there are no existing pending operations, then an error message is displayed.

#### Case 4: selected add to pending operations and there are prior existing pending operations

If you have selected **Add to Pending Operations** and then clicked **Apply**, first the pending operation is created for the selected storage configuration operation.

- If the pending operation is created successfully and if there are existing pending operations, then an information message is displayed:
  - Click **OK** to remain on the page to perform more storage configuration operations.
  - Click **Pending Operations** to view the pending operations for the device.
- If the pending operation is not created successfully and if there are existing pending operations, then an error message is displayed.
  - Click **OK** to remain on the page to perform more storage configuration operations.
  - Click **Pending Operations** to view the pending operations for the device.

#### **NOTE:**

- At any time, if you do not see the option to create a job on the storage configuration pages, go to **Storage Overview > Pending Operations** page to view the existing pending operations and to create the job on the required controller.
- Only cases 1 and 2 are applicable for PCIe SSD. You cannot view the pending operations for PCIe SSDs and hence **Add to Pending Operations** option is not available. Use racadm command to clear the pending operations for PCIe SSDs.

## Blinking or unblinking component LEDs

You can locate a physical disk, virtual disk drive and PCIe SSDs within an enclosure by blinking one of the Light Emitting Diodes (LEDs) on the disk.

You must have Login + Control & Configure System privilege to blink or unblink an LED.

The controller must be real-time configuration capable. The real-time support of this feature is available only in PERC 9.1 firmware and later.

 **NOTE:** Blink or unblink is not supported for servers without backplane.

## Blinking or unblinking component LEDs using web interface

### About this task

To blink or unblink a component LED:

### Steps

1. In the iDRAC web interface, go to any of the following pages as per your requirement:
  - **Storage > Overview > Physical Disks > Status**—Displays the identified Physical Disks page where you can blink or unblink the hard drives and PCIe SSDs.
  - **Storage > Overview > Virtual Disks > Status**—Displays the identified Virtual Disks page where you can blink or unblink the virtual disks.
2. If you select the physical disk:
  - Select or clear all component LEDs—Select the **Select/Deselect All** option and click **Blink** to start blinking the component LEDs. Similarly, click **Unblink** to stop blinking the component LEDs.
  - Select or clear individual component LEDs—Select one or more components and click **Blink** to start blinking the selected component LED(s). Similarly, click **Unblink** to stop blinking the component LEDs.
3. If you select the virtual disk:
  - Select or deselect all physical disk drives or PCIe SSDs—Select the **Select/Deselect All** option and click **Blink** to start blinking all the physical disk drives and the PCIe SSDs. Similarly, click **Unblink** to stop blinking the LEDs.
  - Select or clear individual physical disk drives or PCIe SSDs—Select one or more physical disk drives and click **Blink** to start blinking the LEDs for the physical disk drives or the PCIe SSDs. Similarly, click **Unblink** to stop blinking the LEDs.

4. If you are on the **Identify Virtual Disk** page:

- Select or clear all virtual disks—Select the **Select/Deselect All** option and click **Blink** to start blinking the LEDs for all the virtual disks. Similarly, click **Unblink** to stop blinking the LEDs.
- Select or deselect individual virtual disks—Select one or more virtual disks and click **Blink** to start blinking the LEDs for the virtual disks. Similarly, click **Unblink** to stop blinking the LEDs.

If the blink or unblink operation is not successful, error messages are displayed.

## Blinking or unblinking component LEDs using RACADM

To blink or unblink component LEDs, use the following commands:

```
racadm storage blink:<PD FQDD, VD FQDD, or PCIe SSD FQDD>
```

```
racadm storage unblink:<PD FQDD, VD FQDD, or PCIe SSD FQDD>
```

For more information, see the **iDRAC RACADM Command Line Reference Guide** available at [dell.com/idracmanuals](https://dell.com/idracmanuals).

## Warm reboot

When warm reboot is performed, the following behaviors are observed:

- PERC controllers in iDRAC UI are unavailable immediately after warm reboot. They are available once reinventory is completed after warm reboot. This is only applicable for PERC controllers and not for NVME/HBA/BOSS.
- Storage files in SupportAssist are empty when PERC controllers are unavailable in the UI.
- LC Logging for PAST event and Critical events are done for PERC during `perc reinventory`. Rest all LCL for PERC components are suppressed. LCL resumes after PERC reinventory finishes.
- You cannot start any Real-time job until the PERC reinventory is completed.
- Telemetry data is not collected until PERC reinventory is finished.
- After the PERC inventory is finished, the behavior is normal.

 **NOTE:** After performing a server warm boot, iDRAC may report `Disk Inserted` message in LC logs for drives that are behind the HBA. Ignore this log entry.

## BIOS Settings

You can view multiple attributes, which are being used for a specific server under the BIOS Settings. You can modify different parameters of each attribute from this BIOS configuration setting. Once you select one attribute, it shows different parameters which are related to that specific attribute. You can modify multiple parameters of an attribute and apply changes before modifying a different attribute. When a user expands a configuration group, attributes are displayed in an alphabetical order.

 **NOTE:** Attribute-level help content is dynamically generated.

### Apply

The **Apply** button remains grayed-out until any of the attributes are modified. After you changed an attribute, click **Apply**. The attribute is modified with the required changes. If the request fails to set the BIOS attribute, an error message is displayed. For more information, see the *Dell Technologies 17G, 16G, 15G, 14G, and 13G PowerEdge Servers Error and Event Messages Reference Guide* available on the Dell Support site.

### Discard changes

The **Discard Changes** button is grayed-out until any of the attributes are modified. If you click **Discard Changes** button, all the recent changes are discarded and restored with the previous or initial values.

### Apply and Reboot

When a user modifies the value of an attribute or boot sequence, the user is presented with two choices to apply the configuration; **Apply and Reboot** or **Apply on Next Reboot**. In either of the apply options, the user is redirected to the job queue page to monitor the progress of that specific job.

A user can view auditing information that is related to BIOS configuration in the LC logs.

If you click **Apply and Reboot**, it restarts the server immediately to configure all the required changes. If the request fails to set the BIOS attributes, an error message is displayed.

### Apply At Next Reboot

When a user modifies the value of an attribute or boot sequence, the user is presented with two choices to apply the configuration; **Apply and Reboot** or **Apply on Next Reboot**. In either of the apply options, the user is redirected to the job queue page to monitor the progress of that specific job.

A user can view auditing information that is related to BIOS configuration in the LC logs.

If you click **Apply At Next Reboot**, it configures all the required changes on the next restart of the server. You will not experience any immediate modifications based on the recent configuration changes until the next reboot session is taking place successfully. If the request fails to set the BIOS attributes, an error message is displayed.

### Delete All Pending Values

**Delete All pending Values** button is enabled only when there are pending values based on the recent configuration changes. In case, user decides not to apply the configuration changes, the user can click **Delete All Pending Values** button to terminate all the modifications. In case, the request fails to remove the BIOS attributes, an error message is displayed.

## Pending Value

Configuration of a BIOS attribute using iDRAC is not applied immediately to the BIOS. It requires a server reboot for the changes to take place. When you modify a BIOS attribute then **Pending Value** gets updated. If an attribute already has a pending value (and that has been configured) it is displayed on the UI.

## Modifying BIOS Configuration

Modifying BIOS configuration results in audit log entries, which are entered in LC logs.

## BIOS Live Scanning

BIOS live scanning verifies the integrity and authenticity of the BIOS image in the BIOS primary ROM when the host is powered ON but not in POST.

### NOTE:

- This feature requires an iDRAC Datacenter license.
- You need to have Debug privilege for operating this feature.

iDRAC performs verification of immutable sections of BIOS image automatically at the following scenarios:

- At AC cycle/Cold boot
- On a schedule determined by user
- On demand (initiated by user)

Successful result of live scanning is logged to LC log. Failure result is logged to both LCL and SEL.

### Topics:

- [BIOS Live Scanning](#)
- [BIOS Recovery and Hardware Root of Trust \(RoT\)](#)

## BIOS Live Scanning

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- At AC cycle/Cold boot
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- On demand (initiated by user)

Successful result of live scanning is logged to LC log. Failure result is logged to both LCL and SEL.

## BIOS Recovery and Hardware Root of Trust (RoT)

For PowerEdge server, it is mandatory to recover from corrupted or damaged BIOS image either due to malicious attack or power surges or any other unforeseeable events. An alternate reserve of BIOS image would be necessary to recover BIOS in order to bring the PowerEdge server back to functional mode from unbootable mode. This alternative/recovery BIOS is stored in a 2nd SPI (mux'ed with primary BIOS SPI).

The recovery sequence can be initiated through any of the following approaches with iDRAC as the main orchestrator of the BIOS recovery task:

1. **Auto recovery of BIOS primary image/recovery image**—BIOS image is recovered automatically during the host boot process after the BIOS corruption is detected by BIOS itself.
2. **Forced recovery of BIOS Primary/recovery image**—User initiates an OOB request to update BIOS either because they have a new updated BIOS or BIOS was just crashing by failing to boot.
3. **Primary BIOS ROM update**—The single primary ROM is split into Data ROM and Code ROM. iDRAC has full access/control over Code ROM. It switches MUX to access Code ROM whenever needed.
4. **BIOS Hardware Root of Trust (RoT)**—This feature is available in servers with model number RX5X, CX5XX, and TX5X. During every host boot (only cold boot or A/C cycle, not during warm reboot), iDRAC ensures that RoT is performed. RoT runs automatically and user cannot initiate it using any interfaces. This iDRAC boot first policy verifies host BIOS ROM contents on every AC cycle and host DC cycle. This process ensures secure boot of BIOS and further secures the host boot process.

 **NOTE:** When powering on the server from the **Off** state, it may take 20 to 30 seconds for iDRAC to report power state as **On**.



## Configuring and using virtual console

iDRAC has added an enhanced HTML5 option in vConsole which allows vKVM (virtual Keyboard, Video, and Mouse) over standard VNC client. You can use the virtual console to manage a remote system using the keyboard, video, and mouse on your management station to control the corresponding devices on a managed server. This is a licensed feature for rack and tower servers. You need iDRAC Configure privilege to access all configurations on the virtual console.

The following are the list of configurable attributes in Virtual Console:

- vConsole Enabled—Enabled, Disabled
- Max Sessions—1-6
- Active sessions—0-6
- Video Encryption—Enabled, Disabled
- Local Server Video—Enabled, Disabled
- Dynamic Action on Sharing Request Timeout—Full Access, Read-Only Access, And Deny Access
- Automatic System Lock—Enabled, Disabled
- Keyboard/Mouse Attach State—Auto-attach, Attached, and Detached

The key features are:

- A maximum of six simultaneous Virtual Console sessions are supported. All the sessions view the same managed server console simultaneously.
- You can launch a virtual console in a supported web browser.

### NOTE:

- Any change in web server configuration results in termination of an existing virtual console session.
  - Even if the Video Encryption option is disabled in the UI, you can still configure the feature using other interfaces, Video encryption is enabled by default.
  - Virtual console link may get interrupted while running video stress in Internet Explorer.
- When you open a Virtual Console session, the managed server does not indicate that the console has been redirected.
  - You can open multiple Virtual Console sessions from a single management station to one or more managed systems simultaneously.
  - You can open up to six virtual console sessions from the management station to the managed server.
  - If a second user requests a Virtual Console session, the first user is notified and is provided an option to refuse access, allow read-only access, or allow full shared access. The second user is notified that another user has control. The first user must respond within thirty seconds, or else access is granted to the second user based on the default setting. If neither the first or second user has administrator privileges, terminating the first user's session automatically terminates the second user's session.
  - Boot logs and crash logs are captured as Video logs and are in MPEG1 format.
  - Crash screen is captured as JPEG file.

 **NOTE:** The number of active virtual-console sessions displayed in the web interface is only for active web-interface sessions. This number does not include sessions from other interfaces such as SSH and RACADM.

 **NOTE:** For information about configuring your browser to access the virtual console, see [Configuring web browsers to use virtual console](#).

### Topics:

- [Supported screen resolutions and refresh rates](#)
- [Configuring a virtual console](#)
- [Launching virtual console](#)
- [Using virtual console viewer](#)

# Supported screen resolutions and refresh rates

The following table lists the supported screen resolutions and corresponding refresh rates for a Virtual Console session running on the managed server.

**Table 42. Supported screen resolutions and refresh rates**

| Screen Resolution | Refresh Rate (Hz)  |
|-------------------|--------------------|
| 720x400           | 70                 |
| 640x480           | 60, 72, 75, 85     |
| 800x600           | 60, 70, 72, 75, 85 |
| 1024x768          | 60, 70, 72, 75, 85 |
| 1280x1024         | 60                 |
| 1920x1200         | 60                 |

It is recommended that you configure the monitor display resolution to 1920x1200 pixels.

Virtual Console supports a maximum video resolution of 1920x1200 at 60 Hz refresh rate. In order to achieve this resolution, following conditions are required:

- KVM / monitor attached to VGA that supports 1920x1200 resolution
- Latest Matrox video driver (for Windows)

When a local KVM / Monitor with maximum resolution below 1920x1200 is connected to either VGA connector, it will reduce the maximum resolution supported in virtual console.

iDRAC virtual console leverages the onboard Matrox G200 graphics controller to determine the maximum resolution of the attached monitor when a physical display is present. When the monitor supports 1920x1200 or greater resolution, the virtual console supports 1920x1200 resolution. If the monitor attached supports lower max resolution (like many KVMs), the virtual console max resolution is limited.

## Maximum virtual console resolutions based on monitor display ratio:

- 16:10 monitor: 1920x1200 will be the max resolution
- 16:9 monitor: 1920x1080 will be the max resolution

When a physical monitor is not connected to either VGA port on the server, the OS installed will dictate the available resolutions for virtual console.

## Maximum virtual console resolutions based on host OS without physical monitor:

- Windows: 1600x1200 (1600x1200, 1280x1024, 1152x864, 1024x768, 800x600)
- Linux: 1024x768 (1024x768, 800x600, 848x480, 640x480)

**NOTE:** If a higher resolution through virtual console is required when physical KVM or monitor is not present, a VGA Display Emulator dongle can be leveraged to mimic an external monitor connection with a resolution up to 1920x1080.

**NOTE:** If you have an active Virtual Console session and a lower resolution monitor is connected to the Virtual Console, the server console resolution may reset if the server is selected on the local console. If the system is running a Linux operating system, an X11 console may not be viewable on the local monitor. Press <Ctrl><Alt><F1> at the iDRAC Virtual Console to switch Linux to a text console.

# Configuring a virtual console

Before configuring the Virtual Console, ensure that the management station is configured.

You can configure the virtual console using iDRAC Web interface or RACADM command-line interface.

# Configuring virtual console using web interface

## About this task

To configure Virtual Console using iDRAC Web interface:

## Steps

1. Go to **Configuration > Virtual Console**. Click **Start the Virtual Console** link, then Virtual Console page is displayed.
2. Enable virtual console and specify the required values. For information about the options, see the **iDRAC Online Help**.  
 **NOTE:** If you are using Nano operating system, disable the **Automatic System Lock** feature on the **Virtual Console** page.
3. Click **Apply**. The virtual console is configured.

# Configuring virtual console using RACADM

To configure the Virtual Console, use the `set` command with the objects in the **iDRAC.VirtualConsole** group.

For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

# Previewing virtual console

Before launching the Virtual Console, you can preview the state of the Virtual Console on the **System > Properties > System Summary** page. The **Virtual Console Preview** section displays an image showing the state of the Virtual Console. The image is refreshed every 30 seconds. This is a licensed feature.

 **NOTE:** The Virtual Console image is available only if you have enabled Virtual Console.

# Launching virtual console

You can launch the virtual console using the iDRAC Web Interface or a URL.

 **NOTE:** Do not launch a virtual console session from a web browser on the managed system.

Before launching the virtual console, ensure that:

- You have administrator privileges.
- Minimum network bandwidth of 1 MB/sec is available.

 **NOTE:** When the custom HTTP port or default port is configured in iDRAC, clear the browser cache, and then launch iDRAC with HTTPS and accept the certificates. After this, log in to iDRAC and launch the virtual console.

 **NOTE:** If the embedded video controller is disabled in the BIOS and if you launch the Virtual Console, the Virtual Console Viewer is blank.

The virtual console has the following Console controls:

1. **General**—Sets Keyboard Macros, Aspect Ratio, and Touch Mode.
2. **KVM**—Shows the values for Frame Rate, Bandwidth, Compression, and Packet Rate.
3. **Performance**—Changes the video quality and video speed.
4. **User List**—Views the list of users who are connected to the console.

You can access Virtual Media by clicking the **Connect to Virtual Media** option available in the virtual console.

 **NOTE:** In iDRAC release 5.10.00.00, if the RFS session is active, the Virtual Media Session is blocked. Hence when you upgrade from the 4.40.00.00 release to the 5.10.00.00 release with RFS Session Active, RFS gets remounted when iDRAC is up. In this case, if you try to launch Virtual Media Session, it fails with an error message `Virtual media already in use`.

## Launching virtual console using web interface

You can launch the virtual console in the following ways:

- Go to **Configuration > Virtual Console**. Click **Start the Virtual Console** link. Virtual console page is displayed.

The **Virtual Console Viewer** displays the remote system's desktop. Using this viewer, you can control the remote system's mouse and keyboard functions from your management station.

Multiple message boxes may appear after you launch the application. To prevent unauthorized access to the application, navigate through these message boxes within three minutes. Otherwise, you are prompted to relaunch the application.

If one or more Security Alert windows appear while launching the viewer, click Yes to continue.

Two mouse pointers may appear in the viewer window: one for the managed server and another for your management station.

## Launching a virtual console using a URL

### About this task

To launch the Virtual Console using the URL:

### Steps

1. Open a supported web browser and in the address box, type the following URL in lower case: **https://iDRAC\_ip/console**
2. Based on the login configuration, the corresponding **Login** page is displayed:

 **NOTE:** Internet Explorer supports Local, Active Directory, and LDAP logins. Firefox supports Local and AD logins on Windows-based operating system and Local, Active Directory, and LDAP logins on Linux-based operating systems.

 **NOTE:** If you do not have Access Virtual Console privilege but have Access Virtual Media privilege, then using this URL launches the Virtual Media instead of the Virtual Console.

## Using virtual console viewer

The Virtual Console Viewer provides various controls such as mouse synchronization, virtual console scaling, chat options, keyboard macros, power actions, next boot devices, and access to Virtual Media. For information to use these features, see the **iDRAC Online Help**.

 **NOTE:** If the remote server is powered off, the message 'No Signal' is displayed.

The Virtual Console Viewer title bar displays the DNS name or the IP address of the iDRAC you are connected to from the management station. If iDRAC does not have a DNS name, then the IP address is displayed. The format is:

- For rack and tower servers: <DNS name / IPv6 address / IPv4 address>, <Model>, User: <username>, <fps>

Sometimes the Virtual Console Viewer may display low quality video. This is due to slow network connectivity that leads to loss of one or two video frames when you start the Virtual Console session. To transmit all the video frames and improve the subsequent video quality, do any of the following:

- In the **System Summary** page, under **Virtual Console Preview** section, click **Refresh**.
- In the **Virtual Console Viewer**, under **Performance** tab, set the slider to **Maximum Video Quality**.

## Using a virtual console

 **NOTE:** By default the virtual console type is set to eHTML5.

You can launch a virtual console as a pop-up window by using one of the following methods:

- From the iDRAC home page, click the **Start the Virtual Console** link available in the Console Preview session.
- From the iDRAC Virtual Console page, click the **Start the Virtual Console** link.
- From the iDRAC login page, type **https://<iDRAC IP>/console**. This method is called as Direct Launch.

In the eHTML5 virtual console, the following menu options are available:

- Power
- Boot
- Chat
- Keyboard
- Screen Capture
- Refresh
- Full Screen
- Disconnect Viewer.
- Console Controls
- Virtual Media

The **Pass all keystrokes to server** option is not supported on the eHTML5 virtual console. Use keyboard and keyboard macros for all the functional keys.

- **General** —

- **Console control** — This has the following configuration options:

- Keyboard Macros—This is supported in eHTML5 virtual console and are listed as the following drop-down options. Click **Apply** to apply the selected key combination on the server.

- Ctrl+Alt+Del
- Ctrl+Alt+F1
- Ctrl+Alt+F2
- Ctrl+Alt+F3
- Ctrl+Alt+F4
- Ctrl+Alt+F5
- Ctrl+Alt+F6
- Ctrl+Alt+F7
- Ctrl+Alt+F8
- Ctrl+Alt+F9
- Ctrl+Alt+F10
- Ctrl+Alt+F11
- Ctrl+Alt+F12
- Alt+Tab
- Alt+ESC
- Ctrl+ESC
- Alt+Space
- Alt+Enter
- Alt+Hyphen
- Alt+F1
- Alt+F2
- Alt+F3
- Alt+F4
- Alt+F5
- Alt+F6
- Alt+F7
- Alt+F8
- Alt+F9
- Alt+F10
- Alt+F11
- Alt+F12
- PrntScrn
- Alt+PrntScrn
- F1
- Pause
- Tab
- Ctrl+Enter
- SysRq
- Alt+SysRq

- Win-P

- **Aspect Ratio**—The eHTML5 virtual console video image automatically adjusts the size to make the image visible. The following configuration options are displayed as a drop-down list:

- Maintain
- Do not Maintain.

Click **Apply** to apply the selected settings on the server.

- **Touch Mode**—The eHTML5 virtual console supports the Touch Mode feature. The following configuration options are displayed as a drop-down list:

- Direct
- Relative

Click **Apply** to apply the selected settings on the server.

- **Virtual Clipboard** - Virtual clipboard enables you to cut/copy/paste text buffer from virtual console to iDRAC host server. Host server could be BIOS, UEFI or in operating system prompt. This is a one-way action from client system to iDRAC host server only. Follow these steps to use the Virtual clipboard:
  - Place the mouse cursor or keyboard focus on the desired window in the host server desktop.
  - Select the **Console Controls** menu from vConsole.
  - Copy the OS clipboard buffer using keyboard hotkeys, mouse, or touchpad controls depending on the Client operating system. Or, you can type the text manually in the text box.
  - Click **Send Clipboard to Host**.
  - Then, the text appears on the host server active window.

#### **NOTE:**

- This feature is available in Enterprise and Datacenter license.
- This feature only supports ASCII text.
- This feature supports only English Language Keyboard.
- Control characters are not supported.
- Characters such as **New line** and **Tab** are allowed.
- Text buffer size is limited to 4000 characters.
- If more than the maximum buffer is pasted, then the edit box in the iDRAC UI truncates it to the maximum buffer size.

- **KVM**—This menu has a list of the following read-only components:
  - Frame Rate
  - Bandwidth
  - Compression
  - Packet Rate
- **Performance**—Use the slider button to adjust **Maximum Video Quality** and **Maximum Video Speed**.
- **User List**—View the list of users that are logged in to the Virtual console.
- **Keyboard**—The difference between physical and virtual keyboard is that the virtual keyboard changes its layout according to the browser language.

#### **NOTE:** Ensure that the vKeyboard language and Host operating system language are the same.

- **Virtual Media** —Click the **Connect Virtual Media** option to start the virtual media session.
  - **Connect Virtual Media**—This menu contains the options for Map CD/DVD, Map Removable Disk, Map External Device, and Reset USB.
  - **Virtual Media Statistics**—This menu shows the Transfer Rate (Read-only). Also, it shows the details of CD/DVD and Removable Disks details such as Mapping details, status (read-only or not), duration, and Read/Write Bytes.
  - **Create Image**—This menu allows you to select a local folder and generate a FolderName.img file with local folder contents.

#### **NOTE:** For security reasons, read/write access is disabled while accessing the virtual console in eHTML5.

## Supported Browsers

The eHTML5 virtual console is supported on the following browsers:

- Microsoft EDGE

- Safari 16.6
- Safari 17.x
- Mozilla Firefox 128
- Mozilla Firefox 129
- Mozilla Firefox 130
- Google Chrome 137
- Google Chrome 138

 **NOTE:** It is recommended to have Mac operating system version 10.10.2 (or onward) installed in the system.

 **NOTE:** If you are using Chrome/Edge browser, you may see the Login denied message.

For more details on supported browsers and versions, see the *Integrated Dell Remote Access Controller User's Guide Release Notes* available on the [iDRAC manuals](#) page.

## Using iDRAC Service Module

The iDRAC Service Module is a software application that is recommended to be installed on the server (it is not installed by default). It complements iDRAC with monitoring information from the operating system. It complements iDRAC by providing additional data to work with iDRAC interfaces such as the Web interface, Redfish, and RACADM. You can configure the features that are monitored by the iDRAC Service Module to control the CPU and memory that is consumed on the server's operating system. The host operating system command-line interface has been introduced to enable or disable the status of Full Power Cycle for all System components except the PSU.

### NOTE:

- Use the iDRAC Service Module only if you have installed iDRAC Express or iDRAC Enterprise/Datacenter license.
- iSM versions older than 4.2 do not support TLS 1.3.
- If the HOST network is not configured properly, the iDRAC SupportAssist page reports the LC Log indicating that there is a problem connecting with iSM.
- If you observe the SRV042 warning in iDRAC LC while creating SupportAssist collections, perform the hard reset of iDRAC to resolve this warning in iDRAC LC.

Before using iDRAC Service Module, ensure that:

- You have login, configuration, and server control privileges in iDRAC to enable or disable the iDRAC Service Module features.
- You do not disable the **iDRAC Configuration using local RACADM** option.
- The operating system to iDRAC pass-through channel is enabled through the internal USB bus in iDRAC.

### NOTE:

If you perform LC wipe, `idrac.Servicemodule` values may still show the old values.

### NOTE:

- When iDRAC Service Module runs for the first time, by default it enables the operating system to iDRAC pass-through channel in iDRAC. If you disable this feature after installing the iDRAC Service Module, then you must enable it manually in iDRAC.
- If the operating system to iDRAC pass-through channel is enabled through LOM in iDRAC, then you cannot use the iDRAC Service Module.

### Topics:

- [Installing iDRAC Service Module](#)
- [Supported operating systems for iDRAC Service Module](#)
- [iDRAC Service Module monitoring features](#)
- [Using iDRAC Service Module from iDRAC web interface](#)
- [Using iDRAC Service Module from RACADM](#)

## Installing iDRAC Service Module

You can download and install the iDRAC Service Module from [dell.com/support](https://dell.com/support). You must have administrator privilege on the server's operating system to install the iDRAC Service Module. For information on installation, see the iDRAC Service Module User's Guide available on the [iDRAC Service Module](#) page.

### NOTE:

If USB NIC is disabled on the iDRAC, iSM installer auto enables USB NIC. Once the installation is complete, disable USB NIC if needed.



# Installing iDRAC Service Module from iDRAC Core

From the **iDRAC Service Module** Setup page, click **Install Service Module**.

### Steps

1. The Service Module Installer is available to the host operating system and a job is created in iDRAC.  
For Microsoft Windows operating system or Linux operating system, log in to the server either remotely or locally.
  2. Find the mounted volume labeled as "**SMINST**" on your device list and run the appropriate script:
    - On Windows, open the command prompt and run the **ISM-Win.bat** batch file.
    - On Linux, open the shell prompt and run the **ISM-Lx.sh** script file.
  3. After the installation is complete, iDRAC displays the Service Module as **Installed** and the installation date.
-  **NOTE:** The installer will be available to the host operating system for 30 minutes. If you do not start the installation within 30 minutes, you must restart the Service Module installation.

# Installing iDRAC Service Module from iDRAC Enterprise

### Steps

1. On iDRAC UI, go to **iDRAC Settings > Settings > iDRAC Service Module Setup**.
  2. On the **iDRAC Service Module Setup** page, click **Install Service Module**.
  3. Click **Launch Virtual Console** and click **Continue** on the security warning dialog box.
  4. To locate the iSM installer file, log in to the server either remotely or locally.
-  **NOTE:** The installer will be available to the host operating system for 30 minutes. If you do not start the installation within 30 minutes, you must restart the installation.
5. Find the mounted volume labeled as "**SMINST**" on your device list and run the appropriate script:
    - On Windows, open the command prompt and run the **ISM-Win.bat** batch file.
    - On Linux, open the shell prompt and run the **ISM-Lx.sh** script file.
  6. Follow the instructions on the screen to complete the installation.  
On the **iDRAC Service Module Setup** page, the **Install Service Module** button is disabled after the installation is complete and the Service Module status is displayed as **Running**.

# Supported operating systems for iDRAC Service Module

For the list of operating systems supported by the iDRAC Service Module, see the iDRAC Service Module User's Guide available on the [iDRAC Service Module](#) page.

# iDRAC Service Module monitoring features

The iDRAC Service Module (iSM) provides the following monitoring features:

**Table 43. iSM supported features**

| Running                                                         | Running (Limited Functionality) |
|-----------------------------------------------------------------|---------------------------------|
| Redfish profile support for network attributes                  | Service on Host OS              |
| iDRAC Hard Reset                                                | Operating system information    |
| iDRAC access using Host operating system (Experimental Feature) | Auto System Recovery            |

**Table 43. iSM supported features (continued)**

| Running                                                                                                                                                                                                                       | Running (Limited Functionality)                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| In-band iDRAC SNMP alerts                                                                                                                                                                                                     | Allow Service Module to perform iDRAC Hard Reset. |
| View operating system information                                                                                                                                                                                             | N/A                                               |
| Replicate Lifecycle Controller logs to operating system logs.                                                                                                                                                                 | N/A                                               |
| Perform automatic system recovery options.                                                                                                                                                                                    | N/A                                               |
| Populate Windows Management Instrumentation (WMI) Management Providers.                                                                                                                                                       | N/A                                               |
| Integrate with SupportAssist Collection.<br> <b>NOTE:</b> This is applicable only if iDRAC Service Module version 2.0 or later is installed. | N/A                                               |
| Prepare to Remove NVMe PCIe SSD.<br> <b>NOTE:</b> For more information, see the <a href="#">Support for Dell iDRAC Service Module</a> page.  | N/A                                               |
| Remote Server Power Cycle                                                                                                                                                                                                     | N/A                                               |

## Redfish profile support for network attributes

iDRAC Service Module v2.3 or later provides additional network attributes to iDRAC, which can be obtained through the REST clients from iDRAC. For more details, see iDRAC Redfish profile support.

## Replicate Lifecycle logs to OS log

You can replicate the Lifecycle Controller Logs to the OS logs from the time when the feature is enabled in iDRAC. This is similar to the System event log (SEL) replication performed by OpenManage Server Administrator. All events that have the **OS Log** option selected as the target (in the **Alerts** page, or in the equivalent RACADM interface) are replicated in the OS log using the iDRAC Service Module. The default set of logs to be included in the OS logs is the same as configured for SNMP alerts or traps.

iDRAC Service Module also logs the events that have occurred when the operating system is not functioning. The OS logging performed by iDRAC Service Module follows the IETF syslog standards for Linux-based operating systems.

 **NOTE:** On Microsoft Windows, if iSM events get logged under System logs instead of Application logs, restart the Windows Event Log service or restart the host OS.

## Automatic system recovery options

The Automatic system recovery feature is a hardware-based timer. If a hardware failure occurs, a notification may not be available, but the server is reset as if the power switch was activated. ASR is implemented using a timer that continuously counts down. The Health Monitor frequently reloads the counter to prevent it from counting down to zero. If the ASR counts down to zero, it is assumed that the operating system has locked up and the system automatically attempts to reboot.

You can perform automatic system recovery operations such as reboot, power cycle, or power off the server after a specified time interval. This feature is enabled only if the operating system watchdog timer is disabled. If OpenManage Server Administrator is installed, this monitoring feature is disabled to avoid duplicate watchdog timers.

## In-band Support for iDRAC SNMP Alerts

By using iDRAC Service Module v2.3, you can receive SNMP alerts from the host operating system, which is similar to the alerts that are generated by iDRAC.

You can also monitor the iDRAC SNMP alerts without configuring the iDRAC and manage the server remotely by configuring the SNMP traps and destination on the host OS. In iDRAC Service Module v2.3 or later, this feature converts all the Lifecycle logs replicated in the OS logs into SNMP traps.

 **NOTE:** This feature is active only if the Lifecycle Logs replication feature is enabled.

 **NOTE:** On Linux operating systems, this feature requires a master or OS SNMP enabled with SNMP multiplexing (SMUX) protocol.

By default, this feature is disabled. Though the In-band SNMP alerting mechanism can coexist along with iDRAC SNMP alerting mechanism, the recorded logs may have redundant SNMP alerts from both the sources. It is recommended to either use the in-band or out-of-band option, instead of using both.

### Command usage

This section provides the command usages for Windows, Linux, and ESXi operating systems.

#### • Linux operating system

- On all iSM supported Linux operating system, iSM provides an executable command. You can run this command by logging into the operating system by using SSH or equivalent.
- Beginning with iSM 2.4.0, you can configure Agent-x as the default protocol for in-band iDRAC SNMP alerts using the following command:

```
./Enable-iDRACSNMPTrap.sh 1/agentx -force
```

If `-force` is not specified, ensure that the net-SNMP is configured and restart the snmpd service.

- To enable this feature:

```
Enable-iDRACSNMPTrap.sh 1
```

```
Enable-iDRACSNMPTrap.sh enable
```

- To disable this feature:

```
Enable-iDRACSNMPTrap.sh 0
```

```
Enable-iDRACSNMPTrap.sh disable
```

 **NOTE:** The `--force` option configures the Net-SNMP to forward the traps. However, you must configure the trap destination.

#### • VMware ESXi operating system

 **NOTE:** You must review and configure the VMware ESXi system-wide SNMP settings for traps.

 **NOTE:** For more details, refer to the **In-BandSNMPAlerts** technical white paper available on the [Dell Support](#) page.

## iDRAC access via Host OS

By using this feature, you can configure and monitor the hardware parameters through iDRAC Web interface and RedFish interfaces using the host IP address without configuring the iDRAC IP address. You can use the default iDRAC credentials if the iDRAC server is not configured or continue to use the same iDRAC credentials if the iDRAC server was configured earlier.

### iDRAC access via Windows Operating Systems

You can perform this task by using the following methods:

- Install the iDRAC access feature by using the webpack.
- Configure using iSM PowerShell script.

#### Installation by using MSI

You can install this feature by using the web-pack. This feature is disabled on a typical iSM installation. If enabled, the default listening port number is 1266. You can modify this port number within the range 1024 through 65535. iSM redirects the connection to the iDRAC. iSM then creates an inbound firewall rule, OS2iDRAC. The listening port number is added to the OS2iDRAC firewall rule in the host operating system, which allows incoming connections. The firewall rule is enabled automatically when this feature is enabled.

Beginning with iSM 2.4.0, you can retrieve the current status and listening-port configuration by using the following Powershell cmdlet:

```
Enable-iDRACAccessHostRoute -status get
```

The output of this command indicates whether this feature is enabled or disabled. If the feature is enabled, it displays the listening-port number.

 **NOTE:** Ensure that the Microsoft IP Helper Services is running on your system for this feature to function.

To access the iDRAC Web interface, use the format `https://<host-name>` or `OS-IP>:443/login.html` in the browser, where:

- **<host-name>**— Complete hostname of the server on which iSM is installed and configured for iDRAC access using the operating system feature. You can use the OS IP address if the host name is not present.
- **443**— Default iDRAC port number. This is called the Connect Port number to which all the incoming connections on listen port number are redirected. You can modify the port number through iDRAC Web interface and RACADM interfaces.

### Configuration by using iSM PowerShell cmdlet

If this feature is disabled while installing iSM, you can enable the feature by using the following Windows PowerShell command provided by iSM:

```
Enable-iDRACAccessHostRoute
```

If the feature is already configured, you can disable or modify it by using the PowerShell command and the corresponding options. The available options are as follows:

- **Status**— This parameter is mandatory. The values are not case sensitive and the value can be **true**, **false**, or **get**.
- **Port**—This is the listening port number. If you do not provide a port number, the default port number (1266) is used. If the **Status** parameter value is FALSE, then you can ignore the rest of the parameters. You must enter a new port number that is not already configured for this feature. The new port number settings overwrite the existing OS2iDRAC in-bound firewall rule and you can use the new port number to connect to iDRAC. The value range is from 1024 to 65535.
- **IPRange**—This parameter is optional and it provides a range of IP addresses that are allowed to connect to iDRAC through the host operating system. The IP address range format is in Classless Inter-Domain Routing (CIDR) format, which is a combination of IP address and subnet mask. For example, 10.94.111.21/24. Access to iDRAC is restricted for IP addresses that are not within the range.

 **NOTE:** This feature supports only IPv4 addresses.

### iDRAC access via Linux Operating Systems

You can install this feature by using the `setup.sh` file that is available with the web-pack. This feature is disabled on a default or typical iSM installation. To get the status of this feature, use the following command:

To install, enable, and configure this feature, use the following command:

```
./Enable-iDRACAccessHostRoute <Enable-Flag> [ <source-port> <source-IP-range/source-ip-range-mask>]
```

**<Enable-Flag>=0**

Disable

**<source-port>** and **<source-IP-range/source-ip-range-mask>** are not required.

**<Enable-Flag>=1**

Enable

**<source-port>** is required and **<source-ip-range-mask>** is optional.

**<source-IP-range>**

IP range in **<IP-Address/subnet-mask>** format. Example: 10.95.146.98/24

# Using iDRAC Service Module from iDRAC web interface

## About this task

To use the iDRAC Service Module from the iDRAC web interface:

## Steps

1. Go to **iDRAC Settings > Overview > iDRAC Service Module > Configure Service Module**.  
The **iDRAC Service Module Setup** page is displayed.
2. You can view the following:
  - Installed iDRAC Service Module version on the host operating system
  - Connection status of the iDRAC Service Module with iDRAC.

**NOTE:** When a server has multiple operating systems and iDRAC Service Module is installed in all operating systems, then iDRAC connects only with the most recent instance of iSM among all operating systems. An error is displayed for all the older instances of iSM on other operating systems. To connect iSM with iDRAC on any other operating system which already has iSM installed, uninstall and reinstall iSM on that particular operating system.
3. To perform out-of-band monitoring functions, select one or more of the following options:
  - **OS Information**—View the operating system information.
  - **Replicate Lifecycle Log in OS Log**—Include Lifecycle Controller logs to operating system logs. This option is disabled if OpenManage Server Administrator is installed on the system.
  - **WMI Information**— Include WMI information.
  - **Auto System Recovery Action**—Perform auto recovery operations on the system after a specified time (in seconds):
    - **Reboot**
    - **Power Off System**
    - **Power Cycle System**This option is disabled if OpenManage Server Administrator is installed on the system.

## Results

- NOTE:** When iSM is in either full or limited mode functionality state and iDRAC is reset to factory settings, there is no way for iDRAC to set the Limited mode functionality state for iSM state.

# Using iDRAC Service Module from RACADM

To use the iDRAC Service Module from RACADM, use the objects in the `ServiceModule` group.

For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

# Using Type-C USB Dual-Mode port for server management

On the 17th generation servers, a dedicated Type-C USB Dual-Mode port is provided in the control panel, which is on the front of the server. By default, the port is in the Operating System mode. There is a System ID (i) button on the control panel. To change the port to iDRAC mode, press and hold the i button for 5 to 10 seconds until the wrench LED turns on. To change the port back to Operating System mode, press and hold the i button again for 5 to 10 seconds, until the wrench LED turns off.

Server Configuration Profiles (SCP) import operations through USB is supported only with Enterprise or Datacenter license.

You can perform the following functions using the Type-C USB Dual-Mode port when the port is in the iDRAC mode:

- Connect to the system using the USB network interface to access system management tools such as iDRAC web interface and RACADM.
- Configure a server by using SCP files that are stored on a USB drive.

**NOTE:** To manage a Type-C USB Dual-Mode port or to configure a server by importing Server Configuration Profile (SCP) files on a USB drive, you must have the System Control privilege.

**NOTE:** An alert is generated when a USB device is inserted.

## Topics:

- [Configuring iDRAC using server configuration profile on USB device](#)

## Configuring iDRAC using server configuration profile on USB device

With the Type-C USB Dual-Mode port in iDRAC mode, you can configure iDRAC on the server. Configure the Type-C USB Dual-Mode port settings in iDRAC, insert the USB device that has the server configuration profile, and then import the server configuration profile from the USB device to iDRAC.

## Configuring the Type-C USB Dual-Mode port settings

### About this task

To configure the Type-C USB Dual-Mode port:

### Steps

1. Enable the server Type-C USB Dual-Mode port in iDRAC UI (**Configuration > BIOS Settings > Integrated Devices**. Ensure that **All Ports On** or **All Ports Off (Dynamic)** is selected as **User Accessible USB Ports**. When **All Ports Off (Dynamic)** is selected, ensure that **Enable Front Ports Only** is **Enabled**.
2. Press and hold the System ID button on the control panel for about 5 to 10 seconds until the wrench LED turns on. The USB LED starts displaying solid Green, and the port is configured as an iDRAC port.
3. In the iDRAC UI, go to **iDRAC Settings > Settings > Management USB Settings**. The **USB Management Port** is set to **Enabled**.
4. From the **iDRAC Managed: USB SCP**, select one of the enabled options:
  - **Disabled**
  - **Enabled only when server has default credential settings**
  - **Enabled only for compressed configuration files**
  - **Enabled**

For information about the fields, see the **iDRAC Online Help**.

**NOTE:** iDRAC10 allows you to protect the compressed file with a password after you select **Enabled only for compressed configuration files** to compress the file before importing. You can enter a password to secure the file by using **Password for Zip file** option.

5. Click **Apply** to apply the settings.

## Accessing iDRAC interface over direct USB connection

### About this task

The iDRAC direct feature allows you to directly connect your laptop to the iDRAC USB port. This feature allows you to interact directly with the iDRAC interfaces such as the web interface, RACADM, and Redfish for advanced server management and servicing.

For a list of supported browsers and operating systems, see the *Integrated Dell Remote Access Controller User's Guide Release Notes* available on the [iDRAC manuals](#) page.

**NOTE:** If you are using Windows operating system, install an RNDIS driver to use this feature.

To access the iDRAC interface over the USB port:

### Steps

1. Turn off any wireless networks and disconnect from any other hard-wired network. Ensure that none of the external or internal networks or settings interfere with the communication.
2. Ensure that the USB port is in the iDRAC mode.
3. Wait for a few seconds for the laptop to acquire the IP address, 169.254.0.4. iDRAC acquires the IP address 169.254.0.3.
4. Start using iDRAC network interfaces such as the web interface, RACADM, and Redfish.  
For example, to access the iDRAC web interface, open a supported browser, and type the address **169.254.0.3**, and then press <Enter>.
5. When iDRAC is using the USB port, the LED blinks indicating activity. The blink frequency is once per two seconds.
6. After completing the required actions, disconnect the USB cable from the system.  
The LED turns back to solid Green.

## Importing Server Configuration Profile from a USB device

### About this task

Create a directory in the root of the USB device called `System_Configuration_XML` that contains both the configuration and control files:

- Server Configuration Profile (SCP) is in the `System_Configuration_XML` sub-directory under the USB device root directory. This file includes all the attribute-value pairs of the server. This includes attributes of iDRAC, PERC, RAID, and BIOS. You can edit this file to configure any attribute on the server. The file name can be `<servicetag>-config.xml`, `<servicetag>-config.json`, `<modelnumber>-config.xml`, `<modelnumber>-config.json`, `config.xml` or `config.json`.
- Control file – Includes parameters to control the import operation and does not have attributes of iDRAC or any other component in the system. The control file contains three parameters:
  - ShutdownType – Graceful, Forced, No Reboot.
  - TimeToWait (in secs) – 300 minimum and 3600 maximum.
  - EndHostPowerState – on or off.

Example of `control.xml` file:

```
<InstructionTable>
  <InstructionRow>
    <InstructionType>Configuration XML import Host control Instruction
    </InstructionType>
    <Instruction>ShutdownType</Instruction>
    <Value>NoReboot</Value>
    <ValuePossibilities>Graceful,Forced,NoReboot</ValuePossibilities>
  </InstructionRow>
```

```

<InstructionRow>
  <InstructionType>Configuration XML import Host control Instruction
</InstructionType>
  <Instruction>TimeToWait</Instruction>
  <Value>300</Value>
  <ValuePossibilities>Minimum value is 300 -Maximum value is
    3600 seconds.</ValuePossibilities>
</InstructionRow>
<InstructionRow>
  <InstructionType>Configuration XML import Host control Instruction
</InstructionType>
  <Instruction>EndHostPowerState</Instruction>
  <Value>On</Value>
  <ValuePossibilities>On, Off</ValuePossibilities>
</InstructionRow>
</InstructionTable>

```

You must have Server Control privilege to perform this operation.

 **NOTE:** While importing the SCP, changing the USB management settings in the SCP file results in a failed job or job completed with errors. You can comment out the attributes in the SCP to avoid the errors.

To import the server configuration profile from the USB device to iDRAC:

### Steps

1. Configure the USB management port:

- Set **USB Management Port Mode** to **iDRAC**.
- Set **iDRAC Managed: USB SCP to Enabled with default credentials** or **Enabled**.

2. Insert the USB key (that has the `configuration.xml` and the `control.xml` file) to the iDRAC USB port.

 **NOTE:** File name and file type are case sensitive for XML files. Ensure that both are in lower case.

 **NOTE:** USB drive must have the supported FAT32 filesystem only.

3. The server configuration profile is discovered on the USB device in the `System_Configuration_XML` sub-directory under the USB device root directory. It is discovered in the following sequence:

- `<servicetag>-config.xml` / `<servicetag>-config.json`
- `<modelnum>-config.xml` / `<modelnum>-config.json`
- `config.xml` / `config.json`

4. A server configuration profile import job starts.

If the profile is not discovered, then the operation stops.

If **iDRAC Managed: USB SCP Configuration** was set to **Enabled with default credentials** and the BIOS setup password is not null or if one of the iDRAC user accounts have been modified, an error message is displayed and the operation stops.

5. LED, if present, display the status that an import job has started.

6. If there is a configuration that must be staged and the **Shut Down Type** is specified as **No Reboot** is specified in the control file, reboot the server for the settings to be configured. Else, the server is rebooted and the configuration is applied. Only when the server was already powered off, then the staged configuration is applied even if the **No Reboot** option is specified.

7. After the import job is complete, the LED indicates that the job is complete.

8. If the USB device is left inserted on the server, the result of the import operation is recorded in the `results.xml` file in the USB device.

## LC logs and error messages during USB-related operations

When a USB device is connected, the **System Inventory** page displays the USB device information under the Hardware Inventory section.

An event is logged in the Lifecycle Controller logs when:

- The device is in the iDRAC mode, and USB device is inserted or removed.
- USB Management Port Mode is modified.



- Device is switched manually from iDRAC to Operating System mode.
- Device is removed from iDRAC.

When a device exceeds its power requirements as allowed by USB specification, the device is detached and an over-current event is generated with the following properties:

- Category : System Health
- Type: USB device
- Severity: Warning
- Allowed notifications: Email, SNMP trap, and remote syslog
- Actions: None

An error message is displayed and logged to Lifecycle Controller log when:

- When input files for Server Configuration profile (SCP) operation is incorrect.
- When the USB drive has hardware errors or unsupported filesystems.

## LED behavior

The USB LED indicates the status of a server-configuration profile operation being performed using the USB port. The LED may not be available on all systems.

- Off—The Type-C USB Dual-Mode port is in the Operating System mode.
- Solid Green—The USB port is connected to iDRAC or the SCP import job is completed successfully.
- Blinking Green—The SCP import job is in progress or I/O is in progress.
- Solid Amber—The SCP import job has failed.
- Blinking Amber—The USB hardware has errors.

## Logs and results file

The following information is logged for the import operation:

- Automatic import from USB is logged in the Lifecycle Controller log file.
- If the USB device is left inserted, the job results are recorded in the Results file located in the USB key.

A Result file named `Results.xml` is updated or created in the subdirectory with the following information:

- Service tag – Data is recorded after the import operation has either returned a job ID or returned an error.
- Job ID – Data is recorded after the import operation has returned a job ID.
- Start Date and Time of Job - Data is recorded after the import operation has returned a job ID.
- Status – Data is recorded when the import operation returns an error or when the job results are available.

## Using Quick Sync 2

With Dell OpenManage Mobile running on an Android or iOS mobile device, you can easily access the server directly or through OpenManage Essentials or OpenManage Enterprise (OpenManage Enterprise (OME)) console. It allows you to review server details and inventory, view LC, and System Event logs, get automatic notifications on your mobile device from an OpenManage Enterprise (OME) console, assign IP address and modify iDRAC password, configure key BIOS attributes, and take remediation actions as needed. You can also power-cycle a server, access the system console, or access the iDRAC UI.

OMM can be downloaded for free from the Apple App Store, or from the Google Play Store.

You must install the OpenManage Mobile application on the mobile device (supports Android 5.0+ and iOS 9.0+ mobile devices) to manage the server using iDRAC Quick Sync 2 interface.

 **NOTE:** This section is displayed only in those servers that have the Quick Sync 2 module in the left rack ear.

 **NOTE:** This feature is supported on mobile devices with the Android operating system and Apple iOS.

After Quick Sync is configured, activate the Quick Sync 2 button on the Left Control Panel. Ensure that the Quick Sync 2 light turns on. Access the Quick Sync 2 Information using a mobile device (Android 5.0+ or iOS 9.0+, OMM 2.0 or above).

Using OpenManage Mobile, you can:

- View inventory information
- View monitoring information
- Configure the basic iDRAC network settings

For more information about OpenManage Mobile, see the *Dell OpenManage Mobile User's Guide* available on the [OpenManage manuals](#) page..

### Topics:

- [Configuring iDRAC Quick Sync 2](#)
- [Using mobile device to view iDRAC information](#)

## Configuring iDRAC Quick Sync 2

Using iDRAC web interface, RACADM, or iDRAC HII you can configure the iDRAC Quick Sync 2 feature to allow access to the mobile device:

- **Access**—Configure to Read-Write, Read-only, and Disabled. Read-Write is the default option.
- **Timeout**—Configure to Enabled or Disabled. Enabled is the default option.
- **Timeout Limit**—Indicates the time after which the Quick Sync 2 mode is disabled. By default, the Minutes option is selected. The default value is 2 Minutes. The range is 2 to 60 Minutes.
  1. If enabled, you can specify a time after which the Quick Sync 2 mode is turned off. To turn on, press the activation button again.
  2. If disabled, the timer does not allow you to enter a time-out period.
- **Read Authentication**—Configures to Enabled, this is the default option.
- **WiFi**—Configures to Enabled, this is the default option.

You must have Server Control privilege to configure the settings. A server reboot is not required for the settings to take effect. After configuration, you can activate the Quick Sync 2 button on the Left Control Panel. Make sure the Quick Sync light turns on. Then, access the Quick Sync Information using a mobile device.

An entry is logged to the Lifecycle Controller log when the configuration is modified.

## Configuring iDRAC Quick Sync 2 settings using RACADM

To configure the iDRAC Quick Sync 2 feature, use the `racadm` objects in the **System.QuickSync** group. For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

## Configuring iDRAC Quick Sync 2 settings using web interface

### About this task

To configure iDRAC Quick Sync 2:

### Steps

1. In the iDRAC web interface, go to **Configuration > System Settings > Hardware Settings > iDRAC Quick Sync**.
2. In the **iDRAC Quick Sync** section, from the **Access** menu, select one of the following to provide access to the Android or iOS mobile device:
  - Read-write
  - Read-only
  - Disabled
3. Enable the Timer.
4. Specify the Timeout Limit.  
For more information about the fields, see the **iDRAC Online Help**.
5. Click **Apply** to apply the settings.

## Configuring iDRAC Quick Sync 2 settings using iDRAC settings utility

### About this task

To configure iDRAC Quick Sync 2:

### Steps

1. In the iDRAC GUI, go to **Configuration > Systems Settings > Hardware Settings > iDRAC Quick Sync**.
2. In the **iDRAC Quick Sync** section:
  - Specify the access level.
  - Enable Timeout.
  - Specify the User Defined Timeout Limit (the range is 120 to 3600 seconds.).

For more information about the fields, see the **iDRAC Online Help**.

3. Click **Back**, click **Finish**, and then click **Yes**.  
The settings are applied.

## Using mobile device to view iDRAC information

To view iDRAC information from the mobile device, see the *Dell OpenManage Mobile User's Guide* available on the [OpenManage manuals](#) page. for the steps.

## Managing virtual media

iDRAC provides virtual media with an HTML5-based client with local ISO and IMG file, remote ISO, and IMG file support. Virtual media allows the managed server to access media devices on the management station or ISO CD/DVD images on a network share as if they were devices on the managed server. You need iDRAC Configure privilege to modify the configuration.

Following are the configurable attributes:

- Attached Media Enabled—Enabled or Disabled
- Attach Mode — Auto-attach, Attached, and Detached.
- Max Sessions—1
- Active Sessions—1
- Virtual Media Encryption—Enabled (by default)
- Floppy Emulation—Disabled (by default)
- Boot Once — Enabled or Disabled
- Connection Status—Connected or Disconnected

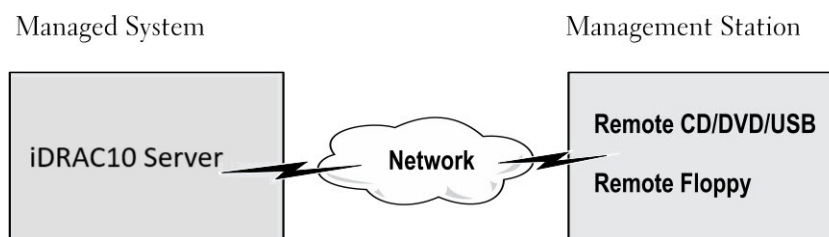
Using the Virtual Media feature, you can:

- Remotely access media that are connected to a remote system over the network
- Install applications.
- Update drivers
- Install an operating system on the managed system.

The key features are:

- Virtual Media supports virtual optical drives (CD/DVD) and USB flash drives.
- You can attach only one USB flash drive, image, or key and one optical drive on the management station to a managed system. Supported optical drives include a maximum of one available optical drive or one ISO image file. The following figure shows a typical Virtual Media setup.
- Any connected Virtual Media emulates a physical device on the managed system.
- On Windows-based managed systems, the Virtual Media drives are auto-mounted if they are attached and configured with a drive letter.
- On Linux-based managed systems with some configurations, the Virtual Media drives are not auto-mounted. To manually mount the drives, use the mount command.
- All the virtual drive access requests from the managed system are directed to the management station across the network.
- Virtual devices appear as two drives on the managed system without the media being installed in the drives.
- You can share the management station CD/DVD drive (read-only), but not a USB media, between two managed systems.
- Virtual media requires a minimum available network bandwidth of 128 Kbps.
- If a LOM or NIC failover occurs, then the Virtual Media session may be disconnected.

After attaching a Virtual Media image through Virtual Console, the drive may not show up in Windows host OS. Check Windows Device Manager for any unknown mass storage devices. Right-click the unknown device and update the driver or choose uninstall driver. The device is recognized by Windows after disconnecting and reconnecting vMedia.



**Figure 2. Virtual media setup**

### Topics:

- [Supported drives and devices](#)

- [Configuring virtual media](#)
- [Accessing virtual media](#)
- [Enabling boot once for virtual media](#)
- [Remote File Share](#)
- [Setting boot order through BIOS](#)
- [Accessing drivers](#)

## Supported drives and devices

The following table lists the drives supported through virtual media.

**Table 44. Supported drives and devices**

| Drive                  | Supported Storage Media                                                                      |
|------------------------|----------------------------------------------------------------------------------------------|
| Virtual Optical Drives | <ul style="list-style-type: none"> <li>• ISO Image File</li> <li>• IMG Image File</li> </ul> |
| USB flash drives       | <ul style="list-style-type: none"> <li>• USB Key image in the ISO9660 format</li> </ul>      |

## Configuring virtual media

### Configuring virtual media using iDRAC web interface

#### About this task

To configure virtual media settings:

 **CAUTION:** Do not reset iDRAC when running a Virtual Media session. Otherwise, undesirable results may occur, including data loss.

#### Steps

1. In the iDRAC Web interface, go to **Configuration > Virtual Media > Attached Media**.
2. Specify the required settings. For more information, see the **iDRAC Online Help**.
3. Click **Apply** to save the settings.

### Configuring virtual media using RACADM

To configure the virtual media, use the `set` command with the objects in the **iDRAC.VirtualMedia** group.

For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

### Configuring virtual media using iDRAC settings utility

#### About this task

You can attach, detach, or auto-attach virtual media using the iDRAC Settings utility. To do this:

#### Steps

1. In the iDRAC Settings utility, go to **Media and USB Port Settings**.  
The **iDRAC Settings Media and USB Port Settings** page is displayed.
2. In the **Virtual Media** section, select **Detach**, **Attach**, or **Auto attach** based on the requirement. For more information about the options, see **iDRAC Settings Utility Online Help**.
3. Click **Back**, click **Finish**, and then click **Yes**.  
The Virtual Media settings are configured.

## Attached media state and system response

The following table describes the system response based on the Attached Media setting.

**Table 45. Attached media state and system response**

| Attached Media State | System Response                                                                                   |
|----------------------|---------------------------------------------------------------------------------------------------|
| Detach               | Cannot map an image to the system.                                                                |
| Attach               | Media is mapped even when <b>Client View</b> is closed.                                           |
| Auto-attach          | Media is mapped when <b>Client View</b> is opened and unmapped when <b>Client View</b> is closed. |

## Server settings for viewing virtual devices in virtual media

You must configure the following settings in the management station to allow visibility of empty drives. To do this, in Windows Explorer, from the **Organize** menu, click **Folder and search options**. On the **View** tab, deselect **Hide empty drives in the Computer folder** option and click **OK**.

## Accessing virtual media

You can access Virtual Media with or without using the Virtual Console. Before you access Virtual Media, make sure to configure your Web browser(s).

Virtual Media and RFS are mutually exclusive. If the RFS connection is active and you attempt to launch the Virtual Media client, the following error message is displayed: **Virtual Media is currently unavailable. A Virtual Media or Remote File Share session is in use.**

Virtual Media and RFS are mutually exclusive. If the RFS connection is active and you attempt to launch the Virtual Media client, the following error message is displayed: Virtual Media is currently unavailable. A Virtual Media or Remote File Share session is in use. If virtual media connected first then user can able to connect RFS1 also.

If the RFS connection is not active and you attempt to launch the Virtual Media client, the client launches successfully. You can then use the Virtual Media client to map devices and files to the Virtual Media virtual drives.

.img file mapped via **Virtual Console > Virtual Media or standalone Virtual Media** does not support write operations in host OS.

## Launching virtual media using virtual console

### About this task

Before you launch Virtual Media through the Virtual Console, make sure that:

- Virtual Console is enabled.
- System is configured to not hide empty drives — In Windows Explorer, navigate to **Folder Options**, clear the **Hide empty drives in the Computer folder** option, and click **OK**.

To access Virtual Media using Virtual Console:

### Steps

1. In the iDRAC web interface, go to **Configuration > Virtual Console**.  
The **Virtual Console** page is displayed.
2. Click **Launch Virtual Console**.  
The **Virtual Console Viewer** is launched.
3. Click **Virtual Media > Connect Virtual Media**.  
The Virtual Media session is established and the **Virtual Media** menu displays the list of devices available for mapping.

 **NOTE:** The **Virtual Console Viewer** window must remain active while you access the Virtual Media.

## Launching virtual media without using virtual console

Before you launch Virtual Media when the **Virtual Console** is disabled, ensure that System is configured to unhide empty drives. To do this, in Windows Explorer, go to **Folder Options**, clear the **Hide empty drives in the Computer folder** option, and click **OK**.

To access Virtual Media when Virtual Console is disabled:

1. In the iDRAC web Interface, go to **Configuration > Virtual Media**.
2. Click **Connect Virtual Media**.

Alternatively, you can also launch the Virtual Media by following these steps:

1. Go to **Configuration > Virtual Console**.
2. Click **Launch Virtual Console**. The following message is displayed:

Virtual Console has been disabled. Do you want to continue using Virtual Media redirection?

3. Click **OK**. The **Virtual Media** window is displayed.
4. From the **Virtual Media** menu, click **Map CD/DVD** or **Map Removable Disk**. For more information, see [Mapping virtual drive](#).
5. **Virtual Media Statistics** shows the list of target drives, their mapping, status (Read-Only or Not), Duration of connection, Read/Write Bytes, and the transfer rate.

**NOTE:** The virtual device drive letters on the managed system do not coincide with the physical drive letters on the management station.

**NOTE:** The Virtual Media may not function correctly on systems running Windows operating system configured with Internet Explorer Enhanced Security. To resolve this issue, see the Microsoft operating system documentation or contact the system administrator.

## Adding virtual media images

### About this task

You can create a media image of the remote folder and mount it as a USB attached device to the server's operating system. To add Virtual Media images:

### Steps

1. Click **Virtual Media > Create Image...**
2. In the **Source Folder** field, click **Browse** and browse to the folder or directory to be used as the source for the image file. The image file is on the management station or the C: drive of the managed system.
3. In the **Image File Name** field, the default path to store the created image files (typically the desktop directory) appears. To change this location, click **Browse** and navigate to a location.
4. Click **Create Image**.

The image creation process starts. If the image file location is within the source folder, a warning message is displayed indicating that the image creation cannot proceed as the image file location within the source folder causes an infinite loop. If the image file location is not within the source folder, then the image creation proceeds.

After the image is created, a success message is displayed.

5. Click **Finish**.

The image is created.

When a folder is added as an image, a **.img** file is created on the Desktop of the management station from which this feature is used. If this **.img** file is moved or deleted, then the corresponding entry for this folder in the **Virtual Media** menu does not work. Therefore, it is recommended not to move or delete the **.img** file while the **image** is being used. However, the **.img** file can be removed after the relevant entry is first deselected and then removed using **Remove Image** to remove the entry.

## Viewing virtual device details

To view the virtual device details, in the Virtual Console Viewer, click **Tools > Stats**. In the **Stats** window, the **Virtual Media** section displays the mapped virtual devices and the read/write activity for each device. If Virtual Media is connected, this information is displayed. If Virtual Media is not connected, the “Virtual Media is not connected” message is displayed.

If the Virtual Media is launched without using the Virtual Console, then the **Virtual Media** section is displayed as a dialog box. It provides information about the mapped devices.

## Resetting USB

### About this task

To reset the USB device:

### Steps

1. In the Virtual Console viewer, click **Tools > Stats**.  
The **Stats** window is displayed.
2. Under **Virtual Media**, click **USB Reset**.  
A message is displayed warning the user that resetting the USB connection can affect all the input to the target device including Virtual Media, keyboard, and mouse.
3. Click **Yes**.  
The USB is reset.

 **NOTE:** iDRAC Virtual Media does not terminate even after you log out of iDRAC Web interface session.

## Mapping virtual drive

### About this task

To map the virtual drive:

 **NOTE:** While using Virtual Media, you must have administrative privileges to map an operating system DVD or a USB flash drive (that is connected to the management station). To map the drives, launch IE as an administrator or add the iDRAC IP address to the list of trusted sites.

### Steps

1. To establish a Virtual Media session, from the **Virtual Media** menu, click **Connect Virtual Media**.  
For each device available for mapping from the host server, a menu item appears under the **Virtual Media** menu. The menu item is named according to the device type such as:

- Map CD/DVD
- Map Removable Disk
- Map External Device

The **Map DVD/CD** option can be used for ISO files and the **Map Removable Disk** option can be used for images with ehtml5 based Virtual Media. The **Map External Device** option can be used for mapping physical USB drives.

 **NOTE:**

- You cannot map physical media such USB-based drives, CD, or DVD by using the HTML5 based virtual console.
- You cannot map USB keys as virtual media disks using Virtual Console/Virtual media over a RDP session.
- You cannot map physical media with NTFS format in ehtml removable media, use FAT or exFAT devices

2. Click the device type that you want to map.

 **NOTE:** The active session displays if a Virtual Media session is currently active from the current Web interface session, from another Web interface session.

3. In the **Drive/Image File** field, select the device from the drop-down list.



The list contains all the available (unmapped) devices that you can map (CD/DVD and Removable Disk) and image file types that you can map (ISO or IMG). The image files are located in the default image file directory (typically the user's desktop). If the device is not available in the drop-down list, click **Browse** to specify the device.

The correct file type for CD/DVD is ISO and for removable disk it is IMG.

If the image is created in the default path (Desktop), when you select **Map Removable Disk**, the created image is available for selection in the drop-down menu.

If image is created in a different location, when you select **Map Removable Disk**, the created image is not available for selection in the drop-down menu. Click **Browse** to specify the image.

 **NOTE:** Floppy emulation is not supported in ehtml5 plugin.

4. Select **Read-only** to map writable devices as read-only.

For CD/DVD devices, this option is enabled by default and you cannot disable it.

 **NOTE:** The ISO and IMG files map as read-only files if you map these files by using the HTML5 virtual console.

5. Click **Map Device** to map the device to the host server.

After the device/file is mapped, the name of its **Virtual Media** menu item changes to indicate the device name. For example, if the CD/DVD device is mapped to an image file named `foo.iso`, then the CD/DVD menu item on the Virtual Media menu is named **foo.iso mapped to CD/DVD**. A check mark for that menu item indicates that it is mapped.

## Displaying correct virtual drives for mapping

### About this task

On a Linux-based management station, the Virtual Media **Client** window may display removable disks that are not part of the management station. To make sure that the correct virtual drives are available to map, you must enable the port setting for the connected SATA hard drive. To do this:

### Steps

1. Reboot the operating system on the management station. During POST, press <F2> to enter **System Setup**.
2. Go to **SATA settings**. The port details are displayed.
3. Enable the ports that are actually present and connected to the hard drive.
4. Access the Virtual Media **Client** window. It displays the correct drives that can be mapped.

## Clearing Java Cache

### About this task

In case of any unexpected errors while using USB, please clear the Java cache. Follow these steps to clear Java cache:

### Steps

1. In the Java Control Panel, under the **General** tab, click **Settings** under the **Temporary Internet Files** section. The **Temporary Files Settings** dialog box appears.
2. Click **Delete Files** on the Temporary Files Settings dialog.  
The **Delete Files and Applications** dialog box appears.
3. Click **OK** on the **Delete Files and Applications** dialog. This deletes all the Downloaded Applications and Applets from the cache.
4. Click **OK** on the **Temporary Files Settings** dialog. If you want to delete a specific application and applet from the cache, click on View Application and View Applet options respectively.

## Unmapping virtual drive

### About this task

To unmap the virtual drive:

### Steps

1. From the **Virtual Media** menu, do any of the following:

- Click the device that you want to unmap.
- Click **Disconnect Virtual Media**.

A message appears asking for confirmation.

2. Click **Yes**.

The check mark for that menu item does not appear indicating that it is not mapped to the host server.

**i** **NOTE:** After unmapping a USB device attached to vKVM from a client system running the Macintosh operating system, the unmapped device may be unavailable on the client. Restart the system or manually mount the device on the client system to view the device.

### Example

**i** **NOTE:** To unmap a virtual DVD drive on Linux OS, unmount the drive and eject it.

## Enabling boot once for virtual media

### About this task

You can change the boot order only once when you boot after attaching remote Virtual Media device.

Before you enable the boot once option, make sure that:

- You have **Configure User** privilege.
- Map the local or virtual drives (CD/DVD, Floppy, or USB flash device) with the bootable media or image using the Virtual Media options
- Virtual Media is in **Attached** state for the virtual drives to appear in the boot sequence.

To enable the boot once option and boot the managed system from the Virtual Media:

### Steps

1. In the iDRAC Web interface, go to **Overview > Server > Attached Media**.
2. Under **Virtual Media**, select the **Enable Boot Once** and click **Apply**.
3. Turn on the managed system and press **<F2>** during boot.
4. Change the boot sequence to boot from the remote Virtual Media device.
5. Reboot the server.  
The managed system boots once from the Virtual Media.

## Remote File Share

This feature is available only with the iDRAC Enterprise or Datacenter license.

RFS mounts are capable of Read Write attribute changes, and it is supported from racadm/redfish only.

**i** **NOTE:** Before using RFS, ensure that you have a minimum network bandwidth of 1 MB/Sec.

### Remote File Share 1 (RFS1)

The Remote File Share 1 (RFS1) feature uses the Virtual Media Implementation in iDRAC.

When an image file is mounted using the RFS1 feature, both the Virtual Media virtual disk drives are visible to the host operating system. If an **.img** file is mapped, and then the floppy/hard disk Virtual drive is used to present the image file to the operating system. If an **.iso** file is mapped, and then the CD/DVD Virtual drive is used to present the image file to the operating system. The unused Virtual drive appears as an empty drive to the operating system. The Virtual Media client can map images or hard drives to both Virtual drives, but RFS can use only one at a time. RFS and Virtual Media features are mutually exclusive.

**i** **NOTE:**

- RFS1 appears as Virtual Optical or Floppy Drive when there is no active Virtual Media Session, based on the image attached.

- RFS1 appears as Virtual Network File 1 when there is an active Virtual Media Session, as Virtual Optical Drive and Virtual Floppy Drives are consumed with Virtual Media.

Enter all the required information and click **Connect** to connect to the RFS1. To disconnect from RFS1, click **Disconnect**. To know more about the required field information, see **Online Help** in iDRAC UI.

#### **NOTE:**

- iDRAC timeout for RFS connect is 55 seconds. If the connection takes longer than 55 seconds, Timeout error is displayed.
- Basic auth and Digest auth for HTTP/HTTPS shares are supported.
- **Connect** is disabled if the RFS feature is not licensed. The **Disconnect** option is always available regardless of the license status. Click **Disconnect** to disconnect an existing RFS connection.

#### **Scenarios**

- If the Virtual Media client has not been launched and if you attempt to establish an RFS connection, the connection is established, and the Remote image is available to the host operating system.
- If the RFS connection is not active and if you attempt to launch the Virtual Media client, the client launches successfully. You can then use the Virtual Media client to map devices and files to the Virtual Media virtual drives.
- If the RFS1 Session is active, and you attempt to establish an vMedia connection, then vMedia Connection is denied.
- If the Virtual Media client is active, and you attempt to establish an RFS connection, it is possible that the Virtual Optical Drive/Virtual Floppy assigned to Virtual media and Virtual Network File1 gets assigned to RFS.

#### **Remote File Share 2 (RFS2)**

The Remote File Share 2 (RFS2) is independent of RFS1 and Virtual media. RFS2 has its own copy of attributes independent of RFS1. The RFS2 image option has the same behavior as the existing RFS1 on all the iDRAC interfaces. Both are allowed to connect/disconnect independently. RFS2 is controlled using Enabled/Disabled and Attach Mode RFS2 attributes.

To boot with RFS2 Virtual Network File 2, select **Virtual Network File 2** from boot options. Virtual Media Boot Once has no impact on RFS2 when enabled.

Enter the required information and click **Connect** to connect to RFS2, and click **Disconnect** to disconnect from RFS2.

When you upload/delete HTTPS certificate in RFS1, the certificate is uploaded/deleted in RFS2 as well. Because this certificate is for iDRAC's identity, and it remains the same for multiple RFS or any shared connections.

The connection status for RFS is available in the iDRAC log. Once connected, an RFS-mounted virtual drive does not disconnect even if you log out from iDRAC. The RFS connection is closed if iDRAC is reset or the network connection is dropped.

If you update the iDRAC firmware while there is an active RFS connection and the Virtual Media Attach Mode is set to **Attach** or **Auto Attach**, the iDRAC attempts to reestablish the RFS connection after the firmware upgrade is completed, and the iDRAC reboots.

If you update the iDRAC firmware while there is an active RFS connection and the Virtual Media Attach Mode is set to **Detach**, the iDRAC does not attempt to reestablish the RFS connection after the firmware upgrade is completed, and the iDRAC reboots.

#### **NOTE:**

- CIFS and NFS support both IPv4 and IPv6 addresses.
- While connecting to a remote file share using IPv6 by providing an FQDN, IPv4 must be disabled on the HTTPS server.
- When the iDRAC is configured with both IPv4 and IPv6, the DNS server can contain records associating the iDRAC hostname to both addresses. If the IPv4 option is disabled in iDRAC, then iDRAC may not be able to access the external IPv6 share. This is because the DNS server may still contain IPv4 records, and DNS name resolution can return the IPv4 address. In such cases, it is recommended to delete the IPv4 DNS records from the DNS server, when disabling the IPv4 option in iDRAC.
- If you are using CIFS and are part of an Active Directory domain, enter the domain name with the IP address in the image file path.
- If you want to access a file from an NFS share, configure the following share permissions. These permissions are required because iDRAC interfaces run in non-root mode.
  - Linux: Ensure that the share permissions are set to at least **Read** for the **Others** account.
  - Windows: Go to the **Security** tab of the share properties and add **Everyone** to **Groups or user names** field with **Read & execute** privilege.

- If ESXi is running on the managed system and if you mount a floppy image (**.img**) using RFS, the connected floppy image is not available to the ESXi operating system.
- Only English ASCII characters are supported in network share file paths.
- The operating system drive eject feature is not supported when virtual media is connected using RFS.
- RFS may get disconnected when iDRAC IP is not reachable for more than 1 minute. Try to remount once the network is up.
- While specifying the network share settings, it is recommended to avoid special characters for username, and password or percent encode the special characters.
- The following characters are supported for **User Name**, **Password**, and the **Image File Path** fields:
  - A-Z
  - a-z
  - 0-9
  - Special characters: . \_ - ? < > / \ : \* | @
  - Whitespace
- For HTTP, do not use the following characters: ! @ # % ^. These characters are supported on other share types. However, to maintain compatibility, use the recommended characters.

### Folder Mount through RFS

iDRAC supports folder mount directly through RFS. This feature allows you to attach a folder directly without converting to ISO/IMG File.

#### NOTE:

- This feature is available with the iDRAC Enterprise or Datacenter license.
- Folder attach is possible only through NFS and CIFS share. HTTP/HTTPS share is not supported.
- The size of the NFS/CIFS Folder to be attached is limited to 1 GB and maximum number of sub-folders is limited to 1000.
- It is not possible to map an empty folder.

Following scenarios explain how RFS1 and RFS2 are listed in BIOS Boot order:

#### Scenario 1:

If Virtual media is already attached using virtual console, BIOS boot order reports devices as **Virtual Optical** or **Virtual Floppy Drive** depending on the image type. When RFS 1 device is attached, BIOS boot order reports it as **Virtual Network File 1**. For RFS 2 device, BIOS boot order reports it as **Virtual Network File 2**.

#### Scenario 2:

When no virtual media is attached, and you attach RFS 1 device, BIOS boot order reports it as **Virtual Optical** or **Virtual Floppy Drive** depending on the image type. When you attach an RFS 2 device, the BIOS boot order reports it as **Virtual Network File 2**.

#### Scenario 3

When Virtual Media is not connected, and RFS1 is attached:

- Virtual Optical Drive for ISO image
- Virtual Floppy Drive For IMG image

Virtual Media session will be blocked as RFS1 session is active.

When Virtual Media is connected and RFS1 is attached, RFS1 is listed as Virtual Network File 1 for both ISO/IMG Image. This is to maintain compatibility with existing vMedia and RFS, which allows only one at a time. RFS2 is listed as **Virtual Network File 2** irrespective of Virtual Media and RFS1.

## Setting boot order through BIOS

### About this task

Using the System BIOS Settings utility, you can set the managed system to boot from virtual optical drives or virtual floppy drives.

 **NOTE:** Changing Virtual Media while connected may stop the system boot sequence.

To enable the managed system to boot:

#### Steps

1. Boot the managed system.
2. Press <F2> to enter the **System Setup** page.
3. Go to **System BIOS Settings > Boot Settings > BIOS Boot Settings > Boot Sequence**.  
In the pop-up window, the virtual optical drives, virtual floppy drives, Virtual Network File 1, and Virtual Network File 2 are listed with the standard boot devices.
4. Make sure that the virtual drive is enabled and listed as the first device with bootable media. If required, follow the on-screen instructions to modify the boot order.
5. Click **OK**, navigate back to **System BIOS Settings** page, and click **Finish**.
6. Click **Yes** to save the changes and exit.

The managed system reboots.

The managed system attempts to boot from a bootable device based on the boot order. If the virtual device is connected and a bootable media is present, the system boots to the virtual device. Otherwise, the system overlooks the device—similar to a physical device without bootable media.

## Accessing drivers

Dell PowerEdge servers have all the supported operating system drivers embedded on the system flash memory. Using iDRAC, you can mount or unmount drivers easily to deploy the operating system on your server.

To mount the drivers:

1. On the iDRAC web interface, go to **Configuration > Virtual Media**.
2. Click **Mount Drivers**.
3. Select the OS from the pop-up window and click **Mount Drivers**.

 **NOTE:** The Expose duration is 18 hours by default.

To unmount the drivers post completion of the mount:

1. Go to **Configuration > Virtual Media**.
2. Click **Unmount Drivers**.
3. Click **OK** on the pop-up window.

 **NOTE:** The **Mount Drivers** option may not be displayed if the driver pack is not available on the system. Ensure to download and install the latest driver pack from [Dell Support](#) page.

# Installing operating systems

You can use any of the following utilities to deploy operating systems to managed systems:

- Remote File Share
- Console

## Topics:

- [Deploying an operating system using remote file share](#)
- [Deploying operating system using virtual media](#)
- [Installing Operating System using Driver pack DUP](#)
- [Installing Operating System using LC UI](#)

## Deploying an operating system using remote file share

### About this task

Before you deploy the operating system using Remote File Share (RFS), ensure that:

- **Configure User** and **Access Virtual Media** privileges for iDRAC are enabled for the user.
- Network share contains drivers and operating system bootable image file, in an industry standard format such as **.img**, **.iso**, or **folder path**.

 **NOTE:** While creating the image file, follow standard network-based installation procedures, and mark the deployment image as read-only to make sure that each target system boots and runs the same deployment procedure.

To deploy an operating system using RFS:

### Steps

1. Using Remote File Share (RFS), mount the ISO or IMG image file to the managed system through NFS, CIFS, HTTP, or HTTPs.
2. Go to **Configuration > System Settings > Hardware Settings > First Boot Device**.
3. Set the boot order in the **First Boot Device** list to select a virtual media such as floppy, CD, DVD, ISO, Virtual Network File 1, and Virtual Network File 2.
4. Select the **Boot Once** option to enable the managed system to reboot using the image file for the next instance only.
5. Click **Apply**.
6. Reboot the managed system and follow the on-screen instructions to complete the deployment.

## Managing remote file shares

Using Remote File Share (RFS) feature, you can set an ISO or IMG image file on a network share and make it available to the managed server's operating system as a virtual drive by mounting it as a CD or DVD using NFS, CIFS, HTTP or HTTPs. RFS is a licensed feature.

Remote file share supports only **.img** and **.iso** image file formats. A **.img** file is redirected as a virtual floppy and a **.iso** file is redirected as a virtual CDROM.

You must have Virtual Media privileges to perform an RFS mounting.

This feature is available only with the iDRAC Enterprise or Datacenter license.

# Configuring remote file share using web interface

## About this task

To enable remote file sharing:

## Steps

1. In iDRAC web interface, go to **Configuration > Virtual Media > Attached Media**. The **Attached Media** page is displayed.
2. Under **Attached Media**, select **Attach** or **Auto Attach**.
3. Under **Remote File Share**, specify the image file path, domain name, user name, and password. For information about the fields, see the **iDRAC Online Help**.

Example for image file path:

- CIFS — `//<IP to connect for CIFS file system>/<file path>/<image name>`
- NFS — `< IP to connect for NFS file system>:/<file path>/<image name>`
- HTTP — `http://<URL>/<file path>/<image name>`
- HTTPs — `https://<URL>/<file path>/<image name>`

- NOTE:** To avoid I/O errors when using CIFS shares hosted on Windows 7 systems, modify the following registry keys:
- Set `HKLM\SYSTEM\CurrentControlSet\Control\Session Manager\Memory Management\LargeSystemCache` to 1
  - Set `HKLM\SYSTEM\CurrentControlSet\Services\LanmanServer\Parameters\Size` to 3

- NOTE:** Both '/' or '\' characters can be used for the file path.

CIFS supports both IPv4 and IPv6 addresses but NFS supports only IPv4 address.

If you are using NFS share, make sure that you provide the exact `<file path>` and `<image name>` as it is case-sensitive.

- NOTE:** For information on recommended characters for user names and passwords, see [Recommended characters in user names and passwords](#).

- NOTE:** The characters allowed in user names and passwords for network shares are determined by the network-share type. iDRAC supports valid characters for network share credentials as defined by the share type, except `<`, `>`, and `,` (comma).

4. Click **Apply** and then click **Connect**.

After the connection is established, the **Connection Status** displays **Connected**.

- NOTE:** Even if you have configured remote file sharing, the Web interface does not display user credential information due to security reasons.

- NOTE:** If the image path contains user credentials, use HTTPS to avoid credentials from displaying in the GUI and RACADM. If entering the credentials in the URL, avoid using "@" symbol, because it is a separator character.

For Linux distributions, this feature may require a manual mount command when operating at runlevel init 3. The syntax for the command is:

```
mount /dev/OS_specific_device / user_defined_mount_point
```

Where, `user_defined_mount_point` is any directory you choose to use for the mount similar to any mount command.

For RHEL, the CD device (**.iso** virtual device) is `/dev/scd0` and floppy device (**.img** virtual device) is `/dev/sdc`.

For SLES, the CD device is `/dev/sr0` and the floppy device is `/dev/sdc`. To make sure that the correct device is used (for either SLES or RHEL), when you connect the virtual device, on the Linux OS you must immediately run the command:

```
tail /var/log/messages | grep SCSI
```

This displays the text that identifies the device (example, SCSI device `sdc`). This procedure also applies to Virtual Media when you are using Linux distributions in runlevel init 3. By default, the virtual media is not auto-mounted in init 3.

# Configuring remote file share using RACADM

To configure remote file share using RACADM, use:

```
racadm remoteimage
```

```
racadm remoteimage <options>
```

Options are:

-c : connect image

-d : disconnect image

-u <username>: username to access the shared folder

-p <password>: password to access the shared folder

-l <image\_location>: image location on the network share; use double quotes around the location. See examples for image file path in Configuring Remote File Share Using Web Interface section

-s : display current Remote Image status

## Usage Examples

- CIFS based RFS:

```
racadm remoteimage -c -u "user" -p "pass" -l //shrloc/foo.iso
```

- NFS Based RFS:

```
racadm remoteimage -c -u "user" -p "pass" -l <nfs ip>:/shrloc/foo.iso
```

- HTTP/HTTPS Based RFS:

```
racadm remoteimage -c -u "user" -p "pass" -l http://url/shrloc/foo.iso
```

```
racadm remoteimage -c -l https://url/shareloc/foo.iso
```

- Disconnect from the remote image:

```
racadm remoteimage -d
```

- Display current Remote Image status:

```
racadm remoteimage -s
```

**NOTE:** This command supports both IPV4 and IPV6 formats. IPV6 is applicable for CIFS and NFS type remote shares.

**NOTE:** -u and -p options are mandatory if the share type is cifs.

**NOTE:** All characters including alphanumeric and special characters are allowed as part of user name, password, and image\_location except the following characters: ' (single quote), "(double quote), ,(comma), < (less than), and > (greater than).

**NOTE:** To avoid I/O errors when using CIFS shares hosted on Windows 7 systems, modify the following registry keys:

- Set HKLM\SYSTEM\CurrentControlSet\Control\Session Manager\Memory Management\LargeSystemCache to 1
- Set HKLM\SYSTEM\CurrentControlSet\Services\LanmanServer\Parameters\Size to 3

For help on viewing the properties of a group, run the command - racadm help get.

For help on configuring the properties of a group, run the command - racadm help set.

```
racadm>>help remoteimage2
```

**NOTE:** remoteimage2 -- makes a remote ISO image available to the server. It requires Remote File Share license.

## Usage

```
racadm remoteimage2 -c -u <user> -p <pass> -l <image_location>
```



```
racadm remoteimage2 -d
racadm remoteimage2 -s
```

Options are:

- c : connect image
- d : disconnect image
- u <username>: username to access the shared folder
- p <password>: password to access the shared folder
- l <image\_location>: image location on the network share; use double quotes around the location. See examples for image file path in Configuring Remote File Share Using Web Interface section
- s : display current Remote Image status

### Usage Examples

- CIFS based RFS:

```
racadm remoteimage2 -c -u "user" -p "pass" -l //shrloc/foo.iso
```

- NFS Based RFS:

```
racadm remoteimage2 -c -u "user" -p "pass" -l <nfs ip>:/shrloc/foo.iso
```

- HTTP/HTTPS Based RFS:

```
racadm remoteimage2 -c -u "user" -p "pass" -l http://url/shrloc/foo.iso
```

```
racadm remoteimage2 -c -l https://url/shareloc/foo.iso
```

- Disconnect from the remote image:

```
racadm remoteimage2 -d
```

- Display current Remote Image status:

```
racadm remoteimage2 -s
```

 **NOTE:** This command supports both IPV4 and IPV6 formats. IPV6 is applicable for CIFS and NFS type remote shares.

 **NOTE:** -u and -p options are mandatory if the share type is cifs.

For help on viewing the properties of a group, run the command - `racadm help get`.

For help on configuring the properties of a group, run the command - `racadm help set`.

## Deploying operating system using virtual media

### About this task

Before you deploy the operating system using Virtual Media, make sure that:

- Virtual Media is in **Attached** state for the virtual drives to appear in the boot sequence.
- If Virtual Media is in **Auto Attached** mode, the Virtual Media application must be launched before booting the system.
- Network share contains drivers and operating system bootable image file, in an industry standard format such as **.img** or **.iso**.

To deploy an operating system using Virtual Media:

### Steps

1. Do one of the following:
  - Insert the operating system installation CD or DVD into the management station CD or DVD drive.
  - Attach the operating system image.

2. Select the drive on the management station with the required image to map it.
3. Use one of the following methods to boot to the required device:
  - Set the boot order to boot once from **Virtual Floppy** or **Virtual CD/DVD/ISO** using the iDRAC Web interface.
  - Set the boot order through **System Setup > System BIOS Settings** by pressing <F2> during boot.
4. Reboot the managed system and follow the on-screen instructions to complete the deployment.

## Installing operating system from multiple disks

### Steps

1. Unmap the existing CD/DVD.
2. Insert the next CD/DVD into the remote optical drive.
3. Remap the CD/DVD drive.

## Installing Operating System using Driver pack DUP

Installing the Operating System on the system using Driver pack DUP is supported on the iDRAC UI with Enterprise or Datacenter license. You can also install the Operating System using Redfish or RACADM or Lifecycle Controller UI with Core license.

### Steps

1. Download and install the Driver pack Dell Update Package (DUP) on your system.
2. Insert the operating system media into the system or attach the operating system ISO file using the Virtual Media feature in iDRAC.
3. Mount the driver pack using one of the following interfaces:
  - a. RACADM—See [Mount driver pack using RACADM](#).
  - b. Redfish API—See [Mount driver pack using Redfish](#).
  - c. iDRAC UI—See [Mount drivers using user interface](#).
4. Boot the system using the operating system installation media.
5. Reboot the system.

## Mount driver pack using RACADM

Run the RACADM commands to mount the driver pack.

### Steps

1. Run the following command to list all the supported operating systems along with their index:

```
racadm driverpack getinfo
```
2. Run the following command to attach the driver pack with the specific operating system index number.

```
racadm driverpack attach -i
```
3. Run the following command to verify that the job OSD: UnpackAndAttach has completed successfully.

```
racadm jobqueue view
```

## Mount driver pack using Redfish

Run the RACADM commands to mount the driver pack.

### Steps

1. Perform a POST operation on the following Redfish URI to check the supported operating system list for the driver pack:

```
https://<iDRAC IP>/redfish/v1/Systems/System.Embedded.1/Oem/Dell/  
DelloSDeploymentService/Actions/DelloSDeploymentService.GetDriverPackInfo
```

2. Perform a POST operation on the following Redfish URI to run the **UnpackAndAttach** operation:

```
https://<iDRAC IP>/redfish/v1/Systems/System.Embedded.1/Oem/Dell/  
DelloSDeploymentService/Actions/DelloSDeploymentService.UnpackAndAttach  
  
BODY:  
{ "OSName": "OS Name" }
```

3. Perform a GET operation on the following Redfish URI to check the status of the installation:

```
https://<iDRAC IP>/redfish/v1/TaskService/Tasks/OSDeployment
```

## Mount drivers using user interface

You can mount the drivers using the iDRAC UI.

### Steps

1. Go to **Configuration > Virtual Media**.
2. Click **Mount Drivers**.
3. Select the operating system from the **Operating System to Install** list, and click **Mount Drivers**.

## Installing Operating System using LC UI

The **Deploy OS** feature in the LifeCycle Controller (LC) user interface simplifies the installation of the operating system. LC user interface is available only in English and does not support other languages.

### Prerequisites

- Install the iDRAC operating system deployment driver pack.  
 **NOTE:** For information about the supported operating systems for each iDRAC release, see the iDRAC Release Notes.
- If required, configure the RAID volume. RAID volume configuration is performed by using either iDRAC interfaces or F2 System Setup.

### Steps

1. Press F10 during POST to log in to the Lifecycle Controller (LC) user interface.  
 **NOTE:** Even when the **System Lockdown** mode is enabled, you can log in to the LC user interface.
2. Select **OS Deployment** from the left pane.
3. Click **Deploy OS**.
4. Select **Directly to OS Deployment** to install the operating system.

5. Select the operating system from the **Operating System** list.
6. Select either **Manual** or **Unattended/Automatic** install.  
 **NOTE:** The **Unattended** method of Operating System installation is supported only using the USB media.
7. Select the **OS Installation Media** (virtual media attached using iDRAC interfaces).
8. Follow the steps in the **OS Installation Wizard** (if **Manual install** is selected).

# Troubleshooting managed system using iDRAC

You can diagnose and troubleshoot a remote managed system using:

- Diagnostic console
- Post code
- Boot and crash capture videos
- Last system crash screen
- System event logs
- Lifecycle logs
- Front panel status
- Trouble indicators
- System health

## Topics:

- [Using diagnostic console](#)
- [Viewing post codes](#)
- [Viewing boot and crash capture videos](#)
- [Viewing logs](#)
- [Viewing last system crash screen](#)
- [Viewing System status](#)
- [Hardware trouble indicators](#)
- [Viewing system health](#)
- [Restarting iDRAC](#)
- [Erasing system and user data](#)
- [Resetting iDRAC to factory default settings](#)

## Using diagnostic console

### About this task

iDRAC provides a standard set of network diagnostic tools that are similar to the tools included with Microsoft Windows or Linux-based systems. Using iDRAC Web interface, you can access the network debugging tools.

To access Diagnostics Console:

### Steps

1. In the iDRAC Web interface, go to **Maintenance > Diagnostics**.  
The **Diagnostics Console Command** page is displayed.
2. In the **Command** text box, enter a command and click **Submit**. For information about the commands, see the **iDRAC Online Help**.  
The results are displayed on the same page.

## Reset iDRAC and Reset iDRAC to default

### Steps

1. In the iDRAC Web interface, go to **Maintenance > Diagnostics**.  
You have the following options:

- Click **Reset iDRAC** to reset the iDRAC. A normal reboot operation is performed on the iDRAC. After reboot, refresh the browser to reconnect and log in to iDRAC.
- Click **Reset iDRAC to Default Settings** to reset the iDRAC to the default settings. After you click **Reset iDRAC to Default Settings**, **Reset iDRAC to factory default window** is displayed. This action reset the iDRAC to the factory defaults. Chose any of the following options:
  - a. Preserve user and network settings.
  - b. Discard all settings and reset users to the shipping value (root/shipping value).
  - c. Discard all settings and reset username and password.

2. A warning message is displayed. Click **Ok** to proceed further.

## Scheduling remote automated diagnostics

You can remotely invoke automated offline diagnostics on a server as a one-time event and return the results. If the diagnostics require a reboot, you can reboot immediately or stage it for a subsequent reboot or maintenance cycle (similar to updates). When diagnostics are run, the results are collected and stored in the internal iDRAC storage. You can then export the results to an NFS, CIFS, HTTP, or HTTPs network share using the `diagnostics export racadm` command.

You must have iDRAC Express license to use remote automated diagnostics.

You can perform the diagnostics immediately or schedule it on a particular day and time, specify the type of diagnostics, and the type of reboot.

For the schedule, you can specify the following:

- Start time – Run the diagnostic at a future day and time. If you specify TIME NOW, the diagnostic is run on the next reboot.
- End time - Run the diagnostic until a date and time after the Start time. If it is not started by End time, it is marked as failed with End time expired. If you specify TIME NA, then the wait time is not applicable.

The types of diagnostic tests are:

- Express test
- Extended test
- Both in a sequence

The types of reboot are:

- Power cycle system
- Graceful shutdown (waits for operating system to turn off or for system restart)
- Forced Graceful shutdown (signals operating system to turn off and waits for 10 minutes. If the operating system does not turn off, the iDRAC power cycles the system)

Only one diagnostic job can be scheduled or run at one time. A diagnostic job can complete successfully, complete with error, or is unsuccessful. The diagnostic events including the results are recorded in Lifecycle Controller log. You can retrieve the results of the last diagnostic execution using remote RACADM.

You can export the diagnostic results of the last completed diagnostics that were scheduled remotely to a network share such as CIFS, NFS, HTTP or HTTPS. The maximum file size is 5 MB.

You can cancel a diagnostic job when the status of the job is **Unscheduled** or **Scheduled**. If the diagnostic is running, then restart the system to cancel the job.

Before you run the remote diagnostics, make sure that:

- Lifecycle Controller is enabled.
- You have Login and Server Control privileges.

## Scheduling remote automated diagnostics and exporting the results using RACADM

- To run the remote diagnostics and save the results on the local system, use the following command:

```
racadm diagnostics run -m <Mode> -r <reboot type> -s <Start Time> -e <Expiration Time>
```

- To export the diagnostic results, ensure that the server is in the **Out of POST** state, and LC is in the **Ready** state. To check the status of LC and server, run the following command:

```
racadm getremoteservicesstatus
```

- To export the last run remote diagnostics results, use the following command:

```
racadm diagnostics export -f <file name> -l <NFS / CIFS / HTTP / HTTPS share> -u  
<username> -p <password>
```

For more information about the options, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

## Viewing post codes

Post codes are progress indicators from the system BIOS, indicating various stages of the boot sequence from power-on-reset, and allows you to diagnose any faults related to system boot-up. The **Post Codes** page displays the last system post code prior to booting the operating system.

To view the Post Codes, go to **Maintenance > Troubleshooting > Post Code**.

The **Post Code** page displays the system health indicator, a hexadecimal code, and a description of the code.

## Viewing boot and crash capture videos

You can view the video recordings of:

- **Last three boot cycles**—A boot cycle video logs the sequence of events for a boot cycle. The boot cycle videos are arranged in the order of latest to oldest.
  - **Last crash video**— A crash video logs the sequence of events leading to the failure.
- NOTE:** Crash video feature is enabled by default. You can enable or disable based on the requirement. This is a licensed feature..iDRAC records fifty frames during boot time. Playback of the boot screens occurs at a rate of 1 frame per second. If iDRAC is reset, the boot capture video is not available as it is stored in RAM and is deleted.

### **NOTE:**

- You must have Access Virtual Console or administrator privileges to playback the Boot Capture and Crash Capture videos.
- The video capture time that is displayed in the iDRAC UI video player may differ from the video capture time that is displayed in other video players. The iDRAC UI video player displays the time in the iDRAC time zone while all other video players display the time in the respective operating system time zones.

### **NOTE:**

- The reason for the delay in boot capture file availability is because the boot capture buffer is not full after the host boot.
- Default /inbox SLES/RHEL video players do not support the MPEG-1 video decoder. You must install an MPEG decoder supported video player and play the files.
- MPEG-1 format videos are not supported in MAC operating system native player.

To view the **Boot Capture** screen, click **Maintenance > Troubleshooting > Video Capture**.

The **Video Capture** screen displays the video recordings. For more information, see the **iDRAC Online Help**.

- NOTE:** When an embedded video controller is disabled and the server has an add-on video controller, then certain latency is expected regarding boot capture. Hence, the End of Post Messages of a video will be recorded in the next capture.

## Configuring video capture settings

### About this task

To configure the video capture settings:

### Steps

1. In the iDRAC Web interface, go to **Maintenance > Troubleshooting > Video Capture**. The **Video Capture** page is displayed.
2. From the **Video Capture Settings** drop-down menu, select any of the following options:
  - **Disable** — Boot capture is disabled.
  - **Capture until buffer full** — Boot sequence is captured until the buffer size has reached.
  - **Capture until end of POST** — Boot sequence is captured until end of POST.
3. Click **Apply** to apply the settings.

## Viewing logs

You can view System Event Logs (SELs) and Lifecycle logs. For more information, see [Viewing System Event Logs](#) and [Viewing Lifecycle Logs](#).

## Viewing last system crash screen

### About this task

The last crash screen feature captures a screenshot of the most recent system crash, saves, and displays it in iDRAC. This is a licensed feature.

To view the last crash screen:

### Steps

1. Make sure that the last system crash screen feature is enabled.
2. In iDRAC Web interface, go to **Overview > Server > Troubleshooting > Last Crash Screen**.

The **Last Crash Screen** page displays the last saved crash screen from the managed system.

Click **Clear** to delete the last crash screen.

 **NOTE:** Once iDRAC is reset or an AC power cycle event occurs, then the crash capture data is cleared.

 **NOTE:** Last crash screen resolution is always 1024x768 irrespective of host OS resolution.

## Viewing System status

The System Status summarizes the status of the following components in the system:

- Summary
- Batteries
- Cooling
- CPUs
- Front Panel
- Intrusion
- Memory
- Network Devices
- Power Supplies
- Voltages

## Viewing system front panel LED status

To view the current system ID LED status, in iDRAC web interface, go to **System > Overview > Front Panel**. The **Front Panel** section displays the current front panel status:



- Solid blue — No errors present on the managed system.
- Blinking blue — Identify mode is enabled (regardless of managed system error presence).
- Solid amber — Managed system is in failsafe mode.
- Blinking amber — Errors present on managed system.

When the system is operating normally (indicated by blue Health icon on the LED front panel), then both **Hide Error** and **UnHide Error** is grayed-out. You can hide or unhide the errors only for rack and tower servers.

To view system ID LED status using RACADM, use the `getLed` command.

 **NOTE:** During hot removal of M.2 drive for BOSS N-1 controller, iDRAC dashboard health status turns amber, but server front/back health indicator LED stays blue.

For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

## Hardware trouble indicators

The hardware related problems are:

- Failure to power up
- Noisy fans
- Loss of network connectivity
- Hard drive failure
- USB media failure
- Physical damage

Based on the problem, use the following methods to correct the problem:

- Reseat the module or component and restart the system
- Replace hard drives or USB flash drives
- Reconnect or replace the power and network cables

If problem persists, see the *Installation and Service Manual* available on the [PowerEdge Manuals](#) page for specific troubleshooting information about the hardware device.

 **CAUTION:** You should only perform troubleshooting and simple repairs as authorized in your product documentation, or as directed by online or telephone service and support team. Damage due to servicing that is not authorized by Dell is not covered by your warranty. Read and follow the safety instructions that came with the product.

## Viewing system health

You can view the status for the following components on iDRAC:

- Batteries
- CPUs
- Cooling
- Intrusion
- Memory
- Power Supplies
- Removable Flash Media
- Voltages
- Miscellaneous

Click any component name in the **Server Health** section to view details about the component.

## Restarting iDRAC

You can perform a hard or soft iDRAC restart without turning off the server:

- Hard restart — On the server, press and hold the LED button for 15 seconds.

- Soft restart — Using iDRAC Web interface or RACADM.

## Resetting iDRAC using RACADM

To restart iDRAC, use the **racreset** command.

For Reset to default operations, use the following commands:

- Upload Custom Defaults file — `racadm -r <iDracIP> -u <username> -p <Password> set -f <filename> -t xml --customdefaults`
- Save Current Settings as Default settings — `racadm -r <iDracIP> -u <username> -p <Password> set --savecustomdefaults`
- Download Custom Default settings — `racadm -r <iDracIP> -u <username> -p <Password> get -f <filename> -t xml --customdefaults`
- Reset to Custom Defaults — `Racadm -r <iDracIP> -u <username> -p <Password> racresetcfg -custom`

## Resetting iDRAC using iDRAC web interface

To reset iDRAC, do one of the following in the iDRAC Web interface:

- Upload Custom Defaults file:
  - Go to **Configuration > Server Configuration Profile > Custom Defaults > Upload Custom Defaults**
  - Upload the customized **CustomConfigured.xml** file from Local Share path
  - Click **Apply**. New Upload Custom Defaults Job is created.
- Reset to Custom Defaults:
  - When Upload Custom Defaults job is successful, go to **Maintainance > Diagnostics**, click **Reset iDRAC to Factory Defaults** option.
  - Select **Discard all settings** and set to **Custom default configuration**.
  - Click **Continue** to initiate Reset to customs Defaults configuration.

## Erasing system and user data

 **NOTE:** Erasing system and user data is not supported from iDRAC UI.

You can erase system components and user data for the following components:

- BIOS reset to default
- Embedded Diagnostics
- Embedded operating system Driver Pack
- Lifecycle Controller Data
- iDRAC reset to default
- Overwrite hard drives that do not support Instant Secure Erase (ISE)
- Reset controller cache
- Erase Hard Drives, SSDs, and NVMeS that support ISE.
- Clear all operating system applications.

Before performing system erase, ensure that:

- You have iDRAC Server Control privilege.
- Lifecycle Controller is enabled.

The Lifecycle Controller Data option erases any content such as the LC Log, configuration database, rollback firmware, factory as-shipped logs, and the configuration information from the FP SPI (or management riser).

 **NOTE:** The Lifecycle Controller log contains the information about the system erase request and any information that is generated when iDRAC restarts. All previous information is removed.

You can delete individual or multiple system components using the **SystemErase** command:

```
racadm systemErase <BIOS | DIAG | DRVPACK | LCData | IDRAC >
```

Where,

- bios—BIOS reset to default
- diag—Embedded Diagnostics
- drvpack—Embedded OS Driver Pack
- ldata—Clear the Lifecycle Controller Data
- iDRAC—iDRAC reset to default
- overwritepd—Overwrite hard drives that do not support Instant Secure Erase (ISE)
- percnvcache—Reset controller cache
- secureerasepd—Erase Hard Drives, SSDs, and NVMe that support ISE
- allapps—Clears all operating system applications

**NOTE:** Secure erase does not erase **iDRAC rollback firmware** from the partition when the command `racadm systemerase ldata` is used.

**NOTE:** If SEKM is enabled on the server, then disable SEKM using the `racadm sekm disable` command before using this command. This can avoid any storage devices being locked out which are secured by iDRAC, if SEKM settings are erased from iDRAC by performing this command.

For more information, see the [Integrated Dell Remote Access Controller RACADM CLI Guide](#).

**NOTE:** The Dell tech center link appears on the iDRAC UI on Dell branded systems.

**NOTE:** After you run System Erase, the VD's may still appear. Run CSIOR after System Erase is completed and iDRAC is rebooted.

**NOTE:** In the latest iDRAC releases, the `LCWipe` command is deprecated. To perform the system erase operation, run the `systemerase` command.

**NOTE:** In RACADM and Redfish, the hash values of the components are erased from the software inventory when system erase or decommission operation is performed. Perform the firmware update of the respective components to display the hash values of all the components. The hash value of iDRAC remains as NA until an upgrade or downgrade of the firmware is performed.

## Resetting iDRAC to factory default settings

You can reset iDRAC to the factory default settings using the iDRAC Settings utility or the iDRAC Web interface.

### Resetting iDRAC to factory default settings using iDRAC web interface

#### About this task

To reset iDRAC to factory default settings using the iDRAC Web interface:

#### Steps

1. Go to **Maintenance > Diagnostics**.  
The **Diagnostics Console** page is displayed.
2. Click **Reset iDRAC to Default Settings**.  
The completion status is displayed in percentage. iDRAC reboots and is restored to factory defaults. The iDRAC IP is reset and is not accessible. You can configure the IP using the front panel or BIOS.

# Resetting iDRAC to factory default settings using iDRAC settings utility

## About this task

To reset iDRAC to factory default values using the iDRAC Settings utility:

## Steps

1. Go to **Reset iDRAC configurations to defaults**.  
The **iDRAC Settings Reset iDRAC configurations to defaults** page is displayed.
2. Click **Yes**.  
iDRAC reset starts.
3. Click **Back** and navigate to the same **Reset iDRAC configurations to defaults** page to view the success message.

# SupportAssist Integration in iDRAC

SupportAssist allows you to create SupportAssist collections and utilize other SupportAssist features to monitor your system and datacenter. iDRAC provides an application interfaces for gathering platform information that enables support services to resolve platform and system problems. iDRAC helps you to generate a SupportAssist collection of the server and then export the collection to a location on the management station (local) or to a shared network location such as FTP, Trivial File Transfer Protocol (TFTP), HTTP, HTTPS, Common Internet File System (CIFS) or Network File Share (NFS). The collection is generated in the standard ZIP format. You can send this collection to technical support for troubleshooting or inventory collection.

## Topics:

- [SupportAssist](#)
- [SupportAssist](#)
- [Collection Log](#)
- [Generating SupportAssist Collection](#)
- [Collection Log](#)

## SupportAssist

 **NOTE:** iDRAC10 does not support SupportAssist registration. You can use OpenManage Enterprise or Secure Connect Gateway for the same.

You can generate and save a collection locally or to a network.

## SupportAssist

Once SupportAssist is configured, you can check the SupportAssist dashboard to view the **Collection log**. Registration is not required to view or send the Collection log.

## Collection Log

**Collection Log** shows the details of **Collection Date and Time**, **Collection Type** (Manual), **Data Collected** (Custom Selection, All Data), **Collection Status** (Complete with Errors, Complete), and **Job ID**. You can send the last persisted collection in iDRAC to Dell.

 **NOTE:** Once generated, the Collection Log Details can be filtered to remove the Personally Identifiable Information (PII) based on the user selection.

## Generating SupportAssist Collection

For generating the OS and Application logs, iDRAC Service Module must be installed and running in Host Operating System.

 **NOTE:** SupportAssist Collection takes more than 10 minutes to complete when performed from OS/iDRAC while OMSA 10.1.0.0 is running with it.

If you have to work with Tech Support on an issue with a server but the security policies restrict direct internet connection, then you can provide Tech Support with necessary data to facilitate troubleshooting of the problem without having to install software or download tools from Dell and without having access to the Internet from the server operating system or iDRAC.

You can generate a health report of the server and then export the Collection log:

- To a location on the management station (local).

- To a shared network location such as Common Internet File System (CIFS) or Network File Share (NFS). To export to a network share such as CIFS or NFS, direct network connectivity to the iDRAC shared or dedicated network port is required.
- To Dell.

The SupportAssist Collection is generated in the standard ZIP format. The collection may contain the following information:

- Hardware inventory for all components (includes system component configuration and firmware details, Motherboard System Event Logs, iDRAC state information and Lifecycle Controller logs).
- Operating system and application information.
- Storage Controller logs.
- iDRAC Debug Logs.
- It contains an HTML5 viewer, that can be accessed once the collection is complete.
- The collection provides a massive amount of detailed system information and logs in a user friendly format that can be viewed without uploading the collection to the Tech Support site.

After the data is generated, you can view the data which contains multiple XML files and log files.

Each time the data collection is performed, an event is recorded in the Lifecycle Controller log. The event includes information such as the user who initiated the report, interface used, and the date and time of export.

On Windows, If WMI is disabled, OS Collector collection stops with an error message.

Check the appropriate privilege levels and make sure there is no firewall or security settings that may prevent from collecting the registry or software data.

Before generating the health report, make sure:

- Lifecycle Controller is enabled.
- Collect System Inventory On Reboot (CSIOR) is enabled.
- You have Login and Server Control privileges.

## Generating SupportAssist Collection manually using iDRAC web interface

### About this task

To generate the SupportAssist collection manually:

#### Steps

1. In the iDRAC Web interface, go to **Maintenance > SupportAssist**.
2. Click **Start a Collection**.
3. Select the datasets to be included in the Collection.
4. You can opt to filter the collection for PII.
5. Select the destination where the Collection must be saved.
  - a. **Save locally** option allows you to save the generated Collection in the local system.
  - b. **Save to Network** option saves the generated Collection to user defined CIFS or NFS share location.

 **NOTE:** If **Save to Network** is selected, and no default location is available, the provided network details will be saved as default location for future collections. If the default location already exists, then the collection uses the details that are specified once only.

If the **Save to Network** option is selected, the user-provided network is saved as default (if no prior network share location is saved) for any future collections.

6. Click **Collect** to proceed with Collection generation.
7. If prompted, accept the **End User Level Agreement (EULA)** to continue.
 

Operating system and Application Data option is unavailable and not selectable if:

  - iSM is not installed or running in the Host operating system, or
  - Operating system Collector has been removed from iDRAC, or
  - OS-BMC pass through is disabled in iDRAC, or
  - Cached OS Application data is not available in iDRAC from a previous collection

# Collection Log

**Collection Log** shows the details of **Collection Date and Time**, **Collection Type** (Manual), **Data Collected** (Custom Selection, All Data), **Collection Status** (Complete with Errors, Complete), and **Job ID** You can send the last persisted collection in iDRAC to Dell.

 **NOTE:** Once generated, the Collection Log Details can be filtered to remove the Personally Identifiable Information (PII) based on the user selection.

## Frequently asked questions

This section lists the frequently asked questions for the following:

- [System event log](#)
- [Network security](#)
- [Active Directory](#)
- [Virtual console](#)
- [Virtual media](#)
- [SNMP authentication](#)
- [Storage devices](#)
- [iDRAC Service Module](#)
- [RACADM](#)
- [Miscellaneous](#)

### Topics:

- [Active Directory](#)
- [iDRAC Service Module](#)
- [Network security](#)
- [RACADM](#)
- [Custom sender email configuration for iDRAC alerts](#)
- [SNMP authentication](#)
- [Storage devices](#)
- [System Event Log](#)
- [Virtual console](#)
- [Front Ports](#)
- [Virtual media](#)
- [Miscellaneous](#)
- [Proxy server settings](#)
- [Permanently setting the default password to calvin](#)

## Active Directory

### Active Directory login failed. How to resolve this?

To diagnose the problem, on the **Active Directory Configuration and Management** page, click **Test Settings**. Review the test results and fix the problem. Change the configuration and run the test until the test user passes the authorization step.

In general, check the following:

- While logging in, make sure that you use the correct user domain name and not the NetBIOS name. If you have a local iDRAC user account, log in to iDRAC using the local credentials. After logging in, make sure that:
  - The **Active Directory Enabled** option is selected on the **Active Directory Configuration and Management** page.
  - The DNS setting is correct on the **iDRAC Networking configuration** page.
  - The correct Active Directory root CA certificate is uploaded to iDRAC if certificate validation was enabled.
  - The Group Name and Group Domain Name matches the Active Directory configuration if you are using standard schema.
  - If the user and the iDRAC object is in different domain, then do not select the **User Domain from Login** option. Instead select **Specify a Domain** option and enter the domain name where the iDRAC object resides.
- Check the domain controller SSL certificates to make sure that the iDRAC time is within the valid period of the certificate.

### Active Directory login fails even if certificate validation is enabled. The test results display the following error message. Why does this occur and how to resolve this?

```
ERROR: Can't contact LDAP server, error:14090086:SSL
routines:SSL3_GET_SERVER_CERTIFICATE:certificate verify failed: Please check the correct
```



Certificate Authority (CA) certificate has been uploaded to iDRAC. Please also check if the iDRAC date is within the valid period of the certificates and if the Domain Controller Address configured in iDRAC matches the subject of the Directory Server Certificate.

If certificate validation is enabled, when iDRAC establishes the SSL connection with the directory server, iDRAC uses the uploaded CA certificate to verify the directory server certificate. The most common reasons for failing certification validation are:

- iDRAC date is not within the validity period of the server certificate or CA certificate. Check the iDRAC time and the validity period of your certificate.
- The domain controller addresses configured in iDRAC does not match the Subject or Subject Alternative Name of the directory server certificate. If you are using an IP address, read the next question. If you are using FQDN, make sure you are using the FQDN of the domain controller and not the domain. For example, **servername.example.com** instead of **example.com**.

#### **Certificate validation fails even if IP address is used as the domain controller address. How to resolve this?**

Check the Subject or Subject Alternative Name field of your domain controller certificate. Normally, Active Directory uses the host name and not the IP address of the domain controller in the Subject or Subject Alternative Name field of the domain controller certificate. To resolve this, do any of the following:

- Configure the host name (FQDN) of the domain controller as the **domain controller address(es)** on iDRAC to match the Subject or Subject Alternative Name of the server certificate.
- Reissue the server certificate to use an IP address in the Subject or Subject Alternative Name field, so that it matches the IP address configured in iDRAC.
- Disable certificate validation if you choose to trust this domain controller without certificate validation during the SSL handshake.

#### **When to configure Global Catalog Address(es)?**

If the users and role groups are from different domains, Global Catalog Address(es) are required. In this case, you can use only Universal Group.

If the users and role groups are in the same domain, Global Catalog Addresses are not required.

#### **How does standard schema query work?**

iDRAC connects to the configured domain controller address(es) first. If the user and role groups are in that domain, the privileges are saved.

If Global Controller Address(es) is configured, iDRAC continues to query the Global Catalog. If additional privileges are retrieved from the Global Catalog, these privileges are accumulated.

.

#### **Does iDRAC always use LDAP over SSL?**

Yes. All the transportation is over secure port 636 and/or 3269. During test setting, iDRAC does an LDAP CONNECT only to isolate the problem, but it does not do an LDAP BIND on an insecure connection.

#### **Why does iDRAC enable certificate validation by default?**

iDRAC enforces strong security to ensure the identity of the domain controller that iDRAC connects to. Without certificate validation, a hacker can spoof a domain controller and hijack the SSL connection. If you choose to trust all the domain controllers in your security boundary without certificate validation, you can disable it through the Web interface or RACADM.

#### **Does iDRAC support the NetBIOS name?**

Not in this release.

**The Active Directory is configured for a domain present in Windows Server 2008 Active Directory. A child or sub domain is present for the domain, the user and group is present in the same child domain, and the user is a member of that group.**

This may be because of an incorrect group type. There are two kinds of Group types in the Active Directory server:

- Security—Security groups allow you to manage user and system access to shared resources and to filter group policy settings.
- Distribution—Distribution groups are intended to be used only as email distribution lists.

Always make sure that the group type is Security. You cannot use distribution groups to assign permission on any object, however use them to filter group policy settings.

# iDRAC Service Module

## iSM details are missing / not updated correctly in iDRAC GUI page of some PowerEdge servers

When a user adds SUB NIC under teaming, the configuration is invalid. This causes iSM to not to communicate with iDRAC properly.

### How to check whether iDRAC Service Module is installed in the host operating system?

To know if the iDRAC Service Module is installed on the system,

- On systems running Windows: Open the **Control Panel**, verify if iDRAC Service Module is listed in the list of installed programs displayed.
- On systems running Linux: Run the command `rpm -qi dcism`. If the iDRAC Service Module is installed, the status displayed is **installed**.
- On systems running ESXi: Run the command `esxcli software vib list|grep -i open` on the host. iDRAC Service module is displayed.

**i** **NOTE:** To check if the iDRAC Service Module is installed on Red Hat Enterprise Linux 7, use the `systemctl status dcismeng.service` command instead of the `init.d` command.

### How to check the version number of the iDRAC Service Module installed in the system?

To check the version of the iDRAC Service Module in the system, do any of the following:

- Click **Start > Control Panel > Programs and Features**. The version of the installed iDRAC Service Module is listed in the **Version** tab.
- Go to **My Computer > Uninstall or change a program**.

### What is the minimum permission level required to install the iDRAC Service Module?

To install the iDRAC Service Module, you must have administrator level privileges.

**The following message is displayed in the OS log, even when the OS to iDRAC Pass-through over USBNIC is configured properly. Why?**

### The iDRAC Service Module is unable to communicate with iDRAC using the OS to iDRAC Pass-through channel

iDRAC Service Module uses the OS to iDRAC pass-through over USB NIC feature to establish the communication with iDRAC. Sometimes, the communication is not established though the USB NIC interface is configured with the correct IP endpoints. This may happen when the host operating system routing table has multiple entries for the same destination mask and the USB NIC destination is not listed as the first one in routing order.

**Table 46. Example of a routing order**

| Destination | Gateway     | Genmask       | Flags | Metric | Ref | Use Iface      |
|-------------|-------------|---------------|-------|--------|-----|----------------|
| default     | 10.94.148.1 | 0.0.0.0       | UG    | 1024   | 0   | 0 em1          |
| 10.94.148.0 | 0.0.0.0     | 255.255.255.0 | U     | 0      | 0   | 0 em1          |
| link-local  | 0.0.0.0     | 255.255.255.0 | U     | 0      | 0   | 0 em1          |
| link-local  | 0.0.0.0     | 255.255.255.0 | U     | 0      | 0   | 0 enp0s20u12u3 |

In the example **enp0s20u12u3** is the USB NIC interface. The link-local destination mask is repeated and the USB NIC is not the first one in the order. This results in the connectivity issue between iDRAC Service Module and iDRAC over the OS to iDRAC Pass-through. To troubleshoot the connectivity issue, make sure that the iDRAC USBNIC IPv4 address (by default it is 169.254.1.1) is reachable from the host operating system.

If not:

- Change the iDRAC USBNIC address on a unique destination mask.
- Delete the entries that are not required from the routing table to make sure that USB NIC is chosen by route when the host wants to reach the iDRAC USB NIC IPv4 address.

### Where is the Replicated Lifecycle log available on the operating system?

To view the replicated Lifecycle logs:

**Table 47. Lifecycle logs location**

| Operating System                                                   | Location                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Microsoft Windows                                                  | <p><b>Event viewer &gt; Windows Logs &gt; System.</b> All the iDRAC Service Module Lifecycle logs are replicated under the source name <b>iDRAC Service Module</b>.</p> <p><b>NOTE:</b> In iSM version 2.1 and later, Lifecycle logs are replicated under the Lifecycle Controller Log source name. In iSM version 2.0 and earlier, the logs are replicated under iDRAC Service Module source name.</p> <p><b>NOTE:</b> The location of the Lifecycle log can be configured using the iDRAC Service Module installer. You can configure the location while installing iDRAC Service Module or modifying the installer.</p> |
| Red Hat Enterprise Linux, SUSE Linux, CentOS, and Citrix XenServer | /var/log/messages                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| VMware ESXi                                                        | /var/log/syslog.log                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

### What are the Linux-dependent packages or executables available for installation while completing the Linux installation?

To see the list of Linux-dependent packages, see the **Linux Dependencies** section in the *iDRAC Service Module User's Guide* available on the [iDRAC manuals](#) page..

## Network security

### While accessing the iDRAC Web interface, a security warning is displayed stating that the SSL certificate issued by the Certificate Authority (CA) is not trusted.

iDRAC includes a default iDRAC server certificate to ensure network security while accessing through the Web-based interface and remote RACADM. This certificate is not issued by a trusted CA. To resolve this, upload a iDRAC server certificate issued by a trusted CA (for example, Microsoft Certificate Authority, Thawte or Verisign).

### Why the DNS server not registering iDRAC?

Some DNS servers register iDRAC names that contain only up to 31 characters.

### When accessing the iDRAC Web-based interface, a security warning is displayed stating that the SSL certificate hostname does not match the iDRAC hostname.

iDRAC includes a default iDRAC server certificate to ensure network security while accessing through the Web-based interface and remote RACADM. When this certificate is used, the web browser displays a security warning because the default certificate that is issued to iDRAC does not match the iDRAC hostname (for example, the IP address).

To resolve this, upload an iDRAC server certificate issued to the IP address or the iDRAC hostname. When generating the CSR (used for issuing the certificate), ensure that the common name (CN) of the CSR matches the iDRAC IP address (if certificate issued to IP) or the registered DNS iDRAC name (if certificate is issued to iDRAC registered name).

To make sure that the CSR matches the registered DNS iDRAC name:

1. In iDRAC Web interface, go to **Overview > iDRAC Settings > Network**. The **Network** page is displayed.
2. In the **Common Settings** section:
  - Select the **Register iDRAC on DNS** option.
  - In the **DNS iDRAC Name** field, enter the iDRAC name.
3. Click **Apply**.

### Why am I unable to access iDRAC from my web browser?

This issue may occur if HTTP Strict Transport Security (HSTS) is enabled. HSTS is a web security mechanism which allows web browsers to interact using only the secure HTTPS protocol, and not HTTP.

Enable HTTPS on your browser and login to iDRAC to resolve the issue.

## Why am I unable to complete operations that involve a remote CIFS share?

Import/export or any other remote file share operations that involve a CIFS share fail if they use only SMBv1. Ensure that the SMBv2 protocol is enabled on the server providing SMB/CIFS share. Refer to the Operating System documentation on how to enable the SMBv2 protocol.

## RACADM

**After performing an iDRAC reset (using the `racadm racreset` command), if any command is issued, the following message is displayed. What does this indicate?**

```
ERROR: Unable to connect to RAC at specified IP address
```

The message indicates that you must wait until the iDRAC completes the reset before issuing another command.

**When using RACADM commands and subcommands, some errors are not clear.**

You may see one or more of the following errors when using the RACADM commands:

- Local RACADM error messages—Problems such as syntax, typographical errors, and incorrect names.
- Remote RACADM error messages—Problems such as incorrect IP Address, incorrect user name, or incorrect password.

**During a ping test to iDRAC, if the network mode is switched between Dedicated and Shared modes, there is no ping response.**

Clear the ARP table on your system.

**Remote RACADM fails to connect to iDRAC from SUSE Linux Enterprise Server (SLES) 11 SP1.**

Make sure that the official openssl and libopenssl versions are installed. Run the following command to install the RPM packages:

```
rpm -ivh --force < filename >
```

where, `filename` is the openssl or libopenssl rpm package file.

For example:

```
rpm -ivh --force openssl-0.9.8h-30.22.21.1.x86_64.rpm
rpm -ivh --force libopenssl10_9_8-0.9.8h-30.22.21.1.x86_64.rpm
```

**Why are the remote RACADM and web-based services unavailable after a property change?**

It may take a while for the remote RACADM services and the Web-based interface to become available after the iDRAC web server resets.

The iDRAC web server is reset when:

- The network configuration or network security properties are changed using the iDRAC web user interface.
- Any of the following attributes in the iDRAC.Webserver group is changed:
  - BlockHTTPPort
  - CustomCipherString
  - HostHeaderCheck
  - Http2Enable
  - HttpPort
  - HttpsPort
  - HttpsRedirection
  - SSLEncryptionBitLength
  - TLSProtocol
- The `racresetcfg` command is used.
- iDRAC is reset.
- A new SSL server certificate is uploaded.

**Why is an error message displayed if you try to delete a partition after creating it using local RACADM?**

This occurs because the create partition operation is in-progress. However, the partition is deleted after sometime and a message that the partition is deleted is displayed. If not, wait until the create partition operation is completed and then delete the partition.

## Custom sender email configuration for iDRAC alerts

**Alert generated email is not from Custom sender email set on Cloud based email service.**

You need to register your cloud email through this process : [Support.google.com](https://support.google.com).

## SNMP authentication

**Why is the message 'Remote Access: SNMP Authentication Failure' displayed?**

As part of discovery, IT Assistant attempts to verify the get and set community names of the device. In IT Assistant, you have the get community name = public and the set community name = private. By default, the SNMP agent community name for iDRAC agent is public. When IT Assistant sends out a set request, the iDRAC agent generates the SNMP authentication error because it accepts requests only from community = public.

To prevent SNMP authentication errors from being generated, you must enter community names that are accepted by the agent. Since the iDRAC only allows one community name, you must use the same get and set community name for IT Assistant discovery setup.

## Storage devices

**OpenManage Storage Management displays more storage devices than iDRAC, why?**

iDRAC displays information only for the Comprehensive Embedded Management (CEM) supported devices.

**For External JBODs/Insights behind the HBA, EEMI Message for the SAS Connector/IOM removal is generated with the EEMI message ID ENC42. However, EEMI message ENC41 and ENC1 for the SAS Connector/IOM restoration is not generated.**

To confirm the restoration of the IOM in the iDRAC web interface:

1. Go to **Storage > Overview > Enclosures**.
2. Select the enclosure.
3. Under **Advanced Properties**, ensure that the value for **Redundant Path** is set to **Present**.

**Why is there a difference in the serial number printed on the PCIe device and in the iDRAC GUI?**

Devices based out of PCIe Base Class can be of different types and form factors. In such scenarios, the serial numbers of the device form may not be the same as that of a base PCIe device. For example, derived forms of PCIe devices such as NVMe Drives, NIC cards, and so on.

## System Event Log

**While using iDRAC Web interface through Internet Explorer, why does SEL not save using the Save As option?**

This is due to a browser setting. To resolve this:

1. In Internet Explorer, go to **Tools > Internet Options > Security** and select the zone you are attempting to download in. For example, if the iDRAC device is on the local intranet, select **Local Intranet** and click **Custom level...**
2. In the **Security Settings** window, under **Downloads** make sure that the following options are enabled:
  - Automatic prompting for file downloads (if this option is available)
  - File download

 **CAUTION:** To make sure that the computer used to access iDRAC is safe, under **Miscellaneous**, do not enable the **Launching applications and unsafe files** option.

# Virtual console

**Can a new remote console video session be started when the local video on the server is turned off?**

Yes.

**Why does it take 15 seconds to turn off the local video on the server after requesting to turn off the local video?**

It gives a local user an opportunity to take any action before the video is switched off.

**Is there a time delay when turning on the local video?**

No, after a local video turn ON request is received by iDRAC, the video is turned on instantly.

**Can the local user also turn off or turn on the video?**

When the local console is disabled, the local user cannot turn off or turn on the video.

**Does switching off the local video also switch off the local keyboard and mouse?**

No.

**Does turning off the local console turn off the video on the remote console session?**

No, turning the local video on or off is independent of the remote console session.

**What privileges are required for an iDRAC user to turn on or turn off the local server video?**

Any user with iDRAC configuration privileges can turn on or turn off the local console.

**How to get the current status of the local server video?**

The status is displayed on the Virtual Console page.

To display the status of the object `iDRAC.VirtualConsole.AttachState`, use the following command:

```
racadm get iDRAC.VirtualConsole.AttachState
```

Or, use the following command from a SSH or a remote session:

```
racadm -r (iDRAC IP) -u (username) -p (password) get iDRAC.VirtualConsole.AttachState
```

The status is also seen on the Virtual Console OSCAR display. When the local console is enabled, a green status is displayed next to the server name. When disabled, a yellow dot indicates that iDRAC has locked the local console.

**Why is the bottom of the system screen not seen from the Virtual Console window?**

Make sure that the management station's monitor resolution is set to 1280 x 1024.

**Why is the Virtual Console Viewer window garbled on Linux operating system?**

The console viewer on Linux requires a UTF-8 character set. Check your locale and reset the character set if required.

**Why does the mouse not synchronize under the Linux text console in Lifecycle Controller?**

Virtual Console requires the USB mouse driver, but the USB mouse driver is available only under the X-Window operating system. In the Virtual Console viewer, do any of the following:

- Go to **Tools > Session Options > Mouse** tab. Under **Mouse Acceleration**, select **Linux**.
- Under the **Tools** menu, select **Single Cursor** option.

**How to synchronize the mouse pointers on the Virtual Console Viewer window?**

Before starting a Virtual Console session, make sure that the correct mouse is selected for your operating system.

Make sure that the **Single Cursor** option under **Tools** in the iDRAC Virtual Console menu is selected on iDRAC Virtual Console client. The default is two cursor mode.

**Can a keyboard or mouse be used while installing a Microsoft operating system remotely through the Virtual Console?**

No. When you remotely install a supported Microsoft operating system on a system with Virtual Console enabled in the BIOS, an EMS Connection Message is sent that requires that you select **OK** remotely. You must either select **OK** on the local system or restart the remotely managed server, reinstall, and then turn off the Virtual Console in BIOS.

This message is generated by Microsoft to alert the user that Virtual Console is enabled. To make sure that this message does not appear, always turn off Virtual Console in the iDRAC Settings utility before remotely installing an operating system.

**Why does the Num Lock indicator on the management station not reflect the status of the Num Lock on the remote server?**

When accessed through the iDRAC, the Num Lock indicator on the management station does not necessarily coincide with the state of the Num Lock on the remote server. The state of the Num Lock depends the setting on the remote server when the remote session is connected, regardless of the state of the Num Lock on the management station.

**Why do multiple Session Viewer windows appear when a Virtual Console session is established from the local host?**

You are configuring a Virtual Console session from the local system. This is not supported.

**If a Virtual Console session is in-progress and a local user accesses the managed server, does the first user receive a warning message?**

No. If a local user accesses the system, both have control of the system.

**How much bandwidth is required to run a Virtual Console session?**

It is recommended to have a 5 MBPS connection for good performance. A 1 MBPS connection is required for minimal performance.

**What is the minimum system requirements for the management station to run Virtual Console?**

The management station requires an Intel Pentium III 500 MHz processor with at least 256 MB of RAM.

**Why does Virtual Console Viewer window sometimes displays No Signal message?**

You may see this message because the iDRAC Virtual Console plug-in is not receiving the remote server desktop video. Generally, this behavior may occur when the remote server is turned off. Occasionally, the message may be displayed due to a remote server desktop video reception malfunction.

**Why does Virtual Console Viewer window sometimes display an Out of Range message?**

You may see this message because a parameter necessary to capture video is beyond the range for which the iDRAC can capture the video. Parameters such as display resolution or refresh rate too high causes an out of range condition. Normally, physical limitations such as video memory size or bandwidth sets the maximum range of parameters.

**Why is the Virtual Console Viewer window blank?**

If you have Virtual Media privilege, but not Virtual Console privilege, you can start the viewer to access the virtual media feature, but the managed server's console is not displayed.

**Why doesn't the mouse synchronize in DOS when using Virtual Console?**

The Dell BIOS is emulating the mouse driver as a PS/2 mouse. By design, the PS/2 mouse uses relative position for the mouse pointer, which causes the lag in syncing. iDRAC has a USB mouse driver that allows absolute position and closer tracking of the mouse pointer. Even if iDRAC passes the USB absolute mouse position to the Dell BIOS, the BIOS emulation converts it back to relative position and the behavior remains. To fix this problem, set the mouse mode to USC/Diags in the Configuration screen.

**After launching the Virtual Console, the mouse cursor is active on the Virtual Console, but not on the local system. Why does this occur and how to resolve this?**

This occurs if the **Mouse Mode** is set to **USC/Diags**. Press **Alt + M** hot key to use the mouse on the local system. Press **Alt + M** again to use the mouse on the Virtual Console.

**Why does GUI session time out after launching a virtual console from the iDRAC interface that is launched from CMC?**

When launching the Virtual Console to iDRAC from the CMC web interface a popup is opened to launch the Virtual Console. The popup closes shortly after the Virtual Console opens.

When launching both the GUI and Virtual Console to the same iDRAC system on a management station, a session time-out for the iDRAC GUI occurs if the GUI is launched before the popup closes. If the iDRAC GUI is launched from the CMC web interface after the popup with the Virtual Console closed, this issue does not appear.

**Why does Linux SysRq key not work with Internet Explorer?**

The Linux SysRq key behavior is different when using Virtual Console from Internet Explorer. To send the SysRq key, press the **Print Screen** key and release while holding the **Ctrl** and **Alt** keys. To send the SysRq key to a remote Linux server through iDRAC, while using Internet Explorer:

1. Activate the magic key function on the remote Linux server. You can use the following command to activate it on the Linux terminal:

```
echo 1 > /proc/sys/kernel/sysrq
```

2. Activate the keyboard pass-through mode of Active X Viewer.

3. Press **Ctrl+Alt+Print Screen**.
4. Release only **Print Screen**.
5. Press **Print Screen+Ctrl+Alt**.

 **NOTE:** The SysRq feature is currently not supported with Internet Explorer and Java.

### Why is the "Link Interrupted" message displayed at the bottom of the Virtual Console?

When using the shared network port during a server reboot, iDRAC is disconnected while BIOS is resetting the network card. This duration is longer on 10 Gb cards, and is also exceptionally long if the connected network switch has Spanning Tree Protocol (STP) enabled. In this case, it is recommended to enable "portfast" for the switch port connected to the server. In most cases, the Virtual Console restores itself.

## Front Ports

### Why is the Front USB Port setting not updated immediately after changing it to Enabled or Disabled?

The **Front USB Port** field in the iDRAC UI is updated only after the BIOS POST stage is complete.

## Virtual media

### Why does the Virtual Media client connection sometimes drop?

When a network time-out occurs, iDRAC firmware drops the connection, disconnecting the link between the server and the virtual drive.

If you change the CD in the client system, the new CD may have an autostart feature. In this case, the firmware can time out and the connection is lost if the client system takes too long to read the CD. If a connection is lost, reconnect from the GUI and continue the previous operation.

If the Virtual Media configuration settings are changed in the iDRAC web interface or through local RACADM commands, any connected media is disconnected when the configuration change is applied.

To reconnect to the Virtual Drive, use the Virtual Media **Client View** window.

### Why does a Windows operating system installation through Virtual Media take an extended amount of time?

If you are installing the Windows operating system using the **Dell Systems Management Tools and Documentation DVD** and the network connection is slow, the installation procedure may require an extended amount of time to access iDRAC web interface due to network latency. The installation window does not indicate the installation progress.

### How to configure the virtual device as a bootable device?

On the managed system, access BIOS Setup and go to the boot menu. Locate the virtual CD or virtual floppy, and change the device boot order as required. Also, press the "spacebar" key in the boot sequence in the CMOS setup to make the virtual device bootable. For example, to boot from a CD drive, configure the CD drive as the first device in the boot order.

### What are the types of media that can be set as a bootable device?

iDRAC allows you to boot from the following bootable media:

- CDROM/DVD Data media
- ISO 9660 image
- 1.44 Floppy disk or floppy image
- A USB key that is recognized by the operating system as a removable disk
- A USB key image

### How to make the USB key a bootable device?

You can also boot with a Windows 98 startup disk and copy system files from the startup disk to the USB key. For example, from the DOS prompt, type the following command:

```
sys a: x: /s
```

where, x: is the USB key that is required to be set as a bootable device.

**The Virtual Media is attached and connected to the remote floppy. But, cannot locate the Virtual Floppy/Virtual CD device on a system running Red Hat Enterprise Linux or the SUSE Linux operating system. How to resolve this?**



Some Linux versions do not auto-mount the virtual floppy drive and the virtual CD drive in the same method. To mount the virtual floppy drive, locate the device node that Linux assigns to the virtual floppy drive. To mount the virtual floppy drive:

1. Open a Linux command prompt and run the following command:

```
grep "Virtual Floppy" /var/log/messages
```

2. Locate the last entry to that message and note the time.
3. At the Linux prompt, run the following command:

```
grep "hh:mm:ss" /var/log/messages
```

where, hh:mm:ss is the time stamp of the message returned by grep in step 1.

4. In step 3, read the result of the grep command and locate the device name that is given to the Virtual Floppy.
5. Make sure that you are attached and connected to the virtual floppy drive.
6. At the Linux prompt, run the following command:

```
mount /dev/sdx /mnt/floppy
```

where, /dev/sdx is the device name found in step 4 and /mnt/floppy is the mount point.

To mount the virtual CD drive, locate the device node that Linux assigns to the virtual CD drive. To mount the virtual CD drive:

1. Open a Linux command prompt and run the following command:

```
grep "Virtual CD" /var/log/messages
```

2. Locate the last entry to that message and note the time.
3. At the Linux prompt, run the following command:

```
grep "hh:mm:ss" /var/log/messages
```

where, hh:mm:ss is the timestamp of the message returned by grep in step 1.

4. In step 3, read the result of the grep command and locate the device name that is given to the **Dell Virtual CD**.
5. Make sure that the Virtual CD Drive is attached and connected.
6. At the Linux prompt, run the following command:

```
mount /dev/sdx /mnt/CD
```

where: /dev/sdx is the device name found in step 4 and /mnt/floppy is the mount point.

### **Why are the virtual drives attached to the server removed after performing a remote firmware update using the iDRAC web interface?**

Firmware updates cause the iDRAC to reset, drop the remote connection, and unmount the virtual drives. The drives reappear when iDRAC reset is complete.

### **Why are all the USB devices detached after connecting a USB device?**

Virtual media devices are connected as a composite USB device to the Host USB BUS, and they share a common USB port. Whenever any virtual media is connected to or disconnected from the host USB bus, all the Virtual Media devices are disconnected momentarily from the host USB bus, and then they are reconnected. If the host operating system uses a virtual media device, do not attach or detach one or more virtual media devices. It is recommended that you connect all the required USB devices first before using them.

### **What does the USB Reset do?**

It resets the remote and local USB devices connected to the server.

### **How to maximize Virtual Media performance?**

To maximize Virtual Media performance, launch the Virtual Media with the Virtual Console disabled or do one of the following:

- Change the performance slider to Maximum Speed.
- Disable encryption for both Virtual Media and Virtual Console.

 **NOTE:** In this case, the data transfer between managed server and iDRAC for Virtual Media and Virtual Console will not be secured.

- If you are using any Windows server operating systems, stop the Windows service named Windows Event Collector. To do this, go to **Start > Administrative Tools > Services**. Right-click **Windows Event Collector** and click **Stop**.

**While viewing the contents of a floppy drive or USB key, a connection failure message is displayed if the same drive is attached through the virtual media?**

Simultaneous access to virtual floppy drives is not allowed. Close the application used to view the drive contents before attempting to virtualize the drive.

**What file system types are supported on the Virtual Floppy Drive?**

The virtual floppy drive supports FAT16 or FAT32 file systems.

**Why is an error message displayed when trying to connect a DVD/USB through virtual media even though the virtual media is currently not in use?**

The error message is displayed if Remote File Share (RFS) feature is also in use. At a time, you can use RFS or Virtual Media and not both.

## Miscellaneous

For High-Bandwidth Memory (HBM) CPUs in HBM mode, MemoryRollupStatus is shown as Unknown.

For HBM-only mode, memory-related data reported in HW inventory, sensors, telemetry, and so on, are not available. You should not consider it as a faulty configuration. For those interfaces that report all individual DIMM slot sensors, they are reported with Unknown state. Similarly, Max DIMM Temperature sensor can also continue to be reported with Unknown state.

When attempting to connect the iDRAC to a different network, the iDRAC does not get a different IP address from the new subnet.

Ensure that the network cable is disconnected from the iDRAC for at least 5 s.

After an iDRAC reset, the iDRAC UI may not display all the values.

 **NOTE:** If you reset the iDRAC for some reason, ensure that you wait for at least two minutes after resetting iDRAC to access or modify any settings in iDRAC.

When an operating system is installed, the hostname may or may not appear/change automatically.

There are two scenarios:

- Scenario 1: iDRAC is not showing the latest hostname once you install an operating system. You need to install OMSA or iSM along with the iDRAC to get the hostname reflected.
- Scenario 2: iDRAC had a hostname for a specific operating system and another different operating system has been installed and still the hostname is appearing as the old hostname without overwriting the hostname. The reason behind, hostname is an information which is coming from the operating system, iDRAC only saves the information. If there is a new operating system has been installed, iDRAC does not reset the value of the hostname. However, newer versions of the OSs are capable to update the hostname in iDRAC during the first OS startup.

iDRAC network connection is not working.

For rack and tower servers:

- In shared mode, ensure that the LAN cable is connected to the NIC port where the wrench symbol is present.
- In Dedicated mode, ensure that the LAN cable is connected to the iDRAC LAN port.

- Ensure that NIC settings, IPv4, and IPv6 settings and either Static or DHCP is enabled for your network.

## iDRAC not accessible in shared LOM

iDRAC may be inaccessible if there are fatal errors in the host operating system such as BSOD error in Windows. To access iDRAC, reboot the host to recover the connection.

## Shared LOM not functional after enabling Link Aggregation Control Protocol (LACP).

The host operating system driver for the network adapter must be loaded before LACP is enabled. However, if a passive LACP configuration is in use, the shared LOM may be functional before the host operating system driver is loaded. See the switch documentation for LACP configuration.

 **NOTE:** Shared LOM IP of iDRAC is not accessible in pre-boot state when the switch is configured with LACP.

## How to retrieve an iDRAC administrative username and password?

You must restore iDRAC to its default settings. For more information, see [Resetting iDRAC to factory default settings](#).

## When attempting to boot the managed server, the power indicator is green, but there is no POST or no video.

This happens due to any of the following conditions:

- Memory is not installed or is inaccessible.
- The CPU is not installed or is inaccessible.
- Video riser card is missing or not connected properly.

Also, see error messages in iDRAC log using iDRAC web interface.

## Unable to log in to iDRAC web interface using Firefox browser on Linux or Ubuntu. Unable to enter the password.

To resolve this issue, reinstall or upgrade the Firefox browser.

## Unable to access iDRAC through USB NIC in SLES and Ubuntu

 **NOTE:** In SLES, set the iDRAC interface to DHCP.

In Ubuntu, use the Netplan utility to configure the iDRAC interface into DHCP mode. To configure the DHCP:

1. Use `/etc/netplan/01-netcfg.yaml`.
2. Specify Yes for iDRAC DHCP.
3. Apply the configuration.

```

# This file describes the network interfaces available on your system
# For more information, see netplan(5).
network:
  version: 2
  renderer: networkd
  ethernets:
    eno1:
      dhcp4: yes
      idrac:
        dhcp4: yes

```

"/etc/netplan/01-netcfg.yaml" 10L, 221C

**Figure 3. Configuring iDRAC interface to DHCP mode in Ubuntu**

Model, Manufacturer, and other properties are not listed for Embedded Network Adapters in Redfish

FRU details for embedded devices will not be displayed. There will not be any FRU object for devices which are embedded on Motherboard. Hence, dependent property will not be there.

## Proxy server settings

### How to configure proxy server settings in RACADM CLI and Redfish API?

In RACADM, the following LC attributes must be set to configure the proxy server:

- LifecycleController.LCAttributes.UserProxyPassword
- LifecycleController.LCAttributes.UserProxyPort
- LifecycleController.LCAttributes.UserProxyServer
- LifecycleController.LCAttributes.UserProxyType

- LifecycleController.LCAttributes.UserProxyUserName

For more information about how to run these commands, see the **Integrated Dell Remote Access Controller CLI guide**.

When using HTTP with a proxy, the connection between iDRAC and the proxy is not as secure as the connection between iDRAC and the HTTPS server. The `UserProxyServer` is an important attribute. If it is not set, the other attributes cannot be used. The benefit of using RACADM and Redfish API is that it is not required to provide the password each time you use the proxy server.

In the Redfish API, perform a patch operation using the URI `/redfish/v1/Managers/<Manager-ID>/Oem/Dell/DellAttributes/<Dell-attributes-ID>` to configure the proxy server.

### How to configure proxy server settings in the iDRAC UI?

In the iDRAC UI, you can update the proxy settings on all the pages where the proxy server is required. Even if you have set up the proxy settings using RACADM and Redfish API, you can still update the proxy settings in the iDRAC UI. To configure the proxy settings in the **System Update** page:

1. Go to **Maintenance > System Update > Manual Update**.
2. Under **Manual Update**, select **HTTPS** in the **Location Type**.
3. Select **Enabled** in the **Enable Proxy Server**.
4. Enter **Server**, **Port**, **User Name**, and **Password**.
5. Select **Type** and click **Save Proxy Settings as Default**.

 **NOTE:** You can configure proxy settings in any of the pages such as **Export Lifecycle Log**, **Automatic Update**, **SupportAssist Collection Settings**, **Server Configuration Profile Import**, and **Server Configuration Profile Export**.

 **NOTE:** By default, **Enable Proxy Settings** in the iDRAC UI is disabled. It will reset to a disabled state after each use. For more information, see the **Integrated Dell Remote Access Controller iDRAC Online Help**.

## Permanently setting the default password to calvin

If your system shipped with a unique default iDRAC password but you want to set **calvin** as the default password, you must use the jumpers available on the system board.

 **CAUTION:** Changing the jumper settings permanently changes the default password to calvin. You cannot revert to the unique password even if you reset iDRAC to factory settings.

For information about the jumper location and the procedure, see the documentation for your server at [Dell Support](#) page.

## Use case scenarios

This section helps you in navigating to specific sections in the guide to perform typical use case scenarios.

### Topics:

- [Troubleshooting an inaccessible managed system](#)
- [Obtaining system information and assess system health](#)
- [Setting up alerts and configuring email alerts](#)
- [Viewing and exporting System Event Log and Lifecycle Log](#)
- [Interfaces to update iDRAC firmware](#)
- [Performing a graceful shutdown](#)
- [Creating new administrator user account](#)
- [Launching servers remote console and mounting a USB drive](#)
- [Installing bare metal OS using attached virtual media and remote file share](#)
- [Managing rack density](#)
- [Installing new electronic license](#)
- [Applying IO Identity configuration settings for multiple network cards in single host system reboot](#)

## Troubleshooting an inaccessible managed system

After you receive alerts from OpenManage Essentials, Dell Management Console, or a local trap collector, five servers in the data center became inaccessible due to issues in the server or unresponsive operating system. Identify the cause to troubleshoot and bring up the server using iDRAC.

Before troubleshooting the inaccessible system, ensure that the following prerequisites are met:

- Enable last crash screen
- Alerts are enabled on iDRAC

To identify the cause, check the following in the iDRAC web interface and reestablish the connection to the system:

- Server's LED status—Blinking amber or Solid amber.
- Operating system image is seen in the Virtual Console. If you can see the image, reset the system (warm boot) and log in again. If you can log in, the issue is fixed.
- Last crash screen.
- Boot capture video.
- Crash capture video.
- Server Health status—Red **x** icons for the system components with issues.
- Storage array status—Possible array offline or failed
- Lifecycle logs for critical events that are related to system hardware and firmware and the log entries that were logged at the time of the system crash.
- Generate a Tech Support report and view the collected data.
- Use the monitoring features provided by iDRAC Service Module

## Obtaining system information and assess system health

To obtain system information and assess system health:

- In iDRAC Web interface, go to **Overview > Summary** to view the system information and access various links on this page to assess system health.
- If iDRAC Service Module is installed, the operating system host information is displayed.

# Setting up alerts and configuring email alerts

## About this task

To set up alerts and configure email alerts:

## Steps

1. Enable alerts.
2. Configure the email alert and check the ports.
3. Perform a reboot, power off, or power cycle the managed system.
4. Send test alert.

# Viewing and exporting System Event Log and Lifecycle Log

## About this task

To view and export lifecycle log and system event log (SEL):

## Steps

1. In iDRAC Web interface, go to **Maintenance > System Event Logs** to view SEL and **Lifecycle Log** to view lifecycle log.  
 **NOTE:** The SEL is also recorded in the lifecycle log. Using the filtering options to view the SEL.
2. Export the SEL or lifecycle log in the XML format to an external location (management station, USB, network share, and so on). Alternatively, you can enable remote system logging, so that all the logs written to the lifecycle log are also simultaneously written to the configured remote server(s).
3. If you are using the iDRAC Service Module, export the Lifecycle log to OS log.

# Interfaces to update iDRAC firmware

Use the following interfaces to update the iDRAC firmware:

- iDRAC Web interface
- Redfish API
- RACADM CLI
- Dell Update Package (DUP)
- Dell Remote Access Configuration Tool (DRACT)

# Performing a graceful shutdown

The software initiates a graceful shutdown by turning off the server, allowing the operating system to safely shutdown processes. It also powers off the PCIe slots. As a result, the adapters in PCIe slots do not respond to NC-SI control commands.

If the commands do not respond, the NIC-CEM module treats the adapters as nonresponsive and it logs HWC8607 Lifecycle Controller logs (LCLOGs) to indicate that communication with the adapters is lost.

To perform graceful shutdown, in the iDRAC Web interface, go to one of the following locations:

- On the **Dashboard**, select **Graceful Shutdown** and click **Apply**.

 **NOTE:** After the request is sent to the host, it is up to the host to honor that request and run it. Success of Graceful shutdown depends on the state of the host.

For more information, see the **iDRAC Online Help**.

# Creating new administrator user account

You can modify the default local administrator user account or create a new administrator user account. To modify the local administrator user account, see [Modifying local administrator account settings](#).

To create a new administrator account, see the following sections:

- [Configuring local users](#)
- [Configuring active directory users](#)
- [Configuring generic LDAP users](#)

# Launching servers remote console and mounting a USB drive

## About this task

To launch the remote console and mount a USB drive:

## Steps

1. Connect a USB flash drive (with the required image) to the management station.
2. Use the following method to launch virtual console through the iDRAC Web Interface:
  - Go to **Dashboard > Virtual Console** and click **Launch Virtual Console**.  
The **Virtual Console Viewer** is displayed.
3. From the **File** menu, click **Virtual Media > Launch Virtual Media**.
4. Click **Add Image** and select the image that is located on the USB flash drive.  
The image is added to the list of available drives.
5. Select the drive to map it. The image on the USB flash drive is mapped to the managed system.

# Installing bare metal OS using attached virtual media and remote file share

See the [Deploying operating system using remote file share](#) section.

# Managing rack density

## About this task

Before you install additional servers in a rack, you must determine the remaining capacity in the rack.

To assess the capacity of a rack to add additional servers:

## Steps

1. View the current power consumption data and historical power consumption data for the servers.
2. Based on the data, power infrastructure and cooling system limitations, enable the power cap policy and set the power cap values.  
 **NOTE:** It is recommended that you set a cap close to the peak, and then use that capped level to determine how much capacity is remaining in the rack for adding more servers.



# Installing new electronic license

See [License operations](#) for more information.

## Applying IO Identity configuration settings for multiple network cards in single host system reboot

### About this task

If you have multiple network cards in a server that is part of a Storage Area Network (SAN) environment and you want to apply different virtual addresses, initiator and target configuration settings to those cards, use the I/O Identity Optimization feature to reduce the time in configuring the settings. To do this:

### Steps

1. Make sure that BIOS, iDRAC, and the network cards are updated to the latest firmware version.
2. Enable IO Identity Optimization.
3. Export the Server Configuration Profile (SCP) file from iDRAC.
4. Edit the I/O Identity optimization settings in the SCP file.
5. Import the SCP file to iDRAC.

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